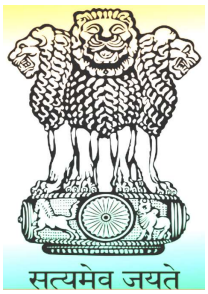
DRAFT

INDIAN RAILWAYS SIGNAL ENGINEERING MANUAL

VOLUME 1

JULY 2020



भारत सरकार

GOVERNMENT OF INDIA

रेल मंत्रालय

MINISTRY OF RAILWAYS

भारतीय रेल

सिगनल इंजीनियरी नियमावली

भाग - 1

INDIAN RAILWAYS
SIGNAL ENGINEERING MANUAL

VOLUME 1

VERSION 0.2 (Draft)

जुलई 2020

JULY 2020

CHALAMALASETTY MOHAN

Digitally signed by CHALAMALASETTY MOHAN
DN: c=IN, st=Telangana, 2.5.4.20=22f2b7f6cfecaa15b6172e7fd882d3a9977340ca67094e845befc318cedffd46, postalCode=500017, street=IRISET TARNAKA ROAD, ou=S AND T, o=I R I S E T, cn=CHALAMALASETTY MOHAN
Date: 2020.07.11 20:20:01 +05'30'

Digitally Signed

Director / IRISET

Signal Engineering Manual Issue History

S No	Signal Engineering Manual	Year
1	First Printed	1955
2	Second Printed – Part I	1988
3	Third Printed - Part II	2001
4	Fourth Printed Volume - 1 Volume - 2	2020 2020

FOREWARD

Intentionally left blank now

PREFACE

Intentionally left blank now

CONTENTS

Chapter No.	Section / Annexure	Chapter Name	Page No.
1		**Organisation of Signal & Telecommunication Department**	
	Section 1	Signal and Telecommunication Department- An Overview	1
	Section 2	Strength of Officers	3
	Section 3	Technical Staff	3
	Section 4	Strength, Charges, Designations	5
	Section 5	Drawing & Design Staff	6
	Section 6	Misc Equivalent Designations/Organisations	6
2		**Duties of Signal & Telecommunication Engineers**	
	Section 1	Duties of Signal and Telecommunication Engineers in charge of Maintenance or Construction	8
	Section 2	Additional Duties of Signal and Telecommunication Engineer in charge of Maintenance	9
	Section 3	Additional Duties of Dy CSTE in charge of Construction/RE/Project/Works	11
		Annexures	
	Annexure 2/1	*Engineer's Progress Report*	13
3		**Duties of Signal Supervisors & Technical Staff**	
	Section 1	Common Duties of SSE Signal In-charge, Sectional SSE/JE Signal	14
	Section 2	Duties Of SSE Signal In-charge	17
	Section 3	Inspection certificates	18
	Section 4	Accidents	19
	Section 5	General responsibilities and duties	20
	Section 6	Additional Instructions for SSE Signal In-charge of Construction	23
	Section 7	Datalogger management centre	25
	Section 8	Duties of signal technicians	25
	Section 9	Duties of helpers / assistants	29
		Annexures	
	Annexure 3/1	*Failure Report of Signalling & Interlocking, Electrical Signalling and Block Signalling Appliances*	31
	Annexure 3/2	*Classification Codes*	33
	Annexure 3/3	*Foot Plate Inspection of Signals*	34
	Annexure 3/4	*Inspection Report*	36

Chapter No.	Section / Annexure	Chapter Name	Page No.
	Annexure 3/5	*Site of Accident*	37
	Annexure 3/6	*Hours of Employment Regulations*	39
	Annexure 3/7	*Signal Sighting Committee Report*	44
	Annexure 3/8	*JE/SSE'S(Signal) Progress Report*	45
	Annexure 3/9	*JE/SSE's(Signal) Completion Certificate*	46
	Annexure 3/10	*Technician's (Signal) Report*	47
	Annexure 3/11	*Disconnection/Reconnection Notice*	48
	Annexure 3/12	*Situations in which disconnection Notice need not be issued provided suitable precautions are taken*	49
	Annexure 3/13	*Competency Certificate cum Training History Book.*	51
4		**Budget, Estimates, Expenditure and Budgetary Control and Completion Reports**	
	Section 1	Introduction	53
	Section 2	Estimates For Working Expenses or Revenue Budget	54
	Section 3	Works, Machinery and Rolling Stock Budget	54
	Section 4	Administering The Budget	55
	Section 5	Maintenance of Accounts	55
	Section 6	Control Over Expenditure	56
	Section 7	Completion Reports	56
		Annexures	
	Annexure 4/1	*Powers of Railway Board*	58
	Annexure 4/2	*Review of Expenditure*	60
	Annexure 4/3	*Revenue Expenditure*	62
	Annexure 4/4	*Revised and Budget Estimates*	64
	Annexure 4/5	*Works Expenditure*	66
	Annexure 4/6	*Capital and Works Expenditure*	67
	Annexure 4/7	*Distribution of Funds*	68
	Annexure 4/8	*Revenue Allocation Registers*	70
	Annexure 4/9	*Expenditure Register of Works*	71
	Annexure 4/10	*Revenue and Works Expenditure*	73
	Annexure 4/11	*Completion Report*	77
5		**Planning, Estimating & Survey**	
	Section 1	Planning	80
	Section 2	Estimates	80
	Section 3	Surveys	82
		Annexures	
	Annexure 5/1	*Preparation of the Preliminary Works Programme*	86
	Annexure 5/2	*Test of remuneratiteness*	89

Chapter No.	Section / Annexure	Chapter Name	Page No.
	Annexure 5/3	*Code Rules for Estimates*	90
	Annexure 5/4	*Grouping of works*	99
6		**Signalling Stores and Accountal**	
	Section 1	Imprest Stores	100
	Section 2	Tools and Plants	100
	Section 3	Protective Clothing	101
	Section 4	Books of Reference	101
	Section 5	Office Furniture	101
	Section 6	Stationery and Forms	101
	Section 7	Stores obtained for works including Special Revenue Works	102
	Section 8	The Daily Material Transactions	102
	Section 9	Returned Stores	103
	Section 10	Requisitions	104
	Section 11	General Instruction	105
		Annexures	
	Annexure 6/1	*Imprest Stores*	107
	Annexure 6/2	*Tools and Plants - Extracts from Engineering Code*	112
	Annexure 6/3	*Guidelines for preparation of Indents*	113
	Annexure 6/4	*MATERIALS AT-SITE ACCOUNT- Extracts from the Engineering Code*	114
	Annexure 6/5	*The Daily Material Transaction Register*	119
	Annexure 6/6	*Returned Stores - Extracts from Stores Code*	120
	Annexure 6/7	*Accounts (EDP) 1*	122
	Annexure 6/8	*General instructions for proper custody and Maintenance of Stores*	126
7		**Essentials of Signalling**	
	Section 1	Signals in generally	127
	Section 2	Warning Boards	132
	Section 3	Indication Boards	132
	Section 4	Points, Point Indicators, Isolation, SPI & Spring Loaded Points	133
	Section 5	Signal Cabins and Interlocking	137
	Section 6	Interlocking of catch siding and slip sidings	139
	Section 7	Operation of points and signals by electrical means	140
	Section 8	Relay Interlocking Installations—Route Setting Type	142
	Section 9	Requirements of signalling circuits using Electronic Equipment & other References	143
		Annexures	
	Annexure 7/1	Extract of S.O.D	146

Chapter No.	Section / Annexure	Chapter Name	Page No.
	Annexure 7/2	*Extract of Rules for opening of railway for Isolation*	147
8		**Drawings, Specifications and Books of reference**	
	Section 1	Plans for New and working installations	148
	Section 2	Indian Railway Standard Drawings	157
	Section 3	Standard Specifications	158
	Section 4	Books of References	158
	Section 5	Maintenance of Drawings, Specifications and Book of Reference	159
	Section 6	Indian Railway Standard Equipments	161
		Annexures	
	Annexure 8/1	*List of Indian Railway Standard Signal Specification*	162
	Annexure 8/2	*List of Important Indian Standard Specification user for Signalling Purpose*	163
	Annexure 8/3	*Statement showing the various Books of Reference and their distribution among the staff.*	165
	Annexure 8/4	*Typical Signalling & Interlocking Plan of Double line*	168
	Annexure 8/5	*Typical Control Table of Double Line*	169
	Annexure 8/6	*Typical Signalling & Interlocking Plan of single line*	173
	Annexure 8/7	*Typical Control Table of Single Line*	174
	Annexure 8/8	*VDU layout for E.I*	176
9		**Procedure for Commissioning of Signalling Installations**	
	Section 1	General Instructions	177
	Section 2	Obtaining Sanction for Works on lines Opened to Traffic	179
	Section 3	New Lines & Electrification	183
	Section 4	Introduction of New Types of Locomotives or Rolling Stock for Increasing Speed	184
		Annexures	
	Annexure 9/1	*Application for Sanction*	186
	Annexure 9/2	*Safety Certificate*	188
	Annexure 9/3	*Station Machinery*	190
	Annexure 9/4	*Station Equipment*	191
	Annexure 9/5	*Safety Certificate for S & T works for Introducing Electrification*	192
	Annexure 9/6	*Joint Safety Certificate*	193
	Annexure 9/7	*Handing over / Taking over of Signaling Assets*	194
	Annexure 9/8	*Application for PCSTE Sanction*	197
	Annexure 9/9	*PCSTE Sanction*	198
	Annexure 9/10	*Check List for CRS/PCSTE application for sanction*	199

Chapter No.	Section / Annexure	Chapter Name	Page No.
10		**Contracts for Works**	
	Section 1	General	200
	Section 2	Tenders/E-Tenders	202
	Section 3	Measurements and Measurement Books	205
		Annexures	
	Annexure 10/1	*Instructions for Securing Measurements*	211
11		**Reliability, Availability, Maintainability, Safety (RAMS)**	
	Section 1	Introduction	213
	Section 2	Terms and Definitions	213
	Section 3	Examples of RAMS Parameters for Railways	216
	Section 4	Safety Integrity Level (SIL)	217
12		**Mechanical Signalling – Installation, Testing & Maintenance**	
	Section 1	Installation	220
	Section 2	Point rodding and fittings	222
	Section 3	Point operation in mechanical signalling	223
	Section 4	Facing point locks, switch & lock movements and spring point levers	224
	Section 5	Location of signals in mechanical signalling	224
	Section 6	Signal wire run	229
	Section 7	Painting of mechanical signalling	229
	Section 8	Maintenance of mechanical signalling	232
	Section 9	Maintenance of lever frame	233
	Section 10	Maintenance of lead outs (rodding)	234
	Section 11	Maintenance of point rodding and fittings	234
	Section 12	Maintenance of cranks, compensators, screws, bolts, nuts, pins and cotters	234
	Section 13	Maintenance of points	235
	Section 14	Maintenance of facing point locks, switch & lock movements and spring point levers	236
	Section 15	Maintenance of lock bars	237
	Section 16	Maintenance of mechanical detectors	238
	Section 17	Maintenance of signal wire run (single wire)	238
	Section 18	Maintenance of pulley stakes and pulleys	239
	Section 19	Maintenance of electrical repeaters and signal lamps/leds	239
	Section 20	Maintenance of double wire installations	239
	Section 21	Maintenance of signals and fittings	241
	Section 22	Maintenance of point, trap and shunting permitted	243

Chapter No.	Section / Annexure	Chapter Name	Page No.
		indicators	
	Section 23	Maintenance schedule	243
	Section 24	Overhauling and testing of interlocking mechanical / electromechanical)	243
	Section 25	Overhauling of interlocking frames, interlocking key boxes and station master's slide control frames.	244
	Section 26	Testing of interlocking frames, interlocking key boxes and station master's slide control frames.	248
	Section 27	Records of testing and overhauling	253
	Section 28	Testing and overhauling of electromechanical installations	254
		Drawings Mechanical Signalling Gears – Semaphore Signal Arms , Point Indicators, Boards & Markers. (Annexure 12/2)	
		Annexures	
	Annexure 12/1	*Essential Requirements before interlocking of a point*	257
	Annexure 12/2	*Colouring scheme*	258
	Annexure 12/3	*Maintenance schedule for mechanical signalling*	264
	Annexure 12/4	*Certificate of Overhauling of Interlocking Frame*	272
	Annexure 12/5	*Certificate of Testing of Locking*	273

CHAPTER 1 : ORGANISATION OF THE SIGNAL & TELECOMMUNICATION DEPARTMENT

Section 1 : Signal and Telecommunication Department - An Overview

1.1.1.0 The efficient upkeep of Signal and Telecommunication assets of the Railways is the responsibility of the Open Line Organization of the Signal and Telecommunication Department. Major construction activities of the Railways may be under the administrative control of a Chief Signal and Telecommunication Engineer (Construction) reporting directly to the General Manager of an Open Line or under administrative control of a General Manager (Construction)/ Chief Administrative Officer (Construction).

1.1.2.0 **Structure of Open Line Organization**

1.1.2.1 The Principal Chief Signal and Telecommunication Engineer is the Administrative and Principal Head of the Signal and Telecommunication Department and is responsible to the General Manager for the efficient and economical working of the Department.

1.1.3.0 The Principal Chief Signal and Telecommunication Engineer may be assisted in his work by one or more heads of the department (HODs), wherever provided HODs shall be, the Chief Signal Engineer and Chief Communication Engineer, Chief Signal and Telecommunication Engineer/Planning, Chief Signal and Telecommunication Engineer/Design& drawing, Chief Signal and Telecommunication Engineer/projects or works and CWM/S&T/Workshop etc., in independent charge of their respective areas or control. Their work is, however, co-ordinated by the Principal Chief Signal and Telecommunication Engineer who is responsible to the General Manager of the Railway.

1.1.4.0 Heads of the Department (HODs) in turn, is assisted at the Headquarters by a team of Officers as required, who may consist of Deputy Chief Signal and Telecommunication Engineers, Senior/Executive Signal and Telecommunication Engineers and Assistant/Assistant Executive Signal and Telecommunication Engineers. HODs shall provide necessary direction and control for the efficient and economical execution of all works and duties under their charge.

1.1.4.1 The Principal Chief Signal and Telecommunication Engineer will lay down the duties of each of the Officers of the Headquarters and Divisions.

1.1.5.0 The expression "Chief Signal and Telecommunication Engineer" wherever used in this Manual includes Chief Signal Engineer, Chief Communication Engineer and Chief Signal and Telecommunication Engineer (Construction), Chief Signal and Telecommunication Engineer/Planning, Chief Signal and Telecommunication Engineer/Works, Chief Signal and Telecommunication Engineer/D&D, Chief Signal and Telecommunication Engineer/ Projects unless the contrary is clear from the context.

1.1.6.0 The Principal Chief Signal and Telecommunication Engineer shall be over all in charge of staff cadre — Gazetted and non-Gazetted.

Construction organisation/Project Organisation/RE Organisation

1.1.7.0 The Chief Signal and Telecommunication Engineer (Construction/Project) shall provide necessary direction and control for the efficient and economical execution of all works under his charge.

1.1.8.0 The Chief Signal and Telecommunication Engineer (Construction/Project) is assisted at the Headquarters by a team of Officers as required who may consist of Deputy Chief Signal and Telecommunication Engineers, Senior Signal and Telecommunication Engineers and Executive Signal and Telecommunication, Assistant Executive Signal and Telecommunication, Assistant Signal and Telecommunication Engineers etc.,.

1.1.9.0 The Chief Signal and Telecommunication Engineer (Construction) shall maintain liaison with the Open Line Organisation and shall follow the general policies and procedure laid down for the execution of works. In cases where a different policy or procedure becomes necessary to be followed, he shall do so after due consultation with the Open Line Organisation to adopt an accepted policy or procedure.

1.1.10.0 The Chief Signal and Telecommunication Engineer (Construction) shall exercise necessary budgetary control within the allotment of funds at his disposal and shall co-ordinate with the Principal Chief Signal and Telecommunication Engineer for compilation of budget at every budgetary stage.

1.1.11.0 The Chief Signal and Telecommunication Engineer (Construction) shall co-ordinate with the Principal Chief Signal and Telecommunication Engineer with regards to the placement of personnel for the various assignments under his charge keeping in view the overall career development of such personnel.

1.1.12.0 In the case of certain major joint projects for Survey, or construction, the Signal and Telecommunication unit may be constituted as a self contained unit under the administrative control of the head of the Project. In such cases, the Head of the Signal and Telecommunication Branch shall be responsible for co-ordination with the Principal Chief Signal and Telecommunication Engineer as mentioned in paragraphs 1.1.9.0. and 1.1.11.0.

Note: - These instructions for construction organization with suitable modification will apply to the other Heads of the department (HODs) in Signal & Telecom departments when entrusted with specific execution of works.

1.1.13.0 **Divisional Officers**

(a) A Signal and Telecommunication Engineer of appropriate grade is in charge of the Signal and Telecommunication Branch on each Division, and is a Branch Officer (BO) of S&T in division. He will be responsible to the Divisional Railway Manager for the maintenance as well as correct functioning of the installations under his charge.

(b) He is under the technical control of the Principal Chief Signal and Telecommunication Engineer to whom he is responsible for all technical matters. He may be assisted by a team of officers of the same grade or in lower grades. In case more than one officer, in the highest grade is available

on a Division, Senior most officers may be nominated as the coordinating officer who would be branch officer (BO). The Principal Chief Signal and Telecommunication Engineer will lay down the duties of the various officers of the Signal and Telecommunication Branch on the Division.

1.1.14.0 The term “Senior Divisional Signal and Telecommunication Engineer" includes Senior Divisional Signal and Telecommunication Engineer in Junior Administrative Grade/Selection Grade and Divisional Signal and Telecommunication Engineer (Senior Scale) holding charge of S&T in a division.

1.1.15.0 **Extra Divisional Organization**

1.1.15.1 The following are the extra Divisional Organizations of the Signal and Telecommunication Department on the Railways :-

(a) Signal and Telecommunication Workshops ;

(b) Signal and Telecommunication Training School/ Centre;

(c) Field Units of Construction Organisation ;

(d) Any other unit designated by the Railway.

1.1.16.0 Each of these Organisations is under the control of an officer of appropriate grade who is responsible for its efficient functioning to the Principal Chief Signal and Telecommunication Engineer/Chief Signal and Telecommunication Engineers (Construction) directly or through an officer of the Headquarters Office in the appropriate grade. The controlling officer may be assisted by a team of officers in various grades as required.

Section 2 – Strength of Officers

1.2.17.0 **Strength of Officers**

Strength of officers in the Headquarters Office of Principal Chief Signal and Telecommunication Engineer, Chief Signal and Telecommunication Engineer (Construction), Divisional offices and in extra divisional organizations shall generally be in accordance with the studies conducted by the Efficiency & Research Directorate of the Railway Board.

Section 3 : Technical Staff

1.3.18.0 **Technical Staff**

1.3.18.1 The following are the technical staff of the Signal and Telecommunication Department for the purpose of this Manual.

(a) Senior Section Engineer (SSE)/Signal, Junior Engineer (JE)/Signal and, artisan staff under their charge.

(b) Senior Section Engineer/Design & Drawing (SSE/SIG/D&D), Junior Engineer/ (JE/SIG/D&D) and staff; and

(c) Senior Section Engineer (SSE), Junior Engineer (JE) and, artisan staff of Signal and Telecommunication Workshop.

1.3.18.2 The Signal supervisors, are classified as Junior Engineer (JE) Signal, Sectional Senior Section Engineer (Sectional SSE) Signal, Senior Section Engineer In-charge (SSE In-charge) Signal, Senior Section Engineer SSE Stores (If SSE is posted for looking after stores only and SSE/JE/Headquarter (Posted in HQ).

1.3.18.3 The duties and responsibilities of Senior Section Engineer/Sectional and Junior Engineer would be same, unless specifically mentioned and is clear from the context. The Senior Section Engineer In-charge, Section Senior Section Engineer and Junior Engineer Signals are directly in charge of installation and maintenance of signaling on a section.

1.3.18.4 The Sectional SSE/Signal are directly in charge of installation and maintenance of more important sections.

1.3.18.5 The SSE/Signal In-charge supervises the work of one or more Sectional Senior Section Engineers and Junior Engineer Signals and are responsible to the Assistant/Divisional Signal and Telecommunication Engineer in appropriate grade.

1.3.18.6 The Technical Supervisors are also headquartered in the Office of the Principal Chief Signal and Telecommunication Engineer/Chief Signal and Telecommunication Engineer/ Construction/Chief Signal and Telecommunication Engineer/Projects/Dy Chief Signal and Telecommunication Engineer /Construction or Projects/Divisional Signal and Telecommunication Engineer and assist in discharging his technical duties.

Note (i) :- If the divisional officer is in Selection grade/JA grade, the prefix senior (Sr) is added to DSTE

(ii) :- Dy. Chief Signal and Telecommunication Engineer/Construction shall be read as Dy. Chief Signal and Telecommunication Engineer/Project/Work/as per executing unit as applicable

1.3.19.0 **Staff under SSE/JE Signal**

1.3.19.1 Each Sectional SSE/JE Signal shall be assisted by one or more Signal Technicians and maintenance gangs.

1.3.19.2 Each SSE/Signal (In-charge), Sectional SSE/JE/Signal shall be provided with one or more special gang/ gangs consisting of Signal Technicians, Artisan staff and Assistants for attending to maintenance requirements that may be considered normally beyond the scope of maintenance by Technician of particular station.

1.3.19.3 Each SSE Signal In-charge shall be provided with office/stores clerks and stores helpers depending on the quantum of office/stores work. Adequate number of chowkidars for guarding railway material in his custody shall also be provided where necessary. Pick up van, utility vehicle, Material handling machines, Material trolleys etc., shall also be provided at stores.

1.3.19.4 SSE(sig)/(In charge) of section shall be provided with pickup van.

1.3.19.5 Each Sectional SSE/JE Signal shall be assisted by appropriate staff as provided for a SSE Signal In-charge.

Section 4 – Strength, Charges, Designations

1.4.20.0 **Strength and Charges of staff :**

1.4.20.1 The strength and charges of SSE/JE, Signal Technicians, Sectional gangs and office staff shall be generally in accordance with an approved yard stick framed by the Efficiency & Research Directorate of the Railway Board. Where such a yard stick is not available, the Principal Chief Signal and Telecommunication Engineer shall frame a suitable one with due consideration to the standard of signalling, type of installation, size of yards, density of train services and general importance of the section.

1.4.21.0 **Designation and Code initials**

1.4.21.1 The designations and code initials of Officers, Supervisors and Signal Technicians shall be as under

1.4.21.2 **Officers**

Post	Designation
(a) Principal Chief Signal and Telecommunication Engineer Zonal Railway/RE :	P.C. S.T. E
(b) Chief Signal Engineer :	C. S. E.
(c) Chief Telecommunication Engineer:	C. C. E.
(d) Chief Signal and Telecommunication Engineer/Construction:	C.S.T.E/C
(e) Chief Signal and Telecommunication Engineer/projects:	C.S.T.E/P
(f) Chief Signal and Telecommunication Engineer/Planning :	C.S.T.E/ Planning
(g) Chief Signal and Telecommunication Engineer/Design & Drawing:	C.S.T.E/ D&D
(h) Chief Signal and Telecommunication Engineer/ Works:	C.S.T.E/ W
(i) Chief Workshop Manager :	CWM/ S&T/ Workshop
(j) Deputy Chief Signal and Telecommunication Engineer:	Dy. C. S. T. E
(k) Senior Divisional Signal and Telecommunication Engineer:	Sr. D. S. T. E
(l) Divisional Signal and Telecommunication Engineer:	D. S. T. E
(m) Senior/Executive Signal and Telecom Engineer (In HQ) :	SSTE/ESTE
(n) Executive Signal & Telecom Engineer (in Extra Divisional Office) :	XSTE
(o) Assistant Signal and Telecommunication Engineer (In HQ and Extra divisional office) :	A.S.T.E
(p) Assistant Divisional Signal and Telecommunication Engineer (In Division) :	ADSTE

1.4.21.2 **Signal Supervisors**

(a) Senior Section Engineer In-charge Signal: SSE In-charge Signal

(b) Sectional Senior Section Engineer Signal: Sectional SSE Signal

(c) Junior Engineer Signal: JE Signal

(d) Senior Section Engineer/Junior Engineer - Head quarter : SSE/JE HQ

Drawing Office

(e) Senior Section Engineer (Design and Drawing): SSE(D&D)

(f) Junior Engineer (Design and Drawing): JE(D&D)

1.4.21.3 Signal Technician

Section 5- Drawing & Design Staff

1.5.22.0 **Drawing Office Organisation**

(a) Drawing and design office of Zonal Railway for open line & Construction organization is generally situated at Zonal HQ. This will be handled by JAG/SG officer who will be assisted by Sr. Scale/Jr. Scale officers & SSE's/D&D.

(b) This office is primarily responsible for preparation and checking of various designs and documents such as Signal Interlocking Plan, Selection Table, Panel/VDU Diagram, Wiring Diagrams, Application logic Circuits, Interface circuits etc., as per latest policies and guidelines of Railway Board and HQ office.

(c) All drawings shall be prepared in Auto CAD or similar software. While proposing alteration in yards, all new work to be shown in red, deletion work to be shown in green or dotted green,. Future works to be shown in yellow colour and retention work to be shown in black or as per zonal railway practice. All SSE posted in drawing office to be given professional training of Auto CAD or similar software.

(d) Checking at ASTE/AESTE & ESTE/SSTE level is required to ensure documents are correctly prepared and are complete in every aspect.

(e) Organization for design and drawing with division and other field units of construction, works, project etc., shall be manned by JE/SIG/D&D or SSE/SIG/D&D.

Section 6 - Misc : Equivalent Designations/Organisations

(a) PCSTE/Co-ordinating CSTE

(b) Construction Organisation/Project Organisation/RE organisation /works organisation.

(c) Sr Divisional Signal and Telecommunication Engineer (Works),

Sr.DSTE(Works), Dy.CSTE/Works, Dy.CSTE/Con, Dy.CSTE/Proj, Dy.CSTE/RE

(d) Sr Divisional Signal and Telecommunication Engineer, Sr.DSTE

(e) Dy Chief Signal and Telecommunication Engineer, Dy.CSTE/Planning, Dy.CSTE/Tele, Dy.CSTE/Signal, Dy.CSTE/Hq, Dy.CSTE/D&D etc.,

(f) DSTE/XSTE/SSTE/ESTE etc.,

(g) ASTE/ADSTE/AXSTE/AESTE etc.,

(h) Supervisor, Inspector, SSE/JE

(i) Maintainer, Technician

(j) Drivers, Loco Pilot

CHAPTER 2 - DUTIES OF SIGNAL AND TELECOMMUNICATION ENGINEERS

Section 1 – Duties of Signal and Telecommunication Engineers in Charge of Maintenance or Construction

2.1.1.0 Duties of Signal and Telecommunication Engineer in charge of Maintenance or Construction

2.1.1.1 The Sr.DSTE/Dy.CSTE (C) In-charge of Maintenance or Construction is generally responsible for :

(a) The installation and maintenance of all Signalling and Telecommunication equipments under his charge in a satisfactory and safe condition.

(c) Preparation of plans, drawings, diagrams, estimates and tender schedules and safe execution of works in his charge.

(d) Ensuring that all inspection notes of higher authorities receive prompt action and submit compliance report

(e) Co-ordination with Engineering and other branches in case of combined works ; obtaining sanction of Commissioner of Railway Safety/PCSTE for new signalling works or alterations and additions to the existing signalling installations, either separately for purely signalling works or jointly with other departmental officers in the case of combined works.

(f) Obtaining TSAA from RDSO/PCSTE as per requirement.

(g) Co-ordination with concerned departments in case of accidents for speedy restoration of traffic and for investigation into the causes of accidents.

(h) Co-ordination with officers and staff of other departments in all other matters to ensure smooth functioning of Signalling and Telecommunication system.

(i) Ensuring supply of approved quality materials and tools for the installation and maintenance of the equipment.

(j) Control over expenditure in relation to budget allotments and sanctioned estimates.

(k) Submission of proposals for Revenue and Works Budget, and for periodic reviews.

(l) Exercise of such powers as may be delegated to him/her in Establishment and other matters.

(m) Ensuring strict discipline amongst his/her staff within the frame work of the Rules.

(n) Dealing promptly with appeals and representations from and looking after the welfare of staff.

(o) Issue of special and specific maintenance schedules for SSE/signal In-charge, Sectional SSE/JE/Signal and Technicians as and when necessary.

2.1.2.0 Transfer of Charge

2.1.2.1 Instructions on "Transfer of Charge" are contained in Chapter I of Indian Railway Code for Engineering Department.

2.1.2.2 Senior Divisional Signal and Telecommunication Engineer/Dy. Chief Signal and Telecommunication Engineers handing over and taking over charge of a Division or of a work shall carry out joint inspection of such works or important sections as necessary.

2.1.2.3 The "Transfer of Charge" statement shall be prepared in adequate number of copies, signed by both and one copy sent to the Principal Chief Signal and Telecommunication Engineer

2.1.3.0 Applicability to other Signal and Telecommunication Engineers

2.1.3.1 These instructions with suitable modifications will apply to the other Signal and Telecommunication Engineers posted to a Division or extra Divisional units for executions of works.

Section 2 - Additional Duties of Signal and Telecommunication Engineer In charge of Maintenance

2.2.4.0 Sr DSTE is generally responsible for :

(a) Ensuring that no alteration to an installation is made, which is a deviation from the original approved plan diagram or specification, without the authority of the, Principal Chief Signal and Telecommunication Engineer

(b) Ensuring inspection of all Signalling and Telecommunication installations & Stores under his/her charge by ASTE/ADSTE/DSTE, once in a year as per the schedule. Every Year, he shall inspect 20 % of Installations in his Jurisdiction and every section must be covered. The inspection shall be intensive with reference to the prescribed schedule of maintenance. A monthly report of inspections so made shall be submitted to the Head quarters Office.

(c) Ensuring of Inspection of Signals within his jurisdiction from footplate of a locomotive or a driving cab both by day and by night in both Up and Down direction once in a year, by ASTE/ADSTE/DSTE preferably jointly with officers of Mechanical/ Electrical/ Operating Departments.

(d) Having in his possession the under mentioned drawings and registers as required when called upon to accompany the inspection of superior officers like Divisional Railway Manager, Principal Chief Signal and Telecommunication Engineer, General Manager, Commissioner of Railway Safety or an officer of the Railway Board etc.

(i) Interlocking plans of the section ;

(ii) Foot plate/station inspection details ;

(iii) Previous inspection reports of the section by HODs/ Principal Chief Signal and Telecommunication Engineer, General Manager, Commissioner of Railway Safety etc.,

(iv) Any other books/papers/documents as per instructions issued by the Headquarters office of the Zonal Railway.

(e) Inspection of office and stores of SSE signal (Stores/In-charge) once in a year. During the inspection a 10% check of some of the stores items, that are costly (Rs 20,000 and above) shall be made and Take action for Proper storage of Materials to avoid deterioration/ Damage/Pilferage/Fire accidents

(f) Keeping watch on inspection of Signal and Telecommunication installations and foot plate inspection by the DSTE/ASTE/SSE/JE/signals under his/her control.

(g) Analysing the Signalling incidences/Data logger exceptional reports, failures from the reports submitted by the Sectional SSE/JE/Signal and taking remedial measures to eliminate recurrence of failures. The reports shall be examined in a meeting jointly with all SSE/Signal In-charge of the division once in 3 months to improve the standard of maintenance. He shall ensure Timely renewal of AMC/Service Contracts, where applicable.

Note:- Each Sr.Divisional Signal and Telecommunication Engineer must maintain a record showing the number of failures and number of trains detained every month over the jurisdiction of each JE/SSE

(h) Drawing out a programme of Integrity Test of PI/RRI/EI installations, overhauling and/or testing of interlocking frames, interlocking key boxes, Station Master's slide control frames, block instruments, relays, cables, point and signal machines, etc., as per instructions contained in Chapter 19.

(i) Reviewing the position with regard to supply of stores on the Division periodically.

(j) Planning replacements of worn out installations, as per codal life or on condition basis and additional signalling inputs necessary to improve the working and submission of proposals for the same.

(k) (i) Reviewing the staff position periodically to ensure that the strength is neither in excess nor short of requirements of Rest Giver, Leave Reserve and Training Reserve to be fulfilled.

(ii) Arranging to obtain timely sanction for additional maintenance staff before new works or additions/alterations to existing installations involving increased workload are commissioned.

(l) Ensure that staff adopt safe working Practices & Arrange for Periodical Technical briefings and update them of Latest Procedures/Circulars

(m) Ensuring sending periodical returns to Headquarters Office as well as

replies to letters from Headquarters within the time specified.

Note: - The rules in Section '2' will also apply to a Dy. Chief Signal and Telecommunication Engineer (Construction/Project/ Works), who is entrusted with the maintenance of any installation as for the time being.

Section 3 – Additional Duties of Dy CSTE In charge of Construction/RE/Project/Works

2.3.5.0 Dy. CSTE In-charge is responsible for

(a) The accuracy, quality and progress of the works entrusted to him and for ensuring that "each work is efficiently organised and so programmed that it progresses speedily and is completed within the time specified". If a work is to be executed through a contractual agency, he shall define Tender Schedules accurately as per actual requirements at site , without any ambiguity , incorporate Special conditions where ever required and ensure quality execution confirming to laid down Quality standards

(b) Ensuring that all works are carried out strictly in accordance with the approved plans, Typical drawings issued by PCSTE, standard drawings, specifications and conforming to the provisions of Manuals. Deviations, if any, shall have the prior approval of the PCSTE. The Signal and Telecommunication Engineer in charge of Construction is responsible for :

(c) (i) Ensuring that traffic notices & Programme of Pre –NI and NI shall be jointly issued by Sr.DSTE/Sr.DOM/Sr.DEN/Sr.DEE(TRD) etc., are issued in consultation with other Divisional officers before any existing installation is altered or any new installation is introduced which affects the safe working of any Signal, points or interlocking :

(ii) Ensuring issue of temporary working instructions/Rules for working of traffic/Green Notice, where necessary.

(d) Furnishing relevant information to the operating department to help them in the preparation of working Rules and temporary working instructions/Rules.

(e) Arranging for obtaining the sanction of Commissioner of Railway Safety/ PCSTE where required as per instructions in Chapter 9.

(f) Advising Commissioner of Railway Safety & PCSTE by a message after bringing a new installation or modification to the existing installation into use and submission of a Safety Certificate as per instructions in Chapter 9.

(g) Submitting progress reports on Form S&T/PR1 (Annexure '2/1') to Headquarters every month.

(h) Periodical verification of the materials at site as per Chapter 6.

Note 1:- The rules in Section '3' will also apply to a Sr. Divisional Signal and Telecommunication Engineer normally in change of maintenance, who is entrusted with the execution of certain specific works.

Note 2 :- The CRS/PCSTE sanction of works executed by PSU to be

processed by Sr.DSTE as decided by PCSTE of Zonal Railway.

Note 3 :- Where ever "His/he " is mentioned in this chapter , it also applies to "Her/She"

☞ *Note :- This Chapter has under mentioned Annexures for further study*

S.no	***Annexure no***	***Description***
1	*2/1*	*Engineers Progress Report*

ANNEXURE '2/1' Form No.S&T.PR/2.1

…….. RAILWAY

SIGNAL & TELECOMMUNICATION DEPARTMENT

…….Division/District

Engineer's Progress Report

Monthly Progress Report for the month ending……. 20

S.No.	Name of work	Cost and Allocation	Estimate No. and date of sanction quoting authority	Date funds allotted quoting authority	Reference to commissioner Railway Safety's approval	Percentage of material received up to end of the month	Date work started	Percentage of work done during the month	Total percentage of work done up to the end of last month	Remarks

Signature ……..

Designation ……..

CHAPTER 3 - DUTIES OF SIGNAL SUPERVISORS AND TECHNICAL STAFF

Section 1 – Common duties of SSE (Signal)/in-charge, Sectional SSE/JE (signal)

3.1.1.0 **Technical Duties**

3.1.1.1 The duties of a Sectional SSE/JE(Signal) of a section are detailed in various Chapters of this Manual, the most essential being :

(a) Efficient and proper maintenance of all signalling and interlocking equipment under his charge in accordance with the provisions of the Manual, Rules and Regulations in force and instructions issued from time-to-time.

(b) Assist the SSE(Signal)/In-charge in execution of works incidental to the maintenance of equipment under his charge, additions and alterations to existing installations and new works in accordance with approved plans and circuit diagrams under Open Line working conditions.

(c) Assist the SSE(Signal)/In-charge in overhauling and carrying out alterations to the existing locking of interlocking frames in accordance with approved interlocking tables and interlocking charts, as also carrying out alterations to electrical signalling and interlocking systems in accordance with approved interlocking and selection tables and circuit diagrams when authorised to do so in writing by the Sr.Divisional Signal and Telecommunication Engineer.

Note:- The term interlocking frames 'includes' Station Master's Slide Control Frames and Interlocking key boxes.

(d) Carrying out works in an emergency on his own initiative and responsibility. In such cases, intimation must be given to his SSE(Signal)/In-charge or Fault control by a message on control or phone or suitable means of communication.

3.1.2.0 **Equipment Inspection**

3.1.2.1 The Sectional SSE/JE Signal shall report to SSE(Signal)/In-charge for his work and route his inspection reports to the Divisional Signal and Telecommunication Engineer through the SSE(Signal)/In-charge.

3.1.2.2 The Sectional SSE/JE Signal shall carry out inspection and testing of all the equipments in his/her charge at intervals not exceeding one month in accordance with the detailed instructions contained in this Manual and Maintenance Schedule issued by PCSTE. Telecommunication equipments entrusted to his maintenance shall be inspected in accordance with instructions contained in the Telecommunication Manual.

3.1.2.3 While carrying out inspections, the Sectional SSE/JE Signal shall take the Technician (Signal) of the section with him/her whenever possible.

3.1.2.4 Special attention shall be paid during inspections to the work of each Technician (Signal) in respect of adjustments, cleaning and lubricating of moving parts of all points and signal equipment to ensure that they work smoothly. Any fault detected shall be arranged to be rectified at the earliest. He shall ensure disconnection of unsafe gear, failure rectification only after receipt of written memo.

3.1.2.5 The inspection shall be thorough with reference to the prescribed Schedule of Maintenance and purposeful to ensure that equipments function satisfactorily, safely and with minimum liability to failures.

3.1.2.6 The Interlocking Plan, Cabin Diagrams, Working Rules etc. shall be inspected to see that they are up-to-date. Any discrepancy noticed shall be brought to the notice of the SSE(Signal)/In-charge where ever applicable.

3.1.2.7 A record shall be made in the Signal Failure and Inspection Book at each station during monthly inspection.

Note: Para 3.1.2.0 is also applicable to all Sectional SSE/JE (Signal) & Technician(Signal).

3.1.2.8 All interlocked points and crossings must be jointly inspected by SSE(Signal)/In-charge with SSE P way/In-charge and SSE/JE(Signal)/section with SSE/JE(Pway)/section alternately and duration between two joint inspections shall not exceed 3 months.

3.1.3.0 **Foot-plate Inspection**

The SSE(Signal)/In-charge, Sectional SSE/JE(Signal) shall carry out foot-plate inspection of all signals by day and night in both Up and Down directions once a month or as specified by Sr DSTE/PCSTE, over his entire jurisdiction. During the inspection he shall take special notice of the following:

3.1.3.1 All signals have adequate visibility as specified in para 7.7 preferably from the Loco pilot side of the foot-plate. If the visibility of a signal is affected by growing tree branches or other obstructions, the matter shall be specially noted. Cases requiring special attention, such as trees in private lands shall be referred to the Sr. DSTE with full particulars of the topography of the area for his information, necessary action and instructions. In other cases the supervisors shall arrange to remove the obstructions as early as possible. P-Way Staff as per IR P-Way Manual Para-222''

3.1.3.2 Anything which may endanger safety or interfere with signalling gears, the matter shall be brought to the notice of the concerned department. If immediate action is necessary for reasons of safety, information to be conveyed to the Station Master and control room.

3.1.3.3 All Signals are correctly focused and are lit brightly. Where a signal light is not lit bright enough to be visible from an adequate distance, the matter shall be reported to the SM concerned.

No fixed light is interfering with the sighting of the signals.

3.1.3.4 A suitable light is displayed at the token delivery and pick up area for the driver to clearly drop the incoming token and pick up the outgoing token.

3.1.3.5 All Warning Boards and other boards maintained by Signal department are in good condition.

3.1.3.6 Signals that are to be replaced to ' ON ' by the passage of trains are being so replaced.

3.1.4.0 **Attending to Failures /Incidences**

3.1.4.1 The Sectional SSE/JE(Signal) shall monitor daily all failures on his section. During his periodical inspection he shall check up the cause indicated as well as repairs carried out to ensure that similar faults do not recur. In case of failures of a serious nature, the investigation shall be carried out immediately, and the SSE(Signal)/In-charge advised.

3.1.4.2 Each Sectional SSE/JE(Signal), shall at the end of every month, summaries and classify the failure reports received from each Technician(Signal) and submit monthly failure report on Form S&T/FS2/(Annexure '3/1') with his remarks to the Sr Divisional Signal and Telecommunication Engineer in duplicate. Failures may be classified as suggested in Annexure '3/2'. Failures pertaining to other departments shall be promptly brought to the notice of the concerned departments.

3.1.4.3 SSE(Signal)/In-charge, Sectional SSE/JE(Signal) shall monitor data Logger exception reports on daily basis and take corrective actions. Where required, he shall take assistance from Data Logger Management Centre (DLMC) in Divisional headquarters. He shall take corrective actions on such unusual incidences.

3.1.5.0 **Competency in Rules and Technical Work :-** Sectional SSE/JE Signal shall be

3.1.5.1 Fully conversant with the rules and regulations, instructions, procedures and practices of installation, operation and maintenance of all installations under his charge.

3.1.5.2 Competent to carry out under open line working conditions

(a) Additions and alterations to existing installations and circuits.

(b) New installations.

(c) Testing, overhauling and carrying out alterations to the existing locking of lever frames and wiring of Electrical Interlocking systems.

(d) Installation, adjustment, maintenance and testing of Electronic Signal and Interlocking equipments.

3.1.6.0 The Sectional SSE/JE(Signal) shall ensure that only competent staff are permitted to work on or make adjustments to any of the signalling gears which are connected with the safety of trains. The supervisors shall further ensure that these instructions are clearly understood by all the staff

concerned.

3.1.6.1 The Sectional SSE/JE(Signal) should be conversant with all the new equipment & technology and shall ensure proper training to subordinate staff for trouble shooting of the equipment under the charge.

3.1.7.0 **Reports of Technicians**

3.1.7.1 The Sectional SSE/JE(Signal) shall ensure that the Technicians are regular in their maintenance programme and their reports in Form S&T/MR (Annexure '3/10') are correctly maintained.

3.1.7.2 Technician(Signal)s programmes shall be prepared in terms of the days of the week. The programme shall be so prepared that it is convenient to the Technician(Signal)s and provides for maximum time at every station with due regard to the rostered hours of the duty. As far as practicable the programme shall include allowance for picking up programme missed during the week on account of failures and other exigencies.

Section 2 - Duties of SSE(Signal)/In-charge

3.2.8.0 **Technical Duties**

3.2.8.1 The duties of SSE(Signal)/In-charge, assisted by one or more Sectional SSE/JE Signal, are detailed in various Chapters, the most essential being :

(a) Exercising supervision over the work done by the Sectional SSE/JE Signal and maintenance staff in accordance with instructions contained in this Manual.

(b) Testing, overhauling and carrying out alterations to the existing signal and interlocking installations in accordance with approved plans and instructions.

(c) Testing any telecommunication equipment specifically entrusted to his maintenance in accordance with instructions in the Telecommunication Manual or other executive instructions.

(d) Carrying out works in an emergency on his own initiative and responsibility. In such an emergency, intimation shall be given to his controlling officer by a message on control or phone.

(e) Providing assistance to the Sectional JE/SSE(Signal) to attend to works which are normally beyond the scope of the maintenance staff under the Sectional JE/SSE(Signal).

3.2.8.2 The SSE(Signal)/In-charge shall see that the fire-fighting appliances at various locations under his charge are kept in a condition fit for immediate use. He shall also see that the staff under him are conversant with the method of operation. He shall arrange actual trials periodically to test appliances for fitness and the staff in the knowledge of the working of the fire-fighting appliances and the method of their operation. The appliances shall be recharged immediately after they have been used.

3.2.8.3 In his work, the SSE(Signal)/In-charge shall be responsible to the Sr. DSTE/DSTE/ASTE.

3.2.8.4 Each SSE(Signal)/In-charge shall submit an annual return in duplicate of all apparatus in service on his section to the Sr.DSTE in prescribed forms.

3.2.9.0 **Inspections**

3.2.9.1 The SSE(Signal)/In-charge shall carry out all the inspections prescribed in Para 3.1.2.0 and 3.1.3.0 over his entire jurisdiction at intervals not exceeding three months.

3.2.9.2 He shall carry out the quarterly foot-plate inspection preferably jointly with the SSE/Loco and Traffic Inspector by day and by night.

3.2.9.3 Joint Certificate of Foot-Plate Inspection shall be submitted to the Sr. DSTE/DSTE/ASTE at the end of every quarter in Form No. S&T/FP (Annexure '3/3'). A copy of the relevant portion of the joint report of Foot-Plate Inspection shall be sent to the station concerned for the reference of inspecting officers and also for necessary action, if any, by the Station Master.

3.2.9.4 He shall also check for Condition of Building and structures in which equipments are housed; Proper Lighting/Ventilation/Air-conditioning arrangements (where existing) and take appropriate action.

3.2.9.5 Exception report for a period from last visit to present day visit shall be taken from Datalogger and action shall be taken against each item to avoid recurrence

3.2.9.6 He shall be a Member of Signal Sighting Committee to ensure proper visibility of Signals

3.2.9.7 A record shall be made in the Signal Failure and Inspection Book at each station during quarterly inspection.

3.2.10.0 **Section Gang and Maintenance Work**

3.2.10.1 Each SSE(Signal)/In-charge shall maintain a register in which all works that are beyond the capacity of the Technician(Signal)s shall be entered. Execution of such works in the order of their importance shall be entrusted to the section gang.

3.2.10.2 The SSE(Signal)/In-charge shall prepare a programme of work for the section gang. The programme shall be so prepared as to reduce travelling time to a minimum and allow the gang to return to Headquarters on rest days.

3.2.10.3 It is desirable that when the gang is sent away from the Headquarters for a number of days a tool van equipped with all materials and tools for the work to be done is provided.

Section 3 - Inspection Certificates

3.3.11.0 **Inspection and Testing Certificates**

3.3.11.1 Each SSE(Signal)/In-charge shall submit an Inspection and Testing Certificate monthly for each station/ installation inspected by him during the month in Form S&T/IC (Annexure '3/4') to the Sr.DSTE indicating the condition of gear inspected. Brief remarks shall be given in regard to any assistance required in regard to labour or stores or any other important matter.

3.3.11.2 The Certificate of Inspection and Testing is an important document and it forms a record of the nature of inspections carried out and the condition of the gear reported upon from time-to-time. SSE(Signal)/In-charge, Sectional SSE/JE(Signal) shall, therefore, make a careful record of all the inspections and the repairs, special or ordinary required on the section. The report shall not be used for representing their general difficulties.

3.3.11.3 The report on each foot-plate inspection must be promptly submitted to the Sr. DSTE in Form. S&T/FP (Annexure '3/3') every month.

Note:- This form may be modified by Zonal Railways to include additional information like condition of speedometer, etc.

3.3.11.4 At the end of each quarter, the SSE(Signal)/In-charge shall certify that all signals on his section have been jointly inspected both by day and by night. He shall also highlight defects and deficiencies noticed during these inspections which could not be rectified, indicating assistance required. This certificate shall be kept on record in the office of the Sr. DSTE.

3.3.11.5 The inspection, testing (including testing of cables) and overhauling certificates, shall be filled in progressively at the stations immediately an inspection or testing or overhauling has been done and not left to be completed at the Headquarters station.

3.3.11.6 Each SSE(Signal)/In-charge, Sectional SSE/JE(Signal) shall send a separate certificate that all the stations under his charge have been inspected as required under Para 3.1.2.0 or 3.2.9.0, as the case may be. If all the stations have not been inspected, the names of stations left out and the reasons why these could not be inspected shall be furnished.

3.3.11.7 Periodicity of inspection of signalling Installations is once in a month Sectional JE/SSE and once in a quarter for SSE incharge

Section 4 - Accidents

3.4.12.0 SSE(Signal)/In-charge, SSE/JE(Signal) and Technicians(Signal) are responsible for supervision of labour and shall take all precautions to ensure safety of staff and equipment, prevent accidents to the staff or damage to apparatus and see that-

(a) Instructions contained in I.R.C.A. Safety First Pamphlet as well as those issued from time-to-time are observed.

(b) Use is made of the safety appliances provided such as Helmets,

Retro reflective Jackets, Safety shoes, belts, pulley blocks, etc.

(c) That all ropes, lifting tackles and staging used for erection of signals, etc., are adequate and in good condition.

(d) That staff working on the line are vigilant and where necessary employ look out men

(e) All accidents are promptly reported.

3.4.13.0 The rules for ensuring safety are contained in General Rule 2.11. The procedures to be followed in regard to accidents are contained in Chapter VI of General Rules and in the Accident Manual. Each Inspector shall be well acquainted with the rules concerned. A few salient points in regard to SSE/JE(Signal) are listed in Annexure ' 3/5'.

Section 5 - General Responsibilities and Duties

3.5.14.0 **Tools and Plant:**

Each SSE(Signal)/In-charge shall be responsible for the issue of proper tools and other equipments to the Technicians (Signal) and artisans under him and for ensuring that these are kept by them in proper working order. Unserviceable and defective tools shall be replaced.

3.5.15.0 **Books of Reference and Acquaintance with Rules:**

3.5.15.1 Each SSE(Signal)/In-charge, Sectional SSE/JE Signal shall have in his possession when on duty an up-to-date copy of the following books of reference :-

(a) General and Subsidiary Rules,

(b) Signal Engineering Manuals & Handbooks

(c) Schedule of Standard Dimensions,

(d) Current Working Time Tables.

(e) Maintenance Manuals for Special equipment under his charge,

(f) Telecommunication Manual,

(g) Accident Manual,

(h) Block Working Manual,

(i) A. C. Traction Manual (in A. C. electrified territory).

(j) Station working rules.

(k) Compendium of Technical Circulars and joint Procedures orders & Typical Drawings.

3.5.15.2 Each SSE(Signal)/in charge shall be well acquainted with all the rules concerned in these publications and circulars issued from time-to-time which concern his work and duties. These books shall be kept up-to-date with all the addenda and corrigenda slips issued from time-to-time. He shall ensure that all staff under his charge are well acquainted with these rules and regulations to the extent applicable to them.

3.5.16.0 **Accompanying Important Inspections:**

Each SSE(Signal) while accompanying an inspection such as that of General Manager or Commissioner of Railway Safety or Divisional Railway Manager or the Principal Chief Signal and Telecommunication Engineer or any other Signal and Telecommunication Engineer or any officer of the Railway Board, shall in addition to the books of reference mentioned in Para 3.5.15.1 have the following in his possession :

(a) Signaling Interlocking Plan, Interlocking and yard diagrams, interlocking tables and selection tables.

(b) Ensuring availability of following documents at respective stations in the inspected section: Cable Route Plans, Cable Core Plans, JB details, wiring diagram, testing reports, Interface Circuits, Application Logic Circuits, Bit Charts, track circuit bonding plan, Video Display Unit Diagram/Panel Diagram whichever is applicable to the inspected section.

(c) Plans of proposed and sanctioned works, Traction Bonding Diagram, Equipment Replacement Register, Asset Register, Joint Inspection Register.

(d) Point test gauges, Multi-meter, TSR insulation test equipment, Earth resistance measuring test equipment, Track circuit test equipment, Cable Fault Locator, Cable Route Locators, a portable telephone to suit overhead lines or cable circuits depending on the section being inspected or any other equipment required to test the signalling gears provided in the section under inspection.

(e) Other tools such as True RMS meter, Clamp meter, measuring tape, foot rule, sealing plier etc.,

(f) Any other items specified by Sr. Divisional Signal and Telecommunication Engineer/ Principal Chief Signal and Telecommunication Engineer.

3.5.17.0 **Establishment matters:**

The SSE(Signal)/In-charge shall ensure that the Rules laid down in Establishment Code, Acts and Regulations and Local circulars issued from time-to-time regarding general service conditions, conduct, discipline, passes, leave, service Records, medical examinations, etc., are strictly complied with, records maintained and returns submitted as laid down. Certain salient points in regard to Payment of Wages Act, Workmen's Compensation Act and Hours of Employment Regulations are listed in

Annexure '3/6'. These only list the broad features and a reference to the Act and Rules should be made while deciding cases.

3.5.18.0 **Muster Sheets**

3.5.18.1 The SSE(Signal)/In-charge shall arrange to have the blank muster sheets of staff top-initialed by the Signal and Telecommunication Engineer under whom he works and issue them to the staff in time. At the end of each wage period, the muster sheet shall be collected and fresh ones issued.

3.5.18.2 Each SSE(Signal)/In-charge shall inspect and initial the muster sheets of staff each time that he is on their section. The presence or absence of staff shall be noted on the spot. The time of inspection and the number of staff present should be recorded in the muster sheet in ink.

3.5.19.0 **Correspondence and Records:**

The SSE(Signal)/In-charge shall keep his correspondence up-to-date. He is responsible to see that his office records, registers, stores ledgers and accounts are maintained correctly and stores ledgers are posted up-to-date. Periodical returns shall be submitted within the dates as per the standing instructions.

Note :- These records include Electronic forms of Records/Registers/Documents

3.5.20.0 **Relinquishment of charge**

3.5.20.1 When relinquishing charge of a section, the SSE(Signal)/In-charge shall prepare a handing over charge statement in duplicate, which shall briefly contain the following :-

(a) The section particulars with staff, their service and leave records.

(b) Details of sanctioned and proposed works and their progress.

(c) General Notes regarding painting, renewals and replacements, etc.,

(d) Overhauling and testing of interlocking frames, block instruments, point machines, signal machines, relays, etc.,

(e) Notes regarding materials, stores, important notes and documents.

(f) Position of stock sheets, Accounts notes, Audit and Accounts Inspection Reports, Completion reports.

(g) The detailed status of the Scheduled Inspections, officers Inspections and compliance, other Organisations work's status and compliance of deficiencies.

(h) Any other important matters.

3.5.20.2 The SSE(Signal)/In-charge taking over and handing over shall inspect all important works in progress, check the staff and their tools and plants and

materials-at-site.

3.5.20.3 The relieving SSE(Signal)/In-charge shall examine all Office records and ledgers and initial them with date.

3.5.20.4 Handing over charge statement shall be jointly signed by the supervisors taking over and handing over and a copy submitted to the Sr. Divisional Signal and Telecommunication Engineer. Errors and discrepancies noted during handing over and taking over, shall be recorded on the statement for the information of the Sr. Divisional Signal and Telecommunication Engineer.

Section 6 - Additional Instructions for SSE (Signal)/In-charge of Construction

3.6.21.1 Code Rules.-The rules for the execution of works are contained in Chapter 9 of this Manual and Chapter XI of the Indian Railway Code for the Engineering Department.

3.6.22.0 **Applicability of Rules :**

3.6.22.1 The instructions for SSE(Signal)/In-charge of construction will also apply to SSE(Signal)/In-charge normally In-charge of maintenance when entrusted with the execution of certain specific works.

3.6.22.2 Similarly the instructions for SSE(Signal)/In-charge of maintenance will apply to SSE(Signal)/In-charge of Construction when entrusted SSE(Signal) with the maintenance of equipments at a station on a section.

3.6.23.0 **Commencement/ Execution of Works :**

3.6.23.1 The instructions contained in Para 9.5 shall be observed.

3.6.23.2 When the estimate for a new work is sanctioned, the SSE (Signal)/In-charge shall submit requisitions for the materials without delay and ensure that all the required materials are made available. No work shall be physically commenced until it is authorised by the concerned officer.

3.6.23.3 No work affecting an existing signalling and interlocking system shall be commenced unit such portions of the work as can be carried out without affecting the working of the existing installation have been completed so that the period of disconnection will be as short as possible.

3.6.23.4 All new signals shall be inspected and passed by a Signal Sighting Committee consisting of (i) SSE(Signal)/In-charge, (ii) SSE/Loco and (iii) Traffic Inspector Sighting Certificates should be promptly submitted on form S&T/SC (Annexure '3/7') to the Dy.CSTE (Construction)/ Sr.Divisional Signal and Telecommunication Engineer.

3.6.23.5 Introduction of a new signal or alteration to an existing signal which would require to be passed by a Sighting Committee shall not be taken on hand without the publication of the Traffic Notice, which shall not be operative for a period longer than three months.

3.6.24.0 **Progress Report of Works**:- The SSE(Signal)/In-charge of works shall submit a Progress Report, every month of all the works in his charge to the Dy.CSTE (Construction) on Form S&T./PR/3.2.9.0 (Annexure ' 3/8') which shall include

(a) The progress of work and probable date of completion;

(b) Reference to requisitions or work orders against which supply has not been made; also details of items which are particularly required and on account of which works are being delayed ;

(c) Delays arising from other causes

3.6.25.0 **Opening of works**

3.6.25.1 The SSE(Signal)/In-charge of works shall advise the Dy.CSTE (Construction), when a work will be ready for opening, in order that the latter may arrange for fixing of a date and for the issue of the necessary notices.

3.6.25.2 When a new work or alteration to an existing work is about to be brought into service, the SSE(Signal)/In-charge shall see that he obtains new or corrected interlocking and yard diagrams, circuit diagrams, etc., before the date of opening. If there are minor alterations, the existing diagrams shall be corrected and endorsement made by the SSE(Signal)/In-charge.

3.6.26.0 **Completion Certificate and Completion Report**

3.6.26.1 The SSE(Signal)/In-charge shall submit a Completion Certificate to the Dy.CSTE (Construction) on Form S&T/CC/3.3.10.0 (Annexure '3/9') immediately copy to Sr.DSTE after a new work/alteration to an existing installation has been brought into service.

3.6.26.2 The SSE(Signal)/In-charge of the works shall, immediately after the completion of a work submit a Completion Report to the officer In-charge which shall contain the following :-

(a) List of all materials received from the stores and other sources with all particulars.

(b) List of all materials used for the work.

(c) List of materials released and returned to stores with particulars.

(d) List of excess materials returned to stores with particulars.

(e) Details of labour with period for which they were employed for the completion of work.

(f) Special problems or difficulties experienced during the execution of the work; if any and

(g) Transfer unutilised materials from 'Works Registers' of that work to other works to get the credit for that work.

Section 7 – Datalogger Management Centre (DLMC)

3.7.27.0 **At Divn HQ**

3.7.27.1 Datalogger Management Centre generally situated at Divisional Headquarters shall be extensively used for monitoring of signalling system in the entire division. DLMC Signal SSE/JE(Signal) at HQ shall continuously or regularly monitor Data Logger exception reports and inform concerned SSE(Signal)/In-charge, Sectional SSE/JE(Signal) for necessary action.

3.7.27.2 When a new station is commissioned or alterations are carried out in existing station, SSE(Signal)/In-charge shall ensure that all specified relays and gears are monitored by data logger at the station.

3.7.27.3 SSE(Signal)/In-charge/DLMC shall ensure that SMS alerts are generated for Safety related faults & Maintenance related faults.

3.7.27.4 SSE/JE(Signal)/DLMC shall ensure monitoring of Relay Room door opening, fire alarm, Power supply/Battery voltages, ELD and other specified gears through SMS alert generated by Data logger.

3.7.27.5 SSE/JE(Signal)/DLMC at divisional HQ shall ensure data availability and network connectivity continuously to zonal HQ/Railway Board.

3.7.27.6 SSE/JE(Signal)/DLMC at divisional HQ shall ensure Prompt and regular validation of Datalogger inputs in coordination with SSE(Signal)/In-charge of the Station/IB/LC

Section 8 - Duties of Technician(Signal)s

3.8.28.0 **Technical duties of Technician(Signal)s**

3.8.28.1 The duties of Signal Technicians are detailed in various Chapters in this Manual, the most essential being :

(a) Efficient maintenance and testing of all equipments under his charge such as Mechanical Signalling equipments, Electrical and Electronic Signalling equipments, Telecommunication equipments etc., so as to keep them properly maintained and in good working condition, in accordance with instructions contained in this Manual and such Circulars or Instructions, as may be issued from time to time. (Note:- The term Electrical and Electronic Signalling equipments includes all types of Block Instruments.)

(b) Carrying out works and alterations to the existing installations under the instructions of the SSE/JE (Signal) or S&T officers.

(c) Bringing to the notice of the SSE/JE Signal any emergency and situation that may be beyond his competence or control by a message on

control or phone or other means of communication.

(d) Ensuring that the safety appliances like Safety Belts, lifting tackles and staging etc., are in good condition and are always made use of in order to ensure his safety and the safety of staff working under him.

(e) Deploying men for look out duties as and when necessary.

(f) Before starting maintenance at station, Exception report for a period from last visit to present day's visit shall be taken from Datalogger where PC or printer is available at station, Action shall be taken against each item to avoid recurrence

3.8.29.0 **Locking of Interlocking frames and Interlocking circuits**

3.8.29.1 The Technician (Signal) shall not remove the covers of interlocking frames/Control panel or disconnect or alter any connection in the interlocking circuits of Electrical Interlocking systems except with the approval of and in the presence of his SSE/JE/Signal. Any action, which will vitiate interlocking, shall be done only after disconnection notice has been issued and accepted.

3.8.29.2 A Technician (Signal) shall, however, attend to the signalling failures promptly to permit safe passage of trains till the arrival of the SSE/JE/Signal. If any unsafe failure/gear is noticed, he shall suspend signaling movement over it.

3.8.29.3 A Technician(Signal) shall also ensure that, once the signals have been suspended as per para 3.8.29.1, normal working shall not be restored until the faulty gear has been attended to, tested and certified by the SSE/JE Signal and the Relay Room, Block Instrument have been closed, pad locked and sealed.

3.8.30.0 **Maintenance Programs**

3.8.30.1 Each Technician(Signal), as far as possible, adhere to the programme laid down for him by his SSE/JE Signal and shall maintain a record of his visits on Form S&T/MR (Annexure-3/10).

3.8.30.2 Technician(Signal) shall not normally leave any gear in disconnected condition and in unavoidable eventuality; a specific advice to ASM, duly acknowledged by him shall be obtained.

3.8.30.3 Each Technician(Signal) shall maintain and test all the equipments under his charge schedule specified for various gears.

3.8.30.4 Each Technician(Signal) shall submit a fortnightly report of his maintenance rounds to his Sectional SSE/JE Signal on Form S&T/MR (Annexure-3/10).

3.8.30.5 Technician(Signal) before leaving his Headquartered Station, shall record inform his movements in the Movement Board kept at the Station Master office to section in charge or signal control.

3.8.31.0 **Disconnection of Apparatus**

3.8.31.1 Each Technician(Signal) shall have in his possession a book of Disconnection Notices- Form S&T/DN (Annexure 3/11). A Technician(Signal) who is in possession of a Competency Certificate cum Training History book (Annexure 3/13) only shall independently undertake works necessitating issues of Disconnection Notices.

3.8.31.2 Disconnection Notices need not be issued in situations as listed in Annexure-3/12 provided suitable precautions are taken. In other situations, when it is necessary to disconnect any equipment in his charge for repairs, replacement or adjustment, the Technician(Signal) shall advise the Station Master on duty in writing on Form S&T/DN (Annexure-3/11) and obtain the latter's signature before work is started and after it has been completed.

3.8.31.3 When it is necessary to disconnect point equipment switches or signals for repairs, replacement or alteration, Warner/Distant and Stop Signals governing the lines in question shall be kept in the 'ON' position and made in-operative until the work is completed.

3.8.31.4 The Technician(Signal) must seal the equipment opened by him under his competence.

3.8.31.5 The Signal Technicians shall possess valid Competency Certificate for handling Signalling equipments like Block instruments, EI, MSDAC etc. The Competency Certificates are issued to Technician(Signal)s by Training Centres after successful completion of the training.

3.8.32.0 **Attending to Failures/Incidences**

3.8.32.1 A Technician(Signal) shall attend to all failures in his section promptly proceeding by the first feasible means on receipt of information. Before taking up work, he shall first obtain failure report/message from SM/ASM in writing in accordance with provision of G.R.3.68 for each failure recorded in the signal failure register and then issue disconnection notice as per Para 3.8.29.1. He shall make every endeavour to rectify the failures expeditiously and take all possible steps to prevent recurrence ensuring proper rectification method. If a gear has failed on the unsafe side and the SM/ASM has been unable to put the relevant signal to 'ON', the Technician(Signal) shall take steps to disconnect/disable the relevant signal and bring it to 'ON'.

3.8.32.2 All failures which are beyond his competence or control must be brought to the notice of the SSE/JE/Signal section by a message on control or phone or other means of communication.

3.8.32.3 Record of the date and time of rectification and the nature of the fault removed must be recorded in the Signal Incidence and Inspection Register provided at each interlocked station and same will be informed to JE/SSE(Signal) and signal control also.

3.8.33.0 ACCIDENTS : On receipt of advice about any accident in his jurisdiction, the Technician shall proceed to the site of accident by first available means. He shall not interfere with any equipment on his own but shall act upon the orders given by the senior most officials at the site of accident.

3.8.34.0 **General Duties of Signal Technicians**

3.8.35.0 Knowledge of Rules and Instructions

3.8.35.1 A Technician(Signal) shall be conversant with rules, regulations and instructions concerning his work contained in the following books of reference as well as other instructions issued from time to time.

a) General and Subsidiary Rules;

b) Signal Engineering Manual;

c) Block Working Manual

3.8.35.2 A copy of those portions of each of the books mentioned in Para 3.8.35.1 as also all circulars and instructions concerning his work, shall be maintained by him for his reference and information. He shall keep them up to date in respect of Correction Slips issued from time to time

3.8.36.0 A Technician(Signal) shall not permit any Helper or artisan staff to do any adjustment to the gear in use except under his personal supervision and he shall ensure that the staff under him clearly understand this rule.

3.8.37.0 **Maintenance of Muster Sheets etc**

3.8.37.1 A Technician(Signal), when posted at station where his seniors are not posted/not available, then, he shall mark his own attendance and that of his staff on the muster sheets received from the SSE/JE Signal before starting his work. Erasing and over-writing is not permitted. Loss of muster sheet shall be brought to the notice of the SSE/JE Signal at the earliest.

3.8.37.2 A Technician(Signal) shall work to the duty rosters provided and see that the staff under him also works according to the roster.

3.8.38.0 **Materials and Tools and Plant**

3.8.38.1 A Technician(Signal) shall, where necessary, get his requirements of materials for maintenance and repair work from the SSE/JE Signal, giving full particulars of station, location and the gears to be replaced. Released materials shall be returned to the SSE/JE Signal immediately. There shall absolutely be no wastage of any material.

3.8.38.2 A Technician(Signal) shall always take his tools with him when on duty. All tools shall be kept in a condition fit for immediate use.

3.8.39.0 **Co-operation between Technician(Signal)s**

3.8.39.1 Technician(Signal) shall co-operate in testing of all the signals operated

mechanically and controlled electrically.

3.8.39.2 Technician(Signal) shall extend full co-operation to each other in their day-to-day work.

Section 9: Duties of Helpers/Assistants

3.9.40.0 **Duty list of Helpers/Assistants**

3.9.40.1 (a) The general duties of Helpers/Assistants are as follows :

(i) Cleaning of Glued Joints/Block joints.

(ii) Cleaning of all S&T gears and their upkeepment.

(iii) Painting of all S&T gears and their upkeepment.

(iv) Cleaning of garbage/vegetation under Point rodding and surrounding of all signaling gears/locations.

(v) Cleaning of chair plates and maintaining their lubrication.

(vi) Reporting of any infringement of signal/Ladder etc., due to dashing.

(vii) Upkeep of Relay room, Power room, Battery room, Maintenance room, Store room etc.

(vii) Maintaining minimum essential tools with him for day to day works as per instructions of Technician/JE/SSE.

(ix) Loading/unloading/Handling of T&P items and all materials as per instructions of Supervisor.

(x) Manning and protection of installations/S&T gears on need basis as guided by Supervisor.

(xi) Reporting of rubbing of stretcher bar with Stock Rail bottom to be ensured.

(xii) Assisting Supervisor/Technician during testing/Maintenance/ Inspections, Carrying Tools & Plants for Maintenance/Failures

(xiii) Handling/topping-up of Distilled water, Electrolyte, HSD oil and applying Petroleum Jelly, Lubricants etc., as per instructions of Technician/JE/SSE.

(xiv) Assisting Supervisor in documentation works.

(xv) Accompanying/Assisting Technician/JE/SSE and Higher officials in their day to day works.

(b) Technical duties of Helper are as under

(i) Removal of obstruction to points.

(ii) Removal of excess bars formation at insulation joints.

(iii) Be aware of working in RE area.

(iv) Check RE bonds, their intactness.

(v) Check water logging in track circuited areas.

(vi) Any other duty assigned by SSE/JE(Signal) or Technician

☞ *Note :- This Chapter has under mentioned Annexures for further study*		
S.no	*Annexure no*	*Description*
1	*3/1*	*Failure Report of Signalling & Interlocking, Electrical Signalling and Block Signalling*
2	*3/2*	*Classification Codes*
3	*3/3*	*Foot Plate Inspection of Signals*
4	*3/4*	*Inspection Report*
5	*3/5*	*Proceeding to Site of Accident:*
6	*3/6*	*Hours of Employment Regulations*
		Workmen's Compensation Act
		Payment of Wages Act
7	*3/7*	*Signal Sighting Committee Report*
	3/8	*JE/SSE's (Signal) Progress Report*
9	*3/9*	*JE/SSE's(Signal) Completion Certificate*
10	*3/10*	*Technician's (Signal) Report*
11	*3/11*	*Disconnection/Reconnection Notice*
12	*3/12*	*Situations in which disconnection notice need not be issued provided suitable precautions are taken*
13	*3/13*	*Competency Certificate cum training History book*

Annexure '3/1'

Form No.S.&T./FS.3.2

RAILWAY

SIGNAL & TELECOMMUNICATION DEPARTMENT

Failure Report of Signalling & Interlocking, Electrical Signalling and Block Signalling appliances for the month of ……..……Sections ……

Details of failures for which Signal Branch is responsible

S.No.	Station name (Code) / Date Time and No. of S.Ms. message	Date and Time of failure	Gear at Fault	Date and time failure message received by Technician	Date, time and Train No. Technician arrived.	Date and time failure rectified	Duration of failure	Trains / Train No.	Detained / Time lost	Classification of failure	Remarks of Inspector briefly indicating cause of failure, how rectified and any special repairs necessary to stop recurrence	Orders of Sr.D.S.T.E
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)

No…….. Date……..

Note:- For further details of failures for which Signal Branch is responsible, please see on the reverse.

Forwarded to …….

Date ……..... Signature……..

Station……

Date…… Designation……..

S.No. of failure (as given on the reverse side)	Date of last visit by Signal Technician	Name of Signal Technician and Head quarters	Date of Last Inspection by JE/Signal	Name of JE/Signal	Date of last inspection by SSE/Signal	Name of SSE/Signal	Description of failure	Remarks of SSE/JE/ Signal
(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)

Annexure '3/2'

S.No.	Classification Code	Department and nature
		(a) 'S' – Signal Department
1	S.M.	Signal maintenance inefficient.
2	S.E.	Signal Equipment-Defective Design, manufacture or excessive wear.
3	S.I.	Signal Installation incorrect.
4	S.O.	Signal-other failures.
		Total 'S' – Signal Department
		(b) 'L' – Electrical Department
5	L.P.	Electrical-Power supply failure.
6	L.F.	Electrical-Fluctuation in voltage.
7	L.O.	Electrical-Other acts of employees of this Department.
		Total 'L' – Electrical Department
		(c) 'B' – BSNL Department
8	B.T.	BSNL.-Theft of line wire.
9	B.C.	BSNL.-Contact of line.
10	B.B.	BSNL.-Break of line.
11	B.E.	BSNL-Earth on line.
12	B.P.	BSNL.-Party working on line.
13	B.T.	BSNL.-Insulation Imperfect.
14	B.C.	BSNL.-Leakage of extraneous current.
15	B.B.	BSNL.-Other acts of employees of this Department.
		Total 'B' – BSNL Department
		(d) 'M' – Miscreants
		Theft.
16	M.T.	Damages.
17	M.D.	Interference.
18	M.I.	
		Total 'M' –Miscreants.
		(e) 'X' – Other Classifications
		Weather extremes like Storms, etc.
19	X.W.	Unknown cause –Not established.
20	X.X	Fire
21	X.F	
		Total 'X' –Other classification.

Annexure '3/3' Form No.S.&T./FP.3.3

……RAILWAY

Foot Plate Inspection of Signals

RECORD OF OBSERVATION

1. Date :	Loco Pilot Equipment Spectacles
2. Section :	Detonators
3. Train No.:	14. First Aid Box
4. Engine No.:	Other Equipments
5. Loco pilot Name :	15. Headlight
6. Asst. Loco Pilot Name :	16. ……. Observance of Speed Restrictions and Caution Orders
7. Guard's Name:	17. …… …… Placing of Engineering Signals
8. Weather : Clear/Cloudy/Foggy	18. Whistling while approaching Level Crossing Gate and 'W' Board
9. Visibility : Good/Fair/Poor	
10. Long/Short Hood Leading:	Remarks:
11. Speedometer	
12. ALP/LP calling out signals	
13. Vigilance Control Device	Note Yes No

Further Remarks :

Annexure '3/3'

Page 2

Station																					
Time Arrival/ Departure/Passing																					
Signal		D W	O	H	S	A S	D W	O	H	S	AS	D W	O	H	S	A S	D W	O	H	S	A S
Aspect	Y																				
	G																				
	Y																				
	R																				
Visibility	Good																				
	Adequate																				
	Insufficient																				
	Reason (Code)																				
Light	Good																				
	Dim																				
	Out of focus																				
	Out																				
Interference from fixed light/Tree branches/other structures																					
Loco Pilot following PSR/TSR																					
Lamp Torch displayed																					
Flag/ Light Showing	Loco Pilot Station																				
	Off																				
	Guard																				
	Cabin																				
	LC gate																				
Adequacy of lights in cabins																					
Lighting Board																					

*Code B-Background, O-Obstruction, T-Tree, L-Light.

Annexure '3/4' Form No.S.&T./IC.3.4

…….RAILWAY
SIGNAL & TELECOMMUNICATION DEPARTMENT

Month……..……20

To,

The Sr DSTE,……

Through……..

This is to certify that I have personally inspected and tested all the Signals and Telecommunication gears at the following stations/section on my jurisdiction. The details of inspection carried out are shown below-

Date	Station/ Location Visited	Nature of Inspection and Time Spent	Gear		Repairs done during the month	Repairs required	Remarks
			Type	Condition			

Signature ……...

Designation…..…...……..

Date…….……

Annexure '3/5'

1. Proceeding to Site of Accident:

1.1. On receipt of information of the occurrence of an accident, the SSE/JE/Signal shall proceed by the Quickest-available means to the scene of the accident, with staff and tools and portable control telephones and establish immediate communication with control.

1.2. He shall make a special note of the exact time of the accident, besides other important details The SE/JE/Signal shall check his watch, if available with Guard's time, station time and control time, as the exact time of an accident is often an important clue in determining the cause of the accident.

2. Action at Site.—If the Signal Inspector is not the senior-most official present at the site of the accident he shall report to the senior-most official and render all possible assistance to him in taking action detailed in Paras 2.1 to 2.8. If however he is the senior-most official on the spot he shall himself initiate such action.

2.1. Check that the obstructed line including adjacent lines on a multiple line section is protected as per rules ;

2.2. Arrange for medical aid to the injured taking the help of medical practitioners, if available;

2.3. After a rapid survey of the position, send brief particulars to the nearest SM to enable him to issue the all concerned message;

2.4. Record the statements of staff and take whatever steps may be necessary to record or preserve the evidence which subsequently may not be available. This shall be done by an officer or SSE/JE/Signal present at the site of the accident even if he is off duty. Evidence shall be collected and recorded with due care and attention, and later submitted to the Enquiry committee.

2.5. Preserve all clues and record the character and relative position of markings to enable reconstruction of the scene of the accident. Where immediate repairs are necessary to pass trains, only such parts of the interlocking gear shall be disconnected as are necessary to carry out the repairs to the road;

2.6. Scrutinise and initial jointly with the concerned officials noting time and date the Train Register Books or Train Message Books, Private Number Books, Caution Orders, S. & T./MR, S. & T./DN forms concerned and any other relevant records. Secure and seal immediately the records directly connected with the cause of the accident.

2.7. Devote special care and attention in the collection and recording of evidence as prescribed and in submitting the evidence with all details to the Enquiry Committee;

Note:- The object of recording all available data at site and presenting the evidence at the enquiry is to ascertain the cause of the accident with a view to prevent its recurrence and, where the cause is due to negligence, to fix responsibility

2.8. Report the position at site of the accident to the Control Office and to the Sr. DSTE by the quickest possible means.

3. Special action by SSE/JE/Signal — The SSE/JE/Signal shall pay special attention to the following:—

3.1. Where there is a possibility of signalling being involved, he shall record the following details which shall be attested by officials of other Departments concerned who are present at the site of the accident:

Position of levers/Knobs, VDU condition, SM control slides, indication of various points, track circuits, routes, slot indicators, etc., position of block instrument handles and indications, if any, aspects of the concerned signals, reading of the counters where provided and conditions of seals of appliances which are normally kept sealed and any other detail relevant to the accident.

3.2. The Signal Inspector shall along with the other officials at site seal up affected interlocking gear such as lever frames, cabins, relay rooms, signal location boxes etc ., immediately an accident occurs in order that these equipments are not interfered with, until necessary tests have been carried out. The seals shall not be removed until authorised by senior-most official present at site after conducting necessary tests and recording all necessary facts.

3.3. Furnish a full report to his Sr. DSTE by the quickest possible means.

3.4. After getting permission to restore signalling, every effort shall be made to complete the restoration as expeditiously as possible.

Annexure '3/6'

Hours of Employment Regulations

1. Application. - Hours of Employment Regulations will apply to all (non-gazetted) railway servants including casual labour and substitutes except those governed under the Factories Act, the Mines Act, Minimum Wages Act and the Merchant Shipping Act.

2. Classification.— Railway servants governed under Hours of Employment Regulations are classified as under—

(i) Intensive;
(ii) Essentially intermittent;
(iii) Excluded; and
(iv) Continuous.

2.1. Intensive.- Die employment of a railway servant is said to be Intensive when it has been declared to be so by the prescribed authority on the ground that it is of a strenuous nature involving continued concentration or hard manual labour with little or no period of relaxation.

The work of a railway servant shall be held to be intensive if it is so declared by the Head of the Railway on the ground that it involves sustained and strenuous attention or physical exertion—

Such that periods of rest, inaction or relaxation do not aggregate to 6 hours or more in a cycle of 24 hours, and

Such that, in any shift of 8 hours he does not get periods of inaction, rest or relaxation of at least one hour in the aggregate.

2.2. Essentially Intermittent,— The employment of a railway servant is said to be Essentially Intermittent when it has been declared to bo so by the prescribed authority on the ground that the daily hours of duty of the railway servant normally include periods of inaction aggregating 6 hours or more (including at least one such period of not less than one hour or two such periods of not less than half an hour each) during which the railway servant may be on duty but is not called upon to display either physical activity or sustained attention.

2.3. Excluded.—The employment of a railway servant is said to be ' excluded ' if he belongs to one of the following categories namely :—

(i) Staff employed in a confidential capacity;
(ii) Armed guards or other personnel subject to discipline similar to that of the armed police forces;
(iii) Staff of Railway Schools imparting technical training or academic education;
(iv) Staff classified as supervisory; and
(v) Such categories of staff of Health and Medical Department as may be specified by the prescribed authority.

3. Main provisions as regards Hours of Employment are as follows :—

The staff governed by Hours of Employment Regulations will be employed as per the rostered hours of duty which will in no way exceed the statutory limit of hours of employment. The statutory limit of hours of employment and the rostered hours of duty will be asunder :—

Statutory limit of hours of employment	Rostered hours of duty
(a) A railway servant whose employment is Intensive shall not be employed for more than 45 hours a week and he shall be granted each week commencing an Sunday, a periodical rest of not less than 30 consecutive hours.	(i) The hours of work shall consist of standard hours of duty as under- (a) 42 hours a week in case of Intensive workers, (b) 48 hours a week in case of Continuous Workers (c) In case of E. I. Workers 48 hours a week plus additional hours of work per week which will be subject to the prescribed maximum.
(b) A Railway servant whose employment is Essentially Intermittent shall not be employed for more than 75 hours in any week and he shall be granted each week, commencing on Sunday, a periodical rest of not less than 24 consecutive hours including a full night.	(ii) In addition to working hours for the three classifications of workers as in (i) above, the time required for Preparatory and/or Complementary work which includes the work, if any, of handing and/or taking over charge and has necessarily to be carried on outside the standard duty hours vide (i) above
(c) A railway servant whose employment is continuous shall not be employed for more than 54 hours a week and he shall be granted each week commencing on Sunday, a periodical rest of not less than 30 consecutive hours.	(d) Office staff are rostered for 8 hours a day including half-an-hour's break, i.e., 7-1/2 hours a day. They enjoy weekly rest on Saturdays and Sundays and holidays as declared by the administration
(d) The staff treated as excluded shall have no limitation of hours of employment and there is no statutory provision for their periodic rest.	

Annexure '3/6' - contd.

Hours of Employment Regulations

4. **Preparation of Duty Rosters. :**

4.1. Hours of duty will be exhibited in the duty rosters. (Duties may be shown as running at a stretch or these may be covered in split shifts). (No duty roster should offend the principle of hours of work prescribed for concerned workers or should offend the rules against' long ons' and ' short offs'). Every roster must provide periodic rest as per rules to be enjoyed within the week commencing from Sunday midnight and ending with Saturday midnight. Weekly rest must nor be spread over two weeks. Duty rosters for Continuous workers, covered by the principle of averaging, should be prepared on the basis of 48 hours a week on the average in two weeks plus time for comp/prep. work, if any. Rosters for those Continuous workers who are not covered by the principles of averaging should be framed in such a way that their maximum daily statutory hours are not exceeded. Duty Rosters for Intensive and E. I. workers should also be framed in the same way in conformity with the two-weekly/weekly average number of hours as prescribed. As fas as possible, duty rosters should be framed avoiding overtime work as a regular feature. Duty Rosters of Rest Givers should also be framed in the same way.

4.2. Short off.—A 'Short off' means a short period of rest between the end of one rostered duty and the beginning of the following rostered duty. It is a period of rest where it is less than 12 hours in a roster of 6 hours' duty and less than 14 hours in a roster mixed of 6 and 8 hours' duty in the case of Intensive workers, less than 10 hours in the case of Continuous workers and less than 8 hours in the case of E. I. workers (No railway servant should normally be called on duty unless he has enjoyed rest as above).

4.3. Long on.—A 'long o' is a period of duty (exclusive of the period or periods of breaks) for long hours and spread over 8 hours in the case of Intensive workers, over 10 hours in the case of Continuous workers and over 12 hours if the ease of E. I. workers.

4.4. Split shifts.—Split shift means a shift which is punctuated in the daily tour of duty by breaks during which the employee is free to leave his place of work. It is a shift which is split up into two or more parts by intervals (not periods of inaction).

4.5. Night duty.— Night duty means employment during any part of the night from 10.00 p. m. to 6.00 a. m. Full night means the period between 10.00 p. m. and 6.00 a. m. Employment of railway servants on continuous night duty should be avoided.

4.6. Work beyond ceiling limits.- All Railway workers governed by Hours of Employment Regulations can be called upon to render duty beyond statutory limits applicable to them by an order of temporary exemption made under Section 71-C of the Indian Railway Act.

Workmen’s Compensation Act

1. **Application:**

1.1. The Workmen's Compensation Act applies to all railway servants as also labourers employed by contractors for the purpose of carrying out railway work but excluding those employed in an Administrative, District or Sub-Divisional Office and those mentioned in Schedule II of the Act whose monthly wages exceed Rs. 500.

1.2. The Workmen's Compensation Act does not apply to staff employed hi foreign territories and no compensation under the Act is, therefore, legally admissible in respect of accidents occurring in those foreign territories.

2. Liability for compensation.—Compensation under Workmen's Compensation Act is payable only when a personal injury is caused to a workman by accident arising out of and in the course of his employment. The Administration shall not, however, be liable to pay any compensation.

2.1. In respect of any injury which does not result in the total or partial disablement of the workman for a period exceeding three days;

2.2. In respect of any injury not resulting in death caused by an accident which is directly attributable to—

2.2.1. the workman having been at the time thereof under the influence of drink or drugs, or

2.2.2. the wilful disobedience of the workman to an order expressly given or to a rule expressly framed, for the purpose of securing the safety of a workman, or

2.2.3. the wilful removal or disregard by the workman of any safety guard or other device which he knew to have been provided for the purpose of securing the safety of workmen.

3. Notice to the Commissioner—

3.1. In the case of accidents, which result hi the death of railway servants or Contractor's servants doing work for the railway, employed otherwise than in clerical capacity, within railway premises, the Divisional Officer concerned shall, within seven days of the date of death, send a report to the Commissioner in the prescribed form. In the case of staff governed by the Factories Act, etc, the forms laid down in those Acts for this purpose shall be used for reporting the fact to the Commissioner.

3.2. This notice shall always be sent irrespective of whether the deceased workman, was or was not on duty at the time-In the latter case the Commissioner's attention shall be drawn to it specifically.

3.3. Copies of such reports shall be sent to the General Manager and the Head of the Department.

Annexure '3/6' - contd.

Payment of Wages Act

1. Application.—The Payment of Wages Act applies to the payment of wages to persons employed in any factory and to persons employed (other than in a factory) upon any railway by a Railway Administration or, either directly or through a subcontractor, by a person fulfilling a contract with a Railway Administration and employing or having employed twenty or more persons on any day of the preceding twelve months.

2. Wages.—Means all remuneration (whether by way of salary allowance or otherwise) expressed in terms of money or capable of being so expressed which would, if the terms of employment, express or implied, were fulfilled, be payable to a person employed in respect of his employment or of work done in such employment.

3. Main provisions of the Act—

3.1. Responsibility far Payment of Wages.—Every employer shall be responsible for the payment to persons employed by him of all wages to be paid under this Act. In factories, the person named as the Manager of the factory will be responsible for the payment.

3.2. Fixation of wage periods.—Every person responsible for the payment of wages to persons employed by him shall fix periods in respect of which such wages shall be payable and no wage period shall exceed one month.

3.3. Time of payment of wages.—Wages shall be paid on a working day within 7 days of the last day of the wage period in an establishment where less than 1000 men are employed and in other establishments before the expiry of the tenth day.

When the employment of any person is terminated by or on behalf of the employer, the wages earned by him shall be paid before the expiry of the second working day from the day on which his employment is terminated.

3.4. Deductions which may be made from wages.- The wages of an employed person shall be paid to him without deductions of any kind except those authorised by or under this Act.

3.5. Deductions for absence from duty.—Deductions may be made only on account of the absence of an employed person from the place or places where he is required to work, such absence being for the whole or any part of the period during which he is so required to work.

3.6. Deductions for damage or loss.—A deduction for damage to or loss of goods not exceeding the amount of the damage or loss caused to the employer by neglect or default of the employed person. Deductions under this clause shall not be made unless it is established—

3.6.1. that the money or goods lost or damaged was expressly entrusted to the custody of the employees concerned, and

3.6.2. that the cause of such loss or damage is directly attributable to his neglect or default

3.7. Display of notices.—The person responsible for the payment of wages to men employed in a factory shall cause so be displayed in such a factory a notice containing such abstract of this act and of the rules made thereunder in English and in the language of the majority of persons in the factory as may be. prescribed by the Government.

Form No.S&T./SC 3.7

Annexure '3/7'

…….RAILWAY

SIGNAL & TELECOMMUNICATION DEPARTMENT

Signal Sighting Committee Report

Division..……...

Name of Work..……..

Sighting of..……

We, the undersigned members of the Signal Sighting Committee having met at ……..……...and sighted the view of the Signals noted, below recommend that their dimensions and locations should be given against each -

Name of the station and Ref. of Signal (giving Up or Down)	Height of existing Signal above rail level	Height of proposed Signal above rail level	Distance in meters of the existing Signal from the adjacent track	Distance in meters of the proposed Signal from the adjacent track	Distance in meters of the existing Signal from a fixed track	Distance in meters of the proposed Signal from a fixed track	Is the existing Signal on the left hand or right hand side of the track to which it refers	Is the proposed Signal on the left hand or right hand side of the track to which it refers	Visibility of the proposed Signals from Drivers point of view in normal weather (Meters)	Are light indicator and arm repeater provided to the existing signal	Are light indicator and arm repeater required for the new signal	Remarks
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)

Signature …….. Signature …….. Signature …….

Designation …….. Designation …….. Designation …….

Date …….. Date …….. Date ……..

Annexure '3/8' Form No.S.&T./PR/3.8

…….RAILWAY

SIGNAL & TELECOMMUNICATION DEPARTMENT

…….Division

JE/SSE's (Signal) Progress Report

Monthly Progress Report of…….JE/SSE's (Signal)……..for the month ending……20

S.No.	Name of work	Reference to		Date work started	Percentage of work done during the month	Total percentage of work done upto the end of last month	Remarks
		Authority	Sanctioned estimate				

Signature ……...

Designation…..……..

Annexure '3/9'

…….RAILWAY

SIGNAL & TELECOMMUNICATION DEPARTMENT

JE/SSE's(Signal) Completion Certificate

No…….

To,

The Sr. DSTE,……

This is to certify that work at ……..in connection with…… …….. has been carried out correctly in accordance with-

(a) Approved plans as under-
   i) Interlocking and Yard Plan No…….……. dated……
   ii) Interlocking Table No.....……. dated……
   iii) Interlocking Chart No…..……..…..……. dated……
   iv) Wiring Diagram No……..……. dated……

(b) Instructions contained in the Signal Engineering Manual and Schedule of Standard Dimensions except deviations as sanctioned . *Vide* your letter No……. dated..……..

2. The work was brought into use on…..…….. 20…….…….

Dated……..20 ..……..…..(Supervisor).

..……..…..(Supervisor).

FORM NO. S&T/MR

Annexure '3/10'

SIGNAL & TELECOMMUNICATION DEPARTMENT

Technician's (Signal) Report

Day/Night Duty for Week/Fortnight/Month ending.....

Name of Technician (Signal)..... Hqrs. of Technician Signal.....

Section.....

Date of Visit	Name of Station/Cabin	Train No. and time arrived	Train No. and time departed	Maintenance Work Done
1	2	3	4	5

Signature of Technician/Signal

Date.....

Annexure 3/11

Disconnection / Reconnection Notice

FORM NO. S&T/DN

_____ RAILWAY

Signal & Telecommunication Department

Acknowledgement of Disconn/ Reconn Notice

No.

To

The Station Master or Cabinman on duty at

..... Station/cabin

For disconnecting the following gear

*.....

on ..... at ..... hrs.

Tech./JE/SE(Sig)

Notice about disconnection received at ..... hrs.

on 20

SM/Cabinman

Disconnection allowed/not allowed** at ..... hrs.

on 20

SM/Cabinman

Disconnection will be allowed at ..... hrs.

on 20

SM/Cabinman

Reconnected at .....hrs. on 20

Tech./JE/SE(Sig)

Notice about reconnecting received at ..... hrs.

on 20

SM/Cabinman

FORM NO. S&T/DN

_____ RAILWAY

Signal & Telecommunication Department

Reconnection Notice

No.

Notice for reconnecting Signalling Gear already disconnected

Division/District .....

To

The Station Master or Cabinman on duty at

..... Station/Cabin

Please note that the disconnected gear referred to in Notice No ..... has since been reconnected on .... at ..... hrs.

Date

Signature

Designation

FORM NO.S&T/DN, Annexure 3/11
Para No. 3.8.31.1

FORM NO.S&T/DN

_____ RAILWAY

Signal & Telecommunication Department

Disconnection Notice

No.

Notice to Transportation Staff for disconnecting Signalling gear

Division/District .....

To

The Station Master or Cabinman on duty at

..... Station/Cabin

Please note that the following gear will be disconnected on ..... at ..... hrs.

*

Date

Signature

Designation

* Fill in details of Gear to be disconnected, * * Reason for not allowing disconnection to be recorded.

Annexure '3/12'

Situations in which disconnection notice need not be issued provided suitable precautions are taken:

A.3.1 Cleaning and/or Lubricating /Graphiting without affecting any Physical Electrical Disconnection

i) Lever frame basement.

ii) Relay Room & Battery Room.

iii) Apparatus case/battery boxes/Goomty.

iv) Lifting barrier mechanism including winch.

v) Lever lock and circuit controller without opening cover.

vi) Inspection and cleaning of inside equipment by opening the covers of point machines, signal motors, reversers, apparatus cases and detectors.

vii) Point, facing point lock and lock bar.

viii) Cranks, compensators, pullies, wheels, roller stands, counterweights & levers.

ix) Power supply equipment.

x) Insulation joints.

xi) Lens/roundels of signal, point indicator, trap indicator & shunt permitting indicator provided phantom indication to driver of an incoming train is prevented by covering lenses.

A.3.2 Testing of

i) Track locking, approach locking, back locking, indication locking, route release.

ii) Checking various parameters of axle counter without disconnecting the equipment.

iii) Power supply equipment.

iv) Lever frame, SM's control frame, signal operation, point operation, level crossing gate & slot circuits.

v) Point by obstruction test for lock only.

vi) Checking and testing of track circuit parameters when the track is unoccupied,

vii) Focussing of colour light signal provided phantom indication to Loco pilot of an incoming train is prevented by covering lenses.

A.3.3 Adjustment of

i) Wire transmission except double wire operated point transmission, where existing

ii) Tightening of terminals using insulated tools without causing any shorts on adjacent terminals.

A.3.4 Replacement of

i) Electric signal lamp provided phantom indication to driver of an incoming train is prevented by covering lenses.

ii) Bond wire, one at a time,

iii) Plug-in relay in case traffic condition permits,

iv) Indicator lamps,

v) Push button/switch/key of panel,

vi) Lock bar clips/bar stop, one at a time,

vii) G. D. tubes,

viii) Fuse, one at a time.

ix) Pulley, bottom roller/ top roller in wire and rod transmission one at a time and split pins,

x) Batteries /cells without affecting disconnection of supply to main equipment.

Annexure '3/13'

.....Railway

SIGNAL AND TELECOMMUNICATION DEPARTMENT
Competency Certificate cum training History book

Photograph (Attested by Gazetted Officer)

Certificate number.....

This is to certify that

Shri	.
S/o Shri	.
Designation	.
Staff No.	.
Station	.
Division	.

who has undergone training in.....has been examined in the relevant rules and instructions pertaining to his duties stipulated in the Signal Engineering Manual & General and subsidiary rules and he is competent to undertake independently such work which may necessitates disconnection of signalling and interlocking gear in service under open line conditions.

This certificate is valid upto.....

He is also qualified to work on the following block instruments and is competent to disconnect and restore working (Tick the appropriate)

1	Double line block instrument	.
2	Single line token block instrument	.
3	Single line push button block instrument	.
4	Single line handle type block instrument	.
5	.	.
6	.	.

Date:.....
Place:.....

Principal
S& T Training centre

.....Railway

RENEWALS OF COMPETENCY CERTIFICATE

Date	Renewed upto	Block Instruments		Signature
		Added	Deleted	

REFRESHER COURSE

S.No	Course No.	Last attended		Next Due	Signature
		From	To		

EQUIPMENT AND OTHER COURSE ATTENDED

S.No.	Date	Name of Course	From	To	Signature

CHAPTER- 4 BUDGET ESTIMATES, EXPENDITURE AND BUDGET CONTROL AND COMPLETION REPORTS

Section 1 : Introduction

4.1.1.0 **Budget Terms**

4.1.1.1 **Budget Estimates :-** Every Railway Administration has to prepare estimates of expenditure expected to be incurred by it in a year and submit them on prescribed dates well in advance of the beginning of that year to the Railway Board for obtaining the sanction of the Parliament or the President. Such estimates are called "Budget Estimates".

4.1.1.2 **Demands for Grants :-** On the basis of the Budget Estimates received from the various Railway Administrations and other spending units, the Railway Board prepare their " Demand for Grants " and present them to the Parliament or the President, as the case may be, for sanction.

4.1.1.3 **Grants and Appropriations :-** The 'Demands for Grants' as finally approved by the Parliament are called the Budget 'Grants' and those sanctioned by the President without reference to the Parliament are called 'Appropriations '

4.1.1.4 **Budget Orders and Allotments.**

(a) The Railway Grants and Appropriations for a year are distributed by the Railway Board to the various Railway Administrations and other spending units directly under the Railway Board through what are known as 'Budget Orders'. The Budget Orders are accompanied by the Budget Documents.

(b) The Grants and Appropriations distributed to the various Divisions and spending units are called Budget ' Allotments

4.1.1.5 **Budget Documents :-** The complete Budget, is printed in the form of books as under :

(a) Demands for Grants

(b) Explanatory Memorandum showing the financial results of the year

(c) Works, Machinery and Rolling Stock Programme of Railways.

4.1.1.6 **Re-appropriation :-** The transfer of funds originally assigned for expenditure on a specific object to supplement the funds sanctioned for another object is called "Re-appropriation". The powers of the Railway Board and Railway Administration in respect of re-appropriation are detailed in Chapter III of the Indian Railway Financial Code Volume I (extracts at Annexure '4/1').

4.1.1.7 **Review of Expenditure :-** The sufficiency or otherwise of the sanctioned budget allotments shall be reviewed periodically by each Railway Administration on the basis of expenditure actually incurred upto the time of review and the anticipated expenditure during the remaining portion of the financial year. The detailed procedure in regard to review of Expenditure is contained in Chapter III of the Indian Railway Financial Code, Volume I (extracts at Annexure '4/2 ').

4.1.2.0 **Demands for Grants :-** All revenue working expenses of the Railway are classified under 13 sub major heads with separate abstract for each sub major head. The sub major heads are divided into minor, sub and detailed heads as detailed in Volume II of the Indian Railway Financial Code (Minor, sub and detailed heads, relevant to Signaling and Telecommunication are at Annexure ' 4/3')

4.1.3.0 **Revenue and Works Budgets :-** Budget estimates of the Signal and Telecommunication Department comprise mainly of the following :-

(a) Estimates of Working Expenses :- These are based on the Working Expenses for the repairs and maintenance of Signaling, interlocking and Telecommunication Services and comprise estimates of expenditure against Demand Nos. 3, 7 and 8 referred to in para 4.1.2.0.

(b) Works Budget :- Open Line Works Budget otherwise known as 'Annual Works Programme' comprises estimates of requirements for Works against Demand No. 16.

(c) Plant and Machinery Programme comprises estimates of requirements for Machinery and Plant against Demand No. 16.

Section 2 : Estimates for Working Expenses or Revenue Budget

4.2.4.0 **Compilation:-**

4.2.4.1 The Revenue Budget or Estimate of Working Expenses is compiled in the prescribed forms in the Headquarters Office on the basis of proposals submitted by the Divisions/Extra Divisional Units in accordance with instructions contained in Chapter III of the Indian Railway Financial Code. Volume I (extracts at Annexure '4/4')

Section 3 : Works, Machinery and Rolling Stock Budget

4.3.5.0 Detailed instructions for preparation and submission of Works, Machinery and Rolling Stock Budget are contained in Chapter III of Indian Railway Financial Code, Volume I (extracts at Annexure '4/5'). The classification of expenditure by primary units of expenditure are contained in Volume II of the Financial Code (Heads relevant to Signal and Telecommunication Department at Annexure '4/6').

4.3.6.0 **Compilation of Plant and Machinery Programme**

4.3.6.1 The detailed instructions for the preparation and submission of the Plant and Machinery Programme are contained in Chapter XV of the Indian Railway Code for the Mechanical Department (Workshops).

4.3.6.2 The requirements of machinery in Signaling and Telecommunication Workshops shall be collected and shown in Plant and Machinery Programme. The description of machinery, number, allocation, cost etc., shall be shown on the prescribed forms and submitted along with the programme. This programme should be as per the schedule of Works Programme.

Section 4 : Administering the Budget

4.4.7.0 Detailed instructions for administering the Budget are contained in Chapter III of the Indian Railway Financial Code (extracts at Annexure '4/7').

Section 5 : Maintenance of Accounts

4.5.8.0 **Code Rules :-** Rules in regard to the maintenance of Open Line accounts are contained in Chapter XIV of the Indian Railway Code for the Engineering Department (extracts at Annexure '-').

4.5.9.0 **Accounting**

4.5.9.1 The expenditure incurred is brought to account through various vouchers. Subject to such powers as may be delegated to him, the Divisional Signal and Telecommunication Engineer in-charge may incur expenditure on all ordinary maintenance works and establishments upto the limits of the provision in the Revenue Budget Estimates, sanctioned for his unit for the year.

4.5.9.2 All vouchers, such as debit notes for stores supplied, freight bills shall be carefully scrutinised before acceptance as to the reasonableness and propriety of the debits raised. Wherever necessary, such debits shall be correlated and checked with the initial records in the office, such as indents and work orders on Workshops, and necessary endorsements made thereon.

4.5.10.0 **Departmental Registers on Extra-Divisional Units**

4.5.10.1 **Revenue Allocation Register :-** The Divisional Signal and Telecommunication Engineer in charge of divisional or an Extra-Divisional Unit shall maintain a Register of Revenue expenditure, if any, incurred by him under the relevant grants in order to keep himself acquainted with the progress of expenditure in his unit as compared with the sanction in the Authorization Rolls. The detailed instructions in regard to the maintenance of Revenue Allocation Register are contained in Para 1469 of Engineering Code (Annexure ‘4/8 ')

4.5.10.2 **Register of Works :-** The object of maintaining the Register of Works and Instructions in regard to maintenance of the same are contained in Paras 1472, 1473, 1474 and 1475 of Engineering Code (Annexure '4/9')

4.5.10.3 **Reconciliation of Registers :-** The registers of the Divisional Office shall be sent monthly to the Accounts Office with a clerk on such dates as mutually agreed upon in order that the entries made therein may be reconciled with similar registers maintained in the Accounts Office.

4.5.11.0 **Departmental Registers on Divisions :-** On Divisions where the initial accounts are maintained by the Accounts Department, neither Allocation Register nor Register of Works need be maintained by the Departmental Officers. There will be only one set of registers and these will be maintained by the Accounts Officer both for his own use and for that of the other Departmental Officers, who will continue to be responsible for the efficient control of expenditure against estimates and allotments.

Section 6 : Control over Expenditure

4.6.12.0 The manner in which the budgetary and expenditure control is to be exercised by the various units is indicated in Chapter V of the Indian Railway Financial Code (relevant extracts at Annexure ' 4/10').

4.6.13.0 **Indents and Work Orders**

4.6.13.1 When passing requisitions for materials and tools, for works or maintenance, and also while sanctioning temporary and seasonal labour, the Sr. Divisional Signal and Telecommunication Engineer in charge shall ensure that the amounts involved are within the allotments and also within provision in the sanctioned estimates.

4.6.13.2 Before execution of agreements and work orders with contractors the Divisional Signal and Telecommunication Engineer shall ensure that these have been carefully checked in every respect. Rates allowed in bills received from firms shall be carefully scrutinized before they are passed for payments.

Section 7 : Completion Reports

4.7.14.0 **Code Rules :-** Rules in regard to the preparation and submission of Completion Reports are contained in Chapter XVII of the Indian Railway Code for the Engineering Department (relevant extracts at Annexure '4/11').

☞ Note :- This Chapter has under mentioned Annexures for further study		
S.no	Annexure no	Description
1	4/1	Powers of the Railway Board
2	4/2	Review of Expenditure
3	4/3	Revenue Expenditure
4	4/4	Preparation of the Budget
5	4/5	Works Machinery and Rolling Stock Budget
6	4/6	Classification of Capital and other Works Expenditure
7	4/7	Distribution of Funds by the Railway Board
8	4/8	Revenue Allocation Registers
9	4/9	Register of works is a collective record of expenditure designed
10	4/10	Revenue and Works Expenditure
11	4/11	Completion Estimates

Annexure '4/1'

(**1) Powers of the Railway Board:** Within the amount of a grant as voted by the Parliament, the Railway Board have full power of transferring the provision from one sub-head to another by a formal order of re-appropriation but re-appropriations, from one grant to another are not permissible. Under Grant No. 16 no re-appropriation of funds is permissible between Capital, Railway Funds and Revenue even though re-appropriation is permissible between the various sub-heads of grant viz., the various plan heads. As regards "Charged" expenditure there are no restrictions on the powers of the Railway Board to transfer provision from one sub-head to another by a formal order of re-appropriation but no re-appropriation is permissible from one grant to another from " Charged " heads to " Voted " heads or Vice Versa. (Para 376)

(2) Powers of Railway Administrations: No re-appropriation is permissible between " Voted " and " Charged " allotments or between the allotments made under one grant and another. In the case of Grant No. 16 no re-appropriation is permissible between the Capital Railway funds and revenue. The re-appropriations. (Para 377)

(i) to and from the provision for the following plan heads under Grant No. 16 :—

(a) New Lines (Construction);
(b) Gauge Conversion;
(c) Electrification Projects;
(d) Track renewals ;
(e) Staff quarters, } Staff Welfare Works.
(f) Amenities for staff } Staff Welfare Works.
(g) Passenger Amenities and other Railway Users' amenities.

(ii) to and from the provision for " Payments to other Railways " in Grant No. 9, and ;

(iii) to and from the provision for " Suspense " in Grant No, 12 require the prior approval of the Railway Board.

(3) The amount allotted to the Railway administrations under the Civil Grant for " Interest on debt and other obligations and reduction or avoidance of debt " is distributed under the following units of re-appropriations, viz.,

(i) State Provident Funds ;
(ii) Railway Staff Benefit Fund ;
(iii) Eastern Group Sleeper Control Provident Fund ;
(iv) Indian Railways Conference Association Employee's Provident Fund ;
(v) Depreciation Reserve Fund, Railways ;
(vi) Revenue Reserve Fund ;
(vii) Railway Development Fund ;
(viii) Accident Compensation, Safety and Passenger Amenities Fund, and
(ix) Pension Fund.
No re-appropriation is permissible from one unit to another.

(4) Other re-appropriations may be sanctioned by the Railway Administrations but no re-appropriations are permissible after the close of the financial year i. e., 31st March. The re-appropriations by Railway administrations should not be made haphazardly on the basis of individual items of expenditure where the original provision is exceeded, nor should they be postponed to be made only towards the end of the year. The Railway administration should review the position as a whole at intervals and carry out the necessary re-appropriations. When funds have to be provided for new expenditure under one of the sub-heads and the administrations are definitely in a position to transfer the grant from another sub-head for this purpose, the appropriations should be made promptly. The whole object is to ensure that, as far as possible, funds which are not required are withdrawn from disbursing officers as soon as it is definitely

known that they are not required, and incidentally to provide that any really unavoidable expenditure is met from such savings as far as possible. (Para 379).

(5) All proposals far re-appropriation of funds which are beyond the General Manager's powers shall be submitted to the Railway Board sufficiently early to admit of action being taken before the close of the financial year. (Para 380)

Annexure '4/2'

Form No. F. 383

(1) Review of Expenditure :—Railway Administrations should review their expenditure in August to see whether any modifications are necessary in the allotments placed at their disposal. The review in respect of each grant should be submitted to the Railway Board in Form No. F-383 so as to reach them not later than 1st September each year. (Para 383)

Statement showing ……..necessary in the allotments under Grant No……..…..during…….

(in thousands of rupees)												
Detailed heads	Actuals last year		Latest Estimate, Current Year					Amount advanced from the Contingency Fund	Total Columns 8 and 9	Variations between Columns (10 and 7)		Brief reasons for large variation
	First Four months	Full year	Actuals first-three month	Approximates for month	Total for first four month	Full year	Budget Current year			Excess (+)	Savings (-)	
1	2	3	4	5	6	7	8	9	10	11	12	13
Detailed heads of the Grant Total…												

(2) The Administrations should review the position in as much detail as possible at the time, and if there is any new expenditure which was definitely not anticipated in the budget and which cannot be postponed without serious detriment to safety or efficiency, they should examine whether such expenditure can be met by saving in expenditure provided for in the budget, when such savings can be foreseen or can be achieved without serious damage. The review should show whether such examination discloses the necessity of additional grants or the possibilities of net savings, in order to enable the Board to set off savings on one railway against excesses on another and to arrive at one estimate of the net additional grant required if any. (Para 384)

(3) **Final Modification Statement.**— The Railway Board should be furnished, for each grant separately, so as to reach them not later than the 21st February each year, with statements showing the additional allotments required (both voted and charged) or surrenders to be made, during the cur rent financial year under each head of appropriation, as prescribed in the budget orders, and requiring the sanction of the President. The variations

between the Final modified allotment required and the revised estimates as fixed by the Board should in all cases be supported by adequate explanations of the reasons for the demand or surrender. (Para 315)

(4) The railways should, however, continue to review the budgetary position further on submission of the Final Modification statements and any modification that may be considered necessary as a result of new factors visualized and the further review of budgetary position should be advised telegraphically to the Board so as to reach them on the 1st of March or any other date in March which would be advised to the Railways by the Board. (Para 386)

(5) Any important changes subsequently found necessary, should be intimated before the 20th March of the year, to enable President to accord sanction to them where possible, so that the Railway Administration may sanction in time, before the 31st March of the year, any re-appropriations to cover the anticipated excess over allotments. (Para 388)

Annexure '4/3'

Revenue Expenditure

S.No	Demand No.	Abstract	Name of Demand	Main Heads		Sub-Heads	
				No.	Name	No.	Name
1	3	'A'	General Superintendance and Services.	800	Signal and Telecommunication Management	810 820 830 840 870 890	Administration. Signalling. Telecommunication Workshop, Drawing and Estimating. Miscellaneous Expenses. Credits and Recoveries.
2	7	'E'	Repairs and Maintenance Plant and Equipment	100	Establishment in offices	110 120 130	Officers and Office Establishment (S&T) Supervisory Staff and their Office Staff / S.&T. Contingent Expenses / S. & T.
				500	Plant and Equipment (Signalling)	510 520 530 540 550	Mechanical Signalling Electrical Block Signalling Electrical Power Signalling Other Repairs and Maintenance. Miscellaneous Expenses.
				600	Plant and Equipment (Tele-communication).	610 620 630 640	Microwave Multichannel Radio Relay System and Wireless System. Railway Telephone Exchange. Line Communication System. Other Expenses.
				700	Rental to P. & T. for Signalling and Telecommunication Circuits.	710 720 730 740	Circuits of Signalling System. Line Wire. Administrative Telephone. V.F.T. Channel.
3	8	'F'	Operating Expenses, Rolling stock and equipment.	700	Signalling and Telecommunication.	730	Other Operating Expenses
4	12	'K'	Miscellaneous Working	500	Cost of Training Staff.	510	Other than Hindi, Medical Health and

			Expenses.				Welfare.
5	14	‘M’	Appropriation to Funds.	100	Appropriation to Depreciation Reserve Fund.	160	Signal and Telecommunication Department.
				500	Appropriation to Accident Compensation, Safety and Passenger Amenity Fund.	510	Appropriation to Accident Compensation, Safety and Passenger Amenity Fund.

Note:- Sub heads are further divided into Detailed Heads. The expenditure is classified into primary Units (objects) of Expenditure. For further details, Revised Classification of accounts of expenditure and earnings may be referred to.

Annexure '4/4'

(1) For the preparation of the Budget by the Railway Board, the Railway Administrations and other authorities empowered to incur expenditure are required to submit to the Railway Board their revised estimates for the current year and budget estimates for the following year on the dates prescribed below— (Para 307)

Revenue Demands 2 to 15 .. .. 1st December.

Works Demand 16 .. .. 23rd December.

The revised estimates are required in respect of the current year and Budget Estimates for the following year.

(2) Responsibility for framing the Estimates.— Preparation of the Revised and Budget Estimates should commence at the 'grass root level' i. e., Division, Workshop, Stores Depot, etc., as the case maybe. The entire responsibility for framing the estimates devolves upon the spending/earning authorities concerned, though the actual work of compilation and scrutiny would rest with the Financial Adviser and Chief Accounts Officer who would also draw the attention of the General Manager to matters of purely financial impart. (Para 309)

(3) The estimates should be as accurate as possible and, to achieve this object, care should be taken to see that the data on which the forecast is based is adequate and reliable and that the conclusions arrived at from the data can be sustained by past experience and future expectations of likely events. (Para 310)

(4) No stereotyped method of estimating will be helpful in forecasting accurately, unless the data on which the forecast is based is selected with due care and all the useful knowledge is brought to bear on the conclusion to be derived from the data. The manner in which the data required for the preparation of the estimate should be collected is, therefore, left to the General Managers, but the general principles on the basis of which the various estimates should be framed are described, wherever necessary, in the following paragraphs. (Para 311)

(5) The Revised and Budget Estimates should be framed by the various concerned authorities separately for—

(i) Ordinary Working Expenses and (Para 312).
(ii) Works Expenditure.

(6) The estimates of expenditure in respect of each demand should be prepared in duplicate, the figure being given in thousands of rupees. In preparing their estimates the Railway Administrations should ensure that the figures of actual expenditure for the preceding year shown in the estimate agree with those reported to the Railway Board. The figures of actual expenditure for the current year should represent the latest data. (Para 325)

(7) **Explanation of variations.**—A brief narrative explanation should be given of the causes (with amounts involved in each case) of substantial differences between the figures adopted for the revised estimates of the current year and (i) the actual of the previous year, and (ii) budget allotment for the current year. Similar explanation should be given for differences between the figures of the budget estimate of the ensuing year and the revised estimate of the current year. Large variations which compensate each other should also be indicated. (Para 326)

(8) The revised estimate for the current and the budget estimate for the next year should be fixed after taking into account the expenditure of the previous year and comparing the expenditure during the first seven months of the year with the corresponding period of the previous year. A full explanation of special features and any exceptional and abnormal adjustments (with amounts involved) included in each period of the pervious and current year as also in the next year, should

be given in the explanatory note accompanying the estimates. The financial effect of variations on account of specific reasons should be clearly brought out under each Demand. (Para 327)

(9) When the expenditure anticipated in the last five months of the year is disproportionate as compared with (i) the first seven months of the year or (ii) the corresponding period of the previous year, reasons for the disproportionate expenditure should be given in the revised estimate. Special and non-recurring items of expenditure in a year should show a corresponding saving in the following year. Likewise, when transfers are made from one demand to another due to changes in classification, the saving shown under one demand should agree with the excess shown under the other demand.

Annexure '4/5 '

1. Works Machinery and Rolling Stock Budget.—The revised and budget estimates for expenditure on construction acquisition, and replacement of assets (briefly known as the Works Budget) are prepared in the form of the Works Machinery and Rolling Stock Programmes. The Works Programme is required to be prepared in accordance with the instructions embodied in Chapter VI of the Indian Railway Code for the Engineering Department. (Para 5.12 may be referred to) (Para 346).

2. There is one single Demand for Grant for Works Budget viz., Demand No. 16—Assets Acquisition, Construction and Replacement ; and expenditure whether met out of loan obtained from the general exchequer or internal resources of Railways viz, ' Revenue', the ' Depreciation Fund ', the ' Development Fund ' and ' Accident Compensation,' Safety and Passenger Amenities Fund in the case of last mamed fund, expenditure on safety works and passenger amenities works only, is included in this Demand. Works Demand provides for booking of expenditure by— various plan heads ; the list of Plan Heads is as follows—

Code No. of Plan Head	**Description of Plan Head**
11	New Lines (construction)
12	Purchase of new Lines
..	.. ..
14	Gauge conversion
15	Doubling
16	Traffic facilities – Yard remodeling and others
29	Road Safety Works ..
33	Signalling and Telecommunication Works
34	Taking over of line wires from P. & T. Departments
35	Electrification projects
..	.. ..
41	Machinery and plant
42	Workshops including production units
..	.. ..
64	Other specified Works (OSW)
65	Training/HRD..

Annexure '4/6'

The sources of financing will be indicated in the beginning by the following alphabets:-

P – Capital.
Q – DRF.
R – OLW (R)
S - DF
T - ACF
U – Revenue.

The next 4 digits will represent the Main, Sub and Detailed Heads of Classification, as illustrated below:-

Classification of Capital and other Works Expenditure

Sl. No.	Demand	Abstract	Name of Demand	Main Heads		Sub-Heads	
				No.	Name	No.	Name
1.	..	..	Assets, Acquisition, construction and other Works Expenditure.	1100	New lines (Construction).	1170	Equipment Plant and machinery*.
						1180	General Charges Establishment. ++
				1400	Gauge Conversation.		Same sub and detailed classification as under no. 1100 with codified digits for the relevant plan Head.
				1500	Doubling.		
				1600	Traffic facilities Yard Remodelling and others.		
				3300	Signalling and Telecommunication works.		Do.
				3400	Taking over of line wires from P. & T. Department.		Do.
				3500	Electrification projects.		Do.

*(Detailed Heads No.1173, 1177 for S.& T. and 1179 for Motor Vehicles).

++ (Detailed Head 1187 for S. & T.).

Annexure '4/7'

(1) **Distribution of Funds by the Railway Board.**—The Grants as voted by the Parliament and the appropriations for the charged expenditure as sanctioned by the President, are distributed by the Railway Board among the Railway Administrations and other authorities subordinate to them, as soon as possible, after the Budget is sanctioned. The sums so distributed are called "Allotments" and the orders by means of which the allotments are made are called "Budget Orders". The allotments made out of funds voted by the Parliament are shown as "Voted" and those fixed by President are shown as "Charged". (Para 361)

(2) The Budget Orders are accompanied by the final issues of " Demands for Grants " and " Works, Machinery and Rolling Stock Programmes" containing the detailed distribution of the Budget allotment made to The Railway administrations for working expenses and Capital, Depreciation Reserve Fund, Development Fund, Open Line Works (Revenue) and Accident Compensation, Safety and Passenger Amenities Fund expenditure. The Budget allotment made to a Railway administration is intended to cover all charges, including the liabilities for past years to be paid during the year or to be adjusted in the accounts for it. It shall be operative until the close of the financial year. Under the ' doctrine of lapse " any unspent balance shall lapse and shall not be available for utilization in the following year. (Para 362)

(3) In the event of the Budget Orders of the Railway Board not being received before the commencement of the financial year, the Railway administrations are empowered to incur expenditure, pending the receipt of the Budget Order, on works which were in progress at the end of the previous financial year. All expenditure incurred under this rule must be treated as a charge, against the allotment eventually made for such works. (Para 363)

(4) **Distribution of Funds by General Managers to Lower Authorities.**— Subject to the other provisions of this Chapter or of any general or specific orders issued by The Railway Board, a General Manager is expected to take steps immediately to distribute the funds, placed at his disposal, to authorities subordinate to him in such manner as he may consider most suitable, provided that the total of the sums so allocated does not exceed that total of the grant placed at his disposal. In making this initial distribution, he may, at his discretion, keep a sum unallotted as a reserve for emergencies that may arise in future. He may also vary the initial distribution as necessity arises during the course of the year. In regard to " Carry over " works, the latest information as to the necessity and extent of funds required for such works should be obtained from the authorities concerned and taken into consideration in making the initial distribution. (Para 365)

(5) No expenditure shall be incurred by an authority without the allotment of necessary funds. The authorities to whom the funds are allotted, shall be responsible to report at once to the next higher authority the probability of any lapses or excesses over the sums placed at their disposal. In exceptional cases, where expenditure is authorized in anticipation of the allotment of funds, or in excess of the existing provision, the authorization should be followed, as soon as possible, by a formal allotment of funds to the extent required. (Para 367)

(6) The expenditure on each work shall be limited to the sum allotted for it. If for exceptional reasons, expenditure in excess of Budget allotment has to be incurred and if the authority incurring the expenditure is either not in a position to find funds by reappropriation or is not empowered to sanction a reappropriation there of, application for additional funds shall be made to the next higher authority slating how the expenditure is proposed to be met. In doing so, it should invariably be explained why the need for the expenditure was not foreseen in time for inclusion in the Budget and why the outlay cannot be postponed to the next financial year. In each case a copy of the order making the allotment or sanctioning a reappropriation shall be sent to the Accounts Officer concerned by the authority issuing the order or sanctioning the reappropriation. (Para 368)

(7) **Responsibility of Railway Administrations in case of Excesses of Lapses.**—The Railway administrations shall be responsible to ensure that no expenditure is incurred in excess of the Budget allotments made to them. Should it become apparent at any time that the grant for the year is likely to be exceeded from any cause whatsoever, the General Manager should report the position to the Railway Board and apply for additional funds. No liability may be incurred in one year against anticipated grants of a succeeding year except that advance commitments for procurement of stores for works may be made as provided in para 812-S to the extent authorised by the Railway Board from time-to-time. (Para 371)

(8) It shall also be the duty of the administrations to see that the allotments made to them are fully expended, in so far as in consistent with economy and the prevention of large expenditure in the last months of the year for the sole purpose of avoiding lapses. They shall be responsible for ensuring that money which is not likely to be needed during the year is promptly surrendered so as to allow of its appropriation for other purposes. (Para 372)

(9) **Powers of Railway Administrations in regard to emergent and inevitable expenditure.**—Not withstanding the provisions of paragraph 7 above, the commencement of works urgently necessary to safeguard life or property or to repair damage to the line caused by flood, accident, earthquake or other unforeseen contingency, so as to restore or maintain through communication may be authorised by the Executive Engineer, but he should at once submit a report through the usual channel to the authority competent to give administrative approval to the work and to allot the required funds. (Para 373)

(10) Expenditure to meet a sudden increase in traffic or for ensuring the safe operation of traffic may be incurred with the personal sanction of the General Manager in anticipation of the allotment of necessary funds by the Railway Board provided that the Financial Adviser and Chief Accounts Officer concurs. This power cannot be delegated and can be exercised by the General Manager only where the expenditure is within his power of sanction. In all cases where this requires an allotment of additional funds, a report showing the expenditure involved and the additional funds required should be submitted to the Railway Board, as soon as possible. (Para 374)

Annexure '4/8 '

Revenue Allocation Registers

The entries in the Revenue Allocation Registers maintained in the Accounts Office shall be sub-divided as under—

Cash,
Transfers,
Stores.

and sufficient space should be left after each head for the entries which normally come under it. Alternatively, separate registers may be maintained to record the expenditure in these three divisions, in which case a separate summary will have to be prepared to arrive at the total expenditure under the various heads of revenue classification. The sanctions by heads of accounts as given in the Authorization Rolls or the Distribution Statement of Budget allotment should be entered in red ink at the head of the several Abstracts so as to form a ready means of comparison and check with the outlay and to admit of this being done the grand total expenditure for each month will be deducted at the foot of the abstract form (Form E. 1469), the sanctioned amount and the balance thus arrived at will be carried forward in red ink from month to month. At the end of the month when the Registers have been totalled, the ' Cash ' totals will be agreed with the debits to ' Demands Payable ' in the Monthly Classified Abstract of Cash transactions and a journal slip will be prepared crediting ' Demands Payable' and debiting the various abstracts of the Revenue Accounts. (Para 1469)

ALLOCATION REGISTER

(Abstract 'A' to 'H' and 'J' to 'N')

For the month of …….19……..Allotment for the year.---

Sl. No.	Particulars of transactions	Voucher		Detailed Heads of Accounts as per revenue Classifications and Demand Heals			Total*	Grand Total	Sl. No.	Remarks
		No.	Date	Rs. P.	Rs. P.	Rs. P.	Rs. P.	Rs. P.		

Total for the month…….
Add total to end of previous month……
Total to end of the month……

---

* In Column (5) under Total should be posted " Wages and Materials" or " Ordinary and special" According to the requirements of the rules.

Note: - The expenditure (Other than on surveys) under the major head 345-A/B Policy formulation direction, research and other Miscellaneous Organizations should be recorded in a separate register in accordance with the prescribed classification.

Annexure '4/9'

(1) Register of works is a collective record of expenditure designed—

(i) for effecting control of expenditure on works with reference to estimates, by facilitating comparison between the expenditure incurred on each work and the detailed provision made in the estimate for work ; (Para 1472)

(ii) for effecting budgetary control, by facilitating a comparison between the budget allotment for the work and the actual expenditure to the end of the month ; and

(iii) to enable any material modification occurring being spotted.

(2) **Single set of works registers.**- Detailed Register of Works in form E-1473 given below should be kept for all sanctioned works including those chargeable to Revenue, 'New minor works', showing the amounts of estimates sanctioned. This register shall be maintained in the Accounts Office both for open line and construction organisation. (Para 1473)

……..RAILWAY **Form E-1473**

WORKS REGISTER

Name of work……

Authority……

Capital……...Rs. Cash……..Rs.

Depreciation Fund…….....Rs. Stores……Rs.

Development Fund……Rs. Total……...Rs.

Accident compensation Passenger

Amenities Fund……

Open Line Works-revenue……..…..

Revenue……Rs

Date of commencement……...

Date of completion…….

Budget Allotment for the year…….....Rs

Original estimate No……

Revised Estimate No……

Annexure – Contd.

Date	No of Voucher	Particular	Items of Estimate				Heads of Accounts				Advance payments for supply of materials	Material at site	Value of material received in advance of payment of contractor	Total charges		
			Amount	Amount	Amount	Amount	Capital	DRF	Revenue	Development Fund				Cash	store	total
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17

(3) This register should show the amount of estimate sanctioned, the budget allotment and details of expenditure on each work by heads of accounts and in addition under sub-heads of estimates in case of track renewal works estimated to cost Rs. 3 lakhs and over and Rs. 1 Lakh over and in the case of other works. In this register sub-heads of estimate estimated, to cost less than Rs. 10,000 need not be separately detailed, but may be grouped together as may be found convenient. The Register may be arranged by detailed heads of classification (for works falling under the demand relating to creation and replacement of assets) separate folios being set apart for each work. At the close of every month the register of works should be totalled up and the monthly, yearly and 'up-to-date' totals for each work struck. The register of works should be preserved for a period of 10 years. (Para 1474)

(4) In posting the Register of works in the case of works, the accounts of which are kept by sub-heads, the last column relating to each work will always show the ' total charges', the last column but one will show ' value of materials received in advance of payment to contractors', the last column but two will show the value of the 'materials-at-site' and the last column but three will show 'advance payment for supply of materials'. In case of works, the accounts of which are not kept by sub-heads, the value of 'materials-at-site' need not be separately shown but the value of materials received in advance of payment to contractor and advance payment for supply of materials, if any, should be shown in all cases. (Para 1475)

Annexure '4/10'

(1) **Revenue and Works Expenditure.**—As all Railway expenditure, other than that ' charged ' on the Consolidated Fund of India, is voted in the form of the Demands for Grants, the budgetary control is intended to ensure that expenditure is incurred for the purposes, and within the limits, as voted by Parliament. (Para 504)

(2) The exercise of authority for incurrence of expenditure carries with it also the responsibility for control of expenditure within the authorised limits. Further, these powers are subject to provisions contained in the various Codes, rules and regulations and also observance of the standards of financial propriety. (Para 505)

(3) **Standards of financial propriety.**—In the exercise of their financial powers, the sanctioning authorities must pay due regard to the following principles :— (Para 116)

(i) The expenditure should not prima facie be more than the occasion demands, and that every Government servant should exercise the same vigilance in respect of expenditure incurred from public moneys as a person of ordinary prudence would exercise in respect of the expenditure of his own money.

(ii) No authority should exercise its powers of sanctioning expenditure to pass an order which will be directly or indirectly to its own advantage.

(iii) Public moneys should not be utilised for the benefit of a particular person or section of the community unless—

(a) the amount of expenditure involved is insignificant ; or

(b) "a claim for the amount could be enforced in a court of law ; or(c) the expenditure is in pursuance of a recognised policy or custom.

(iv) The amount of allowances, such as travelling allowances, granted to meet expenditure of a particular type, should be so regulated that the allowances are not on the whole sources of profit to the recipients.

(4) **Revenue Expenditure.**—The revised and budget estimates of ordinary working expenses submitted by the General Manager to the Railway Board are based on detailed estimates of revenue expenditure obtained by him from the various heads of departments of the railway. On receipt of the allotment from the Railway Board, these detailed estimates may require modification consistent with the allotment sanctioned by the Railway Board or as may be deemed necessary by the General Manager. (Para 506)

(5) Within the amounts of the allotments fixed by General Manager for the various spending authorities in charge of divisions and departments, such authorities have full power to sanction expenditure on revenue account during the year, subject to the condition that the specific sanction of the competent authority must be obtained separately to all items of expenditure on establishment and experimental or special temporary works where under rules or orders such sanction is necessary ; in a case in which the General Manager has prescribed that detailed estimates should be prepared and administrative approval and technical sanction obtained before the incurrence of expenditure, the procedure laid down by the General Manager must be followed. (Para 508)

(6) Revenue Allocation Registers.—All revenue expenditure is to be recorded in registers, which are known as Revenue Allocation Registers, by the various heads of accounts. The object of these

registers is to keep the heads of divisions, and departments informed of the progress of expenditure against the allotments placed at their disposal by the General Manager, which should be entered in red ink in the appropriate columns provided for in these registers, so as to form a ready means of comparsion and check with the outlay. (Para 512)

(7) Monthly Financial Reviews.—The monthly reviews show the expenditure to the end of the previous month, against the allotments placed at the disposal of the controlling authorities under each sub-head of the grant for which they are responsible. The review should be prepared in Form No. 513 and submitted to the controlling authorities every month, by such date as may be fixed in consultation with them. .

Form No.F.51

Sub-heads of grant and heads of account	Budget Allotment for 1974-75	Proportionate Budget allotment to end of June 1974	Actual expenditure to end of June 1974	Actual expenditure to end of June 1973	Expenditure upto June 1974 more (+) or less (%)	
					As compared with column (3)	As compared with column (5)
1	2	3	4	5	6	7

Note- The periods have been shown in the Pro forma for the purpose of illustration only.

(8) **Expenditure on, acquisition construction and replacement of Railway assets.**—Separate allotments are placed at the disposal of Railway Administrations under each Grant for expenditure on works chargeable to Capital, Depreciation Reserve Fund, Development Fund, Accident Compensation, Safety and Passenger Amenities Fund, or Open Line Works Revenue. These allotments are made in lump sums and their distribution over the various sub-heads and over the works for which they are intended are spelt out in the " Works, Machinery and Rolling Stock Programmes " which are furnished to the Railways also with the Budget Orders sanctioning the allotments. These 'Programmes also show, in the case of works costing over Rs.50,005 each, the total estimated cost of each work, and the railway administrations are required to exercise a control over expenditure not only against the allotment sanctioned for the year for each works but also against its total estimated cost as shown in the Programmes'. In regard to works costing not more than Rs. 30,000 each, a lumpsum is allotted to each Railway administration in the ' Programme', before any expenditure is incurred against the lumpsum allotment ; the estimates of such works should be sanctioned by the General Manager or any lower authority empowered by him in this behalf. Thus the control of expenditure on railways is exercised through—

(i) the preparation, in advance, of estimates of the expenditure to be incurred ;
(ii) the allotment of funds through Budget Grants for the year, on the basis of these estimates ; and
(iii) the continuous and concurrent review of the expenditure as incurred against the details of the estimates and against the sanctioned grants, so that revisions of estimates or re-appropriation of funds are arranged for at the earliest possible point of time.

The procedure to be followed in Controlling expenditure against estimates is detailed in Chapter XIV and XII of the Indian Railway Code for the Engineering and Mechanical Departments, respectively. (Para 517)
(9) Watch over Progress of Expenditures.— From the moment expenditure or liability is incurred on works, a check a regular and frequent intervals should be made on its progress, both against

estimates and against funds. The check should originate in the lowest executive unit, Viz., a division. (Para 520)

Note.—Estimates are prepared for New Minor Works costing over Rs. 5,000 each which are chargeable to "Open Line Works-Revenue ". The provisions in paragraph (10) apply equally to such works.

(10) The Works Registers (Form No. E. 1474) maintained in each division enable a running comparison to be made between- (Para 521)

(a) the expenditure incurred on each work and the detailed provision made in the estimate for the work, and
(b) the budget allotment for the work and the actual expenditure to the end of the month.

The executive officer should examine the works registers monthly or at more frequent intervals, and watch the progress of expenditure on each work, so that any tendency towards excess over sanctioned estimates may be investigated and curbed or fresh administrative and technical sanctions obtained in time to cover the anticipated excess.

(11) Review of Expenditure.—The Accounts Officer should prepare every month two reviews, one by sub-heads/Plan heads of the Grant and the other by individual works, and present them to the authorities concerned by such dates as may be fixed in consultation with them. (Para 524)

(12) The review of expenditure chargeable to Capital, Depreciation Reserve Fund, Development Fund, Accident Compensation, Safety and Passenger Amenities Fund Open Line Works—Revenue by sub-heads/Plan Heads of the Grant should be prepared monthly by the Accounts Officer in Form No. F. 525. (Para 525)

Form No.F.525

Review showing Progress of Expenditure chargeable to Capital, Depreciation Reserve Fund, Accident Compensation, Safety and passenger Amenities Fund and Open Line " Works-Revenue to end of ……

Department…….. Office……

(Figures in thousands of rupees)

Sub-head/ Plan head Grant No.16	Budget Allotment for the year	Total Estimates sanctioned during the year	Appropriate Expenditure to end of June 1979						Percentage of expenditure to allotment (Col.9/Col.2 x 100)	Percentage of expenditure to allotment of the corresponding period of last year	Reasons for variations between 1 Col.10 & 1 and brief to remarks as the rate of progress of expenditure
			Capital	DRF	DF	OLW (R)	ACS and PA Fund	Total			
1	2	3	4	5	6	7	8	9	10	11	12

Date…… Signature: ……...

Designation: …….....

The quarterly reviews for period ending June, September and December may be sent to the Board by 25th of the second following month to which the figures relate. The review for the period ending March, will be sent to the Board by 6th August, each year.

Reasons for variations and brief remarks as to the rate of progress of expenditure to be indicated in Col. 12 of the Form, should be meaningful and should be recorded in consultation with the spending authorities.

Annexure '4/11'

(1) Completion Estimates.—In the case of Railway Projects costing over rupees one crore, the Construction or Abstract Estimate should be closed at the end of one of the first three financial half years after the date of " opening " (as defined in para 2) as maybe convenient and a "completion estimate" prepared. The completion estimate is a "stock taking estimate" and all works not started on that date are excluded from the scope of the project and are to be dealt with separately. (Para 1701)

(2) In the case of open line projects the date of "opening " should be held to be the date on which the projects fulfills he purpose for which it was sanctioned. (Para 1702)

(3) Closing the Accounts of a Project.—Those items of work which were in progress at the time of closing of Construction or Abstract should be got completed and the accounts of the project should be closed as soon as possible. Executive Engineers in-charge of project should therefore take prompt action to bring to account all charges and credits pertaining to the projects. They should before closing the accounts of a project, take steps to liquidate all outstanding liabilities, clear suspense balances, pay up all outstanding contractor's claims and dispose of all surplus stores and tools and plant returned from works. After all charges and credits relating to the project have been booked in the accounts of the project a Completion Report of the project should be prepared. (Para 1704)

(4) Form of Completion Report—The Completion Report should be prepared in the following form and brief explanations should be furnished for—

(i) excess of not less than 10 per cent or Rs. 25,000 whichever is less over the estimated provision under each sub-work ;
(ii) saving of not less than 20 per cent or Rs. 1 lakh whichever is less, occurring under any sub-work. (Para 1706)

Note.-The Divisional Railway Manager/Deputy Chief Engineer concerned may be empowered by the General Manager to approve the completion reports of works within the General Manager's competence in the Works Registers if variations are within 5 per cent of the sanctioned estimate.

Completion Report for the work..... **Form No.E.1706**

Particulars (Heads of Account and Description of works)	Amount of Estimate (with reference to authority for sanction)	Actual Expenditure	Difference		Remarks and Explanations
			Excess	Saving	

(5) Completion of Works.—In common with works costing over Rupees one crore, a work or scheme costing less than this amount should be considered as completed when it fulfills the purpose for which it, when sanctioned, was intended, and when there has been no expenditure thereon for three months thereafter. All outstanding debits and credits pertaining to a completed work should as a rule be adjusted in the account of the work within three months of the date of completion. The accounts of a completed work should be closed six months after the date of completion and a completion report of the work drawn. (Para 1708)

(6) A completion report for works costing Rupees one crore and less should subject to the detailed instructions issued by the Railway Administration, be prepared in the same form as that for works costing over Rupees one crore. It should compare the actual expenditure incurred with the sanctioned amount and should give under each sub work brief explanations of all excesses and savings of over 5 per cent or Rs. 10,000 whichever is less. The authority sanctioning the estimate may prescribe any additional information to be submitted along with the completion report. (Para 1709)

(7) A certificate to the effect that necessary addenda or corrigenda to the list of Buildings have been issued should be recorded on all completion reports for staff quarters and other buildings. A certificate to the effect that the necessary entries have been made in the siding registers and the register of deposit works should be recorded on all completion reports relating to assisted sidings and deposit works. (Para 1710)

(8) A completion report duly verified by the Accounts Officer should ordinarily be submitted to the authority who accorded the administrative approval to the work for information or regularisation. In respect of structural and track renewal works costing over Rupees ten lakhs included in the sanctioned budget with the prior approval of the Railway Board, and works sanctioned out of turn with the administrative approval of The Railway Board being beyond the General Manager's power of sanction, the completion reports may be finally dealt with by General Managers except (a) when they involve an excess beyond the General Manager's powers of sanction or (b) when a material modification is involved. For works costing rupees one crore and less the completion report should be submitted within six months of the completion of work. In the case of works on which no expenditure is recorded for three consequtive months, the Accounts Officer should call for the completion reports. If the works are incomplete the executive officer should advise the Accounts Officer of the probable dates of completion and submission of the completion reports. Abnormal delay in the submission of reports by the executive officers should be brought to the notice of the Head of the Railway Administration by the Accounts Officer. (Para 1711)

(9) Completion Reports of Unfinished Works.—If for any reason a work on which expenditure has been incurred is stopped and if there is no reasonable prospect of completing it in near future, the account of the work should as in the case of a completed work, be closed and a completion report drawn and submitted to the authority which accorded the administrative approval to the work, for information. In the case of structural works, other than Track Renewal works, costing more than Rupees one crore each for which administrative approval of the Railway Board is obtained by the submission of separate Abstract Estimates notwithstanding their specific inclusion in the sanctioned budget, the completion reports in such cases should be submitted to the Railway Board, duly verified by the Financial Adviser and Chief Accounts Officer. (Para 1713)

(10) Completion Statements.—As a general rule, a completion report should be submitted in respect of each completed work. Neverthless, in the case of works, the expenditure on which is within the competence of the head of the railway to sanction, a formal completion report on the prescribed from need not be prepared. In such cases all the information required in the Completion Report form, the certificate of the Accounts, Officer and the sanction of the competent executive authority may be recorded in the register of works under the relevant accounts Completion Statements showing the following information being prepared and recorded under the orders of the competent executive authority, after verification by the Accounts Officer:— (Para 1714)

(i) Reference to estimate.
(ii) Amount of sanctioned estimate.
(iii) Actual expenditure as finally booked.
(iv) Brief explanation of excess or saving.

Note.—The provisions of this paragraph do not apply to works, for which detailed estimates are not required to be prepared.

(11) Expeditious closing of Accounts of Works.—The account of a work costing rupees one crore and less should not be kept open beyond three months of the date of its completion. Efforts should be made to bring to account all liabilities and credits pertaining to the work by liquidating the former and realising the latter within that period. If a contractor's claim could not be finally settled within that period whatever the railway considers as legitimately due to the contractors should be adjusted by debit to the work and credit to " Deposit" and the account of the work finally closed. If the disposal of released materials is likely to delay the closing of the account of a work, credit should be afforded to the work on the basis of estimates by the Stores Department in consultation with the Engineer. (Para 1715)

(12) Entry in the Register of Works.—When a work is completed and its accounts closed, a double red ink line should be rated below the last entry in the Register of Works and a note made as under— (Para 1716)

Works completed.

Completion report submitted on.....

CHAPTER - 5: PLANNING, ESTIMATING & SURVEY

Section 1: Planning

5.1.1.0 **Planning General**

5.1.1.1 The process of programming Works, Survey as well as Rolling Stock, Machinery and Plant in advance, deciding their relative priorities, preparing necessary justification and estimates, making available funds for their execution within the ceilings prescribed, and including them in the "Works, Machinery and Rolling Stock Programme" is termed Planning.

5.1.1.2 **Initiation of Works Programme :-** Detailed instructions for preparation and submission of Works Programme are contained in Chapter VI of the Indian Railway Code for Engineering Department (relevant extracts at Annexure ‘5/1’).

5.1.1.3 Programme of Signalling and Telecommunication Works may be initiated as a result of proposals made by the Signal, Operating or other Departments for periodic replacement of worn out assets with or without improvements or for meeting operating requirements or for improving the safety of train operation or for provision of amenities to passengers

5.1.1.4 Works may be initiated either on a Division or at the Railway Headquarters. Works initiated on the Division shall be first examined by the Divisional Officers. Where there is sufficient justification, the proposal after approval by the Divisional Railway Manager shall be forwarded to the Headquarters Office along with the remarks of the Accounts Branch for scrutiny and examination by the Heads of Departments concerned. If the work is approved by them and concurred in by PFA, it is noted for inclusion in the Works Programme of the ensuing year

5.1.1.5 Works initiated by the Headquarters Office shall be sent to the Principal Financial Adviser for obtaining concurrence and thereafter included in the Works Programme. In case it is, however, desired to obtain the estimate from the Division, the Division may be asked to process the work in which case the procedure as in para 5.1.1.4 shall be followed.

5.1.1.6 A work should be considered as remunerative only if it satisfies the test of remunerative ness as contained in Para 204 of the Indian Railway Financial Code (Annexure '5/2 ')

Section 2: Estimates

5.2.2.0 Code Rules :- Instructions regarding the different kinds of estimates and their preparation are contained in Chapter VII of the Indian Railway Code for the Engineering Department (Relevant extracts at Annexure ' 5/3')

5.2.3.0 **General Instructions applicable to Estimates:**

5.2.3.1 While preparing the Abstract Estimate in accordance with the Code Rules, the expenditure for Stores may be grouped under Cabin Equipment, Point gear and connections, Signal fittings and connection, Cable and line wire, Track Circuits, Block Instruments, Station Master's Control Instrument, Control and Cabin telephones, etc., besides Labour, Tools and Plant, Establishment Charges, Freight and Contingencies, Works to be done by other Departments, e. g., construction of cabin and other buildings, construction of staff quarters, provision of sleepers, power supply arrangements, etc., are to be included where necessary.

5.2.3.2 While preparing the detailed estimates in accordance with, the Code Rules, the following guidelines shall be observed:

(a) Details of cost of S&T Building, staff quarters and other buildings and electric power connections shall be included, by obtaining them from the Departments concerned and shall be provided for under separate sub works.

(b) Establishment and Supervision charges, where necessary, shall be included. Additional weightage for works to be done under traffic conditions should be provided.

(c) Provision shall be made for additional quarters required for additional maintenance staff to be posted as a result of the scheme. Provision shall also be made for additional Stores/Office accommodation where necessary.

5.2.3.3 For preparing estimate for Line Capacity Work, the Traffic Department will advise 'the Signal and Telecommunication Department of the detailed traffic projections/requirements. Necessary feasible technical solution will be submitted by Signal & Telecommunication department including justification. If the proposed scheme is agreed to by Traffic department, The estimates shall be prepared on the basis of requirements furnished by the Traffic Department. The justification for the proposed work will be furnished by the Traffic Department together with the anticipated financial implications. A similar procedure shall be followed for works to be executed for other departments, e.g. Electrical, Civil Engineering, etc.

5.2.3.4 In regard to grouping of Works, instructions contained in Para 750 of the Indian Railway Code for Engineering Department and instructions contained in Para 776 of the Indian Railway Code for finance Department in regard to allocation of charges, shall be observed (relevant extracts at Annexure '5/4').

5.2.3.5 Where works are customarily executed through the agency of contractors, each Divisional Office shall maintain the Schedule of Rates, issued under the authority of the Principal Chief Signal and Telecommunication Engineer/ Chief Signal and Telecommunication Engineer (Construction). The Schedule of Rates may be reviewed and revised by the Principal Chief Signal and Telecommunication Engineer/ Chief Signal and Telecommunication Engineer (Construction) every five years or at shorter intervals as considered necessary. No variation in the

Schedule of Rates is permissible without the specific authority of the Principal Chief Signal and Telecommunication Engineer/ Chief Signal and Telecommunication Engineer (Construction).

5.2.3.6 Each Divisional Office shall maintain the Price Lists of Stores as issued by the Stores Department.

5.2.3.7 The Sr Divisional Signal and Telecommunication Engineer in charge shall be in possession of copies of "like authorised Scheduled of Rates and Price Lists of Stores and amendments advised to him from time-to-time.

5.2.4.0 **Reports accompanying Estimates -** The reports shall generally include the following:

5.2.4.1 A clear description of the work to be carried out and the object to be gained by its execution together with such information as will enable the sanctioning authority to appreciate the necessity for the work.

5.2.4.2 Reasons for the adoption of the proposed lay out or design with special reference to any variations from usual practice and its special feature, if any.

5.2.4.3 When the Project is of a nature involving scientific points or other considerations of special character such as a new Signaling or Telecommunication System not tried out before, the reports shall contain a complete account of the basis on which every part of it has been framed, the various considerations in regard to Signaling and Telecommunication details, economy of construction, utility of the practical working of the Project when carried out and the method by which it is proposed to execute any portion of the work involving unusual difficulties of construction, if any.

5.2.4.4 In case of works of important nature like, Centralized Traffic Control, Automatic Signaling, ETCS, TCAS, CBTC, TMS and other Modern Technology, a complete account of the various considerations in regard to engineering details, economy of construction and utility of the Project shall be given. Any local considerations which may affect the Project shall also be fully "detailed.

Section 3: Surveys

5.3.5.0 **Code Rules :-**

5.3.5.1 General instructions for conducting surveys for Railway Projects are contained in Chapter II of the Indian Railway Code for the Engineering Department.

5.3.5.2 The System of Signalling to be provided shall be as in Chapter 7 & other Chapter.

5.3.6.0 Surveys for new lines, line capacity works, electrification, etc. - Engineering-cum-Traffic Surveys shall be carried out for New lines, Conversions, Doublings and other Line Capacity Works costing more than Rs 2.5 Crores before the inclusion of such works in the Works Programme.

5.3.7.0 Survey for Signalling & Telecommunication Works. Field surveys shall be carried out for the following categories of Signal and Telecommunication works costing more than rupees 2.5 crores each before inclusion in the Works Programme:

(a) Electronic/Relay Interlocking of Major yards;

(b) Integrated telephone networks ;

(c) Modern Signalling Schemes including Centralised Traffic Control, ETCS, TCAS, CBTC, TMS etc.

(d) LTE/OFC/MTRC/5G and other latest Telecommunication systems.

(e) IP Networks/Data Security/Web based Applications

5.3.8.0 **Alternative proposals :-** Where alternative proposals of importance have been examined and rejected, particulars shall be suitably indicated in the Plans or in the report and the reasons for rejecting the alternative shall be explained in the report.

5.3.9.0 **Notes to be made in the field :-** During the survey, careful notes with data shall be made at site, from personal enquiry and observation, regarding any information likely to be useful in working out the details of the Projects, and in determining the prospects of the proposals. The following points shall receive special attention:

(a) Availability of required concrete sleepers ;

(b) Availability of power supply at site or sites ;

(c) Whether climatic condition will suit the type of equipment to be installed ;

(d) Special problem like rocky soil, existence of wide rivers which may have to be traversed by cables or overhead lines, etc.

(e) Gradients on the section particularly if the Project is for provision of Automatic Signalling ;

(f) Detention to trains that can be saved ; additional services that can be introduced.

(g) Availability of road approach.

5.3.10.0 **Expenditure on Surveys:**

5.3.10.1 Railway Administrations have no powers to undertake Surveys on their own. After a survey is included in the sanctioned Budget, the General Managers can sanction Survey Estimates as per the powers delegated in SOP.

5.3.10.2 If it is decided to undertake a Survey in connection with any proposal/modification in Signalling or Telecommunication System, an estimate of the cost of the proposed Survey shall be prepared, and the information collected in the course of the preliminary investigation, if any, shall be embodied in the report accompanying the Estimate of the Survey.

5.3.10.3 The proposal for Surveys shall be submitted to the Railway Board well in advance to enable necessary provision being made in the Budget.

5.3.10.4 The Budget allotment for preliminary Survey of a work shall be charged under Demand No. 2 and when final estimate for that particular work is sanctioned, the amount charged under Demand No. 2 shall be transferred to the sanctioned Estimate of the Project.

5.3.11.0 **Imprest Amount :-**

The official in-charge of Survey shall be provided with an adequate Imprest. He shall be responsible for maintaining all accounts with necessary vouchers. The Imprest shall be recouped on need basis, within the validity of the period of Survey.

5.3.12.0 **Survey Teams**

5.3.12.1 A Signalling/Telecommunication team shall be provided in Survey Estimates for Projects like Doublings, Remodeling, Conversions, New Lines, Railway Electrification including Techno-economic Surveys, Feasibility Studies, etc., to carefully study the needs of the Traffic and to make provision for Signaling equipment accordingly duly taking into account the age and condition of existing equipment, if any.

5.3.12.2 The period for which the Signalling/Telecommunication team is to be catered for in the Survey Estimates shall be commensurate with the workload in each case and shall be decided by the Principal Chief Signal and Telecommunication Engineer in consultation with the Principal Financial Adviser of the Railway concerned while finalizing the Survey Estimate.

5.3.12.3 The requirements of the Signal Department shall be obtained from the Principal Chief Signal and Telecommunication Engineer by the Principal Chief Engineer while preparing the Survey Estimates for such Projects.

5.3.12.4 Similarly in Survey Estimates for Signal and Telecommunication Projects, suitable provision shall be made for other Departments, viz., Traffic, Engineering, Accounts, Electrical, etc., as found necessary.

5.3.13.0 **Administrative Control on Survey Team :-** The Signaling /Telecommunication Survey team shall work under the Administrative and Technical Control of the Principal Chief Signal and Telecommunication Engineer or Chief Signal and Telecommunication Engineer (Construction), as the case may be, both for Surveys for Signaling & Telecommunication works and for Signaling portion of the Surveys for Projects mentioned in Para 5.3.12.1.

5.3.14.0 **Progress Reports on Surveys :-** The Official-in-charge shall submit periodical reports of progress in prescribed forms as may be directed by the Principal Chief Signal and Telecommunication Engineer.

5.3.15.0 **Report :-** At the conclusion of the Survey, a report should be formulated by the Officer-in-charge of the Survey. The details of the information collected, calculations and diagrams, shall be embodied in Tables as Annexure to the Report. Inference out of the Survey on the feasibility of the work with clear decision whether to take up the work or drop it shall be brought out at the conclusion of the survey.

5.3.16.0 **Covering Note :-** The Survey Report and Annexures shall be submitted to the Railway Board under a covering note, which shrill have the authority of the Railway Administration submitting the Report. It should provide a summing up and should contain clear recommendation together with the views of the Principal Financial Adviser (PFA).

	Note :- This Chapter has under mentioned Annexures for further study	
S.no	*Annexure no*	*Description*
1	*5/1*	*Preparation of the Preliminary Works Programme*
2	*5/2*	*Test of Remuneratiteness*
3	*5/3*	*Code Rules for Estimates*
4	*5/4*	*Grouping of works*

Annexure '5/1'

Preparation of the Preliminary Works Programme

Extract from the Indian Railways Code for the Engineering Department

609. The Chief Engineer of the Railway will be primarily responsible for ensuring that the proposals prepared by the various departments are complete in all respects and are correctly prepared. The overall priorities within the ceiling given by the Board will also be fixed by him in consultation with the General Manager and other Heads of Departments. He will be responsible for the preparation and timely submission of the Preliminary and the Final Works Programme.

610. In or about June/July each year the Railway Board should convey to each Railway, in respect of each Plan Head, the total outlay within which the Works Programme should be framed by the Railway. A list of the Plan Heads is given in Annexure I. On receipt of this financial ceiling the Railway Administrations should take stock of the schemes already formulated and those under consideration and select for inclusion in the Works Programme within the financial ceiling such works as are expected to yield the maximum benefit to the Railway, preference being given to works in progress. Further necessary changes in the investments schedule may be made in order to work within the financial ceiling for the year such modifications being taken note of in framing the Preliminary Works Programme and revising the financial implications, if necessary.

611. The Preliminary Works Programme for the following year should be submitted by the Railways to the Railway Board by 1st week of September or such earlier date as may be laid down by the Board. Proper financial appraisal of each work should be given in the Preliminary Works Programme together with the comments of the Financial Adviser and Accounts Officer.

612. The project cost should be based on firm data both as to quantity and rates at current price levels, and should any increase occur in prices during the period intervening between the initial preparation of the project estimate and its inclusion in the Works Programme, the estimate should be updated taking into account any significant changes in the wages and material prices as well as increase in freights and fares. No other increase such as on account of change in scope of the project should be allowed without prior reasons being adduced for acceptance by the Railway Board. A sketch showing the proposal should accompany each proposal.

613. Each investment proposal should be accompanied by a detailed plan showing the scheduling of the project to match the traffic requirements and the financial outlay proposed for the year should be in accordance with this project schedule to enable the Railway Board to arrange for a realistic funds allocation for implementation of the programme.

614. In deciding the outlays for the various works Railway Administrations must endeavour to progress all works in progress speedily and bring them into use at the earliest possible date. A work which has been sanctioned and for which funds have been allotted whether in the original or supplementary budget of a year should be treated as a "Work in progress" for the next year and provided for as such in the programme. Such works should be grouped as indicated in para-619.

615. The Railway Administrations should make realistic assessment of the amount required for each work in progress and necessary provision should be made for it in the Works Programme. In estimating the provision for works during the budget year a generous allowance should be made for those delays in execution which though unforeseen are known from experience to be so liable to arise particularly prior to inception and during the initial stages of large projects. The provision made should take into account adjustment of charges on surveys connected with a project.

616. In exhibiting the outlay for the current year against individual works in the works programme, the outlay should be as per Pink Book, and in exceptional cases where the Railways propose any substantial increase in the outlay with corresponding reductions against other works, such revised

outlay may be shown separately in brackets below the outlay as furnished in the Pink Book duly explaining the reasons for doing so in foot notes at the appropriate places. As far as possible only the last sanctioned cost should be exhibited. Wherever it is visualised that the cost would involve an excess over the last sanctioned cost, effective steps should be taken well in time to have the revised estimates prepared and sanctioned by the competent authority before the Works Programme is sent to the Board. In case where the revised estimates are sanctioned subsequent to the despatch of the final Works Programme but before the end of January of the following year, the same should be promptly advised to the Board to enable the latest sanctioned cost being exhibited in the Pink Book to be circulated along with the Budget. In all cases of revised costs sanctioned by the Board, reference to the letter of sanction should invariably be indicated.

617. Works once introduced through a Works Programme (including Track Renewal Programme) and taken up after the estimates have been sanctioned by the competent authority should continue to be included every year till they are finally completed, except in cases where the works have reached the completion stage and where funds required if meager, could be found by reappropriation.

Annexure '5/1' contd.

618, The Works Programs is compiled in the following format —

Work Programme

Demand No……(Figures in thousand of rupess)

Item No.	Authority	Particulars of works	Cost	Expenditure to end of 3/74	Quality		Balance
					1974-75	1975-1976	
1	2	3	4	5	6	7	8

Note—'Years have been shown in the form for the purpose of illustration.

In respect of" Works in Progress "reference to item No. of the current year's Pink Book and also the authority under which the work was first started should be indicated. The works should be arranged as per the Plan Heads.

619. The items in the Works Programe should grouped under the following categories while compiling the works Programms.—

(i) New Works.
(ii) Works in Progress.

(iii) Works approved in earlier years, which have not been actually commenced and on which no expenditure has been incurred till 30th June of the year previous to the Programme year.

(iv) Works approved in the earlier years but estimates for which have not been sanctioned by 30th June of the year previous to the Programme year.

620. The works are further made into sub-groups of (i) Works costing more than Rupees Five Lakhs each, and (ii) works costing upto Rupees Five lakhs each. Under (iii) Works costing upto Rupees two lakhs each in the case of Track Renewal works and for works costing upto Rupees one lakh each in the case of other works, only lumpsum provision should be shown without detailing individual works. Within each sub-group, the works are presented under each Plan Head.

Annexure '5/2'

Test of Remuneratiteness

Extract from Indian Railway Financial Code/Vol. I

204. **Test of Remuneratiteness:** The net financial gain expected to accrue from a project may be either by way of savings in expenditure or increase in the net earnings (i. e., gross earnings less working expenses), or a combination of both. Except in the case of residential buildings, assisted sidings and rolling stock to which special rules are applicable no proposal for fresh investment will be considered as financially justified unless it can be shown that the net gain expected to be realised as a result of the proposed outlay would, after meeting the working expenses (see Para 217), yield a return of not less than 10 per cent on the initial estimated cost.

Note— (1) Interest during construction should be added to the cost (excluding that chargeable to Revenue) of the projects, the construction of which is likely to last for more than one year.

(2) Depreciation should be calculated on the total cost of the scheme and not only on the portion chargeable to Capital, unless the contrary procedure can be justified in any particular case. However, depreciation as an element of working expense is to be ignored for assessing annual cash flows under the B. C. F. method (See Para 228).

(3) In the case of construction of bridges, maintenance charges should include, besides the maintenance charges on the bridges proper, the maintenance charges of the training works also.

Annexure '5/3'

Code Rules for Estimates

Extracts from the Indian Railways Code for the Engineering Department -

701. Kinds of Estimates: All proposals for -

(a) the construction or purchase of new works or assets;
(b) the renewals and replacements of existing works or assets chargeable to Depreciation Fund/Development Fund or Open line Works - Revenue when estimated to cost more than Rs. 10,000 ; or if chargeable to Revenue when estimated to cost more than Rs. 50,000.
(c) the scrapping, dismantlement or abandonment of existing works or assests;
(d) the repairing or reconditioning, if estimated to cost more than Rs. 50,000 of existing works or assets;
(e) temporary and experimental works;
(f) renewals and replacements on worked lines, and
(g) renewals of ballast.

Should, subject to the provisions of paragraph 1103 regarding urgent works, be scrutinized by the authority competent to sanction them before any expenditure or liability is incurred thereon. For the purpose of this scrutiny, all such proposals should be presented in the form of one or other of the following estimates, according to the- circumstances mentioned in paragraphs 702 to 713.

(a) Abstract Estimates.
(b) Detailed Estimates.
(c) Supplementary Estimates.
(d) Revised Estimates.
(e) Project Abstract Estimates.
(f) Construction Estimates.
(g) Completion Estimates.

Note:- (1) for "New Minor Works" costing Rs. 5,000 and less, for renewals and replacement works chargeable to Revenue costing Rs. 50,000 and less and for " Reconditioning Works " costing Rs. 50,000 and less detailed estimates need not be prepared for formal sanction. Rough estimates should, nevertheless, be prepared and kept on record by the Executive Engineer.

702. The Abstract Estimate: An abstract estimate is prepared in order to enable the authority competent to give administrative approval to the expenditure of the nature and the magnitude contemplated, to form a reasonably accurate idea of the Probable expenditure and such other data sufficient to enable that authority to gauge adequately the financial prospects of the proposal. Abstract estimates avoid the expense and delay of preparing estimates for works in detail at a stage when the necessity or the general desirability of the works proposed has not been decided upon by competent authority. An abstract estimate should contain a brief report and justification for the work, specifications and should mention whether funds are required in the current year and to what extent. It should also show the cost sub-divided under main heads and sub-heads or specific items, the purpose being to present a correct idea of the work and to indicate the nature of the expenditure involved. The allocation if each item as between Capital, Development Fund, Open Line Works-Revenue, Depreciation Reserve Fund and Revenue should be indicated.

Note.—(.1) Administrative approval to a work or scheme should be accorded by the authority competent to do so (vide paragraph 748), after a through examination of its necessity, utility and financial propspects. See also rules in Chapter II of Indian Railway Financial Code.

……RAILWAY **Form E.702**

Abstract Estimate

Department…….Division…….Station……..
Estimate No……
Framed By…….Division……
Description of work……
Plan No……...
Reference……..

Cost	Capitals	DF	DRF	OLWR	Ordy. Rev	Deposit	Misc. advances	Total
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Existing sanctioned estimate (if any) Present Estimate ... .. Total ...								

Cost of stores……. To be purchased ..Rs……Rs……..

Funds ... . To stock…….Rs……

Report and Justification--- Specification.
Enclosed- * Details of cost with allocation (Form E.704). Subestimate
Has Accounts Officer agreed to the pro. allocation ?

Station: ……. Signature: ……...

Date: …… Designation: …….....

*Required in case of any existing sanctioned estimate.

703. Detailed Estimates.—On receipt of administrative approval to a project or scheme other than that for which construction estimate in Form E. 553 is prepared and, conveyed through the sanction to the abstract estimate relating thereto detailed estimates for various works should be prepared and submitted for technical sanction of the competent authority. It should be prepared in sufficient detail to enable the competent authority to make sure that the abstract estimate sanctioned by a higher authority is not likely to be exceeded. No work included in an abstract estimate should be commenced till a detailed estimate for the same is prepared and sanctioned and adequate funds are allotted by the competent authority. The detailed estimate of an open line work will comprise (i) statements showing details of estimated cost and (ii) an outer sheet giving the abstract of cost of work, the report, the financial justification and the allocation.

Note:-Technical Sanction - The sanction of the competent authority to the detailed estimate of a work is called the "technical sanction". The authority according technical sanction should satisfy itself that (i) the details of the scheme as worked out are satisfactory, (ii) the methods proposed for the execution of the work are adequate; and (iii) the cost has been estimated from reliable data and is likely to be reasonably accurate.

(2) In the case of works within his power of sanction, the General Manager may, in lieu of the procedure of preparing Abstract Estimates for administrative approval, prescribe that both the administrative approval and the technical sanction should be accorded on the detailed estimates. 707. A supplementary estimate should be prepared for any item of work, which ought to have been included in the first instance in an estimate already sanctioned but has not been so included, or which it is found later, should be considered as being a part or a phase of an estimate already prepared and sanctioned, if it cannot be met out of contingencies (see paragraph 727). Such, a supplementary estimate should be prepared in the same form and the same degree of detail as the main estimate and for all purposes be treated as a part of the main estimate.

708. Revised Estimate.—As soon as it becomes apparent that the expenditure on a work or project is likely to exceed the amount provided therefor in the detailed estimate or construction estimate a revised estimate should (subject to provision of paragraph 1336) be prepared and submitted for the sanction of the competent authority. It should , unless otherwise ordered by the sanctioning authority, be prepared in the same form and the same degree of detail as the original estimate and should be accompanied by a comparative statement showing the excess or saving under each sub-head of account against the latest sanction. In cases where a supplementary estimate; or a previous revised estimate has been sanctioned by the Railway Board, it should be made clear how the original sanction has been modified by such further sanctions.

709. The Project Abstract Estimate.—The abstract estimate of a Construction Project should be submitted for the approval of the Railway Board on Form E. 554 " Abstract cost of Railway " accompanied by (i) an abstract estimate of junction arrangements, (ii) a narrative report explaining the salient features and major items of expenditure (iii) detailed estimates on Form E. 553 prescribed for a construction estimate under the following heads :—

Capital - 1120—Land Structural Eugineerong Works.—1132—Tunnels, 1151 and 1152—Major Bridges, 1153 and 1154 Minor Bridges, 1140—Ballast and Permanent-Way (Detailed estimate for one kilometre). 1180 and 1190-----
General Charges—Establishment and General Charges—Other than Establishment. 2000—Rolling Stock.

Note:- Head shown above are for Plan Head ' New Lines' as an example.

Those detailed estimates should be prepared from an engineering survey report.

Note:- The abstract estimate for an unremunerative project chargeable to Development fund will be prepared in the same detail as the abstract estimate for a construction project chargeable to Capital.

710. The Construction Estimate: When it is decided to undertake the execution of a new line gauge conversion, or doubling of lines a final location survey should be made ; and based on the information collected in that survey detailed estimates of all the works included in the project as a whole should be prepared (of paragraph 540). These detailed estimates are collectively called the " Construction Estimate " of the Project. It should be prepared after a careful examination of the various details of construction involved in the Project. It should be in such detail as to render it possible to dispense with working estimates or any other further estimating after the Construction Estimate has been sanctioned (except when supplementary or revised estimates are necessary). It should provide for the buildings and equipment of the Railway upto a standard that will be sufficient for working such traffic as may be expected during the first year or two after opening of the line. It is the basis on which technical sanction to the various works included in the construction of a Project is accorded.

713. **Completion Estimate:** A Completion Estimate is prepared in super session of a construction estimate as provided in paragraph 1701. It should show in a tabular form (E. 713) the following particulars in respect of all the works included in the construction estimate :—

(i) amount of sanctioned estimate ;
(ii) actual expenditure on all works up to the date of construction estimate ;
(iii) commitments on that date ;
(iv) anticipated further outlay ;
(v) total estimated cost ; and
(vi) difference between the sanctioned estimate and the estimated cost.

An abstract of the completion estimate showing the above particulars against the various heads of capital classification should be submitted for information or sanction, vide paragraph 1703 to the Railway Board together with brief explanations for excesses of not less than Rs. 10,000 or 10 per cent over the provision under sub-heads of account and for savings of 20 per cent or one lakh, whichever is less, occurring under any main head of account. Provision for further outlay should be made in completion estimate only for those works which are in progress or completed on the date of closing of the construction estimate. All works not started on that date should be dealt with separately as open line works both as regards estimate and expenditure. In forwarding, therefore, estimates for sanction for works in connection with new lines opened, it should be clearly indicated whether the cost of the work is chargeable to Capital Construction or Open Line Capital.

Note.—The completion estimate of an unremunerative construction project chargeable to Development Fund will be prepared in the same detail as the original construction estimate using the various heads of capital classification for the purpose

714. Estimates for Railway Project: Scope - A Construction Estimate (Form E. 553) should be prepared in such detail as to reduce to a minimum the probability of omission of any item of expense which is capable of being foreseen. It should be remembered particularly that the provision for contingencies allowed in the estimate is not intended to meet items of expense which can be foreseen and which are reasonably likely to occur. With good estimating it should seldom be necessary to encroach, to any appreciable extent, on the provision for contingencies.

727. Provision for contingencies: Provision for unforeseen contingencies should be made in all estimates at 3 per cent of the total estimated cost. All incidental expenditure which can be foreseen such as works establishment, sheds for workmen and stores should be separately estimated and provided for in the estimates. The provision for contingencies should not be diverted to any new work or repair which is not provided for the estimate, and of which the cost exceeds Rs. 1,000 without the sanction of the authority who sanctioned the estimate.

729. Schedule of Rates: To facilitate the preparation of estimates, a schedule of rates of each kind of work commonly executed should be maintained in each open line division, and it will be the duty of the Chief Engineer when inspecting the divisional offices, to see that correct schedules of the rates at which work is actually being carried out, are invariably recorded in a complete and satisfactory manner. The regulations for the due record of rates in a clear and systematic manner and for their periodical revision to bring them on line with the rates prevailing in the market and those paid by other government departments will be laid down by the General Manager.

Estimates of Deposit Works

732. A Railway Administration is occasionally required to execute works for and at the cost of other Government departments, local bodies, private persons, etc. Such works are referred to in this Code as " Deposit Works " (see also paragraph 1843 for a definition of this term). To meet the cost of plans and estimates of such works as also those to be carried out for other Departments out of Railway funds, which are subsequently not carried out, charges at the following sliding scale shall be levied on the total of the estimate inclusive of departmental charges :-

	Rs.		Rs.	Percent
For works costing over	1,00,000			2
For works costing over	60,000	but not more than	1,00,000	2 ½
For works costing over	30,000	but not more than	60,000	3
For works costing over	20,000	but not more than	30,000	3 ½
For works costing over	10,000	but not more than	20,000	4
For works costing over	1,000	but not more than	10,000	4 ½
For works costing over	1,000	and below 5 percent subject to a minimum of Rs.25		

The acceptance of the government departments or the payment in cash by the local bodies or private individuals concerned should be obtained to the above percentage charges before the work of preparation of plans and estimates is taken in hand In cases where the proposed works are subsequently carried out, these percentage charges should be adjusted against departmental charges.

The levy of the above percentage charge may, at the discretion of the General Manager, be waived in particular case subject to the conditions laid down in paragraph 1138.

Note: The scale of charges prescribed in this rule does not apply to assisted sidings, recovery of preliminary expense in respect of which has been separately provided for under paragraph 1825.

733. In preparing estimates of works for other departments, local bodies, private individuals, & c, particular care should be taken to see that.

(i) Railway freight and carriage charges of materials proposed to be used in the works are provided for at the rates applicable to the public and not at the concessional rates applicable to railway material; (Works of Branch Line Companies are not governed by this rule, but by the relevant contracts).

(ii) departmental charges at the prescribed rates (paragraph 1137) are provided for.

734. All estimates of deposite works should be got accepted by the parties ordering the works before submission to the competent railway authority for sanction. In the case of works, which under the rules (see paragraph 1851) are required to be maintained after completion by the Railway department at the cost of the department, local body, private firms or individuals ordering the work, the acceptance of the party concerned should also be obtained for the recurring expenditure that is likely to be incurred on repairs, maintenance, & c.

735. No work asked for by another government department should be commenced till a detailed estimate for the same has been accepted by the department concerned and sanctioned by the competent railway authority. No work asked for by local bodies, private individuals, and c, should be commenced till a detailed estimate for the same has been sanctioned the competent Railway

authority and the estimated cost thereof deposited with the Railway. The amounts so deposited should be credited to the head "Deposits— Miscellaneous".
Competency of Sanction

748. The previous sanction of an authority higher than the General Managers of Indian Railways is necessary:-

(i) To expenditure on new lines or rolling stock or surveys not provided in the sanctioned budget for the year or carried forward from the sanctioned budget of the previous year;
(ii) To expenditure on other works not provided in the sanctioned budget or carried forward from the sanctioned budget of any previous year except:-

(a) (i) Track renewal works- costing not more than Rupees two lakhs;
(ii) Other works - costing not more than Rupees one lakhs;
(iii) Machinery and Plant - costing not more than Rupees Fifty thousands.

Provided that the total lump-sum provision made in the budget for such works is not exceeded.;

(b) (i) On line capacity works costing above Rupees one lakh but not more than Rupees ten lakhs each;
(ii) On track renewal works costing above Rupees two lakhs but not more than Rupees five lakhs each;
(iii) On other than line capacity and track renewal works costing above Rupees one lakh but not more than Rupees five lakhs each;

Subject to ceiling of Rupees one crore in all in a financial year provided that the sanctioned budget (other than lump-sum) for works in these categories is not exceeded.

Note:- (1) The works thrown forward from previous years may be taken up only if the funds required for them can be found by reappropriation within the sanctioned allotment.

(2) The savings in the lumpsum provision made in the sanctioned budget shall not be utilized for the category of works in (b) above without the prior approval of the Railway Board.

(3) The General Manager may sanction expenditure on new works out of turn in respect of users' amenities including goods shed and booking office not exceeding Rupees one lakh in each case provided the funds required for such works as provided in the sanctioned budget for works in these categories is not exceeded; and expenditure upto Rupees fifty thousand in each case, in respect of existing Railway schools, institutes, hospitals and dispensaries provided the lumpsum provision in the sanctioned budget is not exceeded.

(iii) To expenditure on works provided in the sanctioned budget for the year or carried forward the sanctioned budget of any previous year as follows:-

(a) Works sanctioned under the lumpsum provision - To an excess over the total lumpsum provision in the sanctioned budget for such works ;

(b) Works outside the lumpsum provision - Rolling Stock, Track renewals and other works - to an excess over the estimated cost as entered in the sanctioned budget or sanctioned separately, as follows:-

(i) of more than 25 per cent over the original estimated cost ;
(ii) of more than 15 per cent over the first revised cost ;
(iii) of more than 10 per cent over the second and further revised estimated cost.

Provided further that all codal provisions regarding sanctions for material modification are strictly followed.

(c) Surveys:- To an excess over 10 per cent on original estimate sanctioned by higher authority (General Manager can sanction survey estimates costing upto Rupees one lakh each provided the surveys are included in the sanctioned budget)

Note.—(1) The works thrown forward from previous years may be taken up only if the funds required for them can be found by reappropriation within sanctioned allotment.

(2) The amount first given in the Budget or sanctioned separately by higher authority shall be the original estimated cost and the revised amount given in subsequent Budgets or sanctioned separately by higher authority, the second, third, etc., revised estimated cost.

(3) The powers of sanction to excess over estimated costs allowed in item (b) above shall not make the cost of such works exceed Rupees one crore in any case.

(4) The powers for sanctioning the excess over the estimated costs of works outside the lumpsum provision as in item (b) above shall not be redelegated to lower authority in respect of works, the original estimated cost of which is over Rupees fifty lakhs.

(iv) To the scale of -

(a) a portion of a Railway line;
(b) in item of the authorised Rolling Stock ; or
(c) any other Railway asset costing over Rupees three lakhs.

756. Currency of Sanction: The sanction to an estimate will ordinarily remain current for five years from the date on which it has been accorded unless it has been renewed for a further term by the acceptance of a revised estimate. Acceptance by competent authority however, of a budget estimate which includes specific provision/for expenditure on a work which is in progress, may be regarded as reviving for the year in which provision is made, the sanction to the estimate-regardless of the five years' limit. But if no work has been commenced on a sanctioned scheme within two years of the date on which the sanction was accorded to the estimate, such sanction should be held to have lapsed and fresh sanction should be obtained from the competent authority by the submission of an up-to-date estimate, if necessary.

758. Scope of the sanction to an Estimate: The authority granted by a sanction to an estimate should, on all occasions, be looked upon as strictly limited to the precise objects for which the estimate was intended to provide. Accordingly any anticipated or actual saving on a sanctioned estimate for a definite project should not, without special authority be applied to carry out additional work not contemplated in the original projector fairly contingent on its actual execution. Saving due to the abandonment of a substantial sanction of any project should not be considered as available for work on other sections

The Execution of Works

1102. The ordinary rule is that no work may be commenced and no liability or expenditure incurred on a work until a a detailed estimate for it has been sanctioned and an allotment of the requisite funds made by competent authority. This rule pertaining to preparation of estimates does not apply in the following cases :—

(i) For the construction or purchase of new work or asset under the category of new minor works if estimated to cost upto Rs. 5000.

(ii) For renewals and replacements of existing works and assets chargeable to Open Line Works Revenue as a new minor work if estimated to cost upto Rs. 10,000.

(iii) For renewals and replacements of assets charged to Development Fund and Depreciation Reserve Fund if estimated to cost upto Rs. 10,000.

(iv) For renewals and replacements of assets charged to Revenue if estimated to cost upto Rs. 50,000; and

(v) For repairs and reconditioning of existing assets if estimated to cost upto Rs. 50,000 (of para 701).

1103. Work started on Urgency Certificates: In addition to the exception mentioned in the preceding paragraph the following are the only classes of work on which expenditure or liability may be incurred prior to the receipt of sanction of the authority competent to sanction the estimates under the ordinary rules:-

(i) Works, which are considered to be urgently necessary to safeguard life or property or to repair damage to the line caused by flood, accident or other unforeseen contingency, so as to restore or maintain through communication.

(ii) Works considered urgent but not falling within (i) above, as for instance, works required to meet the immediate needs to traffic, which are considered by the General Manager so urgent that they must be started before the earliest date by which detailed estimates could be prepared.

Note:- Where applicable the instructions in paragraph 1260 should be observed.

1105. In according administrative approval or sanction and in allotting funds, if required, the competent authority will give such instructions as he considers necessary regarding the execution of the work to executive officers subordinate to him and fix a date by which the detailed estimate for the work should be prepared and got sanctioned. He will at the same time furnish a copy of his sanction and instructions to the Accounts Officers.

1109. Material Modifications: No material modification in a work or scheme as sanctioned, should be permitted or undertaken without the prior approval of the authority who sanctioned the estimate. In the case of estimates sanctioned by the Railway Board or higher authority, instances of what will be considered to be a material modifications of a sanctioned project or work are given in paragraph 1110 below.

1110. The following may be taken as material modifications on lines under construction and open line works estimated to cost rupees one crore and over:-

(a) Any change in the alignment likely to affect the facilities offered to the public in the neighbour-hood or likely to increase or decrease the length of the line by over one kilometre.

(b) Introduction of any new station or omission of any station.

(c) Any alteration in the type or number of engines or vehicles provided in an estimate for rolling stock.

(d) A change in the layout of a yard affecting the general method of working or increasing or reducing the number of trains that can be dealt with.

(e) Any departure from the standards of construction as prescribed in Chapter II or as accepted by the Railway Board in the Abstract Estimate or use of any second hand material if it affects the speed of trains or the number of trains to be dealt with than contemplated originally.

(f) The introduction or omission of any work or facility involving a sum of Rs. 5 lakhs and over.

(g) Any modification of a sub-work provided for in the estimate of a sanctioned work involving an additional outlay on that sub-work of more than Rs. 5 lakhs.

(h) The introduction of any new sub-work not provided for in the estimate of a sanctioned work involving an outlay of more than Rs. 5 lakhs.

(i) Any alteration in the standards of interlocking.

1137. Departmental Charges: When work is undertaken by the Engineering Department of a Railway for outside parties, including other Railways, Government Department, Public bodies (e. g., Municipalities, Port Trust, etc.) and employees of the Railway, Departmental charges should be levied to cover the cost of tools and plant and of establishment supervision The charges leviable will be 12-1 /2 per cent on the total cost of the work (wages and materials) including the cost of land, except where a rate higher than 12-1/2 per cant is charged to Governmental Departments on a reciprocal basis. The charge will be levied once only on the total outlay on a work (cash and stores, and will not be addition to a first charge of 12-1/2 per cent on the value of stores, vide paragraph 2329-S.

Note:- (1) The above rules do not apply to works undertaken by Engineering Workshops to which the rules applicable to works undertaken in workshops of the Locomotive and Carriage and Wagon Department apply.

(2) Railway Units of the Territorial Army:- The charges under the foregoing rules may be remitted in the case of works not exceeding Rs. 500 in cost executed by the Engineering Department of a Railway for Railway Units of the Territorial Army provided that no additional establishment is entertained for such works.

1138. Remission of Departmental Charges: The General Manager may, at his discretion, waive wholly or partially the recovery of the Departmental charges leviable under Paragraph 1137 provided that -

(a) the Railway gains some advantage, not necessarily financial, by such remission :
(b) the reasons for the remission are recorded in each case and the remission is allowed with the concurrence of the Financial Adviser and Chief Accounts Officer; and
(c) the non-incurrence of extra charges on accounts of tools and plant and establishment supervision does not in itself constitute sufficient justification for the remission.

Annexure '5/4'

Grouping of works

Extract from the Indian Railways Code for the Engineering Department

750. Grouping of works: When two or more works are so connected either by their situation or by the purpose or purposes which they are designed to serve, that construction of one necessarily involve that of the other or others, the works should be considered as one scheme and the aggregate estimated cost of the work so connected should determine the authority competent to sanction expenditure on the scheme. When the works constituting a connected scheme are situated in more than one executive division separate detailed estimate should be prepared of the cost of the work in each division in order that the Executive Engineer entrusted with the actual construction may be in a position to watch expenditure against a sanctioned estimate of the cost of the work in his charge.

Allocation of Charge

Extract from Indian Railway Financial Code, Vol. I

776. **Cost of Tools and Plants and Special Posts:** No portion of the pay or allowances of permanent open line staff shall be charged to Capital, Depreciation Reserve Fund, Development Fund or Accident Compensation, Safety and Passenger Amenities Fund or Open Line Works- Revenue, as the case may be, when such staff is employed on special works and the vacancies thus caused in the open line cadre remain unfilled. The cost (less return value) of tools and plant specially purchased and the cost of any posts specially created for the supervision or construction of a work chargeable purely to Capital or Depreciation Reserve Fund or Development Fund or Accident Compensation, Safety and Passenger Amenities Fund or Open Line Works- Revenue, as the case may be. When the cost of work is chargeable partly to Capital or Development Fund, Accident Compensation, Safety and Passenger Amenities Fund or Depreciation Reserve Fund or Open Line Works- Revenue, the cost of such posts and tools and plant should be allocated to Capital or Development Fund or Accident Compensation. Safety and Passenger Amenities Fund or Depreciation Reserve Fund or Open Line Works- Revenue in the same proportion in which the estimate for the work excluding the estimated cost of tools and plant and special posts, is chargeable. For working out the proportion, the gross amount of the estimate (ignoring the provision for credits on account of released materials) should be taken into account. The cost of a post, for the purpose of this rule includes the leave salary and contribution towards passages, pensions provident fund, bonus and special contribution to provident fund, which the holder of the post may be entitled to.

Note:- The provision for credits on accounts of released materials to be ignored in working out the proportion of cost of tools and plants and special posts, should be the net credit in terms of paragraph 710(2).

CHAPTER - 6 : SIGNALLING STORES AND ACCOUNTAL

6.0.1.0 **Classification of Stores -** The Stores in the custody of the SSE (Signal) in-charge and SSE(Signal) Stores may be broadly grouped under the following Stock Heads

(a) Imprest Stores - Charged Off' to Revenue

(b) Tools and Plant

(c) Protective Clothing

(d) Books of Reference

(e) Office furniture

(f) Stationery and Forms

(g) Stores obtained for specific works

(h) Returned store

Section 1 : Imprest Stores

6.1.2.0 **Operation of Imprest Store** - The materials classified as Imprest Stores, their procurement and accountal are contained in Chapter XXVIII of Stores Code and XIV of Engineering Code (relevant extracts at Annexure '6/1).

6.1.3.0 **Location and Scale of Imprest -** Imprest Stores shall be provided with each SSE (Signal) in-charge and SSE (Signal) Stores on a Division/ Construction /Project or any other as may be nominated for the purpose. The nature of items of Imprest Stores, location and the scale for each imprest holder shall be fixed by the PCSTE. The quantity of each item of imprest stores which each imprest holder may hold at a time shall be generally based on quarterly average consumption for normal maintenance and upkeep of equipment. Time for procurement in case of non-stock items and also time required for Stores Van movements in the case of stocked items may be taken into consideration for determining the quantity of imprest. The sanctioned scale shall not be increased or decreased without prior approval of the sanctioning authority.

Section 2 : Tools and Plants

6.2.4.0 **Scale** - A scale of Tools and Plant shall be fixed by the Principal Chief Signal and Telecommunication Engineer for each SSE (Signal) in-charge/SSE (Signal) Stores. This scale will include Tools for each maintainer and artesian besides a small reserve with inspector.

6.2.5.0 **Accountal** - Instructions in regard to accountal of Tools and Plant are

contained in Paras 1456 and 1457 of Engineering Code (extracts at Annexure'6/2').

Section 3 : Protective Clothing

6.3.6.0 **Periodical requirements** - The SSE (Signal) in-charge/ SSE (Signal) Stores shall submit to his Sr DSTE/Dy CSTE of work, the periodical requirements of protective clothing for all the staff eligible for the supply of the protective clothing in accordance with instructions issued from time-to-time.

6.3.7.0 **Preparation of Indents** - Guidelines in regard to preparation of indents and accounting of protective clothing are contained in Annexure ' 6/3 '.

6.3.8.0 **Accountal of protective clothing -** Accounts of protective clothing shall be maintained by the SSE (Signal) in-charge and SSE (Signal) Stores on Ledgers similar to the Tools and Plant Ledgers.

Section 4 : Books of Reference

6.4.9.0 **Supply of Books of Reference -** The SSE (Signal) in-charge and SSE Signal Stores shall arrange to get copies of all Books of Reference, in accordance with, instructions in Chapter 8 for the use of his office, for himself and for the use of staff under him by submitting necessary indents to his Sr Divisional Signal and Telecommunication Engineer.

6.4.10.0 **Accountal** - Accounts of Books of reference shall be maintained on Ledgers similar to the Tools and Plant Ledgers.

Section 5 : Office Furniture

6.5.11.1 The SSE (Signal) in-charge and SSE (Signal) Stores shall maintain an account of all office furniture in his charge on Ledgers similar to the Tools and Plant Ledgers.

6.5.11.2 All replacements shall be made on return of unserviceable articles. For any additional item, prior approval of the competent authority shall be obtained as per the SOP before placing the indent.

Section 6 : Stationery and Forms

6.6.12.1 **Indents of Annual requirements:** The SSE (Signal) in-charge and SSE (Signal) Stores shall submit the indents of their annual requirements of stationery to the Sr DSTE/Dy CSTE in-charge of the work in accordance with the scale for the supply of stationery.

6.6.12.2 The Sr DSTE/Dy CSTE in-charge of the work shall see that the sanctioned scale is not exceeded and the items for which no sanctioned scales have been fixed are essential and have necessarily to be supplied.

6.6.12.3 **Accountal** - A numerical accountal of receipt and issue of all items shall

be kept. The office of Sr DSTE/Dy CSTE in-charge of the work shall make checks periodically to see that there is no accumulation of items of stationery and forms in the subordinate offices.

Section 7 : Stores obtained for Works including Special Revenue Works

6.7.13.1 **Requisitioning** - Materials for specific works shall not ordinarily be requisitioned unless the estimate of the work has been sanctioned by the competent authority and funds have been allotted.

6.7.13.2 Requisitions for materials shall show the name and particulars of the estimate and sanctioning authority for the work.

6.7.13.3 Advance procurement of stores may be authorized by the PCSTE or Sr DSTE or Dy CSTE in-charge of the work for long lead vital materials such as relays, cables, point machines, block instruments, etc., which are required for the next three years, and which have to be specially arranged or have to be imported. In such cases, the indents may be placed as soon as a Work appears in the Final Works Programme. These indents shall be prepared on the basis of realistic estimate of quantities of materials and subject to prior scrutiny by Finance Branch in respect of reasonableness of the quantity of materials indented. The PCSTE or Sr DSTE or Dy CSTE in-charge of the work, shall certify the quantities of materials indented for each work. Indents shall specify the delivery schedule for the next three years and funds shall be arranged according to the delivery schedule.

6.7.13.4 The materials on receipt shall be either utilized on the work or if this is not immediately possible, stored carefully.

6.7.14.0 **Materials-at-Site account -** The material received for Works detailed in Para 6.7.13.1 above, if not used up immediately, shall be kept at debit of a numerical account of Materials-at-site of the particular work. Detailed instructions in regard to maintenance of accounts for Works estimated to cost less than Rs. 1 lakh and more than Rs. 1 lakh are contained in Chapter XIV of Engineering Code (extracts at Annexure ' 6/4 ‘)

Section - 8 : The Daily Material Transactions

6.8.15.0 **Daily Material Transactions Register DMTR**

6.8.15.1 A Daily Material Transaction Register shall be maintained by each SSE (Signal) in-charge and SSE (Signal) Stores, similar to Form No. No. S&T/DT. (Annexure ‘6/5’.)

6.8.15.2 All receipts and issue of Stores pending their transfer to their appropriate Ledgers shall be entered in this Register.

6.8.15.3 The Register shall be written up daily. A line shall be drawn across both pages under the last entry of each date to prevent subsequent entries

being made.

6.8.15.4 There shall be no direct posting of materials in Ledgers from Challan. All transactions shall first be shown in the Daily Material Transaction Register.

6.8.15.5 The dates shall be the same in both Ledger and the Register.

6.8.15.6 Issues of materials from outside stocks will be recorded by the SSE (Signal)/in-charge and SSE (Signal)/Stores in their line note books first. These entries will then be transferred to the Daily Material Transaction Register. The date of entry in the Daily Material Transaction Register shall be recorded on the note book.

6.8.15.7 The SSE (Signal)/in-charge and SSE (Signal)/Stores are personally responsible for all the Stores in their custody and shall satisfy themselves that the Daily Material Transaction Register and the Ledgers are being correctly posted. They shall initial the Daily Material Transaction Register at least once a week in token of having verified the entries thereof.

Section 9 : Returned Stores

6.9.16.0 **Instructions** - Instructions regarding returned stores are contained in Chapter XVI of the Indian Railway Code for Stores Department (relevant extracts at Annexure '6/6').

6.9.17.0 **Dispatch of Spare, second hand and scrap materials**

6.9.17.1 Spare and second hand materials sent to Stores Depot shall be carefully loaded to avoid loss or breakage.

6.9.17.2 Material complete but having pins badly worn out and not sufficiently good to be considered second hand, shall be entered up as serviceable scrap and their approximate weight shall be stated. If parts are missing, full details shall be given.

6.9.17.3 In the case of Signals, height condition of post and fittings shall be stated.

6.9.17.4 All cast iron, steel, brass, zinc, copper and lead scrap shall be collected from Sections regularly and sent to Stores Depot, with description and approximate weight.

6.9.17.5 Empties such as tins, drums kegs, barrels and cement bags shall be accounted for along with the materials contained in them and shown in the Returns. They shall not be held longer than necessary and shall be returned to Stores Depot for disposal.

6.9.18.0 **Credit for ‘‘Returned Stores**

6.9.18.1 The credit value allowed in the estimate for the returned Stores which are not likely to be required again shall be kept within the figure likely to be realised for it as an obsolete material or as scrap.

6.9.18.2 For materials likely to be used again, after return credit value proportionate to its further life may, however, be provided in the estimate. The normal life of some of the Signalling equipments as laid down in para 219 of the Indian Railway Financial Code is as follows

(a) Signalling apparatus – both Mechanical and Power - 25 Years

(b) Underground Cables - 30 Years

(c) Block token instruments - 25 years

(d) Electrical power plant - Oil Engine driven - 15 years

(e) Motor Vehicles - Road and Rail Motor Trolley - 10 years

Section – 10 : Requisitions

6.10.19.0 **Preparation of Requisitions**

The following instructions shall be observed in the preparation of requisitions.

6.10.19.1 Separate requisitions shall be prepared for each item of material. Requisition for stock items shall be prepared in Form No. S. 1313 and for Non-Stock items in Form No. S.1302 (samples of Forms at Annexure ' 6/7 ')

6.10.19.2 Nomenclature, price list Nos., and other references shall be correctly reproduced, in the case of special and non-standard items, description with complete specifications and drawings shall be given.

6.10.19.3 Blank space, if any, below the last item shall be crossed.

6.10.19.4 The designation of the consignee shall be written in full. No code abbreviations shall be given.

6.10.19.5 The head chargeable shall be entered on all the requisitions.

6.10.19.6 The requisition for materials for different sanctioned works and revenue maintenance shall be distinguished by a mark or a code letter, as laid down by the Stores Department.

6.10.19.7 In the case of sanctioned works, the number of estimate and the reference of the sanctioning authority shall be given clearly.

6.10.19.8 The quantity of each material shall be given in correct units in words and figures.

6.10.19.9 Corrections, if any, shall be initialed.

6.10.19.10 Availability of funds shall be certified by the Sr DSTE/ Dy CSTE in-charge

of the work.

6.10.20.0 **Requisitioning of Materials**

6.10.20.1 All materials and equipment shall normally be indented in accordance with Indian Railways Standard Drawings and specification. Where any such drawing or specification number is quoted, the latest alteration number as on the date of purchase will automatically apply. For items for which an I. R. S. specification does not exist, an appropriate specification shall be quoted.

6.10.20.2 No alteration or modification to or divergence from I. R. S. drawings shall be permitted without the specific sanctions in writing of the PCSTE. In the case of deviations having been decided upon before the placing of an order, whether direct or through the Director-General, Supplies and Disposals the indenting authority concerned shall quote such sanction in the order.

When deviations are decided upon or desired after the placing of an order, the necessary sanction shall be obtained, in writing, by the Inspectorate concerned in consultation with the indenting authority.

Section – 11 : General Instruction

6.11.21.0 **Stores supplied by firms/Contractors** - Receipt of Stores received direct from firms/Contractors shall be promptly acknowledged. All details of Make, Model Number, S.No, Invoice, PO/Agt No, Inspection Details shall be noted, Discrepancy or defect (if any) shall be brought to the notice of the Asst. Divisional Signal and Telecom Engineer.

6.11.22.0 **Custody and Maintenance:-** The SSE (Signal)/in-charge & SSE (Signal)/ Stores shall be responsible for the maintenance of all Stores and Tools and Plant in their charge in proper condition. Where Watchmen/Private Security is necessary, the SSE (Signal)/in-charge & SSE (Signal)/Stores shall approach the Sr DSTE/DyCSTE in-charge of the work giving full justification. The guidelines laid down in Annexure ‘6/8’ shall be followed as far as practicable.

☞	*Note :- This Chapter has under mentioned Annexure for further study*	
S.no	*Annexure no*	*Description*
1	*6/1*	*Imprest stores*
2	*6/2*	*Tools and Plant*
3	*6/3*	*Guidelines for preparation of Indents*
4	*6/4*	*Materials at-site account*
5	*6/5*	*The Daily Material Transaction Register*
6	*6/6*	*Returned Stores*
7	*6/7*	*Accounts (EDP) Requisition and issue note*
8	*6/8*	*General instructions for proper custody and Maintenance of Stores*

Annexure '6/1'

IMPREST STORES

Extracts from Stores Code

1801. Definition of Imprest Stores: Materials whether stock items or non-stock items, held in stock not for specific works but as a standing advance for the purpose of meeting day to day requirements in connection with the repair and maintenance shall, including consumable stores such as cotton waste, jute, oil, grease etc., be treated as imprest stores, so long as they are under the control of the authority in charge of the executive unit and have not been issued to works.

1802. Object of Imprest System: The main object of the Imprest System is to control the expenditure on materials used each month in the running repairs of rolling-stock, etc. But this involves inter alia, effective financial and departmental control on the stocks of materials held at outstations and their issue and recoupment.

1805. Schedule of Imprest Stores: Schedules of items of such materials showing the quantities of each item which should be held as an imprest should be prepared separately for each imprest holder. This schedule (S. 1805) shall show the following details;—

(1) Class of Stores. **Form No. S.1805**
(2) Price List or part List Number.
(3) Brief description of Stores.
(4) Sanctioned Imprest.
(5) Rate.
(6) Value.

1806. Copies of the schedule and any subsequent modifications of it should be supplied to the DRM, immediate superior of the imprest holders, the Stores Department and Accounts Officer, Stores, who should each keep the same up-to-date.

1811. Control of Imprests: It shall be the duty of the controlling authority in charge of the imprest holders to see that the imprests are revised from time-to-time so as to ensure that they are the lowest possible both as regards quantity and value, consistent with actual working requirements. The imprest holder shall be held responsible to ask for a reduction or deletion of any item of imprest on the basis of the consumption in his shed or station.

1817. Accounting: Each JE/SSE holding imprest stores shall maintain a numerical ledger on Form No. S. 1817 showing receipts, issues and balances of imprest materials. Each item of the imprest stores shall be dealt with on a separate page of the ledger.

1829. Closing of Ledgers and Recoupment of Imprests: The Divisional or District Officers should fix the dates of accounting period for each imprest holder and intimate the same to the Accounts Officer, Stores.

1830. Each Imprest Holder should close his ledgers monthly on the dates fixed for him and check them with the Issue and Recoupment Schedule which should be prepared in the form shown below from the Monthly Summary of Issues (S. 1823).

Monthly Summary of Issues- "Charged OFF" Stores Class- **Form No. S.1823**

Date	P.L. No.			P.L. No.		
	Allocation	Allocation	Allocation	Allocation	Allocation	Allocation
	Quantity	Quantity	Quantity	Quantity	Quantity	Quantity
Total quantity for each head of account under each P.L. No.						
Total Issues						

Note:- The need for the daily abstracts and monthly summaries provided for in paragraphs 1821 to 1823 does not arise in case the Issue Tickets (S. 1819) themselves are valued.

1838. Authority to Sign Indents: Indents for recoupment may be made directly by the imprest holders under their signatures and the counter signature of District or Divisional Officers for such indents is not necessary.

Extracts from Engineering Code

1417. Monthly Stores Returns: At the close of each month, separate returns showing receipts, issues and balances of all imprest and surplus stores operated on during the month should be submitted by each engineering subordinate to the Divisional Officer through his Assistant Engineer. These returns are referred to in this Chapter as Imprest Stores Account and Surplus Stores Accounts and should be submitted in the form printed below (Form No. E. 1417). Both the accounts should as far as possible, be supported by the various receipt and issue vouchers, i. e., Issue Notes Adjustment Mema and Advice Notes. These accounts should bear a certificate from the subordinate concerned that no stores other than those included in the accounts have been received or issued during the month and that the balances of all items of stores not included in the accounts remain the same as at the end of the previous month.

…….RAILWAY
Account for Engineering Materials **Form No. E.1417**

Permanent Way Imprest for the month of ……20……

Division……. Section…….

Section of Material	U.L. No.	Class I or II	Rate	Description of Material	Sanctioned Imprest Quantity	Opening Balance		Receipt			Issues		
						Quantity Number	Value	Source of receipt and particulars of receipt documents	Quantity Number	Value	Works for which issued	Quantity Number	Value
1	2	3	4	5	6	7	8	9	10	11	12	13	14

Closing Balance			Corresponding material released						How disposed off with particular of document reference
Quantity number	Value	Remarks	Description	U.L.No.	Class	Rate	Quality	Value	
15	16	17	18	19	20	21	22	23	24

Forwarded to :……. Signature :…….

Station/Office :…….. Designation:…….

1418. **Check of Monthly Stores Returns:** In the Divisional Office, the Imprest and Surplus Stores Accounts (Form No. E. 1417) received from the subordinates should be checked to see -

(i) that the opening balances of the items appearing in the accounts agree with the closing balances of the accounts in, which the items last appeared ;

(ii) that all receipts have been taken correctly to account as debits ;

(iii) that the issues of materials are in order and reasonable and have been correctly taken to account as credits ;

(iv) that all transfers within the Division agree - i.e., that the issues in the account of one subordinate agree with the receipts in the account of another subordinate, any disagreement noticed being rectified and the subordinates concerned advised of the correction ; and

(v) that, in the case of Imprest Accounts, the sanctioned imprest has not been exceeded without proper authority.

Annexure '6/1' **Form No. S.1817**

……Railway

……..Department

Ledger for Imprest Stores

Reference to Authority:

Authorized Stock…….Unit……Rate……

Class…… P.L.No……..

Nomenclature……

Date of Receipt or issue	Issue or Receipt Notes		Quantity			Remarks	Date of Receipt or issue	Issue of Receipt Notes		Quantity			Remarks
	Number	Date	Receipt	Issue	Balance			Number	Date	Receipt	Issue	Balance	
1	2		3			4	5	6		7			8

Annexure '6/2')

Tools and Plant

xtracts from Engineering Code

1456. Tools and Plant for Maintenance: A scale of Tools and Plant will be fixed for each gang under a Permanent Way Inspector and the total with his gangs plus a small reserve in his own godown will form the scale fixed for each Inspector. A scale for tools and plant will similarly be fixed for other Engineering Supervisors. On issue to the permanent way and other Engineering Supervisors, the cost of these tools and plant is charged off finally. It will, therefore, be necessary for the Supervisors to maintain a tools and plant register (Form E. 1462) and submit it after the close of each financial year to the Divisional Office for check.

…….RAILWAY **Form No. E.1456**

Register of Tools and Plants

Department ……Division……..Office……..Sanctioned Strength……

Authority…….

Class	Name of articles	Sanctioned scale	Balance brought over	Receipts April…… March (to show 12 months separately)				Total receipts and balance	
				Date	Particular of Voucher	Quantity	Cost	Quantity	cost
(1)	(2)	(3)	(4)	(5a)	(5b)	(5c)	(5d)	(6a)	(6b)

Issues April…… March (to show 12 months separately)		Quantity	Total	Balance at the end of year		Explanation of difference Ref. and Remarks
Date	Particular of Voucher			As per the return	As actually counted	
(7a)	(7b)	(7c)	(8)	(9)	(10)	(11)

Place……..…… Signature……..

Dated……..……..…… Designation…….

Annexure '6/3'

Guidelines for preparation of Indents

A consolidated indent for the supply of uniforms for all the eligible sanctioned staff shall be prepared by the Sr. DSTE and submitted to the PCMM for compliance. The indent shall be prepared in accordance with the following guide lines

(1) Clear description of Uniforms with Style number shall be furnished and the category of staff indicated for whom Uniforms are required.

(2) The year for which the supply is meant shall be furnished and fund certification indicated.

(3) Since one P. L. number has been allotted for one Style of all sizes, indents shall be submitted accordingly for one style duly indicating the required sizes for that style category wise, consignee wise for the entire Division and sent to PCMM.

(4) For staff headquartered at places not coming under zones prescribed as " Winter or mild Winter" but who are required to be supplied with Winter uniforms due to their being rostered for travel during the course of their duty to " Mild Winter " and " Winter " places at least five times a month, a certificate to this effect shall be furnished on the reverse of the indent.

(5) Certificate to the effect that the styles and scales furnished are as per standing instructions shall be given in the indent.

(6) It shall also be certified that the size furnished are as per measurement card maintained for the staff concerned.

Annexure '6/4'

MATERIALS AT-SITE ACCOUNT
Extracts from the Engineering Code

1441. **Daily Record:** The estimated value of works for which accounts are kept by such-heads in the Register of Works will be as provided for in Para 1486. A daily numerical record of receipts and issues of materials shall be maintained by the Stock holder in Form E. 1441.

…….RAILWAY **Form No. E.1441**

Materials-at-Site Record Ledger Form

Division…… Section……

Description of material	Unified nomenclature List No.	Classification I or II	Opening Balance	Unit	Receipts				Issues				
					Date	Source and reference	Quantity	Date	Sub-head to which chargeable and particulars of documents under which issued	Quantity	Closing balance	Remarks

Received for the work : B.F…… From page No……...

Carried forward B.F……. To page No……

Corresponding Material Released from the work .. B.F. From page No……..

Carried forward .. .. To page No……..

Return submitted on……..

Stock Holder's Initials……

1442. Materials obtained for the work should on receipt, be entered as such under the 'Receipts' together with the date, quantity, issue notes or other reference. In the case of materials obtained by direct purchase particulars should be entered in a Measurement Book as well.

As materials are issued for consumption on the work, the date, the quantity and the sub-head to which they are chargeable should be recorded under 'Issues'. Materials so issued, but found subsequently to be surplus to requirements should be brought back into record as'Minus Issues'. Such transactions should wherever practicable be avoided by confirming issues to exact and immediate requirements.

Materials released from the work should, on displacement be recorded separately as such and be entered with the date and quantity as 'Receipts'. When subsequently utilized on the work again they should be shown as 'Issues'.

Materials-at-site returned to stores, transferred or otherwise disposed off should be shown in the respective records with date and disposal reference as 'Minus Receipts'.

1446. Quarterly Materials-at-site-Returns. - The Supervisor who is executing the work and is also functioning as stock holder shall prepare a quarterly Materials at site account return (Form E. 1446) in respect of each work valued at Rs. 3 lakhs in the case of Track Renewal Works and Rs. 1 lakh in other works for which Register of Works is maintained by sub-heads of estimates, and send the same to the Divisional Office. This return should include only those items for which there have been any receipt or issues/transactions during the quarter. The quarterly return may be followed by a complete Materials-at-site account return for all the items at the end of half-year ending September and March. A number of works may be included in one form provided that all such works are under the executive charge of the same authority.

1447. The check of Materials at site return: The Materials-at-site returns received in the Divisional Office should be checked in the following respects:-

(a) The Opening balance should be checked with the closing balance of the previous return;
(b) The receipts during the period covered by the returns should be checked with the relevant issue notes, the summary of stores, adjustment mema and other receipt vouchers;
(c) The reasonableness of the issues during the period covered by the return should be checked with reference to the relevant sanctioned estimate, and the progress of work reported during the period;
(d) As regards materials returned to Stores Depots or transferred elsewhere, the correctness of credits to site accounts should be checked with the relevant advice of returned stores or adjustment mema.
(e) The arithmetical accuracy of the returns should be checked.

Annexure '6/4' **Form No. E.1446**

…….Railway

Material-at-Site Record Quarterly Returns

Division…… Section……

Name of work…….……..…..Sanction No…….…….……Date……..

Designation of Subordinate…… Account of Receipts and Issues during the Quarter ending……..…..

Sl. No.	Name of work and description of material	Classification	U.N. List No.	Rate	Unit	Opening balance			Receipts				Issues				Closing balance		
						Date	Quantity	Value	Date	Source and reference	Quantity	value	Date	Sub-head to which chargeable	Quantity	value	Quantity	Value	Remarks
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Forwarded to…….……

Station/office…….

Date……

Signature ……..

Designation……

Annexure '6/4'

1448. Accountal in the Register of Works: All materials obtained specifically for a particular work should be charged off immediately to the head to which the cost of that work as a whole is allocated (or if more than one head is involved to that bearing the greatest cost); but should, so long as they are not consumed on the work be borne under a suspense head "Materials-At-Site" opened under that head. All materials released from a work should also be borne under the same suspense head. The adjustment from the "MAS" suspense to the relevant final detailed heads should be carried out as soon as the materials are shown in the monthly return as having been issued for use on work.

1449. The responsibility for having a daily record maintained, a monthly return prepared, checked and valued and cost adjusted as prescribed in these rules devolves on authority in executive charge of the works. The Accounts Officer who maintain the works Register is responsible for the final adjustment of all materials-at-site transactions.

1450. Verification of Materials-at-site: The authority in executive charge of works for which material at site accounts are maintained should arrange for a periodical verification of the following materials at site pertaining thereto-

(a) Permanent Way materials.

(b) Other materials at site which can be readily separated and distinguished from any of the same description but of different category.

It is not necessary that all articles should be checked at the same time; but every item should be verified at least once in a year. A certificate by the authority in executive charge of works that such verification has been carried out should, together with a note as to whether or not the materials were found to be unduly depreciated, be furnished on the materials at site return for the month of March, or on the last return on which a balance is shown, submitted in the year. The stock verification prescribed in this paragraph is in addition to the periodical stock verification of engineering stores arranged for by the Accounts Officer.

1452. Daily Record: A daily numerical record in respect of materials-at-site works in this category for which a separate account is not maintained by sub-heads in the Register of Works shall be kept in Form E. 1441.

1453. Monthly Return: At the end of every month, an Excess Materials Return in Form E. 1453 given below should be prepared in respect of all completed works in this category. These Returns should show separately for materials obtained and materials released, the numerical balance only of materials-at-site i. e., those that have neither been consumed on the work nor returned to the stores, transferred or otherwise disposed off and the date of completion of the works to which they relate.

……..RAILWAY **Form No. E.1453**

Materials-at-Site Record excess Materials Return

Division…… Section……

Balance of Materials still “At-Site” on ……

Name of work		Materials Obtained for the work		Materials Released in connection with the work		Date of completion of the work	Remarks
Description of Materials	Classification	Rate	Quantity	Rate	Quantity		

Station/office…… Signature……..

Date…… Designation…….

1454. Check of Excess Materials Returns: The Excess Materials Returns should be checked in the Divisional Office as regards the correctness and reasonableness of the balances shown therein against the various receipt vouchers and the sanctioned estimates and put up to the Executive Engineer for his orders as to the disposal of the balances. If the ' Excess Materials' cannot be utilized on some other works, they should either be returned to the Stores Depot or taken to Engineering Stores Surplus—(i) Permanent Way, (ii) other than Permanent Way ; (iii) Awaiting sale.

1455. Accountal in the Register of works: Materials obtained specifically for particular works in this category should be charged off finally in accordance with their allocation.

ANNEXURE '6/5' **Form No. S&T/DT**

……RAILWAY

SIGNAL AND TELECOMMUNICATION DEPARTMENT

The Daily Material Transaction Register

Receipt							Issues						
Date	Description of material	Quantity	Ledger reference and Page	Challan or Receipt Note No. and date	From whom received	For what work	Date	Description of material	Quantity	Ledger reference and Page	Challan No. and date	To whom issued	For what work

ANNEXURE '6/6 '

Returned Stores

Extracts from Stores Code

1601. All Stores which have been previously issued for the services of the Railway and are no longer required on a work should, in the absence of special instructions to the contrary, be returned to the Stores Depots. If there is more than one Stores Depot on a Railway, the PCMM may nominate the depot or depots to which particular classes of Stores may be returned and advise the various Railway departments accordingly.

1602. Advice Notes.—The Officer Returning Stores to Stores Depots should prepare Advice Notes in Form (S. 1539) in six foils by carbon process. Separate Advice Notes should be prepared for each class and for new, second hand and scrap stores.

1604. Disposal of the Foils of Advice Notes.—One foil of the Advice Note should be retained by the subordinate returning the Stores to the Stores Depot as his office copy, three foils (the 2nd, 3rd and 4th) should be sent to the depot direct along with the Railway Receipt. Of the remaining two copies, one copy (the 5th) should be sent to the Accounts Officer, Stores, and the other (the 6th) to the Divisional or District Officer through the immediate superior of the subordinate returning the Stores.

1605. Departmental Register of Advice Notes.—The Advice Notes (6th foil) should be carefully examined in the division or district office to see that they have been correctly prepared. They should then be listed in a Register of Advice Notes for Returned Stores maintained in the form shown below :—

Form No. S.1605

Departmental Register of Advice Notes for Returned Stores

Designation of the Officer returning the Stores …….

Serial Num-ber	Advice Note		Materials by Returning Officer to have been returned		Date of dispatch of the 6th foil of the Advice Note to the Depot Officer	Materials acknowledged by depot to have been received			No. & date of Advice of credit received from the Stores Accounts Officer	Remarks
	Number	Date	Description of Materials	Quantity		Date of acknowledgements & RO No. & date	Description of Material Acknowledged	Quantity acknowledged		
1	2	3	4	5	6	7	8	9	10	11

Annexure '6/6 '

1606. A separate page of this register should be allotted to each returning officer or subordinate as may be found convenient. If the particulars furnished by the subordinates returning the stores regarding nomenclature, rates, etc., are found to be incomplete, the Advice Notes (S. 1539) (6th foil) should be completed in the divisional or district office in these respects. The Advice Notes (6th foil) should then be forwarded to the depot to which the stores have been returned.

1620. Posting Numerical Ledgers.—The Depot Officer, after having the necessary entries made in the numerial ledgers as well as in in the Depot Register of Advice Notes (S. 1608) should retain one copy of the Advice Note (2nd foil) as his office record and send the two copies (3rd and 4th foil) duly completed in all respects to the Accounts Officer, Stores.

1621. Disposal of Receipt (6th) Foil.—The sixth foil of the Advice Note should also be sent by the Depot officer duly receipted to the divisional or district officer concerned to be retained as the latter's office record after making the entries in columns 7 to 9 of the Departmental Register of Advice Notes (S. 1605). He should take up with the depot and returning officers any discrepancies in quantities (vide columns 5 and 9 of the Register) if the reasons given on the Advice Notes are not satisfactory.

1626. Advice of Credits.—From the 3rd foil will be prepared the Advice of Credits (S. 2705) according to the departments from which the stores have been received. The number and date of the Advice of Credit should be noted in the Accounts Register of Advice Notes (S. 1622) against the entries of the Advice Notes included in the Advice of Credits . The Advice of Credits with the 3rd foils as supporting vouchers and the corresponding 5th foils should then be to sent to the divisional or district officer concerned.

1628. Procedure in the District or Divisional Office.—On receipt of the copies of the Advice Notes along with the Advice of Credit from the Stores Accounts Officer in the division or district office, the 5th foil should be transmitted to the subordinate concerned after noting on it the fact of credit having been received. The Advice of Credit supported by the 3rd foil should then be carefully checked with and noted in the Departmental Register of Advice Notes (S. 1605).

1607. The Register of Returned Stores (S.1605) should be inspected frequently to see that there is little avoidable delay on the part of the stores depots in acknowledging the stores. Any cases of omission to acknowledge the stores and/or to afford credit for the same within one month of the Advice Notes should be taken up with the Depot and Accounts Officers.

Annexure '6/7'

……..RAILWAY **ACCOUNTS (EDP) 1**

REQUISITION AND ISSUE NOTE **S.1313/S.1319**

To be filled by Indentor To be filled by Store Depot CD Code **56**

Indentors	Consignee	Supplying Depot	Depot	Ward	Issue Note No.	Date

Consignee code	Allocation	Quantity Issued in Words
Requisition No.	Date	

Qty. Demanded in figures	P.L. No.	Category	Unit code	Quantity Issued in Figures

Qty. Demanded in words	Unit	Rate Rs.	P.	Value Rs.	P.

Description	No. of Packages	Wagon No.
	R.R.No.	Date

Signatures

Indenting Official Approving Officer	Issuing Official Receiving Official

……..RAILWAY **DEPOT 2**

REQUISITION AND ISSUE NOTE **S.1313/S.1319**

To be filled by Indentor To be filled by Store Depot CD Code **56**

Indentors	Consignee	Supplying Depot	Depot	Ward	Issue Note No.	Date

Consignee code	Allocation	Quantity Issued in Words
Requisition No.	Date	

Qty. Demanded in figures	P.L. No.	Category	Unit code	Quantity Issued in Figures

Qty. Demanded in words	Unit	Rate Rs.	P.	Value Rs.	P.

Description	No. of Packages	Wagon No.
	R.R.No.	Date

Signatures

Indenting Official Approving Officer	Issuing Official Receiving Official

Annexure '6/7'

……..RAILWAY **INDENTOR 3**

REQUISITION AND ISSUE NOTE **S.1313/S.1319**

To be filled by Indentor To be filled by Store Depot CD Code **56**

Indentors	Consignee	Supplying Depot	Depot	Ward	Issue Note No.	Date

Consignee code	Allocation	Quantity Issued in Words
Requisition No.	Date	

Qty. Demanded in figures	P.L. No.	Category	Unit code	Quantity Issued in Figures

Qty. Demanded in words	Unit	Rate Rs.	P.	Value Rs.	P.

Description	No. of Packages	Wagon No.
	R.R.No.	Date

Signatures

Indenting Official Approving Officer	Issuing Official Receiving Official

……..RAILWAY **RECEIPT 4**

REQUISITION AND ISSUE NOTE **S.1313/S.1319**

To be filled by Indentor To be filled by Store Depot CD Code **56**

Indentors	Consignee	Supplying Depot	Depot	Ward	Issue Note No.	Date

Consignee code	Allocation	Quantity Issued in Words
Requisition No.	Date	

Qty. Demanded in figures	P.L. No.	Category	Unit code	Quantity Issued in Figures

Qty. Demanded in words	Unit	Rate Rs.	P.	Value Rs.	P.

Description	No. of Packages	Wagon No.
	R.R.No.	Date

Signatures

Indenting Official Approving Officer	Issuing Official Receiving Official

Annexure '6/7'

……..RAILWAY **BILL 5**

REQUISITION AND ISSUE NOTE **S.1313/S.1319**

To be filled by Indentor To be filled by Store Depot CD Code **56**

Indentors	Consignee	Supplying Depot	Depot	Ward	Issue Note No.	Date

Consignee code	Allocation	Quantity Issued in Words
Requisition No.	Date	

Qty. Demanded in figures	P.L. No.	Category	Unit code	Quantity Issued in Figures

Qty. Demanded in words	Unit	Rate Rs.	P.	Value Rs.	P.

Description	No. of Packages	Wagon No.
	R.R.No.	Date

Signatures

Indenting Official Approving Officer	Issuing Official Receiving Official

……..RAILWAY **BLOCK 6**

REQUISITION AND ISSUE NOTE **S.1313/S.1319**

To be filled by Indentor To be filled by Store Depot CD Code **56**

Indentors	Consignee	Supplying Depot	Depot	Ward	Issue Note No.	Date

Consignee code	Allocation	Quantity Issued in Words
Requisition No.	Date	

Qty. Demanded in figures	P.L. No.	Category	Unit code	Quantity Issued in Figures

Qty. Demanded in words	Unit	Rate Rs.	P.	Value Rs.	P.

Description	No. of Packages	Wagon No.
	R.R.No.	Date

Signatures

Indenting Official Approving Officer	Issuing Official Receiving Official

Annexure '6/7'

RAILWAY

REQUISITION FOR STORES

DEPOT 1 **S.1302**				Requisition No.	Date
Indentor	Consignee	Consignee Code		Matrls Reqd. at	Depot
Description		P.L No.	Cat.	Qty. Demanded in Figs	Unit
				Qty. Demanded in Words	
				Allocation	
Last purchase particulars		Rate	Value	Purpose	

Funds Availability Certified. Originating Official. Approving Officer.

RAILWAY

REQUISITION FOR STORES

DEPOT 1 **S.1302**				Requisition No.	Date
Indentor	Consignee	Consignee Code		Matrls Reqd. at	Depot
Description		P.L No.	Cat.	Qty. Demanded in Figs	Unit
				Qty. Demanded in Words	
				Allocation	
Last purchase particulars		Rate	Value	Purpose	

Funds Availability Certified. Originating Official. Approving Officer.

Annexure '6/8'

General instructions for proper custody and Maintenance of Stores

(1) All materials shall be kept clean and free from dust. Materials liable to get rusted shall be kept free from rust by oiling/greasing whenever required.

(2) Stores enclosures shall be provided with weather proof protections where necessary.

(3) Materials shall be kept properly sorted and neatly stacked. Racks or other suitable arrangements shall be provided for storing any tools and petty consumable stores.

(4) Parts and small items of materials shall be kept in properly labelled bins or on shelves.

(5) Clutch Resetting Handles/Crank Handles/Electric key Transmitter keys, master keys and spare keys shall be kept locked up with the key in the custody of the JE/SSE.

(6) Bags of cement shall not be stored in too large a quantity nor for too long a period. Bags shall be issued strictly in rotation, the first bags is being the first to be issued out. Bags of cement damaged by rain or moist air shall not be issued.

(7) Point rodding shall be loaded in open wagons.

(8) Inflammable materials such as oils, spirits, petrol, etc., shall be stored separately and away from cotton waste, stationery and furniture. Naked lights and smoking shall not be allowed inside such godowns. Adequate arrangements shall be made for fire protection, i. e., fire extinguishers shall be kept handy at easily accessible spots and fire buckets shall be kept full of water/or sand where inflammable materials are stocked.

(9) Cables should be stored properly without damage to the cable drum as far as practicable. Cable shall not be removed from drums until ready to install. When a length of cable is cut off, the exposed end shall be effectively sealed.

(10) Signalling relays shall be kept separately at a place where environment is not likely to be damp and free from chemical pollution due to acid fumes, etc. Relays shall be transported from one place to another carefully in box with "thermocole" packing to avoid damage during transportation. Similar care should be taken for upkeep of tokenless block instruments and other sophisticated materials like Electronic Interlocking Cards, IPS cards. Axle Counters Relay Groups, etc.

(11) Paints shall be stored in a cool dry place away from flame or naked light. All containers shall be kept tightly closed to avoid loss of material due to skinning and contamination caused by open atmosphere.

CHAPTER 7 : ESSENTIALS OF SIGNALLING

Scope: - This Chapter deals with Essentials of Signalling concerned to Multiple Aspect Colour Light Signalling. Please refer to Previous Issue of SEM Part 1, 1988 for information on Semaphore signalling (LQ, MLQ, MAUQ) or Two aspect Colour Light Signalling, Mechanical operation of points.

Please refer to Concerned Chapters in this Manual for Topics on LC Gates, Block Instruments, Automatic Signalling, 25 KV RE, CTC

Section 1 - Signals generally

7.1.1.0 **Multiple Aspect Colour Light Signalling:-**

7.1.1.1 All signals shall be Electrically Lit

7.1.1.2 The signals shall be free from the possibility of phantom indication.

7.1.2.0 **Location of Signals**

7.1.2.1 Signals shall be so located and aligned as to;

(a) display the best possible view of their aspects to Loco Pilot/Motor man of approaching trains.

(b) avoid, as far as possible the possibility of the aspect of one signal being mistaken for the aspect of another, and

(c) avoid confusion between the lights of running signals and the lights of subsidiary signals or any other lights.

7.1.2.2 Signals shall normally be on the left of, or above the line to which they apply, unless authorized by special instructions to the contrary.

7.1.3.0 **Size of Lenses/LED Signals -** The size of lenses/LED Signals CLS of subsidiary signals shall be of smaller diameter than those of running signals. Subsidiary signals shall not be brighter than running signals.

7.1.4.0 **Number of Signals and height -** The number of signals and their height shall be limited to what is necessary for Safety and operational requirements.

7.1.5.0 **Outside interference -** All apparatus shall be so designed, located and secured as to obviate, as far as possible, outside interference.

7.1.6.0 **Signals controlled through slots -** Where a signal is controlled through slots, it shall not be possible to take 'off' a signal unless all the controls have been exercised by the controlling agencies. It shall be possible for any one of the controlling agencies to replace the signal to ' ON ' position by withdrawal of the control.

7.1.7.0 **Standard and type of Signalling on a section -** It is desirable that the signalling at all block stations and interlocked level crossings in the same section conform to the same type & standard

7.1.8.0 **Visibility of signals -** The minimum visibility distance of various signals shall be as under;

7.1.8.1 **Distant Signal -** 400 Metres.

Note: (1) In exceptional Circumstances, where it is not possible to have 400 meters visibility even by relocation of signal (due to permanent constraints like gradient, curve, terrain, tunnel, bridges etc) , then imposition of suitable speed restriction may be considered, based on local conditions

(2) An inner Distant Signal where provided, shall be visible from a minimum distance of 200 metres.

7.1.8.2 **All Stop Signals** - 200 metres.

If it is not possible to ensure 200 metres continuous visibility of any stop Signal while approaching it, a suitable speed restriction shall be imposed.

7.1.9.0 **Aspects of Signals -**

7.1.9.1 **Aspects of Distant Signal**s

(a) **In Single Distant Territory** : The Distant signal shall display a yellow light (bottom) in the ' ON ' position for the ' Caution' aspect, two yellow lights in a vertical line (Distinguishably apart) for the ' Attention ' aspect and one Green light for the ' Proceed 'aspect.

(b) In Second Distant Territory:-

(i) The Distant signal shall display, two yellow lights in a vertical line (Distinguishably apart) for the 'ON' aspect and one green light for the 'Proceed 'aspect.

(ii) The Inner Distant signal shall display a yellow light (bottom) in the ' ON ' position for the ' Caution' aspect, two yellow lights in a vertical line (Distinguishably apart) for the ' Attention ' aspect and one green light for the ' Proceed 'aspect. i.e Inner Distant signal shall have the same aspects as that of Distant signal in single Distant territory.

Note :- The Colour Light Distant Signal shall be provided with a marker consisting of a white coloured disc with letter ' P ' in black.

7.1.9.3 **Aspects of Stop Signal :-**

The signal shall display a ;

Red light in the 'ON' position, An yellow light for the 'Caution' aspect, Two yellow lights in a vertical line (Distinguishably apart) for the 'Attention' aspect and One green light for the 'Proceed' aspect

Note :- Number of aspects & their indications may vary from two to four as per type of signal & it`s location.

7.1.10.0 **Aspects Sequence**

7.1.10.1 When a Signal is displaying the 'Stop' aspect, the signal next in rear shall not display a less restrictive aspect than ' Caution '.

7.1.10.2 Every Multiple Aspect Signal, whether automatic, semi-automatic, or manually operated which, is required to display 'caution' aspect to a train shall normally be placed at an adequate distance from the next signal in advance. This adequate distance shall not be less than 1 kilometre. Where the adequate distance cannot be provided and the 'caution' aspect is being displayed, the next signal in rear shall display the 'attention'/'caution' aspect

7.1.10.3 If necessary, the 'Attention'/'Caution' aspect shall be repeated back on successive signals, in rear until the adequate distance is obtained.

7.1.10.4 In all cases the signal next in rear of a Signal protecting diverging lines shall display 'Attention'/'Caution' aspect in the 'OFF' position, when the points are set for any line other than the line over which the higher speed is permitted. When this signal cannot be placed sufficiently in rear of the 'STOP' signal protecting the Points to permit the Loco pilot of an approaching train to reduce to the sanctioned speed, the 'Attention'/'Caution' aspect shall be repeated back on successive signals in rear until the required distance is obtained.

7.1.10.5 For Metro Services, every Multiple Aspect Signal, whether automatic, semi-automatic, or manually operated which, is required to display 'Caution' aspect to a train, it shall normally be placed at an adequate distance from the next signal in advance. This adequate distance shall not be less than the maximum emergency braking distance of the trains running on the section at the maximum permissible speed on the ruling gradient. Where the adequate distance cannot be provided and the 'caution' aspect is being displayed, the next signal in rear shall display the 'attention/'caution' aspect. The 'attention'/'caution' aspect shall be repeated back on successive signals in rear until the adequate distance is obtained".

7.1.11.0 **Placement of Signals**

7.1.11.1 **Distant Signal –**

(a) Distant Signal shall be placed at an adequate distance in rear of the first Stop Signal.
(b) Such an adequate distance shall not be less than 1 kilometre in Single Distant Territory
(c) In Second Distant territory, Distant signal shall be placed at an adequate distance of 2 kms in rear of First Stop signal and Inner Distant Signal shall be placed at 1 km in rear of First Stop signal.

7.1.12.0 **Placement of Stop Signals**

7.1.12.1 **Home Signal :- T**he Home Signal shall be placed in rear of all connections, if any, on the line to which it refers. The Home Signal shall be placed not less than 180 metres in rear of the points up to which the line may be obstructed after line clear has been granted to the station in rear.

Note:-The distance shall be increased to 300 metres, where there is a continuous falling gradient of 1 in 260 towards station, between Distant and Home signal in single distant territory and between Inner Distant and Home signal in second distant territory.

7.1.12.2 **Routing Signal –** A Routing Signal shall be placed in the rear of the Points, which it protects.

7.1.12.3 **Starter Signal -** When one Starter Signal is provided for each converging line, it shall be so placed as to protect the adjacent running line or lines.

7.1.12.4 **Intermediate Starter Signal -** An Intermediate Starter Signal shall be placed clear of fouling marks in rear of the points, if any, which it protects.

7.1.12.5 **Advanced Starter Signal - U**nless approved under special instructions, an Advanced Starter Signal shall be placed outside all connections on the line to which it applies;

It shall be located at a distance of not less than 120 meters from the outermost points in the case of Single Line section. This distance shall be

reckoned from the starter in case of Double Line Section.

Note:- The portion between the fouling mark to block section limit must be track circuited.

7.1.13.0 **Combined Signals -** When owing to their location, it is necessary to combine two stop signals, one Stop Signal only may be provided under approved special instructions, capable of displaying any or all the following aspects :-

(a) Danger
(b) Caution
(c) Attention
(d) Clear

7.1.14.0 **Distinction between Signals - For Diverging Lines -** Requirements are as under

7.1.14.1 Unless otherwise permitted by approved special instructions, where two or more lines diverge, the signals shall be fixed with a route indicator of approved design. In special circumstances or where the number of signals is considerable, they may be fixed on bridge or gantries, with the approval of PCSTE & PCE/PCEE as the case may be

7.1.14.2 Where Signals are fixed on bridge or gantries, left hand signal in each case shall refer to the left hand line, the second signal from the left shall refer to the next line from the left and so on. The Signal for the main line shall be placed at a higher level than the signal or signals referring to the other running line or lines.

7.1.15.0 **Distinction between Signals - For converging lines.** Where Signals are fixed on bridge or gantries (with the approval of PCSTE & PCE/PCEE as the case may be), left hand signal in each case shall refer to the left hand line, the second signal from the left shall refer to the next line from the left and so on.

The Signal for the main line shall be placed at a higher level than the signal or signals referring to the other running line or lines.

7.1.16.0 **Route Indicator -** The type of Route Indicator to be provided instead of separate signals shall be as under -

(a) For speeds in excess of 15 kmph on Diverging lines – (i) Direction type Route Indicator for up to 3 diversions in each side. (ii) Direction Type Route Indicator with LED Matrix for more than 3 diversions in each side.
(b) For speeds not exceeding 15 kmph on Diverging lines - Any Route Indicator of approved design.

7.1.17.0 **Subsidiary Signals**

7.1.17.1 **Calling-on Signal:-**

Function :- A Calling-on signal when taken 'OFF' calls on the Loco Pilot of a train to draw ahead with caution, after the train has been brought to a stop even though the Stop Signal above it is at 'ON'. It indicates to the Driver that he should be prepared to Stop short of any obstruction.

7.1.17.2 A Colour Light Calling-on Signal shall display no light in the ON position and a miniature yellow light in the 'OFF' position.

The colour light Calling-on Signal shall be provided with a marker consisting of a white enamelled disc with letter 'C' in black.

7.1.17.3 **Calling On Signal Placing and working:**

(a) Calling-on Signal shall be placed below a Stop Signal governing the approach of a train. Under approved special instructions, a Calling-on Signal may be provided below any other Stop Signal except the last Stop Signal.

(b) The Calling-on Signal shall not be capable of being worked at the same time as the Stop Signal above or shunt Signal below it, if any.

(c) A Calling-On signal shall detect all the points in the route, which the main signal above it detects excluding those in overlap.

(d) At stations where Station Master controls the reception and dispatch of trains, such control shall be extended to Calling-on Signals also.

(e) It is required to provide track circuits at a suitable distance and a time delay circuit to ensure that the Calling-on Signal is taken 'OFF' only after the train has been brought to a stop.

Note: (i) For reception signals, 5 rail length Calling-on track circuit and 60 seconds time delay shall be provided

(ii) For Despatch signals, no such time delay is required. However, Operating staffs have to ensure that the train has stopped before taking Off Calling-on signal, through a provision in Station Working Rules.

7.1.17.4 **Shunt Signal:-**

Shunt Signals shall be of the Position Light type. The lights of a Position Light Signal shall be white. Two lights of a Position Light signal shall be horizontal in the 'ON' position and 45° above the horizontal in the 'OFF' position

Placement Of Shunt Signals -

(a) Shunt Signals may be separately located on posts or close to the ground or may be fixed below Stop Signals other than the First Stop Signal of a station.

(b) Where a Shunt Signal is required to be fitted on a Signal post on which a Calling-on signal is also fitted, the Shunt Signal shall be fitted below the Calling-on Signal.

(c) A Shunt Signal placed below a Stop Signal or a Calling-on Signal shall not be capable of being worked at the same time as the relative Stop Signal or Calling-on signal.

(d) Where a Shunt Signal is fixed below a Stop Signal it shall show no light in the 'ON' position.

7.1.17.5 **Starter Indicator: -** Starter Indicators may be provided to repeat the aspect of the Starter as an aid to the Guard to enable him to know the aspect of the Starter. This repeater may be provided at a convenient place. The Starter Indicator shall exhibit no light when Starter is at 'ON' and yellow light when it is 'OFF'.

Note :- This is typically required for Long/Curved Platforms where guard is unable to see Starter Signal aspects

7.1.17.6 Please refer to Chapter 14 for Gate signals.

Section 2 - Warning Boards

7.2.18.0 A Warning Board to warn the driver of his approaching a Stop Signal shall be provided at a distance of 1.4 km. in rear of the First Stop Signal of a station and Gate Slop Signal. This distance may be increased suitably to cater for gradients. *Note: (a) On Broad Gauge and Metre Gauge, Passenger Warning Board should be provided at not less than 1000 m in rear of the first stop signal of a station, intermediate block stop signal and gate stop signal. The Passenger Warning Board need not be provided where the distance between distant signal and stop signal is 1000 m or more.*

(b) Goods warning board shall be provided at a distance of 1400 m in rear of the first stop signal of a station, intermediate block stop signal and gate stop signal.

(c) Wherever second distant signal has been provided no warning board is required.

7.2.19.0 On sections where emergency braking distance of more than 1.0 km is to be catered for, a second approach signal shall be provided. On 3 aspect automatic territory where emergency braking distance is more than 1.0 km is to be catered for, 4 aspect automatic signalling shall be provided. If not provided, suitable speed restriction shall be imposed".

7.2.20.0 In cases where Gate Signals and station signals happen to be located close to each other, only one warning board shall be provided at a distance of not less than 1.4 km. from the First Stop Signal

7.2.21.0 Where inter-signal distance between two signals is less than the distance required for warning Board, the signals in rear shall either be controlled by the signal in advance or they shall be combined.

7.2.22.0 Self-reflecting sheets of approved type or reflectors shall be provided on the warning board as per approved drawings.

7.2.23.0 The warning board shall normally be on the left of the line to which it applies unless there are special reasons to the contrary.

Section 3: Indication Boards

7.3.24.0 Indication Boards are provided to give warning to Loco Pilots about changes in type of signalling or type of Block Working.

7.3.25.0 Where a Block station, which is provided with two aspect signalling without a Warner signal in rear of the First Stop Signal, is approached from a section provided with Multiple Aspect Signalling or with two aspect signalling with a Warner signal in rear of the First Stop Signal, an Indication Board bearing alternate yellow and black diagonal stripe shall be fixed at not less than 1.4 km. in rear of the First Stop Signal. The Board shall have a legend "Approaching unwarned Stop Signal".

7.3.26.0 When a station provided with single line token block is approached from a Double Line or a Single Line section equipped with token less block working, an indication board with a legend "Entering Token Territory" shall be provided at the First Stop Signal of the station.

7.3.27.0 Where the block working at one end of a station is automatic and the other end is absolute, an indication board with suitable legend shall be provided at the First Stop Signal of the station. The legend on the board shall be "Entering Absolute Block Territory" or "Entering Automatic Block Territory" as appropriate.

7.3.28.0 **Shunting Limit Board -** A Shunting Limit Board shall be provided at Class 'B' stations over a single line section worked on Absolute Block System where shunting in the face of an approaching train is permitted and where an Advanced Starter is not provided. The Shunting Limit Board shall be fixed at a distance of 180 metres in advance of the First Stop Signal.

7.3.29.0 **Misc Indication Boards:**

(a) An indication board shall be provided at the foot of Gate signal if it is protecting more than one mid section LC gate with appropriate legend.
(b) An indication board shall be provided at the foot of the IB signal if it is protecting mid section LC gate with legend" Look out for LC gate when passing IB signal at ON".

7.3.30.0 **Block Section Limit Board (BSLB) :-**

7.3.30.1 At Class 'B' stations worked on Double Line Absolute Block System (with multiple aspect signalling) a Block Section Limit Board shall be provided where there are no points or the Outermost points are trailing.

7.3.30.2 The Block section Limit Board shall be located at 180 metres in advance of the Home Signal and protecting the fouling mark of the trailing points, if any.

7.3.31.0 The Block section Limit Board or the Shunting Limit Board shall be so located that the legend is facing towards the station. They shall be fitted with lamps showing white light in both directions during night Or reflective type coating painted.
Note.-Chapter 12 may be referred to for the painting details.

Section 4 – Points, Point Indicators, Isolation, SPI & Spring Loaded Points

7.4.32.0 **Obstruction test: -** The insertion of a 5 mm. obstruction between the switch rail and the stock rail of a Facing Point, approximately 150 mm. from the toe of the switch shall prevent the Point from being locked and the relative signal or signals, where provided, being taken 'OFF' for movements in the both directions.

7.4.33.0 **Detection by Shunt Signals :-**

7.4.33.1 In all installations, Points on a running line used in the facing direction for shunting movement shall be provided with a Facing Point Lock/Point machine

7.4.33.2 The Shunt Signal shall lock and detect all Points in the route.

7.4.34.0 **Point Indicators**

7.4.34.1 **Type** - Point Indicators shall be of the target type

7.4.34.2 **Provision** - Point indicators shall be provided at all Points on running lines which are not interlocked with Signals, unless the position of Points is otherwise proved.

7.4.34.3 **Indications:**

(a) When the points are set for the straight line, the indicator shall display a white target by day and a white light by night, in each direction.
(b) When the points are set for the diverging line, the indicator shall display the edge of the disc by day and a green light by night, in each direction.
(c) At Points where a green light would give a misleading indication to a Loco

pilot, a red light may be permitted under special instructions.

7.4.35.0 **Trap Indicators**

7.4.35.1 **Type** - Trap indicators shall be of the target type.

7.4.35.2 **Provision :- T**rap indicators shall be provided at all trap Points on running lines which are not interlocked with signals unless the position of trap points is otherwise proved.

Note :- Points leading to a short dead end and used solely for the purpose of trapping the running line or sidings shall be treated as derailing switch.

7.4.35.3 **Indications –**

(a) When the Trap Points or derailing switches are open, the Indicator shall display a red target by day and a red light by night in each direction.

(b) When the trap Points or derailing switches are closed, the indicator shall display the edge of the disc by day and a green light by night, in each direction. At trap points where a green light would give a misleading indication to a Loco pilot, a white light may be permitted to indicate trap "CLOSED" position only.

7.4.36.0 **Size of lights and placing:-** Points and trap Indicators shall be provided with miniature lights and as a rule be placed as close to the ground as circumstances permit.

7.4.37.0 **Shunting Permitted Indicator**

7.4.37.1 **Type -** Shunting Permitted Indicator may be of the light type.

7.4.37.2 **Provision -** Shunting Permitted indicator may be provided to permit uninterrupted to and fro movements towards shunting neck or other connected lines.

7.4.37.3 **Indications -** Day and night indications of the Shunting Permitted Indicator shall be as follows :

Type	**Indication when shunting is permitted in the direction to which it refers**		**Indication when shunting is not permitted in the direction to which it refers**	
	Day Indication	**Night Indication**	**Day Indication**	**Night Indication**
Disc Type	Black disc with a yellow cross painted on it	Yellow cross light	Edge of disc	No light
Light Type	Yellow cross light	Yellow cross light	No light	No light

7.4.37.4 **Working -** Shunting permitted Indicator may be operated by a ground lever/EKT. SPI will work in conjunction with stop signals and is provided for shunting movement in either direction in the non-interlocked portion of the yard after being isolated from interlocked portion. Either shunting permitted indicator or the

associated shunt signal signal/stop signal can be taken off at a time.

7.4.38.0 **Spring Points :-**

7.4.38.1 Spring points shall be used only in exceptional cases where the same purpose cannot be achieved conveniently by other means.

7.4.38.2 Spring Points if used in the facing direction must be fitted with facing point locks. They may be operated from a Cabin or from a ground lever. In the latter case, the ground lever shall be of such a type that it will not move when the Points are being trailed through.

7.4.39.0 **Means of Isolation :-**

7.4.39.1 Sand humps, trap points, or other approved means of isolation shall be provided on all goods lines and sidings at their junctions with passenger lines, the normal setting being such as to prevent the passenger lines from being fouled.

7.4.39.2 At interlocked layouts, the means of isolation shall be interlocked with the relative signals.

7.4.39.3 In order to maintain safety for through running, Points for trap sidings shall not be inserted in the main or through line. An exception may, however be allowed under approved special instructions in cases where, owing to grades in or near stations, it is necessary to prevent -

(a) trains being brought to a stand at a Stop Signal on a rising grade, or

(b) Vehicles running away from the station.

(c) An exception may also be allowed under approved special instructions, where it is considered necessary in the interest of working to receive trains from opposite directions at the same time.

Note:- At stations where trap siding are inserted on Main lines, through running shall be permitted only under approved special instructions.

7.4.40.0 **Provision of Isolation**

7.4.40.1 At block stations where trains are permitted to run at speed in excess of 50 kilometres an hour, the line on which that speed is permitted, shall be isolated from all connected lines during the passage of the train.

7.4.40.2 Such isolation need not be provided at Block stations where trains are permitted to run at a speed of 50 kmph. or below provided GR. 4.11(2) is complied with

7.4.40.3 The provisions of Para no. 7.4.40.1 do not apply to

(a) Junctions where two block section lines meet at the same end of a station and system of Block Working with adjacent stations on both lines is done by one of the approved means and the junction is equipped with full complement of signals.

Note (i) At such junctions, the First Stop Signal on Single Line sections shall be placed at an adequate distance from the outermost facing point/fouling mark, the adequate distance not being less than the sum total of the adequate distances prescribed in GR. 8.01 in regard to condition for granting Line Clear and GR. 3.40 in regard to the condition for taking off the Home Signal.

(ii) On Double Line section, the First Stop Signal shall be placed at an adequate distance from the outermost facing point/fouling mark, this distance not being

less than the adequate distance prescribed in GR. 8.01 in regard to the conditions for granting Line Clear.

(b) Block stations where track circuits or other appliances have been provided to prove whether the connected non-isolated lines are clear or occupied and the signalling is such that a distinctive aspect is given to the Loco Pilot of a run through train, restricting the speed to 50 kilometres per hour when a connected line is occupied.

(c) Catch and Slip Sidings and Sidings provided for isolation purposes only.

7.4.41.0 **Arrangements of Sidings**:- Sidings shall be so arranged that shunting operations upon them involve the least possible use of, or obstruction, to running lines.

Siding Points on Passenger running lines outside station limits

7.4.42.0 **Facing Points (Single Line) : -**

7.4.42.1 Where Siding Points take off a running line in a facing direction, the following minimum equipment shall be provided :-

(a) A gauge tie plate where steel sleepers are not provided;

(b) A facing point lock or point machine which shall lock each switch independently

7.4.42.2 The control of the points shall be made by means of a key or other suitable device which shall secure the bolting mechanism of the Points in the plunged or locked positions, i.e., when the Points are set and locked for the running line. The means for control of the Points shall be interlocked with the block system in force.

7.4.42.3 Where Siding Points are provided only with the minimum equipment specified in Para no. 7.4.42.1, an appropriate speed limit over the facing points shall be imposed for all trains passing over such Points in the facing direction only and an 'S' marker at the Points and a speed indicator at not less than 30 metres from the Points shall be provided, neither of which, need be lighted. In addition a caution and termination indicator shall also be provided as for open line speed restrictions. Where the sanctioned speed of the section does not exceed 50 kmph, the 'Marker' at the points and the Indicators need not be provided.

7.4.42.4 The siding points on passenger lines outside the station limit shall be detected for permitting speeds of up to 50 kmph in facing direction. In case the detection is not provided, speed restriction of 15 kmph shall be imposed

7.4.43.0 **Facing Points Double Line:- T**he minimum equipment for Points taking off a running line in the facing direction on Double Line shall be in accordance with the provisions of Para 7.4.42.0.

7.4.44.0 **Trailing Points (Double Line)** - Where Points take off a running line in a trailing direction the following minimum equipment shall be provided for unrestricted speed :—

(a) A gauge tie plate where steel sleepers are not provided.

(b) A suitable type of key lock or equivalent mechanism, the key of which can only be extracted when the Points are set and locked for the running line. The points shall be controlled through the block system in force;
(c) An 'S' Marker at the Points which need not be lighted.

Section 5: Place of Operation and Interlocking

7.5.45.0 **The Place of Operation** shall be so located, as to provide the Operator with sufficient space and to enable him to have an adequate view of signals operated from place of operation as well as of the movements taking place in the portion of the yard controlled by him unless a substitute for direct vision is provided.

7.5.46.0 **Signal and Interlocking apparatus Installation** :- Signal and interlocking apparatus shall be, as far as possible, so installed and the connections and circuits so arranged, that the relevant fixed signal shall remain at or return to its most restrictive aspect in the event of failure of any part of its connection and circuits.

7.5.47.0 **Essentials of Interlocking.—A**pparatus provided for the operation and control of signals, points, etc., shall be so interlocked and arranged as to comply with the following essentials :—

(a) It shall not be possible to take 'OFF' a running signal, unless all points including isolation are correctly set, all facing points are locked and all interlocked level crossing are closed and locked against public road for the line on which the train will travel including the overlap.

(b) After the signal has been taken 'OFF' it shall not be possible to move any points or lock on the route, including overlap and isolation, nor to release any interlocked gates until the signal is replaced to the 'ON' position.

(c) It shall not be possible to take 'OF' at the same time, any two fixed signals which can, lead to any conflicting movements.

(d) Where feasible, points shall be so interlocked, as to avoid any conflicting movement.

7.5.48.0 **Holding of Route :-**

7.5.48.1 (a) For holding the route, it is essential to provide continuous track circuit/Axle Counters from the signal upto the end of the route (including overlap where required) at all class of stations. The route shall be released normally on passage of train

(b) The Stop signals shall be provided with Approach locking Or Dead approach locking.

7.5.48.2 At a station where trains run through at speeds more than 50 kilometres per hour, such arrangements to hold the route are also required in case of trailing points situated more than 180 meters from the signal controlling them.

However, such arrangements are not required if the Points are locked in either position by the signal in advance.

7.5.48.3 Route holding arrangements for facing or trailing points are, however, not necessary, if due to pre-defined system of control, it is impossible under normal working conditions, for the Points to be operated before the train has passed.

7.5.48.4 Overlap shall be released with either the passage of Train or Time delay or Emergency Route Release.

7.5.49.0 **Locking trailing points by Signals in advance :—**

Stop Signals which are next in advance of trailing points operated from the same

panel/VDU shall, when taken off, lock such point in either position unless route locking is provided or the distance between the Points and the signal is such that the locking interferes with traffic movement.

7.5.50.0 **Clearance at Junction Point :-**

7.5.50.1 Where it is difficult under normal conditions of visibility for an Operator to estimate clearance, track circuiting/Axle counter shall be provided in order to define the fouling points of junctions, loops, siding connections, crossings etc.

7.5.50.2 Where the movement of trains over the Points is not visible to the Operator operating the Points— Occupation of the track between Stop Signal reading over the Points up to the fouling mark ahead of such Points shall be electrically indicated at the place of operation;

7.5.51.0 **Track Circuit/Axle Counter Control of Signals** :- Where continuous track circuiting/Axle Counters are provided, the occupation by a vehicle of any track circuited/Axle Counter section shall control the running signal or signals leading to the same line and shall also lock in either position, the Points on the section.

7.5.52.0 **Slot Circuits**:- Slot circuits for Home Signals leading to non-track circuited reception lines shall be so arranged that a slot once given is effective for the reception of only one train and a fresh slot has to be given for a subsequent train.

7.5.53.0 **Station Master's Control**

7.5.53.1 The Station Master shall be provided with interlocked electrical control over the Home and Last Stop Signals. Separate control Calling-on Signal may be provided where required.

7.5.53.2 The Station Master's control over Home Signals can be dispensed with only if all the following conditions are satisfied :-

(a) All reception lines are fully track-circuited from Home Signal to corresponding Last Stop Signal in double line and Home Signal to Home Signal in Single line; or All reception lines are track-circuited from the fouling mark to fouling mark and the non-track circuited portion of the line from the fouling mark to the Home Signal is within the range of visibility of the Cabin Assistant Station Master.

(b) Station Master does not allot the line; and

(c) Cabin Assistant Station Master posted in the Cabins operates the block instruments and controls receptions/despatch signals.

7.5.54.0 **Interlocking of Last Stop Signal and First Stop Signal with Block Instrument:**- Where block instruments are in use;

(a) The Last Stop Signal shall not be capable of being taken off until Line Clear has been obtained from the block station in advance;

(b) It shall not be possible to close the line and grant or receive ' Line Clear unless the 'ON' aspect of the relevant approach signals are proved.

7.5.55.0 **Automatic replacement of Signals:**

7.5.55.1 On Double Line Sections where Double Line Block Instruments have been provided and on Single Line sections where token less block instruments have been provided, the Advanced Starter Signal as well as the Starter Signal shall be automatically replaced to 'ON' position by the entry of a train into the block section.

7.5.55.2 On Single Line Sections where Token instruments are in use, it is desirable to provide such automatic replacement arrangement for the Last Stop Signal.

7.5.55.3 On Double Line Sections where Double Line Block Instruments have been provided and on Single Line sections where token less Block Instruments have been provided, the Home Signal shall be automatically replaced to 'ON' position by the passage of a train in advance of the Home Signal.

Section 6 - Interlocking of Catch Siding and Slip Sidings

At stations where Catch and Slip Sidings are provided in accordance with the Rules for opening of a Railway, interlocking arrangements and other safeguards in accordance with Paragraphs 7.6.57.0 to 7.6.62.0 shall be provided.

7.6.57.0 **Reception of trains:**

7.6.57.1 Either a minimum of two Stop Signals shall be provided in rear of the Catch Siding points or the First Stop Signal shall be at an adequate distance equal to Block overlap from the catch siding points.

7.6.57.2 A train shall first be brought to a stop at the First Stop Signal, before the Catch Siding points are set for the main line and the reception signals taken 'OFF' unless the following conditions are satisfied :—

(a) The line on which the train is to be received is clear and the train is to be received on the main line.

(b) The points leading to the Catch Siding as well as all the Points required for a run through train are set for the main line immediately after granting Line Clear to the Block station in rear.

(c) Line clear has been obtained from the Block section in advance.

(d) The gradients in the block section ahead are such that the train can be brought under control easily.

7.6.58.0 **Setting of Catch Siding/Slip Siding :**—The take off points of a Catch/Slip siding shall normally be set and locked for the siding and interlocking between the points and block instruments shall be provided as in Para 7.6.59.0 and other safeguards as in paragraph 7.6.60.0 to 7.6.62.0

7.6.59.0 **Interlocking of Catch/Slip Siding points with Block Instruments:-** The interlocking shall be such that the key required to set the siding points is released from the instruments in the "Train Going to"/"Train Coming from" position and once the key is removed from the block instruments, the instrument gets locked in the relevant position. The instrument can be normalised only after the points are set for the Slip Siding/Catch siding and the "Train Going To"/"Train Coming from" key is released from the Points and brought back to release the instrument. Where a Slip Siding is located at the departure end of a Double Line Station, the interlocking shall be such that the Points can be set towards the Block section only when the block instrument is set to “Line Clear".

Note:- For more Information on Block working , Please refer to Chapter 18

7.6.60.0 **Audible Indication:-** An audible indication shall be provided at the place of operation of Points as an aid to the Operating staff, indicating that the train has been received or despatched and that the points shall now be reset for the Catch Siding/Slip Siding. This indication would continue till the points are reset for the Catch Siding/Slip Siding.

7.6.61.0 **Safeguards in working.**—In all cases where interlocking arrangements stipulated in Para 7.6.59.0 are provided between the Slip/Catch Siding and the block instrument, the following safeguards shall also, *inter alia* be provided in the Station working Rules :—

(a) Shunting on non-isolated lines shall cease once Line Clear has been granted

(b) Points shall remain set and locked for the Catch Siding until Home Signal has to be taken 'OFF'.

(c) Before normalising the block instrument the Station Master /Assistant Station Master, shall verify the Complete arrival or despatch of the train, even if he is getting an audible warning, which can arise due to reception or despatch of a. complete train or part thereof.

7.6.62.0 **Emergency Key: -** An Emergency Key of the Catch/Slip Siding shall be kept in a sealed box under the custody of the Station Master. This is used for operating the Slip Siding/Catch Siding points when either the block instruments have failed, or when the train is still in the Block section and a train is required to be despatched into the Block section.

Section 7 - Operation of Points and Signals

7.7.63.0 **Operation of Points and Signals by Electrical Means**

The points and Signals shall be operated from a -

(a) Control panel for Relay Interlocking

Or

(b) Control panel and / or VDU for Electronic Interlocking.

Note :- Dual VDU shall be provided in lieu of Control Panel and VDU in all future Installations

7.7.64.0 **Locking and sealing arrangements: -** Locking and sealing arrangements shall be provided for the covers of locking tray of lever frame/power frame/Relay Rooms/Control Panels/Block Instruments etc., to secure against unauthorized opening.

7.7.65.0 **Operation of Signals by Electrical means: T**he aspect of the signal shall be indicated at the place of operation.

7.7.66.0 **Operation of points by Electrical means:**- unless the point mechanism has made the required movement and the points are set and locked in a position corresponding to the switch/Knob , No indication shall be given.

7.7.67.0 Visual indicators shall be provided to show the "Normal" and "Reverse" positions of all points, the condition of all track circuited sections, route setting and to repeat the indications of Colour Light Signals.

7.7.68.0 **Track circuits/Axle Counters as a means for detecting obstructed lines** - Adequate arrangements shall be made where necessary for reminding the Operator of vehicle., which are standing within his control. In case of passenger lines or where light engine crossing and shunt movements are frequent or where stop Signals are at a considerable distance from the Cabin, or where the view of the Operator is likely to be obstructed, the provision of track circuits or other automatic device is essential.

Note —Whenever the use of normally closed type of track circuit is specified in this Chapter, it shall automatically cover the use of axle counter in lieu of track

circuit.

7.7.69.0 **Track Circuiting/Axle Counters :-** At Stations where points and signals are operated from an Assistant Station Master's Office at a central place, track circuiting/Axle Counters of the entire station section including all lines where direct reception is provided , shall be provided .

7.7.70.0 **Point & Signal Circuits :-**

7.7.70.1 The circuits controlling the operation and indication of Points shall be so arranged that, as far as practicable cross connection or a short circuit cannot operate the Point or give a false indication of the position of the points.

7.7.70.2 Circuits controlling the operation and indication of Signals shall be so arranged that, as far as practicable, a cross connection or a short circuit on any of the wires cannot give a false clear indication.

7.7.70.3 The battery or power supply for line circuits, as far as practicable, shall be arranged at the end of the circuit farthest from the operated unit.

7.7.70.4 Where the line supply voltage normally exceeds 125 volts protection arrangements as laid down in the Indian Electricity Rules, 1956, shall be provided.

7.7.71.0 **Crank Handle interlocking :-**

7.7.71.1 At installations with motor operated points, crank handles shall be kept electrically locked and released for every operation requiring a crank handle. The interlocking systems shall be such that the crank handle cannot be released when relevant signals are taken 'OFF' and once the crank handle is released the relevant signals cannot be taken 'OFF' until the crank handle is restored to its electric lock and gets locked therein.

7.7.71.2 In case of major stations, grouping of Points shall be resorted to and non-interchangeable crank handles provided for each group.

7.7.71.3 **Emergency Crank Handle:** To provide for an eventuality **of** interlocking system failure or Crippling Power Failure, resulting in inability to extract the normal crank handle, an Emergency crank handle may be provided, which when extracted, shall cause relevant signals to assume 'ON' aspect and it shall not be possible to take 'OFF' the relevant signals until the Emergency crank handle(s) is (are) restored to its EKT(s).

Note :- (i) At way side stations having motor operated points , where crank handles are located in SM's office, an emergency crank handle which is universal to all the points Or emergency crank handles equal in number and having same ward as normal crank handles , may be kept free in EKT(s) but proved , without lock coil/Pawl in a glass fronted box (duly sealed by signal staff and padlocked with keys in the possession of on-duty SM). Such usage shall be recorded in TSR with reasons there of and monitored through Datalogger.

(ii) the interlocking shall be such that Emergency crank handle(s) EKT key 'IN 'contact is proved in concerned crank handle circuit. The crank handle of the concerned group can be taken out only when emergency crank handle is 'IN

,

Section 8 : Relay/Electronic Interlocking Installations - Route Setting type

7.8.72.0 **Relay & Electronic Interlocking Installations (Route Setting Type)**

Route setting shall be on the basis of "Entrance-Exit" principle. Facility for individual operation of points shall also be provided.

Note :- Where ever Control Panel is mentioned it also applies to Video Display Unit (VDU) used in Electronic Interlocking.

7.8.73.0 Route setting, i.e., setting and locking of the route and clearing of main signals include the following processes: —

(a) Prove that the route selected does not conflict with any other route which has already been set and locked;
(b) Operate the points - in the route, overlap and those required for isolation to the required position, lock and detect them;

(c) lock the route including overlap and isolation;
(d) Prove that the tracks in the route and the overlap are clear;
(e) Prove that all interlocked level crossing gates in the route including overlap are closed and locked against road traffic;
(f) prove that the signals are controlled by the Block System in force as necessary;
(g) Prove that the relevant crank handles are electrically locked in;
(h) Clear the signals after the above conditions are satisfied.
(i) Prove that the relevant emergency crank handle is 'IN'. (where provided)

7.8.74.0 The system shall be so designed that the points in a route once set shall remain in the last operated position until they are required to be operated for the setting up of another route or for individual operation except when stipulated otherwise

7.8.75.0 The route shall ordinarily be released by the passage of the train over the route. Sectional Route Release may be provided where necessary. Emergency route release shall also be provided with appropriate time delay.

7.8.76.0 Approach locking and back locking shall be provided for all routes governed by main signals. Approach and back locking shall be continuously effective from the approach track which shall commence from an adequate distance in rear of the signal.

Note :- In the absence of 'approach track' , Dead Approach Locking shall be provided with suitable Time Delay

7.8.77.0 **Point Operation:**

7.8.77.1 The points shall be operated by electric point machines. Points leading to un signalled lines or sidings may be manually operated with suitable electrical control

7.8.77.2 Where required, provision may be made for emergency operation of points during track circuit failures. Each such individual operation shall be recorded on a suitable Counter.

7.8.78.0 When, required, provision may be made for emergency sub-section route release. Where such release is provided, it shall be possible to release the sub-section only after physical verification of the sub – section by traffic representative and using the emergency sub – route cancellation facility which

shall release the sub – route after a delay of minimum 120 seconds.

7.8.79.0 All running signals shall be colour light multi-unit type and shunt signals shall be of the Position Light type. Calling on signals may be provided as per Operational needs

7.8.80.0 A signal in the 'OFF' position shall be replaced to 'ON' position automatically by the passage of a train. It shall also be possible to replace a manually controlled signal in the 'OFF' position to the 'ON' position manually

7.8.81.0 The Calling-on Signals shall lock and detect points in the Route and prove that the level crossing gates in the Route are closed and locked against road traffic

7.8.82.0 Shunt movements shall, as far as possible, be controlled by Shunt Signals. The Shunt Signals shall lock and detect the points in the Route and prove that the tracks in the route excluding the berthing track are clear and that the level crossing gates in the Route are closed and locked against the road traffic.

7.8.83.0 Where local control panels/VDUs are provided over a portion of the yard, the points and signals of that portion of the yard shall be operated as follows: —
(a) directly from the main control panel ;or

(b) from the local control panel/VDUs after obtaining release from the main control panel; or

(c) from the local control panel/VDUs after operation of an emergency switch. This emergency operation shall be suitably counted in a Counter.

7.8.84.0 Suitable indications shall be provided on the control panel/VDUs for points, track sections, routes signals including Calling-on Signals, 'A' Markers, level crossings, train approach, power supply route cancellation, slot and crank handle interlocking, emergency crank handle (where provided) etc., as required

Note :- Please refer to Chapter 21 for further Details

Section 9: Requirements of Signalling Circuits using Electronic Equipment & other References

The following fail-safe principles shall be incorporated in the design of the equipment

7.8.85.0 Component failure shall be self-detecting by way of causing a signal to display a most restrictive aspect as far as practicable.

7.8.86.0 All fail-safe circuits shall work on continuous energisation principle such that open circuits in wiring, relay contacts, etc., or loss of power supplies shall not cause unsafe conditions.

7.8.87.0 Due consideration shall be given to the effects of faults in fail-safe electronic equipment to allow open or short circuit or earthling conditions and variation in component values due to ageing, replacement of faulty component with new components of specified tolerance, etc. Safety shall not be impaired as a result of multi-terminal devices failing-either open circuit, short circuit or with partial short circuit between any pair of terminals or earthing

7.8.88.0 The physical construction of fail-safe equipment shall be designed to eliminate the possibility of external objects causing short circuits between combinations of terminals in vital circuits. This may be achieved for example, by adequate

separation of terminals, and by the fitting of protective shrouds where necessary.

7.8.89.0 RDSO`s Technical Guidelines shall be followed in the design, Installation, Testing , Commissioning & Maintenance of Electronic Equipments/Systems

7.8.90.0 Please refer to Chapter 18 for Block working & Intermediate Block working

7.8.91.0 Please refer to Chapter 20 for Automatic Block signalling

7.8.92.0 Please refer to Chapter 21 for Relay & Electronic Interlocking

7.8.93.0 Please refer to Chapter 23 for ATP CTC TMS

7.8.94.0 Please refer to Chapter 22 for Requirements in 25 kv A. C. Electrified Sections.

7.8.95.0 **Standards of Signalling and Interlocking**
Minimum signalling equipment which shall be provided at various stations is indicated in the Tables below :

☞ *Note : This Chapter has under mentioned Annexures for Further Study*		
S.no	*Annexure*	*Description*
1	*7/1*	*Extract from Schedule of Dimensions (SOD)*
2	*7/2*	*Extract of Rules for Opening of Railway Lines for "Isolation"*

NDIAN RAILWAY SIGNAL ENGINEEIRNG MANUAL 2020, Chapter 7 Section : Important Minimum Signalling Features - As per para 7.8.95.0

SN	DETAILS	STD I	STD II	STD III	STD IV
1	**Allowable Speed(Kmph)**	Upto 50	Upto 110	Upto 140	Upto 160
2	**Isolation (By approved means)**	Y* See Notes below	Required	Required	Required
3	**Signalling (MACLS)**	**MACLS** Permitted	**MACLS** Permitted	**MACLS** Permitted	**MACLS** With MA (Movement Authority)
4	**Double Distant**	Not Compulsory	Required (on sections where goods Trains have a breaking distance more than 1 Km)	Required Or 4A Automatic signalling	Required Or 4A Automatic signalling
5	**Point Mechanism (Operation , Locking & Detection)**	Point Machine	Point Machine	Point Machine	Point Machine & Clamp type with Direct TWS
6	**Interlocking (Relay/Electronic)**	Permitted	Permitted	Permitted	Permitted
7	**Train Detection (Track Ckt/Axle Counters)**	On All Run through Lines	On All Running Lines	On all Running Lines	On all Running Lines
8	**Block Working (Absolute/Automatic signaling)**	Permitted including Token Working	Permitted (Other than Token)	Permitted (other than Token) Or 4A Automatic signalling	Permitted (other Than Token) Or 4A Automatic signalling
9	**Verification of complete arrival of train (By approved means)**	Not Compulsory	Required At stations with centralized operation or in high density routes	Required	Required
10	**ATP**	Not Compulsory	Desirable	Desirable	Required
11	**Mobile Train radio communication**	Not Compulsory	Desirable	Desirable	Required
12	**CTC**	Not Required	Desirable	Desirable	Desirable

NOTES

(1) ISOLATION

Y* : Isolation is not compulsory , provided that the conditions laid down in the second paragraph of the General Rule 4.11 are complied with “Limits of speed while running through stations:-

1. No train shall run through an interlocked station at a speed exceeding 50 kilometers an hour , or such less speed as may be prescribed by approved special instructions unless the line on which the train is to run has been isolated from all other lines by the setting of points or other approved means and interlocking is such as to maintain this condition during the passage of the train
2. In every case in which trains are permitted to run through on a non isolated line , all shunting shall be stopped and no vehicle unattached to an engine or not properly secured in accordance with General Rule 5.23 may be kept standing on a connected line which is not isolated from the through line”.
3. All Passengers running Lines shall be isolated from Goods lines or Sidings connected thereto . All goods running Lines may be isolated from all sidings connected there to (Ref: Rules for Opening of Railway Lines - para 36(5),(6))

(2) APPLICABILITY

1. The provisions of the above Table shall apply to future Signalling and Interlocking Installations. Wherever existing installations do not fulfill these requirements, existing speed of operation may be permitted to continue.
2. In case Block working is achieved directly through Electronic Interlocking , provision of separate block instrument is not required.
3. This Table does not cover Semaphore Signalling (LQ,MAUQ, MLQ), 2 Aspect Colour Light Signalling , Mechanical Signalling/FPL, Key Locking/Rudimentary Interlocking , where existing equipment & mode of operations shall continue as existing till their replacement.
4. Those existing Installations/Plans showing STD I R , STD II R, STD III R, STD IV R (as per Previous SEM Part 1 , 1988 edition) may continue to show them.

Annexure 7/1

Extract from Schedule of Dimensions (SOD)

S.N	ITEM
1	Minimum distance between center to center line of adjacent track is 5.3 m(SOD CH-1 item 1(ii))
2	Maximum height of the equipment provided between the rails of the track is 64 mm above the rail level (SOD CH-1 item 12).
3	For a distance of 229 mm outside and 140 mm inside the gauge faces of the rail, no gear or track fittings must project above rail level except such parts as are required to be actuated by the wheels or wing rails(SOD CH-1 item 12 Note (a)).
4	Point machine/ electrical point detector should be provided at a Min. distance of **1.6** m from nearest central line of track. (**SOD CH-1 item 12 Note (b)).**
5	Signal wires or supports for signal wires may be allowed at not less than 1600mm or 1830mm in the case of tunnels or through or semi-through girder bridges on either side of the center of track provided that they are not more than 203mm above rail level
6	TLJB should be provided at a Min. distance of 1905 mm from nearest central line of track. (SOD CH-II item 11(B) i CS 26)
7	If the edge of a signal foundation height is within 305 mm from above the rail level, then it should be at a Min. distance 1905 mm from nearest central line of track. (SOD CH-II item 11(B) i CS 26)
8	Location Box should be provided at a Min. distance 2360mm from nearest central line of track.. (SOD CH-1 item 13 ii (c)).
9	Signal post should be provided at a Min. distance 2360mm from nearest central line of track (SOD CH-II item 11(B) iii CS 26)
10	Red aspect of a signal should be at a height 3.65 m from above the rail level.
11	If a signal's post is at a distance 2360mm from nearest central line of track and if its height is 4420mm above the rail level then the Signal unit should be at a Min. distance 2135mm from nearest central line of the track (SOD CH-II item 11(B) v CS 26)
12	Route indicator of a signal (i) if it is at a height between 4420mm to 4610 mm above rail level then it should start from Min. distance 2135 and end at1980 mm from the nearest central line of the track. (ii) if it is at a height 4610mm to 6250 mm above rail level, then it should be at a Min. distance of 1600 mm from the nearest central line of the track. (SOD CH-II item 11(B) vi CS 26)
13	*If a Ladder of signal erected at a distance within 2360 mm from* central line *of adjacent track then it should be blanked off (wrapping MS sheet around ladder) to a height* of 300mm between 2060mm and 2360mm above rail level.(SOD CH-1 item 8 note(c))

Annexure 7/2

Extract of Rules for Opening of Railway Lines for "Isolation"

The term isolation denotes the condition in which a line for a particular movement is separated from all adjoining lines connected to it in such a manner that the isolated line cannot be fouled or interfered with by any movement taking place on the adjoining lines. Rules for isolation are laid down in Chapter VIII, Part III of" Rules for the opening of a Railway or Section of Railway for the Public Carriage of Passengers"

Isolation

Rules of Isolation

There are three rules of isolation as given in Rules for Opening Railways. These rules ensure the protection of important lines. The rules are –

(i) The line on which speed is higher than 50 kmph should be isolated from all other running lines.

(ii) All passenger lines irrespective of the speed should be isolated from goods lines/sidings.

(iii) It is desirable to isolate goods lines from sidings.

* It is not necessary to isolate one goods reception line from another or one passenger line from another when dealing with speed of 50 KMPH and less.

Rules (i) and (ii) are obligatory. Rule (iii) is optional.

Isolation may be accomplished by – (a) Connection to another line or ling siding; (b) The provision of short dead end siding ; or (c) the provision of trap.

CHAPTER - 8: DRAWINGS & SPECIFICATIONS

Section 1 - Plans for New and Working Installations

8.1.1.0 Preparation - General Procedure

8.1.1.1 All plans shall be prepared in accordance with the instructions issued by the Principal (or Coordinating) Chief Signal and Telecommunication Engineer.

8.1.1.2 The names of the junction or terminal stations should be noted on plans on both sides, that on the left hand side being the one from which the kilometre progressively increases. The names of the adjacent block stations and mid-section sidings/interlocked level crossing gates with gradients, IBSs should also be indicated on the plans as also their respective distances from the centre line of the station for which the plan is prepared. Communication to mid-section Manned LC gates shall be shown on the plan of the station to which such communication is established.

8.1.1.3 Standard drawings/Typical drawings shall not be issued without the specific permission of the Principal (or Coordinating) Chief Signal and Telecommunication Engineer.

8.1.1.4 The Signal and Telecommunication Engineer in charge should ensure that drawings submitted with an estimate for a work are complete and include all information necessary -

(a) For the proper understanding of the scheme, relevant notes are being written on the drawings; and include all necessary information

(b) For proper check of the design and estimate. When a part of the work has to be done to a standard plan or to an existing drawing, the fact should be stated. Such drawings need not be reproduced. If a part of the information necessary has to be supplied by another department, the officer-in-charge should obtain and incorporate the details on the drawing.

8.1.1.5 The north direction should be shown on every signalling plan.

8.1.1.6 All drawings should be quoted by number in their proper place in the estimate and in the covering letter accompanying the estimate.

8.1.1.7 Tentative signalling plans prepared for estimating purposes should bear the legend: "Tentative - or estimating purposes only". Tentative signalling plans issued for tender purposes should bear the legend: "Tentative - For Tender purposes only". Draft signalling plan sent to Divisions for comments should bear the legend: "Draft Signalling Plan For comments". All tentative/draft plans should be signed by at least a Senior Scale Officer.

8.1.2.0 Plans for other Departments

8.1.2.1 Divisional/Sr.Divisional Siganl & Telelcom Engineer/Dy. Chief Signal & Telecom

Engineer may at his discretion prepare sketches for officers of other Departments or for Deposit Works. No detailed plans for Deposit Works should be prepared except on receipt of orders from the Principal (or Coordinating) Chief Signal and Telecommunication Engineer.

8.1.2.2 Sr.Divisional Signal & Telecom Engineer/Dy. Chief Signal & Telecom Engineer shall obtain complete details from officers of other Departments when preparing plans that affect those departments and embody their requirements on the plans, if considered necessary. He should arrange for the plans, to be signed by the representative of the department concerned in token of approval.

8.1.3.0 General practice and sizes of Drawings

8.1.3.1 The code of Practice for General Engineering drawings and the standard sizes of drawings - IS. 696 shall be followed as far as practicable.

8.1.3.2 In preparing plans or making ferro prints, wastage of drawing paper, tracing cloth and ferro paper should be avoided. Multiple ferro copying should be avoided by use of scanning from tracing. All field units should be issued soft copies only in pdf or tif or jpg from scanned drawings. Field units to take colour prints from ferro scanned soft copies for application to CRS's/PCSTE's sanction

8.1.3.3 The following considerations govern the size of a drawing-

(a) Folding, approximately to foolscap size to accompany correspondence and reports.

(b) The commercial size of drawings, tracing and paper rolls.

8.1.3.4 The plan sizes given below shall be adopted.

Type of Drawing	Size	Measurement including Border	Border
Sketches	A-4	210 mm. x 297 mm.	10 mm.
	A-3	297 mm. x 420 mm.	
Locking Table	A-3	297 mm. x 420 mm.	10 mm.
Selection Table	A-3	297 mm. x 420 mm.	10 mm.
Circuit Diagram	A-3	297 mm. x 420 mm.	10 mm.
Power Supply Diagram	A-3	297 mm. x 420 mm.	10 mm.
Track Bonding Plan	A-4	210 mm. x Any length	10 mm.
	A-3	297 mm. x Any Length	
Cable Termination Diagram	A-3	297 mm. x 420 mm.	10 mm.
Operating Panel	A-3	297 mm. x Any Length	10 mm.
Signalling Plan	A-3	297 mm. x Any Length	10 mm.
Locking Diagram	A-3	297 mm. x Any Length	10 mm.
Cable Core Chart	A-3	297 mm. x Any Length	10 mm.
Cable Route Plan	A-3	297 mm. x Any Length	10 mm.
VDU Diagram	A-3	297 mm. x 420 mm for Standard Yard	10 mm.
	A-2	420 mm. x 594 mm for Bigger Yard	
Floor Plans for TSAA, Earthing Plan	A-3	297 mm. x 420 mm	10 mm.

8.1.3.5 Drawings for a large project should be bound together, each such drawing should be marked as sheet 1, 2 and so on as also the total number of sheets (e.g., 1/3, 2/3, 3/3) and bear in the proper place, the separate numbers by which each is identified. These numbers should be entered in the Office Register of Drawings and should also be marked on the outside bindings.

8.1.4.0 Titles and numbering of Drawings

8.1.4.1 The title and number may be placed at the bottom right hand corner of the plan, sufficient space being left for "Notes" to be entered as necessary.

8.1.4.2 (a) Drawings pertaining to any station, viz., Signalling plans, Locking Tables/Selection Tables/Route Control Charts, Locking diagrams, wiring diagrams etc. shall bear the same number e.g. SIP.SCR.GTL.VPL.Ø1. Document or Drawing type to have standard abbreviations such as SIP/RSP/PD/ST/LT/RCC/WD etc and for Electronic Interlocking Configuration Diagram (EIC), Relay Disposition Chart (RDC), Inter-Rack Cable Schematic (IRC), Earthing Arrangements (EAR), Equipment Rack Layout (ERL), Signal Equipment Room Layout (SER), Station Master Room Layout (ASM), Power Supply Room Layout (PSR) , Interlocking Logic Circuit Diagrams(ILC), Interface Circuits (IFC), Visual Display Units (VDU) etc. All plans/drawings shall be version controlled & updated in the following manner:-

(i) Any controlled drawing or document needs to be identified uniquely and unambiguously.

(ii) Version number shall indicate document type, zone, division, station and its revision number

(iii) Unique codes are to be assigned for zone, division and station and document type.

(iv) Revision number may take two digit number XX which can cater for 99 versions

(v) Electronic record copies shall have the file names same as Version number.

(vi) Structure of the Version number DocType. ZoneCode. DivCode. StationCode. Version For example, SIP of Venkatampalli (VPL) Station, Guntakal (GTL) division of South Central Railway (SCR) may have a version number -SIP.SCR.GTL.VPL.Ø1

(vii) When drawings are having multiple sheets such as Interlocking Logic Circuits, Interface Circuits etc., shall have an index sheet, as deliberated at para titled Indexing

(viii) Structure of the Version code where multiple drawing sheets are involved - Index Sheet - DocType. ZoneCode. DivCode. StationCode. Version Sheet - Sheet#nnn. Version For example, Internal Circuit diagrams of Venkatampalli (VPL) Station may have a version number ILC. SCR.GTL..VPL.Ø1 Sheet#ØØ1. Ø1

(ix) Version number for Part drawing or document Sometimes, only part of the drawing may be required for illustration or planning purpose. In such cases, drawing may have suffix '.Part'. SIP. SCR. GTL.. VPL. Ø1.Part

(b) Version Update

(i) Whenever changes have been made in the drawing, versioning needs to be revised upwards. If drawing contains more than one sheet, all sheets are required to be updated to the latest version.

(ii) For example, if changes are made in some of the sheets in signal interlocking circuit drawings, rest of the sheets are not required to be versioned upwards. However, version of modified sheets and associated index sheet will versioned upwards

(iii) It is not necessary to change version of other document/ drawing types when one document type undergoes changes and version update. For example, if changes are made to track bonding diagrams, version will be upgraded only to track bonding diagrams.

(iv) A registry shall be maintained to keep track of versions for all document types for a signalling installation.

(v) Each version of production copies viz., copies of drawings and documents when printed for issue, shall be given a change number; starting at “01” for the original version. Subsequent numbers shall be allocated to modified or amended versions.

(vi) Each source record shall bear the current version in the bottom right hand corner. It may be coloured on production copies.

8.1.4.3 If more than one sheet is used for a particular work, each must be distinguished by the sheet number as indicated in Para 8.1.3.5 immediately following the drawing number. The title may be placed below the number. If the drawing cancels the previous one, a note to this effect and the number of the cancelled drawing should be recorded at the right hand top corner of the drawing.

8.1.4.4 Every plan/drawing/circuits should bear in small letters at the lower left hand corner, the name and initials with designation of the JE/SSE(D&D) who prepared and officials who checked& approved the plan/drawings/circuits. If plan/drawing/circuits are prepared by contractor then Contractor's designer name/designation/license number with their initials shall be ensured

8.1.4.5 All signatures in tracings should be in indelible ink. All signature should be dated with date month and year. If feasible, plans/drawings/circuits may be digitally signed.

8.1.5.0 Details on Drawings

8.1.5.1 All dimensions and distances shall be written carefully upon that part of the drawing to which they refer. The distance to be embraced by the figures shall be indicated by arrow heads. Figuring and descriptive matter should be so printed that without moving the plan, it can be read with ease.

8.1.5.2 The following information should invariably be shown on signalling plans:-

(a) Standard of interlocking and Class of station.

(b) Holding capacity of all running lines and sidings without fouling adjacent lines, taken from Fouling Mark/Point SRJ to Fouling Mark/Point SRJ duly considering the length required for Glued joints & placing Signals.

(c) Direction of reception and despatch on running lines and description of sidings,

(d) Restriction on dead-end sidings (e. g. Not For Stabling) if any.

(e) All gradients within the station limits and upto 2.5 kilometres in rear of first stop signal.

(f) Kilometrage and class of level crossings within the station limits, whether interlocked or not. Kilometrage & class of Mid section interlocked level crossings with gradients upto 2.5 kilometres in rear of gate stop signals. Working in brief of Lifting barrier & Sliding boom if level crossings interlocked.

(g) Type of Block Working with adjacent station and location of block Instruments,

(h) Up and Down directions and names of important junctions on either side.

(i) Reference to condonation of gradient infringements, CRS's dispensation for deviations from General Rules/Signal Engineering Manual, if any.

(j) Reference to approved Engineering plan based on which the signalling plan is prepared.

(k) Note regarding telephone communication provided between A.S.M./Cabin man and level crossings within and outside station limits.

(l) Aspect sequence chart for colour light signals.

(m) Whether turnout is 1 in 8-1/2 or 1 in 12 or 1 in 16 or Symmetrical split or Thick web switch etc. with chainage of SRJ’s and Fouling Marks. Crank handle nos. with point nos. and their locations should also be tabulated.

(n) Details of Detection Table etc., which are not apparent in the plan. List of all Main/Calling-On & Shunt Routes should be included in the Signalling Plan with all overlaps.

(o) Details of Track Circuits/Axle Counter along with Dual detection (if any)

(p) Intersignal distances and distance between Warning Boards and Signals.

(q) Details of open bridges/RUB/ROB/LHS

(r) List all Main routes with all overlaps, Calling-On routes & Shunt Routes.

(s) Signal overlap in big yards.

(t) Custody of spare keys.

(u) Date of all previous references of commissioning the installation with their sanction numbers and deviations (if any) and date of commissioning. A sample copy of Signalling plan & Control table are at Annexures

8.1.6.0 Check and issue of Drawings

8.1.6.1 Signalling Plans and Locking Tables/Selection Tables shall be checked in full at two levels which can be AESTE/AESTE or AESTE/ESTE or ESTE/ESTE before they are approved and signed by an officer in Junior Administrative grade or above, authorised by the Chief Signal and Telecommunication Engineer. In case of work done by Railway PSUs then 1st level check should be carried out by PSU's officer equivalent to rank of Executive Engineer or above in railways and 2nd level check should be done by railways AESTE or ESTE as per availability and approved & signed by an officer in JAG or above, authorised by the Principal (or Coordinating) Chief Signal and Telecommunication Engineer. .

8.1.6.2 Locking Diagram for lever frames having more than 50 levers shall be checked and approved as required in Para 8.1.6.1….. Locking Diagram for lever frames having upto 50 levers, cable plans and power supply distribution diagrams shall be checked in full by Assistant Divisional Signal and Telecommunication Engineer before they are approved and signed by Sr.DSTE/Dy.CSTE. In case of work done by Railway PSUs, these diagrams should be checked by PSU's officer equivalent to rank of Executive Engineer or above in railways and approved & signed by Divisional Signal and Telecommunication Engineer. Detailed Wiring diagram shall be approved by an officer in JAG

8.1.6.3 Typical circuit diagrams for inter cabin control and automatic signalling shall be checked in full at two levels which can be AESTE/AESTE or AESTE/ESTE or ESTE/ESTE before they are approved and signed by Deputy Chief Signal and Telecommunication Engineer and to be approved by PCSTE. Detailed wiring diagrams for individual stations prepared on the basis of these typical circuit diagrams should be checked in full by Assistant Signal and Telecommunication Engineer and approved & signed not below Divisional Signal and Telecommunication Engineer. Detailed Wiring diagram not based on typical circuits to be approved by an officer in JAG. In case of work is done by Railway PSUs for circuit diagrams based on railways approved typical circuit diagram, 1st level check should be carried out by PSU's officer equivalent to rank of Executive Engineer or above in railways and approved & signed by executive signal and telecommunication engineer.

8.1.7.0 Completion Drawings

8.1.7.1 The Signal and Telecommunication Engineer-in-charge of Construction/Work executing agency should submit site verified signed paper prints to the Principal Chief Signal and Telecommunication Engineer for works completed. These

should indicate the work as actually carried out including the dimensional details as actually measured at site, location of signals, details of cables laid, wiring diagram, locking diagram etc. Completion of Signalling plan shall be based on completion Engg. scale plan if there is any change in ESP. Completion plan/drawings/circuits original tracings shall be signed by ADSTE/DSTE of executing unit as confirmation of 1st level check. In case of work done by Railway PSUs then 1st level check should be carried out by PSU's officer equivalent to rank of Executive Engineer and signed by divisional officials before submission to zonal HQ. 2nd level check should be done by railways AESTE or ESTE as per availability and approved & signed by an officer in JAG or above, authorised by the Principal (or Coordinating) Chief Signal and Telecommunication Engineer.

8.1.7.2 On receipt of these drawings, the original tracings will be amended in the Office of Principal Chief Signal and Telecommunication Engineer and marked "Completion Drawing". Scanned completion drawings will be sent to division and Requested number of copies of the completion drawing can be printed by Sr. Divisional Signal and Telecommunication Engineer.

8.1.8.0 Supply of Plans

8.1.8.1 Plans of working installations- In addition to the Standard drawings, each Divisional Signal and Telecommunication Engineer/Assistant Signal and Telecommunication Engineer shall have copies of the following diagrams and charts of working installations:-

(a) Engineering Plan and Signalling Plan for each interlocked station including interlocked level crossings situated outside station limits and interlocked mid-section sidings.

(b) Station Working Rule diagram where issued for each station including interlocked level crossings situated outside station limits.

(c) Locking Table and Locking diagrams of each interlocking frame, Station Master's slide control frame, interlocking key box, power frame with mechanical locking.

(d) Selection Tables/Locking Table/Route Control Table for each station.

(e) Panel Diagram /VDU Diagram. VDU diagram shall be submitted to actual size or in proportionate reduced size of the VDU for approval. Track Layout as per approved signalling plan with symbol for direction of traffic for each line of the layout; nomenclature of each line/siding; boards for BSLB/GWB/STOP; SM's key In/Out indications; SI cancellation Key-In indication; provision of all un-blocking; List of all emergency counters; relevant signalling details such as track section nos with route set-clear/set-occupied or occupied lit indications in different colour; main/calling-on/shunt signal nos with timer, locked indication & each aspects indications; points nos. with 'Normal' & 'Reverse' indication slits, locking indications, sub route cancellation symbol in blue colour with white dot; each signal route initiation & signal overlap set/release timer indications; block instruments status LCPR/SR/Block release indications; Crank handle Key-In/Key-Out indications for each crossover; status of active/standby VDU; Link status, health status, OC/CIU status, CRC/Checksum; Date/Time, colour scroll for

flashing indication etc. are the least but not limited, to be shown . Annexure 8/8 for VDU layout may be referred.

(f) Bonding diagrams of track circuits for each yard showing complete layout and diagrams of individual track circuit showing location of insulation joints, jumpers, relay and feed ends, polarity, length, traction bonds etc.

(g) Diagrams of connections at track, line and other relays, as necessary for each track circuited yard.

(h) Diagram of Relay Interlocking, Electronic Interlocking circuits for each yard.

(i) Diagrams of single and double line block instrument circuits.

(j) Diagrams of Location/Junction boxes showing description of wires at terminals for each yard.

(k) Arrangement of relays in relay racks and contact analysis sheet, fuse & terminal analysis sheets for Relay Interlocked/Electronic Interlocked stations

(l) Diagrams showing connections of power supply panels for each power supply installation with perimetric/ring earthing diagram ;

(m) Cable route plan showing disposition of underground cables at OHE mast/regular intervals with respect to nearest track centre, for each yard.

(n) Disposition charts of underground cables for control telephones, block instruments, administrative trunks and other railway circuits.

(o) Disposition charts of underground cables for circuits such as key transmitters and other signalling circuits under the charge of each SSE/JE..

8.1.8.2 Senior Section Engineers (in charge) and Senior Section Engineers/Junior Engineer (Section) SSE/JE shall be supplied with copies of the above mentioned Drawings and charts.

8.1.8.3 At stations provided with centralised operation of points and signals, a set of signalling plans, locking/selection tables/route control chart, locking diagrams and wiring diagram, cable route plan etc. may be kept at the station for reference by the maintenance staff.

8.1.8.4 While signalling plans, Locking table/Section table/Route control chart and wiring diagrams shall be supplied by Headquarters Office, other plans mentioned in Para 8.1.8.0 shall be prepared by Deputy Signal & Telecommunication Engineer (Construction) or Senior Signal & Telecommunication Engineer (maintenance) as the case may be. Signalling plans for non-interlocked stations shall be prepared by Deputy Signal & Telecommunication Engineer (Construction) or Senior Signal & Telecommunication Engineer (maintenance) as the case may be.

8.1.9.0 Care and file of tracings

8.1.9.1 Tracings shall not be used for reference as they are likely to get lost or damaged. Required number of prints taken from scanned tracing prints should be supplied to Officers and JE/SSEs. Each particular file should contain a print taken from scanned tracing print of the works relating to it.

8.1.9.2 Should it be necessary to send a tracing from one office to another, it shall be rolled and inserted in a cardboard cylinder.

8.1.9.3 Print from scanned tracing should be folded in concertina fashion and shall when folded measure approximately 210 mm x 300 mm. The folding should be arranged so as to make visible the title of the plan without unfolding the plan. When prints are rolled for despatch, they should be rolled with the working side outwards.

8.1.9.4 The Records section of each Drawing Branch may file every tracing and original drawing on the basis of the subject classification and the index card filing system. Each drawer of the index card cabinet should be distinguished by a classification number: each card should be complete as regards title of the drawing, other connected drawings, file reference and the drawer number in which the original is stored.

8.1.9.5 The storage-drawers should have placards on the outside indicating the contents in each. The plans should be stored flat in shallow drawers of convenient dimensions. Probability of damage by moths or white ants or fire should be guarded against.

Section 2 - Indian Railway Standard Drawings

8.2.10.0 Signalling and Interlocking Drawings - IRS (S) & Notation

8.2.10.1 Indian Railway Standard Drawings, designated by the code word I.R.S. have been issued by the Director-General, Research Designs and Standards Organisation, Ministry of Railways, Lucknow. The signal drawings are marked IRS (S) -"S" stands for "Signal". The particulars of the drawings and their reference numbers are detailed in an "Index of Indian Railway Standard Signalling and Interlocking Drawings - IRS (S)". This index shows all the IRS (S) drawings arranged alphabetically, as well as serially, in the order of their numbers. Each drawing number is either prefixed with letters 'SA' or letter ‘S’. 'SA ' stands for a signal assembly and ' S ' stands for a part of a signal assembly

8.2.10.2 New designs and drawings which are accepted for adoption as standards have the word "Advance" suffixed to their number e.g. S-8716 (Advance), pending their final adoption as Indian Railway Standard Drawings. For such drawings the manufacturers shall have a sample approved by the purchaser before undertaking the bulk manufacture.

8.2.11.0 Supply of Standard Drawings

8.2.11.1 A set of all I.R.S. (S) drawings shall be supplied to the office of SrDSTE/DyCSTEs, Assistant Signal and Telecommunication Engineer and to each JE/SSE as required.

8.2.11.2 I.R.S. drawings should not be traced by the Zonal Railway. Copies in reproduction tracings should be obtained from the Director General, Research, Designs and Standards Organization (Signal and Telecommunication), Ministry of Railways, Lucknow, whenever required.

Section 3 - Standard Specifications

8.3.12.0 Indian Railway Standard Specifications.

8.3.12.1 Indian Railway Standard Specifications- Specifications for materials used for signalling purpose have been drawn out by the Director, Research General, Designs and Standard Organisation, Ministry of Railways, Lucknow and are titled as "Indian Railways Standards Specifications". These specifications are issued under a fixed serial number e. g. S-12-54, the letter "S" denoting "Signals" the number "12" representing the serial number of the specifications and the final number "54" indicating the year of original adoption as standard, or in the case of revision, the year of last revision. A list of I.R.S. Signal specification is at Annexure "8/1".

8.3.13.0 Other Specifications:-Specifications issued by the British Standards Institution, the Indian Standards Institution and the British Railway Board have also been adopted for items of equipment used for signalling purposes for which no I.R.S. specification exists. Some of these specifications pertaining to items in common use are detailed in Annexure "8/2"

8.3.14.0 Supply of Specifications:- Each Sr Divisional Signal and Telecommunication Engineer and Assistant Signal and Telecommunication Engineer should have a copy of all Indian Railway Standard (Signal) specifications in his office. Copies of such British Standard and Indian Standard specifications that are generally required may also be kept. He should also have a copy of all specifications issued by Principal Chief Signal and Telecommunication Engineer for local use.

8.3.15.0 Availability of Specifications

8.3.15.1 Indian Railway Standard Specifications are obtainable from the Manager of Publications, Civil Lines/Delhi-6.

8.3.15.2 Indian Standard Specifications are obtainable from the Indian Standards Institution, Manak Bhavan, Bhadur Shah Zafar Marg, New Delhi, or its branch offices at Mumbai,Kolkata, Chennai Benguluru, Hyderabad, and Kanpur

8.3.15.3 British Standard Specifications may be purchased from Indian Standard Institution Offices at New Delhi, Mumbai, Kolkata and Chennai

Section 4 - Books of References

8.4.16.0 Books of Reference

8.4.16.1 Books of reference should be supplied to Officers, JE/SSEs and Technicians for their personal use, as well as for use in their offices. A statement showing the various books and their distribution is at Annexure "8/3"

8.4.17.0 Reports of Signal Standards Committee, Technical papers and Journals:- Each Sr Divisional Signal and Telecommunication Engineer should arrange to have the under mentioned technical literature in the Divisional Library-

(a) Reports of the Signal Standards Committee.

(b) Proceedings and Technical papers issued by the Institution of Railway Signal and Telecommunication Engineers, New Delhi.

(c) I.R.S. specifications, British Standard specifications, Indian Standards specifications, specifications of the Association of American Rail Road Signal section relating to Signalling and Telecommunication equipment as required.

(d) Quarterly Technical Bulletin and other technical papers on Signalling and Telecommunication matters published by the Research, Designs and Standards Organization/Lucknow.

(e) Notes published by Indian Railways Institute of Signal, Engineering and Telecommunications, Secunderabad.

(f) Technical books and journals of interest on Signalling and Telecommunication.

(g) Latest Approved vendor lists for Signalling & Telecommunication items issued by RDSO.

(h) Above documents (latest) shall be made available in RDSO website for guidance of all

8.4.17.1 **Few Important International Standards for Reference**

(a) Few International standards of IEC - “International Electrotechnical Commission” and CENELEC - “European Committee for Electrotechnical Standardization” are given below

(i) EN-50126 | IEC-62278 | Railway Applications - The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) (Part-1 & 2)

(ii) EN-50128 | IEC-62279 | Railway applications —Communication, signalling and processing systems — Software for railway control and protection systems.

(iii) EN-50129 | IEC-62425 | Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling.

(iv) EN-50159 | IEC-62280 | Railway applications — Communication, signalling and processing systems — Safety-related communication in transmission systems

(v) IEC 62305 - PER (Perimeter Ring Earth) for S&T buildings

Section 5 - Maintenance of Drawings , Specifications and Book of Reference

8.5.18.0 Folders for Drawings

(a) Typical drawings, plans of working installations and specifications should be maintained in a book form separately bound in suitable folders.

(b) Where Soft Copies are maintained, they shall be securely stored Division Wise, Section wise, Sub section wise, Stn wise, with Version number. A secured Computer

server with UPS back up & Data back up shall be utilised for such storage

8.5.19.0 Issuing of Addenda and Corrigenda

8.5.19.1 **I.R.S.Drawings and Specifications** - A quarterly notification is issued by the Director General, Research, Designs and Standards Organization. It gives the details of the new Indian Railway Standard Specifications and drawings issued and also those which have either been modified or cancelled during the quarter covered by the notification.

8.5.19.2 **British Standard Specifications -** British Standard Year Book is published by the British Standards Institution, Victoria Street, London S. W. I. It contains a list of up-to-date British Standards in numerical order and also gives a brief description of each.

8.5.19.3 **Indian Standards Specifications** - The I.S.I. Hand Book of Publications contains up to-date list of Indian Standards and is available from the Indian Standards Institution, Bahadur Shah Zafar Marg, New Delhi, or any of its branch offices at Mumbai, Kolkata, Chennai, Hyderabad, Kanpur and Benguluru.

8.5.19.4 **Standard Drawings and Plans of working Installations** - Railway should publish lists of standard drawing as also lists of plans of working installations for the information of the staff. Addenda and Corrigenda slips should be issued to these lists regularly once in six months incorporating particulars of drawings and plans issued, modified or cancelled.

8.5.19.5 **Books of reference** - Addenda and Corrigenda slips to the books of reference are issued from time to time by the Railway Board and the Railways, as the case may be.

8.5.20.0 Accountal - All Indian Railway Standard drawings and Specifications, British Standard and Indian Standard Specifications as well as books of reference must be accounted for in the same way as tools and plant items.

8.5.21.0 **Responsibility**

8.5.21.1 All officials to whom books of reference have been supplied shall be responsible for -

(a) their safe custody and good order.

(b) pasting of all addenda and corrigenda slips promptly and seeing that these are up-to-date to the last slip as modified from time-to-time.

(c) returning of all books issued to them for personal use prior to retirement.

8.5.21.2 Each JE/SSE shall be responsible to see that -

(a) The standard drawings and plans of working installations are properly maintained and kept up-to-date in respect of new drawings issued and old ones cancelled;

(b) The staff working under him understand and carry out work in accordance with standard drawings and plans of working installations. Any mistake in drawings and plans that may come to his notice should be promptly intimated to the Divisional Signal and Telecommunication Engineer for arranging correction; and

(c) The staff under them maintain their books of reference up-to-date and in good order.

8.5.21.3 Each Senior Divisional Signal & Telecom Engineer shall be responsible to see that -

(a) the standard drawings, plans and specifications in his custody are kept up-to-date and that the obsolete and cancelled ones are destroyed.

(b) All JE/SSEs keep their standard drawings and plans up-to-date.

(c) the staff properly understand and carry out work in accordance with the standard drawings and plans.

(d) any mistake in the standard drawings, working plans and specifications, which comes to his notice, is promptly intimated to the Principal (or Coordinating) Chief Signal and Telecommunication Engineer for arranging necessary correction.

(e) Each Sr Divisional Signal and Telecommunication Engineer shall make periodical check to see that JE/SSEs maintain their books of reference up-to-date and in good order. He should encourage his JE/SSEs and other staff to study reports, proceedings, papers and journals mentioned in Para 8.4.17.0. So as to enhance their knowledge and to keep themselves informed about the up-to-date developments, methods and technique in Railway Signalling and Telecommunications.

(f) Proper Distribution, Responsibility & Accountal of Documents

Section 6 - Indian Railway Standard Equipments

8.6.22.0 Indian Railway Standard Designs

(a) Where Indian Railway Standard designs exist, they should invariably be followed for all new works and no modification of such designs should be introduced without the previous approval of the Railway Board.

(b) If any defect in Standard designs is noticed under service conditions or if certain modifications to the design are considered desirable, the matter should be brought to the notice of the Director General, Research, Designs and Standard Organisation Signal and Telecommunications for examination in consolation with the Signalling Standard Committee.

Note :- This Chapter has under mentioned Annexures for further study

S.no	*Annexure no*	*Description*
1	*8/1*	*List of Indian Railway Standard Signal Specification*
2	*8/2*	*List of Important Indian Standard Specification user for Signalling Purpose*
3	*8/3*	*Statement showing the various Books of Reference and their distribution among the staff.*
4	*8/4*	*Typical Signalling Plan for Double Line station*
5	*8/5*	*Typical Control Table for Double Line station*
6	*8/6*	*Typical Signalling Plan for Single Line station*
7	*8/7*	*Typical Control Table for Single Line station*
8	*8/8*	*VDU Layout Plan for E.I*

Annexure – '8/1'

List of Indian Railway Standard Signal Specification

Sr. No	Description	Specification No
1	Galvanized solid steel wire for signalling purpose	S1
2	Galvanized steel wire stranded for signaling purpose	S2
3	Galvanized steel wire rope for signaling purpose	S3
4	Galvanized Short link chain for signaling purpose	S4
5	Point Rodding	S5
6	Tubular Steel Signal Poles	S6
7	Mechanical Signalling and interlocking Equipment	S10
8	Channel Pins	S17
9	I.R. S. Post type Electric Signal Reverser	S19
10	Single line token less Block Instrument	S20
11	Electric key transmitter	S21
12	Electrical Signalling and Interlocking Equipment	S23
13	Non trailable Electric Point Machine with plunger type locking	S24
14	Electric Signal Machine	S25
15	Colour light signal multiunit type	S26
16	Key lock ' E' type	S30
17	D.C polarized relay 3 position	S31
18	Push button type single line tokenless block instrument	S32
19	Testing of Railway Signalling Relays General	S34
20	PVC insulated aluminum screened cables for Railway Signalling	S35
21	Relay interlocking system	S36
22	Motors/Electric Point Machine	S37
23	Automatic warning systems	S38
24	Nylon insulated joints and nylon components	S40
25	Electric Lifting Barriers	S41
26	Axle counting equipment	S42
27	Automatic Block Signalling for unidirectional traffic	S43
28	Insertion and testing of slot indicators	S44
29	Earth leakage detector	S45
30	D. C. Neutral line relay (proved)	S46
31	Instruction and testing of Station Master's control frames	S48
32	Electric Point Detector	S49
33	Twilight switches	S51
34	Surge discharger	S52
35	Rubber packing used in point cylinder and retarder cylinder	S55
36	Inverter for Railway Signalling	S58
37	Colour light signal transformer	S59
38	A.C. immunization characteristics of D. C. neutral tractive armature miniature plug in type line relays	S60
39	Electrically operated automatic half barriers	S61
40	P.V.C insulated cables(unscreened)	S63

Annexure - '8/2'

Part-I

List of Important Indian Standard Specifications used for Signalling purpose

Sr. No	Description	Specification No
1	Rubber insulated cables-Part-I with copper conductor. Part II-with aluminum conductor	IS-434
2	Leclanche type dry battery for telecommunication, Signalling and General purpose.	IS-586
3	P.V.C insulated cables (for voltages upto 1100V Part-I) with copper conductors	IS-694
4	P.V.C insulated (heavy duty) electric cables Part-I for working voltages upto and including 1100 volts.	IS-1554
5	Stationary cells and batteries load and type (with tubular positive plate)	IS-1652
6	Stationary cells and batteries lead and type (with planted positive plate)	IS-1652
7	Galvanized steel wire strand for signaling purpose	IS-2265

Part-II

List of Important British Standard /British Railway Standard Specifications used for Signalling purpose

Sr. No	Description	Specification No
1	Railway Signalling Symbols-	BS-376
	Part-I- Schematic symbols	
	Part-II-Wiring symbols and written circuit	
2	Alternating current line relays, double element 3 position for Railways Signalling	BS-561
3	Glossary of terms used in Railway Signalling	BS-719
4	Alternating current relays for Railway Signalling	BS-1745
	(i) Track Relays (double element. 2 position)	
	(ii) Line relays (single element. 2 position)	
5	Miniature tractive armature D. C Neutral Line Relay. Plug-in type, for Railway Signalling purpose	BRS-930
6	Miniature tractive armature A. C immune D.C Neutral Relay , plug-in type, for Signalling purpose	BRS-931
7	Miniature tractive armature A. C immune D. C. biased Neutral line relay, plug-in type for Railway signaling purposes	BRS-932
8	Miniature tractive armature A.C. immune, D. C slow pick-up neutral line relay, plug-in type, for Railway signaling purposes.	BRS-933
9	Miniature tractive armature. A. C immune D.C. slow release neutral line relay, plug-in type for Railway signaling purpose.	BRS-934
10	Miniature tractive armature D. C. magnetically latched neutral line relay plug-in type for Railway signaling purposes.	BRS-935

11	Miniature tractive armature D. C. polarized magnetic stick line relay plug-in type, for Railway signaling purposes.	BRS-936
12	Miniature D. C. Neutral Thermal Time Element Relay. Plug-in type, for Railway signaling purpose.	BRS-937
13	Miniature tractive armature D. C. neutral track relay, plug-in type, for Railway signaling purpose.	BRS-938
14	Miniature tractive armature A. C. immune D. C. Neutral track relay, plug-in type, for Railway signaling purpose.	BRS-939
15	Miniature tractive armature D. C. single wound lamp proving relay, plug-in-type, for Railway signaling purpose.	BRS-940
16	Miniature tractive armature A. C. lamp proving relay, plug-in-type, for Railway signaling purposes.	BRS-941

Annexure - '8/3'

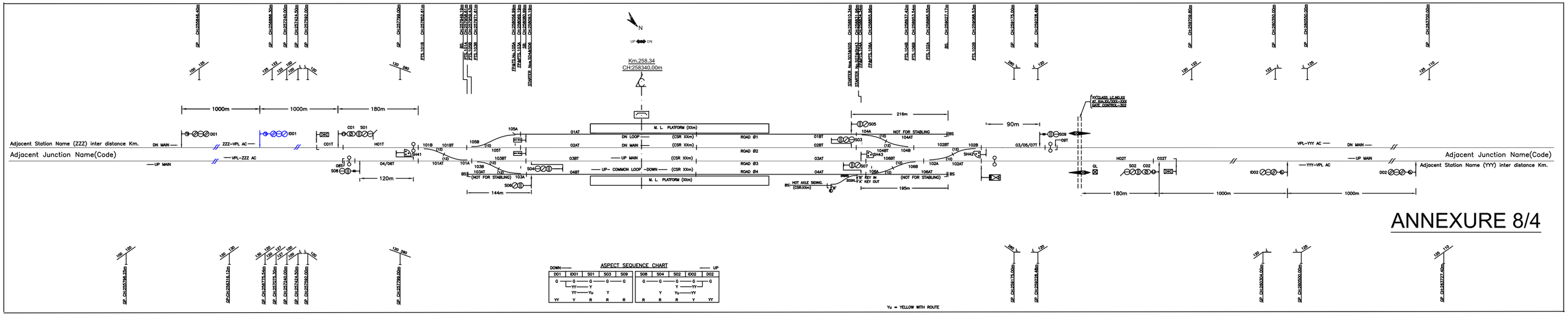
Statement showing the various Books of reference and their distribution among the staff

Sl. No	Publications	Scale						
		Office of			Personal Possession of			
		Sr.Divl. S&T Engineer	Asst. S&T Engineer	SSE-Signal (In-charge)	Sr.Divl./ S&T Engineer	Asst. S&T Engineer	SSE-Signal (In-charge) JE/SSE (Sectional)	Signal Technician
1	2	3	4	5	6	7	8	9
1	Rules for the opening of a Railway for the Public carriage of passengers.	1	1	--	1	1	--	--
2	General and Subsidiary Rules	1	1	1	1	1	1	1
3	Schedule of Dimensions (B. G. M. G/N.G as applicable)	1	1	1	1	1	1	--
4	Signal Engineering Manual	1	1	1	1	1	1	1
5	Telecommunication Manual	1	1	1	1	1	1	
6	Block Working Manual	1	1	1	1	1	1	1
7	Accident Manual	1	1	1	--	--	--	--
8	A.C. traction Manual	1	1	1	--	--	--	--
9	Permanent Way Manual	1	1	--	--	--	--	--
10	Track Manual	1	1	--	--	--	--	--
11	Indian Government Railway Code for the Engineering Department.	1	1	--	--	--	--	--
12	Indian Government Railway Establishment Code.	1	1	--	--	--	--	--
13	Indian Government Railway Financial	1	1	--	--	--	--	--

Sl. No	Publications	Scale						
		Office of			Personal Possession of			
		Sr.Divl. S&T Engineer	Asst. S&T Engineer	SSE-Signal (In-charge)	Sr.Divl./ S&T Engineer	Asst. S&T Engineer	SSE-Signal (In-charge) JE/SSE (Sectional)	Signal Technician
	Code							
14	Indian government Railway Code for Stores Department.	1	1	--	--	--	--	--
15	Indian government Railway code for Accounts Department.	1	1	--	--	--	--	--
16	Schedule of Powers of Divisional/district and Assistant Officers in other than Establishment matters (as published)	1	1	--	--	--	--	--
17	Pass Manual	1	1	1	--	--	--	--
18	The Hours of Employment Regulations with modifications as issued	1	1	1	--	--	--	--
19	Payment of Wages Act with Notifications as issued.	1	1	1	--	--	--	--
20	Working Time Table and Appendix thereto.	1	1	1	1	1	1	1
21	Alphabetical list of Railway stations.	1	1	1	--	--	--	--
22	Catalogue of stores	1	1	1	--	--	--	--
23	Indian Railways Act	1	1	--	--	--	--	--
24	Indian Workmen's Compensation	1	1	--	--	--	--	--

Sl. No	Publications	Scale						
		Office of			Personal Possession of			
		Sr.Divl. S&T Engineer	Asst. S&T Engineer	SSE-Signal (In-charge)	Sr.Divl./ S&T Engineer	Asst. S&T Engineer	SSE-Signal (In-charge) JE/SSE (Sectional)	Signal Technician
	Act.							
25	Indian Electricity Act	1	1	--	--	--	--	--
26	Indian Petroleum Act	1	1	--	--	--	--	--
27	I.R.S Drawings	1	1	*1	--	--	--	--
28	I.R.S. specifications (as required)	1	1	--	--	--	--	--
29	Railway Standard Drawings.	1	1	*1	--	--	--	--

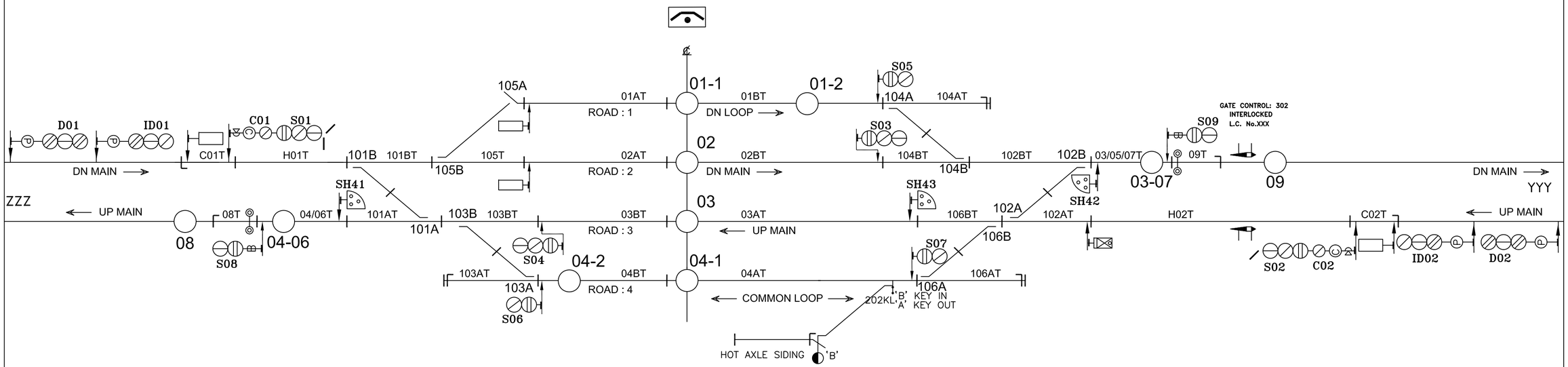
* As Required.


Km.258.34
CH:258340.00m
Adjacent Station Name (ZZZ) inter distance Km.
Adjacent Junction Name(Code)
DN MAIN
UP MAIN
ZZZ-VPL AC
VPL-ZZZ AC
VPL-YYY AC
YYY-VPL AC
Adjacent Junction Name(Code)
Adjacent Station Name (YYY) inter distance Km.
M. L. PLATFORM (XXm)
UP LOOP
DN LOOP
UP MAIN
DN MAIN
UP- COMMON LOOP -DOWN
(CSR XXm)
ROAD Ø1
ROAD Ø2
ROAD Ø3
ROAD Ø4
1000m
1000m
180m
120m
144m
216m
195m
90m
180m
1000m
1000m
(NOT FOR STABLING)
HOT AXLE SIDING. (CSR:XXm)
'A' KEY IN 'A' KEY OUT
GF
GL
*XX=LAND I.C. NO.XX
IF Km.XX.XXX=XXX
GATE CONTROL - XXX
ASPECT SEQUENCE CHART
DOWN
UP
D01 ID01 S01 S03 S09
S08 S04 S02 ID02 D02
G Y YY Yu R

ANNEXURE 8/4

ANNEXURE 8/5

4 ROAD STATION ON DOUBLE LINE



PRODUCTION & CHECKER BLOCK			
COMPANY		RAILWAYS	
DRAWN		DRAWN	(SSE/D)*
CHECKED	(SD)†	CHECKED	(ASTE)*
	(DV)†		(SSTE)*
APPROVED	(CRE)†	APPROVED	(Dy.CSTE)*
CRE: CONTRACTOR'S RESPONSIBLE ENGINEER *SIGNATURE WITH DATE, † SIGNATURE WITH DATE AND LICENSE No			

INTERLOCKING CONTROL TABLE	STATION LAYOUT	
ICT.SCR.GTL.VPL.Ø2	SHEET No:	VERSION
4 RD STATION	ØØ1	Ø1

SNo.	SIGNAL No.	TO DESTINATION	SELECT ROUTE		IN ROUTE			OVERLAP			ISOLATION POINT		CRANK HANDLE NORMAL	FOULING TRACK CIRCUITS	APPROACK LOCK BY TRACK CIRCUITS	BACK LOCK BY TRACK CIRCUITS	SIGNAL REPLACED BY TRACK CIRCUITS	OVERLAP RELEASES AFTER 120 SEC BY		SLOTTED BY GATE	RELEASED BY OTHER CONTROLS &SIDING POINT DETECTION	LOCKS ROUTES	SIGNAL ASPECT CONTROLLED BY				REMARKS
			SIGNAL BUTTON	ROUTE BUTTON	POINT NOR.	POINT REV.	TRACK CIRCUIT	POINT NOR.	POINT REV.	TRACK CIRCUIT	POINT NOR.	POINT REV.						OCCUPIED BY TRACK CIRCUITS	CLEARENCE OF TRACK CIRCUIT				YELLOW WITH ROUTE	YELLOW	DOUBLE YELLOW	GREEN	
1.	D01	ID01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ID01 HHG/ DG	DG CONTROLLED BY ID01 DG/HHG WITH POINTS No. 101N. 105N
2.	ID01	S01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	S01 HG	S01 DG	DG CONTROLLED BY S01 DG WITH POINTS No. 101N.105N
3.	S01	DN MAIN	S01	02	101. 105	—	H01T. 101BT. 105T. 02AT. 02BT	104. 102	—	104BT. 102BT. 03/05/07T	—	—	CH01. CH03. CH04	—	DEAD APPROACH	H01T. 101BT.105T	01T	02AT. 02BT	105T	—	—	C01–02. SH42–02	—	S03 RG/ HG/ DG	—	S03 DG	TIME RELEASE 120 SEC
4.	S01	DN LOOP SET TO BS	S01	01–1	101	105	H01T. 101BT. 105T. 01AT. 01BT	104	—	104AT	—	—	CH01. CH03	—	DEAD APPROACH	H01T. 101BT.105T	01T	01AT. 01BT	105T	—	—	C01–01–1. SH42–02	S05 RG	—	—	—	TIME RELEASE 120 SEC.
5.	S01	DN LOOP SET TO MAIN	S01	01–2	101	105	H01T. 101BT. 105T. 01AT. 01BT	102	104	104AT. 104BT. 102BT. 03/05/07T	—	—	CH01. CH03. CH04	—	DEAD APPROACH	H01T. 101BT. 105T	01T	01AT. 01BT	105T	—	—	C01–01–1. SH42–01–1	S05 RG/ HG	—	—	—	TIME RELEASE 120 SEC.
6.	S01	COMMON LOOP SET TO BS	S01	04–1	—	101. 103	H01T.101BT. 101AT.103BT. 103AT.04BT. 04AT	106	—	106AT	105	—	CH01. CH02. CH03	—	DEAD APPROACH	H01T.101BT. 101AT.103BT. 103AT	01T	04BT. 04AT	103AT	—	—	C01–04–1. C02–03. SH42–02. SH42–03	S07 RG	—	—	—	TIME RELEASE 120 SEC. 'A' KEY IN.
7.	S01	COMMON LOOP SET TO MAIN	S01	04–2	—	101. 103	H01T.101BT. 101AT.103BT. 103AT.04BT. 04AT	—	**106. 102**	106AT. 106BT. 102AT. 102BT. 03/05/07T	105	—	CH01. CH02. CH04. CH03	—	DEAD APPROACH	H01T.101BT. 101AT. 103BT. 103AT	01T	04BT. 04AT	103AT	—	—	C01–04–1. SH42–04–1	S07 RG/ HG	—	—	—	TIME RELEASE 120 SEC. 'A' KEY IN.
8.	C01	DN MAIN	C01	02	101. 105	—	C01T OCCUPIED.	—	—	—	—	—	CH01. CH03	—	DEAD APPROACH	H01T. 101BT.105T	—	—	—	—	—	S01–02. S03–03–07. S05–03–07. S07–03–07. SH42–01–1. SH42–02. SH42–03. SH42–04–1. SH43–03–07	—	—	—	—	TIME RELESE 120 SEC. HG CLEARS AFTER 60 SEC.
9.	C01	DN LOOP	C01	01–1	101	105	C01T OCCUPIED.	—	—	—	(104W SH42–03 SH42–04–1 SH43–03–07)	—	CH01. CH03	—	DEAD APPROACH	H01T. 101BT. 105T	—	—	—	—	—	S01–01–2. S05–03–07. SH42–01–1. SH42–02.	—	—	—	—	TIME RELESE 120 SEC. HG CLEARS AFTER 60 SEC.
10.	C01	COMMON LOOP	C01	04–1	—	101. 103	C01T OCCUPIED.	—	—	—	105	—	CH01. CH02. CH03.	—	DEAD APPROACH	H01T. 101BT. 101AT. 103BT. 103AT	—	—	—	—	—	C02–03. C02–04–1. S01–04–2. S07–03–07. SH42–02. SH42–03. SH42–04–1	—	—	—	—	TIME RELESE 120 SEC. HG CLEARS AFTER 60 SEC. 'A' KEY IN.

PRODUCTION & CHECKER BLOCK			
COMPANY		RAILWAYS	
DRAWN		DRAWN	(SSE/D)*
CHECKED	(SD)†	CHECKED	(ASTE)*
	(DV)†		(SSTE)*
APPROVED	(CRE)†	APPROVED	(Dy.CSTE)*

CRE: CONTRACTOR'S RESPONSIBLE ENGINEER
*SIGNATURE WITH DATE, † SIGNATURE WITH DATE AND LICENSE No

ANNEXURE 8/5

INTERLOCKING CONTROL TABLE	PART I	
ICT.SCR.GTL.VPL.Ø2	SHEET No:	VERSION
VENKATAMPALLE	ØØ2	Ø1

SNo.	SIGNAL No.	TO DESTINATION	SELECT ROUTE		IN ROUTE			OVERLAP			ISOLATION POINT		CRANK HANDLE NORMAL	FOULING TRACK CIRCUITS	APPROACK LOCK BY TRACK CIRCUITS	BACK LOCK BY TRACK CIRCUITS	SIGNAL REPLACED BY TRACK CIRCUITS	OVERLAP RELEASES AFTER 120 SEC BY		SLOTTED BY GATE	RELEASED BY OTHER CONTROLS &SIDING POINT DETECTION	LOCKS ROUTES	SIGNAL ASPECT CONTROLLED BY				REMARKS
			SIGNAL BUTTON	ROUTE BUTTON	POINT NOR.	POINT REV.	TRACK CIRCUIT	POINT NOR.	POINT REV.	TRACK CIRCUIT	NOR.	REV.						OCCUPIED BY TRACK CIRCUITS	CLEARENCE OF TRACK CIRCUIT				YELLOW WITH ROUTE	YELLOW	DOUBLE YELLOW	GREEN	
11.	D02	ID02	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ID02 HHG/ DG	DG CONTROLLED BY ID02 DG/HHG WITH POINTS No. 106N
12.	ID02	S02	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—		—	—	—	—	—	—	S02 HG	S02 DG	DG CONTROLLED BY S02 DG WITH POINTS No. 106N
13.	S02	UP MAIN	S02	03	102. 106	—	H02T.102AT. 106BT. 03AT. 03BT	103. 101	—	103BT. 101AT. 04/06T	—	—	CH04. CH02. CH01	—	DEAD APPROACH	H02T. 102AT. 106BT	02T	03AT. 03BT	106BT	302	—	C02–03. SH41–03	—	S04 RG/ HG/ DG	—	S04 DG	TIME RELEASE 120 SEC
14.	S02	COMMON LOOP SET TO BS	S02	04–1	102	106	H02T.102AT. 106BT. 106AT. 04AT. 04BT	103	—	103AT	—	—	CH04. CH02	—	DEAD APPROACH	H02T. 102AT. 106BT. 106AT	02T	04AT. 04BT	106AT	302	202N	C02–04–1. SH41–03	S06 RG	—	—	—	TIME RELEASE 120 SEC. 'A' KEY IN.
15.	S02	COMMON LOOP SET TO MAIN	S02	04–2	102	106	H02T.102AT. 106BT. 106AT. 04AT. 04BT	101	103	103AT.103BT. 101AT. 04/06T	—	—	CH04. CH02. CH01	—	DEAD APPROACH	H02T. 102AT. 106BT. 106AT	02T	04AT. 04BT	106AT	302	202N	C02–04–1. SH41–04–1	S06 RG/ HG	—	—	—	TIME RELEASE 120 SEC. 'A' KEY IN.
16.	C02	DN MAIN	C02	03	102. 106	—	C02T OCCUPIED.	—	—	—	—	—	CH04. CH02	—	DEAD APPROACH	H02T. 102AT. 106BT	—	—	—	302	—	S01–04–1. C01–04–1. S02–03. S04–04–06. S06–04–06. SH41–03. SH41–04–1	—	—	—	—	TIME RELESE 120 SEC. HG CLEARS AFTER 60 SEC.
17.	C02	COMMON LOOP	C02	04–1	102	106	C02T OCCUPIED.	—	—	—	—	—	CH04. CH02	—	DEAD APPROACH	H02T. 102AT. 106BT. 106AT	—	—	—	302	202N	C01–04–1. S02–04–2. S06–04–06. SH41–03. SH41–04–1	—	—	—	—	TIME RELESE 120 SEC. HG CLEARS AFTER 60 SEC. 'A' KEY IN.
18.	S03	DN MAIN	S03	03–07	104. 102	—	104BT. 102BT. 03/05/07T	—	—	—	—	—	CH03. CH04	—	02AT. 02BT, S01–02	104BT. 102BT	104BT	—	—	—	—	C01–02. SH42–02	—	S09 RG/ DG	—	S09 DG	TIME RELEASE 120 SEC
19.	S04	UP MAIN	S04	04–06	103. 101	—	103BT. 101AT. 04/06T	—	—	—	—	—	CH02. CH01	—	03AT. 03BT, S02–03	103BT. 101AT	103BT	—	—	—	—	C02–03. SH41–03. SH42–03.	—	S08 RG/ DG	—	S08 DG	TIME RELEASE 120 SEC
20.	S05	DN MAIN	S05	03–07	102	104	104AT. 104BT. 102BT. 03/05/07T	—	—	—	—	—	CH03. CH04	—	01AT. 01BT	104AT. 104BT. 102BT	104AT	—	—	—	—	C01–01–1. C01–02. SH42–01–1	—	S09 RG/ DG	—	S09 DG	TIME RELEASE 120 SEC

PRODUCTION & CHECKER BLOCK			
COMPANY		RAILWAYS	
DRAWN		DRAWN	(SSE/D)*
CHECKED	(SD)†	CHECKED	(ASTE)*
	(DV)†		(SSTE)*
APPROVED	(CRE)†	APPROVED	(Dy.CSTE)*

CRE: CONTRACTOR'S RESPONSIBLE ENGINEER
*SIGNATURE WITH DATE, † SIGNATURE WITH DATE AND LICENSE No

ANNEXURE 8/5

INTERLOCKING CONTROL TABLE	PART II	
ICT.SCR.GTL.VPL.Ø2	SHEET No:	VERSION
VENKATAMPALLE	ØØ3	Ø1

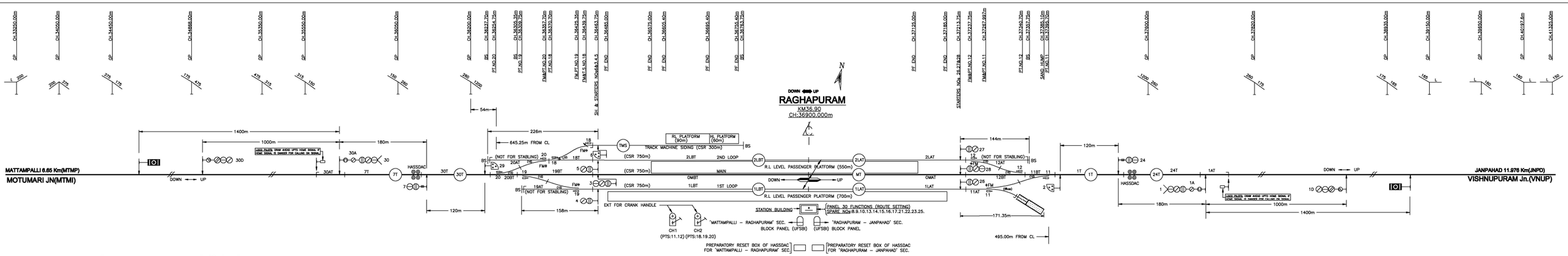
SNo.	SIGNAL No.	TO DESTINATION	SELECT ROUTE		IN ROUTE			OVERLAP			ISOLATION POINT		CRANK HANDLE NORMAL	FOULING TRACK CIRCUITS	APPROACK LOCK BY TRACK CIRCUITS	BACK LOCK BY TRACK CIRCUITS	SIGNAL REPLACED BY TRACK CIRCUITS	OVERLAP RELEASES AFTER 120 SEC BY		SLOTTED BY GATE	RELEASED BY OTHER CONTROLS &SIDING POINT DETECTION	LOCKS ROUTES	SIGNAL ASPECT CONTROLLED BY				REMARKS
			SIGNAL BUTTON	ROUTE BUTTON	POINT NOR.	POINT REV.	TRACK CIRCUIT	POINT NOR.	POINT REV.	TRACK CIRCUIT	NOR.	REV.						OCCUPIED BY TRACK CIRCUITS	CLEARENCE OF TRACK CIRCUIT				YELLOW WITH ROUTE	YELLOW	DOUBLE YELLOW	GREEN	
21.	S06	UP MAIN	S06	04–06	101	103	103AT. 103BT. 101AT. 04/06T	—	—	—	—	—	CH02. CH01	—	04BT. 04AT	103AT. 103BT. 101AT	103AT	—	—	—	—	C02–03. C02–04–1. SH42–03. SH42–04–1. SH41–04–1	—	S08 RG/ DG	—	—	TIME RELEASE 120 SEC. 'A' KEY IN.
22.	S07	DN MAIN	S07	03–07	—	106. 102	106AT. 106BT. 102AT. 102BT. 03/05/07T	—	—	—	104	—	CH02. CH04. CH03	—	04BT. 04AT	106AT. 106BT. 102AT. 102BT	106AT	—	—	—	—	C01–02. C01–04–1. SH42–04–1. SH41–03. SH41–04–1.	—	S09 RG/ DG	—	—	TIME RELEASE 120 SEC. 'A' KEY IN.
23.	S08	UP MAIN	S08	08	101	—	08T	—	—	—	—	—	CH01	—	—	—	08T	—	—	—	—	SH41–03. SH41–04–1	—	—	—	—	CONTROLLED BY UP LINE BLOCK INSTRUMENT 'LINE CLEAR' CONDITION. UP LINE BPAC CLEAR.
24.	S09	DN MAIN	S09	09	—	—	09T	—	—	—	—	—	—	—	—	—	09T	—	—	302	—	SH42–01–1. SH42–02. SH42–03. SH42–04–1. SH43–03–07	—	—	—	—	CONTROLLED BY DN LINE BLOCK INSTRUMENT 'LINE CLEAR' CONDITION. DN LINE BPAC CLEAR.
25.	SH41	UP MAIN	SH41	03	101. 103	—	101AT. 103BT	—	—	—	—	—	CH01. CH02	—	04/06T	101AT. 103BT	101AT	—	—	—	—	S02–03. S02–04–2. C02–03. C02–04–1. S04–04–06. S07–03–07. S08–08. SH42–03	—	—	—	—	TIME RELEASE 120 SEC
26.	SH41	COMMON LOOP	SH41	04–1	101	103	101AT. 103BT. 103AT	—	—	—	—	—	CH01. CH02	—	04/06T	101AT. 103BT. 103AT	101AT	—	—	—	—	S02–04–2. C02–03. C02–04–1. S06–04–06. S07–03–07. S08–08. SH42–04–1	—	—	—	—	TIME RELEASE 120 SEC
27.	SH42	DN LOOP	SH42	01–1	102	104	102BT. 104BT. 104AT	—	—	—	—	—	CH04. CH03	—	03/05/07T	102BT. 104BT. 104AT	102BT	—	—	—	—	S01–01–2. C01–01–1.C01–02. S05–03–07.S09–09	—	—	—	—	TIME RELEASE 120 SEC
28.	SH42	DN MAIN	SH42	02	102. 104	—	102BT. 104BT	—	—	—	—	—	CH04. CH03	—	03/05/07T	102BT. 104BT	102BT	—	—	—	—	S01–01–1. S01–02. S01–04–1. C01–01–1. C01–02. C01–04–1. S03–03–07. S09–09	—	—	—	—	TIME RELEASE 120 SEC
29.	SH42	UP MAIN	SH42	03	106	102	102BT. 102AT. 106BT	—	—	—	104 W C01–01–1	—	CH04. CH02.	—	03/05/07T	102BT. 102AT. 106BT	102BT	—	—	—	—	S01–04–1. C01–02. C01–04–1. S04–04–06. S06–04–06. S09–09. SH41–03. SH43–03–07.	—	—	—	—	TIME RELEASE 120 SEC
30.	SH42	COMMON LOOP	SH42	04–1	—	102. 106	102BT. 102AT. 106BT. 106AT	—	—	—	104 W C01–01–1	—	CH04. CH02.	—	03/05/07T	102BT. 102AT. 106BT. 106AT	102BT	—	—	—	—	S01–04–2. C01–02. C01–04–1. S06–04–06. S07–03–07. S09–09. SH41–04–1	—	—	—	—	TIME RELEASE 120 SEC. 'A' KEY IN.
31.	SH43	DN MAIN	SH43	03–07	106	102	106BT. 102AT. 102BT	—	—	—	104 W C01–01–1	—	CH02. CH04.	—	03BT. 03AT	106BT. 102AT. 102BT	106BT	—		—	—	C01–02. S09–09. SH42–03	—	—	—	—	TIME RELEASE 120 SEC

PRODUCTION & CHECKER BLOCK			
COMPANY		RAILWAYS	
DRAWN		DRAWN	(SSE/D)*
CHECKED	(SD)†	CHECKED	(ASTE)*
	(DV)†		(SSTE)*
APPROVED	(CRE)†	APPROVED	(Dy.CSTE)*

CRE: CONTRACTOR'S RESPONSIBLE ENGINEER
*SIGNATURE WITH DATE, † SIGNATURE WITH DATE AND LICENSE No

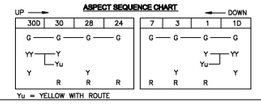
ANNEXURE 8/5

INTERLOCKING CONTROL TABLE	PART III	
ICT.SCR.GTL.VPL.Ø2	SHEET No:	VERSION
VENKATAMPALLE	ØØ4	Ø1



Annexure `8/6'

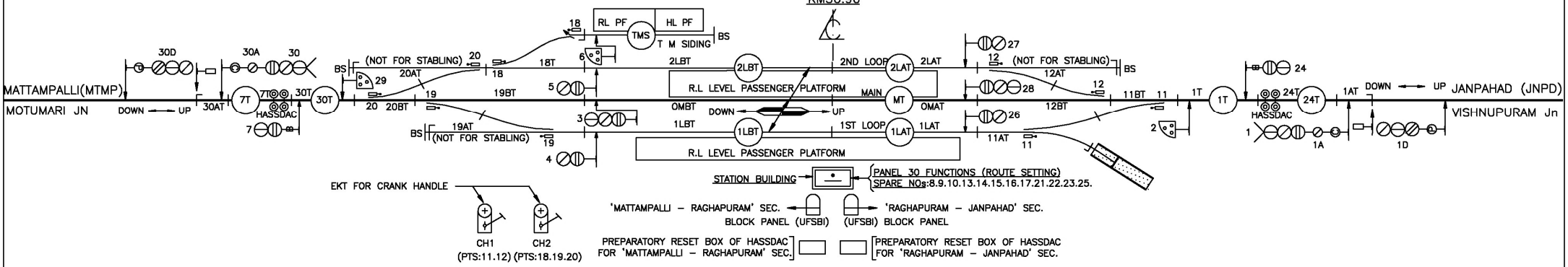
SYSTEM OF BLOCK WORKING				
S.NO	SYSTEM	SECTION	BLOCK INSTRUMENT	LOCATION
1	ABSOLUTE BLOCK	"MATTAMPALLI – RAGHAPURAM"	UFSBI WITH HASSDAC	PANEL ROOM
2	ABSOLUTE BLOCK	"RAGHAPURAM – JANPAHAD"	UFSBI WITH HASSDAC	PANEL ROOM



Annexure `8/7`

RAGHAPURAM

KM36.90



SL.NO	SIG.NO.	LEADING TO	PRESS BUTTON		APPROACH LOCKED BY	BACK LOCKED BY TRACKS	CONTROLLED BY TRACKS	DETECTS POINTS		LOCKS SIGNALS	REMARKS
			SIG.	ROUTE				NORMAL	REVERSE		
1	1D	DOWN HOME	---	---	———————	———————	———————	———	———	———————	DG CONTROLLED BY 1DG. HHG CONTROLLED BY 1HG.
2	1A	1ST LOOP	1A & COGGN	1LAT	DEAD APPROACH	24T. 1T. 11BT. 11AT.	1AT (OCCUPIED).	12.	11.	1. 2. 4. (6 W 19R). 24. 26.(29 – TMS. 2LAT W 19R). (29. 30A – MT. 1LAT). (30 – 1LBT).	APPROACH CLEARED AFTER 60 SEC. TIME RELEASE 120 SEC. EMERGENCY CANCELLATION 240 SEC. CH1.
3	1A	MAIN	1A & COGGN	MT	DEAD APPROACH	24T. 1T. 11BT. 12BT.	1AT (OCCUPIED).	11. 12.	———	1. 2. 3. 4. 5. 6. 24. 28. 29. 30. 30A.	APPROACH CLEARED AFTER 60 SEC. TIME RELEASE 120 SEC. EMERGENCY CANCELLATION 240 SEC. CH1.
4	1A	2ND LOOP	1A & COGGN	2LAT	DEAD APPROACH	24T. 1T. 11BT. 12BT. 12AT.	1AT (OCCUPIED).	11.	12.	1. 2. 5. 6. 24. 27. (29. 30A – 2LAT. MT). (29 – TMS). (30 – 2LBT).	APPROACH CLEARED AFTER 60 SEC. TIME RELEASE 120 SEC. EMERGENCY CANCELLATION 240 SEC. CH1.
5	1	1ST LOOP (SET TO BS)	1	1LAT	DEAD APPROACH	24T. 1T. 11BT. 11AT.	24T. 1T. 11BT. 12BT. 11AT. 1LT. 19AT.	12. 19.	11.	1A. 2. 24. 26. (29. 30A – MT).	TIME RELEASE 120 SEC. 1UG. CH1. CH2. 4RG.
6	1	1ST LOOP (SET TO MAIN)	1	1LBT	DEAD APPROACH	24T. 1T. 11BT. 11AT.	24T. 1T. 11BT. 12BT. 11AT. 1LT. 19AT. 19BT. 20BT. 30T.	12. 20.	11. 19.	1A. 2. 24. 26. 29. 30. 30A.	TIME RELEASE 120 SEC. 1UG. CH1. CH2. 4RG/HG.
7	1	MAIN	1	MT	DEAD APPROACH	24T. 1T. 11BT. 12BT.	24T. 1T. 11BT. 12BT. OMT. 19BT. 20BT. 30T.	11. 12. 19. 20.	———	1A. 2. 24. 28. 29. 30. 30A.	TIME RELEASE 120 SEC. DG CONTROLLED BY 3DG. CH1. CH2. 3RG/HG/DG.
8	1	2ND LOOP (SET TO BS)	1	2LAT	DEAD APPROACH	24T. 1T. 11BT. 12BT. 12AT.	24T. 1T. 11BT. 12BT. 12AT. 2LT. 18T. 20AT.	11. 18. 20.	12.	1A. 2. 24. 27. (29. 30A – MT).	TIME RELEASE 120 SEC. 1UG. CH1. CH2. 5RG.
9	1	2ND LOOP (SET TO MAIN)	1	2LBT	DEAD APPROACH	24T. 1T. 11BT. 12BT. 12AT.	24T. 1T. 11BT. 12BT. 12AT. 2LT. 18T. 20AT. 19BT. 20BT. 30T.	11. 18.	12. 20.	1A. 2. 24. 27. 29. 30. 30A.	TIME RELEASE 120 SEC. 1UG. CH1. CH2. 5RG/HG.
10	2	1ST LOOP	2	1LAT	1T.	11BT. 11AT.	11BT. 12BT. 11AT.	———	11.	1. 1A. 4. 24. 26. (29 – 1LAT). (30 – 1LBT). [30A – (2LAT W 12R). MT. 1LAT].	TIME RELEASE 120 SEC. CH1.
11	2	MAIN	2	MT	1T.	11BT. 12BT.	11BT. 12BT.	11. 12.	———	1. 1A. 3. 4. 5. 24. 28. (29 – MT). 30. 30A.	TIME RELEASE 120 SEC. CH1.
12	2	2ND LOOP	2	2LAT	1T.	11BT. 12BT. 12AT.	11BT. 12BT. 12AT.	11.	12.	1. 1A. 5. 24. 27. (29 – 2LAT). (30A – 2LAT. MT). (30 – 2LBT).	TIME RELEASE 120 SEC. CH1.
13	3	MAIN	3	30T	OMT. (S1 – MT).	19BT. 20BT.	19BT. 20BT. 30T.	19. 20.	———	(1A. 2 – MT). 28. 29. 30. 30A.	TIME RELEASE 120 SEC. DG CONTROLLED BY 7DG. CH2. 7RG/DG.
14	4	MAIN	4	30T	1LT.	19AT. 19BT. 20BT.	19AT. 19BT. 20BT. 30T.	20.	19.	(1A. 2 – 1LAT. MT). 26. 29. 30. 30A.	TIME RELEASE 120 SEC. CH2. 7RG/DG.
15	5	MAIN	5	30T	2LT.	18T. 20AT. 20BT.	18T. 20AT. 19BT. 20BT. 30T.	18. 19.	20.	(1A. 2 – MT. 2LAT). 27. 29. 30. 30A.	TIME RELEASE 120 SEC. CH2. 7RG/DG.
16	6	TO CLEAR OF Sh29	6	30T	DEAD APPROACH	18T. 20AT. 20BT.	18T. 20AT. 19BT. 20BT.	———	18. 20.	[1A – (1LAT W 19R). MT. 2LAT]. 7. 29.	TIME RELEASE 120 SEC. CH2.
17	7	MAIN (MTMP)	7	7T	———————	———————	7T. HASSDAC.	———	———	6. 29. 30. 30A.	CONTROLLED BY MATTAMPALLI–RAGHAPURAM SEC. BLOCK INSTRUMENT (UFSBI)
18	24	MAIN (JNPD)	24	24T	———————	———————	24T. HASSDAC.	———	———	1. 1A. 2.	CONTROLLED BY RAGHAPURAM–JANPAHAD SEC. BLOCK INSTRUMENT (UFSBI)
19	26	MAIN	26	1T	1LT.	11AT. 11BT.	11AT. 12BT. 11BT. 1T.	12.	11.	1. 1A. 2. 4. (29. 30A – MT. 1LAT).	TIME RELEASE 120 SEC. CH1. 24RG/DG.
20	27	MAIN	27	1T	2LT.	12AT. 12BT. 11BT.	12AT. 12BT. 11BT. 1T.	11.	12.	1. 1A. 2. 5. (29. 30A – 2LAT. MT).	TIME RELEASE 120 SEC. CH1. 24RG/DG.

SL.NO	SIG.NO.	LEADING TO	PRESS BUTTON		APPROACH LOCKED BY	BACKLOCKED BY TRACKS	CONTROLLED BY TRACKS	DETECTS POINTS		LOCKS SIGNALS	REMARKS
			SIG.	ROUTE				NORMAL	REVERSE		
21	28	MAIN	28	1T	OMT. (S30 – MT).	12BT. 11BT.	12BT. 11BT. 1T.	12. 11.	–––	1. 1A. 2. 3. (29. 30A – MT).	TIME RELEASE 120 SEC. DG CONTROLLED BY 24DG. CH1. 24RG/DG.
22	29	TO CLEAR OF Sh6	29	TMS	30T.	20BT. 20AT. 18T.	20BT. 19BT. 20AT. 18T.	–––	20. 18.	[1A – (1LAT W 19R). MT. 2LAT]. 6. 7.	TIME RELEASE 120 SEC. CH2.
23	29	2ND LOOP	29	2LAT	30T.	20BT. 20AT. 18T.	20BT. 19BT. 20AT. 18T.	18.	20.	(1 – 2LBT). [1A – (1LAT W 19R). MT. 2LAT]. (2 – 2LAT). 5. 7. 27. 30. 30A.	TIME RELEASE 120 SEC. CH2.
24	29	MAIN	29	MT	30T.	20BT. 19BT.	20BT. 19BT.	20. 19.	–––	1. 1A. (2 – MT). 3. 7. 26. 27. 28. 30. 30A.	TIME RELEASE 120 SEC. CH2.
25	29	1ST LOOP	29	1LAT	30T.	20BT. 19BT. 19AT.	20BT. 19BT. 19AT.	20.	19.	(1 – 1LBT). (1A – 1LAT. MT). (2 – 1LAT). 4. 7. 26. 30. 30A.	TIME RELEASE 120 SEC. CH2.
26	30	2ND LOOP (SET TO BS)	30	2LAT	DEAD APPROACH	7T. 30T. 20BT. 20AT. 18T.	7T. 30T. 20BT. 19BT. 20AT. 18T. 2LT. 12AT.	19. 18. 12.	20.	(1A. 2 – MT). 5. 7. 29. 30A.	TIME RELEASE 120 SEC. 30UG. CH1. CH2. 27RG.
27	30	2ND LOOP (SET TO MAIN)	30	2LBT	DEAD APPROACH	7T. 30T. 20BT. 20AT. 18T.	7T. 30T. 20BT. 19BT. 20AT. 18T. 2LT. 12AT. 12BT. 11BT. 1T.	19. 18. 11.	20. 12.	1. 1A. 2. 5. 7. 29. 30A	TIME RELEASE 120 SEC. 30UG. CH1. CH2. 27RG/HG.
28	30	MAIN	30	MT	DEAD APPROACH	7T. 30T. 20BT. 19BT.	7T. 30T. 20BT. 19BT. OMT. 12BT. 11BT. 1T.	20. 19. 12. 11.	–––	1. 1A. 2. 3. 7. 29. 30A.	TIME RELEASE 120 SEC. DG CONTROLLED BY 28DG. CH1. CH2. 28RG/HG/DG.
29	30	1ST LOOP (SET TO SAND HUMP)	30	1LAT	DEAD APPROACH	7T. 30T. 20BT. 19BT. 19AT.	7T. 30T. 20BT. 19BT. 19AT. 1LT. 11AT.	20. 11.	19.	(1A. 2 – MT). 4. 7. 29. 30A.	TIME RELEASE 120 SEC. 30UG. CH1. CH2. 26RG.
30	30	1ST LOOP (SET TO MAIN)	30	1LBT	DEAD APPROACH	7T. 30T. 20BT. 19BT. 19AT.	7T. 30T. 20BT. 19BT. 19AT. 1LT. 11AT. 12BT. 11BT. 1T.	20.	19. 11.	1. 1A. 2. 4. 7. 29. 30A.	TIME RELEASE 120 SEC. 30UG. CH1. CH2. 26RG/HG.
31	30A	2ND LOOP	30A & COGGN	2LAT	DEAD APPROACH	7T. 30T. 20BT. 20AT. 18T.	30AT (OCCUPIED).	19. 18.	20.	(1 – 2LBT). (1A. 2 – MT. 2LAT). (2 – 1LAT W 12R). 5. 7. 27. 29. 30.	APPROACH CLEARED AFTER 60 SEC. TIME RELEASE 120 SEC. EMERGENCY CANCELLATION 240 SEC. CH2.
32	30A	MAIN	30A & COGGN	MT	DEAD APPROACH	7T. 30T. 20BT. 19BT.	30AT (OCCUPIED).	20. 19.	–––	1. 1A. 2. 3. 7. 26. 27. 28. 29. 30.	APPROACH CLEARED AFTER 60 SEC. TIME RELEASE 120 SEC. EMERGENCY CANCELLATION 240 SEC. CH2.
33	30A	1ST LOOP	30A & COGGN	1LAT	DEAD APPROACH	7T. 30T. 20BT. 19BT. 19AT.	30AT (OCCUPIED).	20.	19.	(1– 1LBT). (1A. 2 – 1LAT. MT). 4. 7. 26. 29. 30.	APPROACH CLEARED AFTER 60 SEC. TIME RELEASE 120 SEC. EMERGENCY CANCELLATION 240 SEC. CH2.
34	30D	UP HOME	---	---	–––––––	–––––––	–––––––––	–––	–––	–––––––––	DG CONTROLLED BY 30DG. HHG CONTROLLED BY 30HG.

NOTE: 1. 1LT DENOTES 1LAT. 1LBT.
2. OMT DENOTES OMAT. OMBT.
3. 2LT DENOTES 2LAT. 2LBT.
4. EMERGENCY CANCELLATION FOR CALLING ON SIGNALS TO BE APPLIED DURING FAILURE OF ANY BACK LOCK TRACK CIRCUIT.

Annexure `8/7'

POINTS CONTROL TABLE

SL.NO.	POINTS No.	TRACK LOCKED BY	LOCKED BY SIGNALS	REMARKS
1.	11.	11AT. 11BT.	1. 1A. 2. 26. 27. 28. (30 – 2LBT. MT. 1LAT. 1LBT).	CH1.
2.	12.	12AT. 12BT.	1. 1A. (2 – MT. 2LAT). 26. 27. 28. (30 – 2LAT. 2LBT. MT).	CH1.
3.	18.	18T.	(1. 30 – 2LAT. 2LBT). 5. 6. (29 – TMS. 2LAT). (30A – 2LAT).	CH2.
4.	19.	19AT. 19BT.	(1 – 1LAT. 1LBT. MT). 3. 4. 5. (29 – MT. 1LAT). 30. 30A.	CH2.
5.	20.	20AT. 20BT.	(1 – 1LBT. MT. 2LAT. 2LBT). 3. 4. 5. 6. 29. 30. 30A.	CH2.

DESIGNED & DRAWN BY
MEDHA
MEDHA SERVO DRIVES PVT.LTD
P-4/5B, I.D.A, NACHARAM, HYDERABAD

REV	DATE	DESIGNED	CHECKED	APPROVED	MODIFICATION
00	14.11.2018	AKM	PSR	GSR	INITIAL DESIGN

M/S KIRAN INFRA ENGG. LTD.
B-141, ROAD No.9D,
VISHWAKARMA INDUSTRIAL AREA,
JAIPUR-302013.

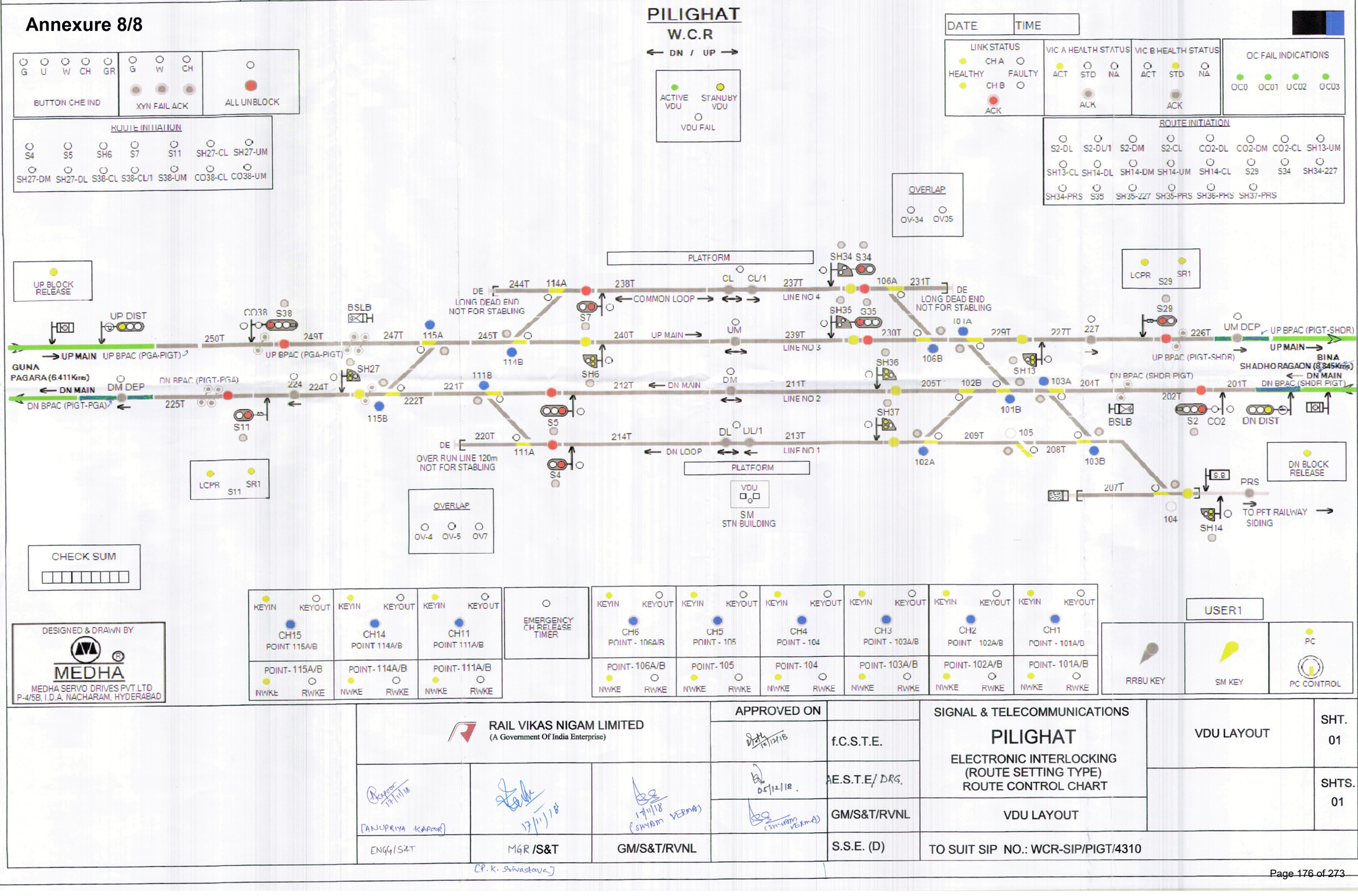
CHECKED

WEST CENTRAL RAILWAY

PILIGHAT STATION

NOTE :-
i) DISPLAY AREA 42 INCHES.
ii) ERRB,COGGZ,OYZ,EBPU,RRBU COUNTERS PROVIDED SEPARATELY IN A SEALED BOX IN SM's ROOM.

Annexure 8/8



DESIGNED & DRAWN BY
MEDHA
MEDHA SERVO DRIVES PVT.LTD
P-4/5B, I.D.A, NACHARAM, HYDERABAD

RAIL VIKAS NIGAM LIMITED
(A Government Of India Enterprise)

APPROVED ON		SIGNAL & TELECOMMUNICATIONS	VDU LAYOUT	SHT. 01
16/12/18	f.C.S.T.E.	PILIGHAT		
05/12/18.	AE.S.T.E/ DRG.	ELECTRONIC INTERLOCKING (ROUTE SETTING TYPE) ROUTE CONTROL CHART		SHTS. 01
(SHYAM VERMA)	GM/S&T/RVNL	VDU LAYOUT		
	S.S.E. (D)	TO SUIT SIP NO.: WCR-SIP/PIGT/4310		

17/11/18 (ANUPRIYA KAPOOR) — ENGG/S&T
17/11/18 — MGR /S&T
17/11/18 (SHYAM VERMA) — GM/S&T/RVNL
[P. K. Srivastava]

CHAPTER - 9: PROCEDURE FOR COMMISSIONING OF SIGNALLING INSTALLATIONS

Section 1: General Instructions

9.1.1.0 Reference to Rules

9.1.1.1 The safety of the travelling public is ensured by the Rules laid down in -

(a) The Indian Railway Act.

(b) The General Rules for all Open Line Railways.

(c) The Rules for Opening of a Railway or section of a Railway for the public carriage of passengers.

(d) The Indian Railway Schedule of Dimensions.

9.1.1.2 The rules provide for the legal authorization that shall be obtained for any work which affects the running line, before the work is started or brought into use and before a new section of a line is opened for public traffic.

9.1.2.0 Notification to Railway officials before opening works. - No signalling work affecting the running of trains or working of traffic at a station shall be brought into use until staff of all concerned Departments have been notified by means of a circular issued by the Operating Department Timely intimation of the date of commencement of work, duration of work, arrangements for working of trains during the progress of the work, date of opening of work, etc., shall be given to the Operating Department, whenever any new or revised traffic working instructions are to be brought into use to enable operating Department to give the station and running staff due notice.

9.1.3.0 Approved plans, Drawings and Specifications. - All works shall be carried out strictly in accordance with the approved plans, standard drawings and specifications and should conform to the provisions of this manual where such are applicable. Deviations, if any, shall have the prior approval of the Chief Signal and Telecommunication Engineer. All deviations shall be marked on the completion drawings and forwarded to the Chief Signal and Telecommunication Engineer for correction of the tracings and issue of revised prints.

9.1.4.0 Responsibilities of Engineer in charge

9.1.4.1 The Signal Engineer-in-Charge shall ensure that no work is commenced without

(a) Proper men and materials being available for its execution.

(b) Approved signalling plans, locking tables and diagrams, selection tables and circuits/Application Logic, standard drawings and specifications.

(c) Sanction of Commissioner of Railway Safety or Principal Chief Signal & Telecom Engineer (as applicable) in the case of works on lines already opened for passenger traffic

(d) Station working Rules and temporary working instructions.

(e) Sanction to the detailed estimate for the work with necessary allotment of Funds. (This does not apply to works started on urgency Certificates).

9.1.4.2 The Engineer-in-Charge shall, besides authorizing commencement of works

(a) Be responsible for proper and efficient execution of works,

(b) Make frequent inspections and issue detailed instructions to the JE/SSE's

(c) See that the progress of work is satisfactory and submits progress reports to the Principal Chief Signal and Telecommunication Engineer.

(d) Be responsible for correct booking and control over expenditure within the funds allotted.

(e) Assist the Operating Department in preparation of the Station working Rules.

(f) Arrange for sighting of new Signals by sighting committee before commissioning the same.

(g) Fix a date for opening jointly with other concerned officials and notify the same to all concerned.

(h) Arrange to issue safety certificate and joint message to the Commissioner of Railway Safety

9.1.4.3 **Responsibilities of the SSE In charge -** The SSE in charge of the work shall, besides ensuring the conditions stipulated in Para 9.1.4.0, be responsible for ensuring that -

(a) a work is not started unless authorized by the concerned Officer;

(b) all necessary steps are taken for the safety of trains movements during the execution of the work;

(c) there is no avoidable detentions to trains and if any detentions do take place particulars are advised promptly to the Engineer-in-Charge ;

(d) the station staff have received the necessary notice and Station working Rules;

(e) the work is tested and found correct by him before requesting the Engineer-in-Charge to test and commission the work ;

(f) all precautions are taken to prevent accidents to staff or damage to equipment ; and

(g) all accidents are promptly reported.

9.1.5.0 Joint Inspection /Handing Over of new assets

9.1.5.1 (a) Joint inspection by Divisional open line and construction/Work Executing

agency both at SSE-In-Charge & officer level shall be undertaken before taking up Non-interlocking of a station in connection with yard re-modelling/interlocking changes or commissioning of Signalling works for a new line/gauge concerted line . (Annexure-9/7)

(b) The non interlocking of the station / commissioning of any signalling work shall be undertaken after the items considered essential for safety of train operation are completed. (Annexure-9/7)

9.1.5.2 Within 15 days of commissioning of Signalling works, another Joint inspection both at SSE-In-Charge & Officer level by Divisional Open line and /Construction /Work Executing agency shall be undertaken. The deficiencies noted during these inspections shall be listed as; (Annexure-9/7)

(a) Category A - Items which shall be complied before handing over

(b) Category B - Petty items (like painting, labling etc.) which may be complied after handing over as per MOU to be entered between open line and construction organization/Work Executing Agency

9.1.5.3 All the new signalling equipments and associated gadgets shall be handed over/taken over within 2 months period after commissioning of the signalling works. After expiry of 2 months period, however the responsibility of maintenance shall devolve on divisional open line organization after commissioning, subject to rectification of category "A" defects.

9.1.5.4 In order to ensure that handing over/ taking over of signalling assets is accomplished in a smooth manner, a specific handing over/ taking over procedure as detailed in Annexure-`9/7` shall be followed.

Section 2 : Obtaining Sanction for Works on Lines Opened to Traffic

9.2.6.1 **Works requiring notice to and sanction of the Commissioner of Railway Safety**

Under Section 20 of the Indian Railways Act and Chapter VI of the "Rules for the Opening of a Railway or Section of a Railway for public Carriage of Passengers" the approval of Commissioner of Railway Safety is required for the execution of any work on the open line which will affect running of passenger trains and any temporary arrangement necessary, for carrying it out except in cases of emergency. The following signal and interlocking works , when they are connected with or form part of a Railway already opened for carriage of passengers requires the sanction of Commissioner of Railway Safety before they are commenced or opened:-

(a) Additions/extensions or alterations to existing Block, Signalling and interlocking installations.

(b) Change in block, Signalling and Interlocking scheme

(c) New stations temporary or permanent.

(d) Interlocking of level crossing, catch siding and slip sidings etc.

9.2.6.2 The following works shall, which enhance level of safety, however, do not require the sanction of the Commissioner of Railway Safety:-

(a) Provision of Telephones at already manned level crossings.

(b) Provision of electrical & mechanical lifting barrier including emergency interlocking arrangements, sliding boom at already interlocked Level Crossing Gates.

(c) Interlocking of existing LC Gates within already existing interlocked station yard by existing signals in the same or shifted location.

(d) Provision of all categories of track circuits, axle counters including dual detection.

(e) Replacement /Modification of signalling assets without any change in the yard layout or signal interlocking scheme either in station or at mid section Level Crossing Gates .

(f) Replacement of Block instruments by any other approved type of instrument or provision of block proving through Axle counters or Track Circuiting using existing Block instruments.

(g) The replacement of block instrument without any change in the system of block working.

(h) Closing of interlocked Level Crossing gates after approval of district administration obtained by Engineering department.

(i) Providing Electronic Interlocking/Panel Interlocking/Route Relay Interlocking in Non passenger Line section

(j) Providing/Removal of key lock Interlocking on siding point within already existing interlocked station yard by existing signals in same location or shifted location.

9.2.6.3 **NOTE:-**

(a) The personal approval of the PCSTE will be required for sanction of the works mentioned at para 9.2.6.2. This will not be re-delegated to lower level. PCSTE of Railway will submit Safety certificates & quarterly statement of all such sanctions accorded by him to the concerned CRS.

(b) CSTE's sanction are not given in a routine manner but following laid down procedure. It is therefore, suggested that while giving sanctions by PCSTEs, following must be ensured:-

(i) All relevant documents that are prepared by field units (for submission to CRS for his sanction) for commissioning of the works would be submitted to PCSTE for his sanction. This includes Engineering scale/Signalling Plans, SWR diagrams and SWR.

(ii) Format as per Annexure '9/8' shall be submitted duly filled up by JA grade

officer, incharge of the work for applying for PCSTE's sanctions in terms of Para 9.6.2 of the IRSEM Part-I(1988 Edition). Similarly, PCSTE's sanction shall also be communicated in the defined format enclosed as Annexure '9/9'.

(iii) Application for CSTE's sanction received in Headquarter's office shall be scrutinised as per the check list laid down as Annexure '9/10' by Dy.CSTE/Incharge of design or CSTE(planning) and put up for the consideration of the PCSTE.

(iv) Personal approval of PCSTEs would be required and it shall not be delegated to any lower level.

(v) Sanctions would be recorded by PCSTEs as a numbered sanction maintaining a register for the same similar to sanction numbers being maintained by CRS.

(vi) The validity of the PCSTE's sanction shall be for a period of six months only. Re-valuation of the sanction after six months may be considered based on unavoidable reasons.

9.2.7.0 Application to the Commissioner of Railway Safety/PCSTE for sanction for works

9.2.7.1 Applications to the Commissioner of Railway Safety/PCSTE for sanction for carrying out works listed in Para 9.2.6.1 shall be submitted in the following manner -

(a) When the work is executed by the Divisional organization , the application shall be made by the Sr.Divisional Signal & Telecom Engineer for Signalling works. For joint works or work by Railway's PSU the application shall be made by Divisional Railway Manager.

(b) When the work is executed by an extra Divisional Organization e. g. Construction Organization, the application shall be made by an officer not below the rank of Deputy Chief Signal and Telecommunication Engineer for signalling works. For joint works the application shall be made by an officer not below Junior Administrative grade of the department which controls the work.

(c) Station Working Rules obtained from the Operating Department shall accompany the application.

(d) The application shall be made on the prescribed form 1606/1 "Application for Sanction" (Annexure '9/1') and shall be complete in respect of all documents mentioned therein.

(e) The application for sanction shall ordinarily be made at least fourteen days in advance of expected commencement of the work. When a sanctioned work is not taken on hand within twelve months of the date of sanction, a fresh sanction shall be obtained.

9.2.7.2 If any material deviation from the plan approved by the Commissioner of Railway Safety which affects the yard layout or signalling and interlocking arrangements or system of train working is found necessary his prior approval to such deviations

shall be obtained with reference to application made initially.

9.2.7.3 The Commissioner of Railway Safety/PCSTE in according his sanction may or may not propose to inspect the works.

9.2.7.4 If the Commissioner of Railway Safety/PCSTE decides to inspect the work prior to opening, he will, after inspection in the company of the officers concerned, communicate in writing his sanction to open the work.

9.2.8.0 Safety Certificate

9.2.8.1 (a) If the Commissioner of Railway Safety decides not to inspect the work prior to the opening, the Engineer-in-Charge shall submit the Safety Certificate on the prescribed form (Annexure ’9/2') signed by him prior to the opening of works and in case of joint works, signed also by Engineering-in-Charge of Civil/Electrical Engineering portion of the work. In case of work by Rly PSU, the Safety Certificate shall also be signed by officials not below the equivalent rank of JAG in Rlys.

(b) Part Safety certificate can be issued for partially completed works with clear mention of incomplete work. Final Safety certificate shall be issued separately for whole work on completion of left out work. In case of work by Rly PSU, the Safety Certificate shall also be signed by officials not below the equivalent rank of JAG in Rlys. Safety Certificate should be signed by executing officers.

9.2.8.2 Before signing the Safety Certificate the Engineer-in-Charge shall

(a) Test the new works and see that the installation has been carried out correctly.

(b) Fully satisfy himself, that the work can be opened for the public carriage of passengers without endangering the safety of the traveling public or of the employees of the Railway.

In case of joint works, Safety certificate shall also be signed by the Engineer-in-Charge of the concerned Departments.

In case of work by Rly PSU, the Safety Certificate shall also be signed by officials not below the equivalent rank of JAG in Rlys.

9.2.8.3 (a) After fully satisfying himself he shall bring the installation into use by issuing a message in the form given on the sanction Application (Annexure '9/1') to the Commissioner of Railway Safety with copy to all concerned.

(b) In case of joint works the message shall also be signed by the Engineer-in-Charge of the concerned Departments for their portion of the work.

9.2.8.4 The Safety Certificate shall be countersigned by the Officers who applied for the sanction of Commissioner of Railway Safety/PCSTE. A certificate from Operating Department, stating that the necessary Station Working Rules have been issued and giving reference to sanction of deviations (if any) from General and Subsidiary Rules, shall also be attached.

9.2.8.5 Copies of the Safety Certificates shall be sent to the Divisional Railway Manager and the Principal Chief Signal and Telecommunication Engineer and also to the Principal Chief Engineer/Principal Chief Electrical Engineer if the

Engineering/Electrical Department is/are involved in the work.

Section 3 : New Lines & Electrification

9.3.9.0 **Application to the Commissioner of Railway Safety**

9.3.9.1 In the case of new lines or electrification, prior sanction of the Commissioner of Railway Safety for commencement of the work is not necessary but when it is proposed to open the line for passenger traffic or to initiate electric traction on a line already opened, the following documents are to be furnished by the General Manager of Railway to the Commissioner of Railway Safety at least one month before the date by which the line is expected to be ready for opening. These documents collectively are termed the Opening Documents ;—

(a) Tabulated details.

(b) Index plan and section of Railway,

(c) Drawings of works.

(d) List of questions and answers duly filled in.

(e) Certificates.

(f) List of infringements of maximum and minimum dimensions,

(g) Diagram of proposed testing train

(h) Station working rules to be enforced at each station.

9.3.9.2 Of the above only the following items concern signalling and interlocking ;—

(a) Tabulated details Form No. VIII of Rules for Opening of a Railway. (An example at Annexure '9/3').

(b) Drawings of works, plans of station yard showing gradients, the layout of tracks and signals and interlocking including Locking and Selection tables/Route Control chart.

(c) List of Questions and Answers (Annexure '9/4').

(d) Station working Rules (assisting the Operating Branch in their preparation).

(e) Certificate for introducing or extending electric traction, containing details of maximum value of Traction current (Annexure '9/5')

9.3.9.3 The new line shall be opened for passenger traffic or electric traction shall be initiated only after the Commissioner of Railway Safety authorises the Railway to do so after his inspection.

Note: If any part of the work relating to a new line or electrification affects any existing signalling or interlocking installation, prior sanction of the Commissioner of Railway Safety shall be obtained for that part of the work and the work executed on the basis of instructions contained in Section '1' of this Chapter.

Section 4 : Introduction of New types of Locomotives or Rolling Stock for increasing Speed

9.4.10.0 Sanction of the Commissioner of Railway Safety is necessary for the running of new types of locomotives or rolling stock or for increasing the maximum permissible speed on a specified section or of locomotives or rolling stock. The application for this purpose shall be made by the Principal Chief Engineer/Principal Chief Electrical Engineer/Principal Chief Mechanical Engineer as the case may be and it shall be accompanied by the following documents :-

(a) Load diagram.

(b) Certificate for Track strength.

(c) Certificate for strength of girders.

(d) Certificate of test runs (if required by the Commissioner of Railway Safety) obtained from Operating Department.

(e) Certificate on Form Annexure '9/6' signed jointly by the Principal Chief Mechanical Engineer, Principal Chief Engineer, Principal Chief Signal and Telecommunication Engineer, Principal Chief Electrical Engineer and Principal Chief Operating Manager.

Note :- Joint Safety Certificate shall mention maximum traction current permitted in the section.

(f) A statement (in Form XI of the" Rules for Opening of a Railway or section of a Railway for the Public carriage of passengers") detailing any "infringement of maximum and minimum dimensions" involved in running of the locomotive or rolling stock.

(g) Certificate of Emergency Braking Distance of the locomotive or rolling stock for the ruling gradient in the section

(h) Joint Safety Certificate should certify that maximum traction current permitted in the section is suitable to signalling installations.

(i) On receipt of such an application, the Commissioner of Railway safety if he so desires, will inspect and/or try out the new locomotives and/or rolling stock and the Railway Administration will provide him the necessary assistance to do so.

☞	*Note :- This Chapter has under mentioned Annexures for further study*	
S.no	*Annexure no*	*Description*
1	*9/1*	*Application for Sanction*
2	*9/2*	*Safety Certificate*
3	*9/3*	*Station Machinery*
4	*9/4*	*Station Equipment*
5	*9/5*	*Safety Certificate for S&T works for introducing Electrification*
6	*9/6*	*Joint Safety Certificate*
7	*9/7*	*Handing over / Taking over of Signalling Assets*
8	*9/8*	*Application for PCSTE Sanction*
9	*9/9*	*PCSTE Sanction*
10	*9/10*	*Check List for CRS/PCSTE application for sanction*

Annexure '9/1'

Application for Sanction

……RAILWAY Office…….

Department…… Dated:……..

No.....

From:

The……

Designation……..

To

The Commissioner of Railway Safety,

……

Sir

I hereby apply for your sanction to

*……
……
……..

being commenced and opened for the public carriage of passengers when ready.

2. With reference to Chapter VI of the Rules for the Opening of a Railway, I have to enquire whether you wish to inspect the work prior to its opening for the public carriage of passengers in which case intimation will be given of the date of completion.

3. In the event of your deciding not to inspect the work prior to opening, the Engineer-in-charge will, on completion of the work, submit the Safety Certificate, duly signed by him, prior to the opening of the work for public carriage of passengers and when required, also dispatch a telegram ** to your address intimating that the work has been opened and the Safety Certificate has been signed by him.

4. The application for the use of Locomotives and Rolling-stock to be drawn or propelled thereby on the proposed line, in accordance with Section 16 (1) of the Indian Railway Act, 1890 (IX of 1890), is sent herewith/not required.

5. The following documents @ are enclosed:-

(I) Temporary works:-

(a) Description of proposed works.
(b) Drawing of Temporary work.
(c) List of infringements to Schedule of dimensions.
(d) List of deviations from the Signal Engineering Manual.
(e) List of deviations from General and Subsidiary Rules.
(f) Restrictions
(g) Rules for Traffic Working.
(h) Documents for Bridges as per Chapter-VII of the Rules for the Opening of Railways.

(II) Permanent works:-

(a) Description of proposed works
(b) Drawing of Permanent work.
(c) List of infringements to Schedule of dimensions.

(d) List of deviations from the Signal Engineering Manual.
(e) List of deviations from General and Subsidiary Rules.
(f) Restrictions
(g) Rules for Traffic Working.
(h) Documents for Bridges as per Chapter-VII of the Rules for the Opening of Railways.

6. Certified that a detailed examination of the strength and arrangement of the materials to be used in the temporary/permanent works in above connection, have been made and that the design and the materials to be used are upto the loads, which they will be required to carry and that their opening for public carriage of passengers will not be attended with any danger.

(Delete permanent/temporary work, as the case may be).

Yours faithfully

Signature……..

Designation……

Dated……..

No…….

From

The Commissioner of Railway Safety.

……..

Sir,

Your No……

Sanction is accorded to the above work being carried out. % I do not propose to inspect the work prior to its opening for the carriage of passengers. When ready, it may be opened on a safety Certificate (vide paragraph 3 of your letter) which should be submitted to me direct without any delay.

% I propose to inspect the work prior to its opening for public carriage of passengers. Advice of the date, when the work will be ready for inspection should be intimated at least 14 days before it is proposed to open it.

……..……..

Commissioner of Railway Safety

*Here enter the name of work and mention whether permanent or temporary.

Form of telegram “Reference Sanction No. _____ dated _____ work open for public traffic on _____ First train to pass _____ No Danger to public. Certificate signed.

@If any of the document are not sent, then “NIL” to be written against such items. Working rules for extensive remodelling scheme may be sent in not later than one month before the date on which the work is to be brought into use, and in such cases “will follow” should be written instead in “NIL”.

% Strike out paragraph not applicable.

Annexure '9/2'

……RAILWAY

Safety Certificate

When the Commissioner of Railway Safety does not inspect the work prior to opening, this certificate must be signed before opening temporary or new works.

From,

The Divisional Railway Manager/Deputy Chief Engineer (Construction)/Deputy Chief Signal and Telecommunication Engineer (Construction)……..Division.

To,

The Commissioner of Railway Safety,
……..

Description of work……

……

Reference: Divisional Railway Manager
Chief Engineer/Deputy Chief Engineer (Construction),
Deputy Chief Signal and Telecommunication Engineer (Construction).

Application……..……..……dtd……sanctioned under Commissioner of Railway Safety No……..dtd…….to commence and open the above work.

Following permanent /temporary work has been done-

……

……..

I/We do hereby certify that, in the work above mentioned-

(i) The schedule of dimensions has not been infringed * except in regard to the items sanctioned under letter * No……..dated…….……..
(ii) Engineering work has been carried out in accordance with Plan No……*except in regard to the alterations sanctioned under letter No…….dated…….
(iii) The weight of rails, strength of bridges and general structural character of the works are such as have been prescribed under the rules.
(iv) The *Signalling and *Interlocking and *Block Signalling has been carried out in accordance with Signalling Plan No…….. and the requirements and instructions laid down in the Signal Engineering Manual have been completed * except in regard to the items sanctioned under letter No……..dated…….……..

The work has been carried out in accordance with the documents already supplied.

2. A certificate from the Divisional Safety Officers/……..stating that the necessary working rules have been issued and giving reference in regard to sanction to deviation (if any) from General and Subsidiary rules is attached/not required
   * To be scored out if not applicable.

3. I/We hereby certify that on ……20……. I/We have carefully inspected and tested the above work and that I/We have satisfied myself/ourselves that it has been properly completed and is in good working 'order' *and that the work can be opened for Public carriage of passengers without endangering the safety of the travelling public, or of the employees of the Railway, subject to the following speed restrictions:-

@ Temporary……..Km. p.h……due to ……..
……..…
@Permanent……..Km. p.h……..due to……..….
……..

4. The work is being opened on…….……..…….
   *Necessary in case of signaling and interlocking works only.
   @To be scored out if not applicable.

	Countersigned by:-
	…….
Assistant Engineer	……
Dated…….	Dated……..

	Countersigned by:-
	…….
Assistant Signal & Telecommunication Engineer	……
Dated…….	Dated……..

	Countersigned by:-
	…….
	……
	Dated……..

No……..Dated……..……..

Countersigned and forwarded to Commissioner of Railway Safety for information:-

1.
2.
3.
4.
5.

Divisional Railway Manager
Chief Engineer (Construction)
Deputy Chief Signal and
Telecommunication Engineer
(Construction)

Annexure '9/3'

AMINABAD-BISHANPUR RAILWAY

(5'- 6' Gauge)

(62.464 Kilometers)

Station Machinery

Name of Station	Interlocking				Warner/ Distant signal	Outer Signal	Home Signal	Starter Signal	Advance Starter Signal	Types/ block working in use
	Inter-locked	Semi-inter-locked	Non-inter-locked	Isolated						
Sogaria	Yes	--	--	Yes	3	--	3	6	3	S/Line Daido and D/Line SGE
Chandresal (Flag) (a)	--	--	Yes	--	--	--	--	--	--	--
Digod	Yes	--		Yes	2	--	2	5	2	D/Line SGE
Srikalyanpura	Yes	--	--	Yes	2	--	2	5	2	Do
Bhonra	Yes	--	--	No	2	--	2	8	2	Do
Antah	Yes	--	--	Yes	2	--	2	4	2	Do
Bijora	Yes	--	--	Yes	2	--	2	4	2	Do
Sundlak	Yes	--	--	--	2	--	2	6	2	Do
Baran	Yes	--	--	Yes	2	--	2	9	2	Do

(a) No points in the main line

Note: - This is a sample. Specific details to be furnished as per section required.

Annexure '9/4'

Station Equipment

		ROR Form IX Question No.
(i)	Are the clocks provided at all block station?	36
(ii)	Have the requirements and recommendation for signaling and interlocking installations vide Chapter VIII of Rules for Opening of the Railway and the Appendix thereto, been fully complied with, according to the class if signaling and interlocking installed?	37
(iii)	(a) have the requirements for single line token instrument and for their installation vide Chapter IX of Rules for Opening of the Railway been fully complied with? (b) Have the requirements for double line block instruments and for their installation vide Chapter IX of Rules for Opening of the Railway been fully complied with? If not, in what respect do the arrangements provided fall short of them?	38
	(c) Have the requirements for tokenless block instrument and for their installation been fully complied with? If not, in what respect do the arrangements provided fall short of them?	
(iv)	Are these stations situated on a sleeper grade than 1 in 400 If so, what special safety devices are adopted and/or special rules enforced at those stations?	43
(v)	What safety sidings are providing? Are they suitable? Are any other necessary?	44

Annexure '9/5'

Safety Certificate for S&T works for introducing Electrification

It is hereby certified that the Signalling and Telecommunication works to be executed for the purpose of introduction of 25KV single phase, 50 cycles A.C. traction between……..and ……on …… Railway have been carried out properly and that electric traction can be introduced for public carriage of passengers without endangering the safety of the travelling public or of the employees of the Railway, **subject to the maximum value of traction current not exceeding…….Amps.**

Name:

Designation:

Annexure '9/6'

Joint Safety Certificate

Certified that it is safe to run……..(Particulars of locomotives and rolling stock proposed to run) not exceeding……..units in the case of locomotives) coupled together on the section (station)…….to …….. (station) from……..(km.) to…….(km.) of the …….Railway at a maximum speed of …….(kmph) against a maximum speed of ……..(kmph.) certified by R. D. S. O. **subject to the maximum value of traction current not exceeding…….Amps. and** subject to the following speed restrictions and conditions:-

(a) Speed Restrictions:-

SL. No	From Km to Km	Nature of speed restriction	Brief reason for restriction

(b) Special Conditions:-

P.C.M.E. P.C.E.

P.C.E.E P.C.S.T.E P.C.O.M

Note: (i) When the speed of a loco/rolling stock is to be increased the P.C.O.M. and P.C.S.T.E. should be associated when the increase in speed is contemplated over the maximum sanctioned speed for that category of train (Passenger or Goods) over a particular section.

(ii) P.C.E.E. should countersign wherever electric traction is involved.

Annexure '9/7'

Handing over / Taking over of Signalling Assets

Procedure Order for Handing Over /Taking over of Newly Created Signalling Assets

1.	Dy. CSTE/C or Rly PSUs in-charge of the work in field shall submit the programme of commissioning the works during the year to Sr.DSTE of the division in the beginning of the financial year.
2.	The works shall be carried out as per the standard drawing, extant instructions and the quality manual/guidelines. Any deviation from the extant instructions will require the approval of PCSTE.
3	The Station Working Rules (SWR) shall be prepared by the concerned Construction Officer or Rly PSUs Officer and signed by the Divisional S&T and Operating Officers.
4	All the works to the extent possible should be completed during pre non-interlocking period ((Pre NI). Outdoor works such as preparation of ground connections and fixing of point machines on points and its testing, re-allocation and termination of cables, re-allocation of outdoor signaling gears etc. shall invariably be completed by the construction wing or Rly PSUs after taking disconnection/block. The reconnection of the existing gears shall be given only after the gears have been fully tested by the concerned maintenance staff of open line so as to ensure that the gears are in proper safe working condition.
5	CSTE/C or Rly PSUs in-charge of the work shall discuss the modalities with PCSTE/OL before non- interlocking of major station yards for signalling works, similarly for medium and wayside stations the Dy.CSTE/C or Rly PSUs in-charge of the work shall discuss the modalities of the works to be done during NI period with Sr. DSTE in-charge of the concerned division. Joint inspection by Divisional open line and construction or Rly PSUs both at SSE-In-Charge & officer level shall be undertaken before taking up Non-interlocking of a station in connection with yard re-modelling / interlocking changes for commissioning of Signalling works.
6	No major work shall be undertaken by the S&T construction organization or Rly PSUs in-charge of the work inside the relay room/cabin/cabin basement at any station unless the **maintenance of signaling equipment at the station is taken over by construction organization**. The maintenance in such cases shall extend to all signaling gears in the yard. However, minor alterations inside the relay room/cabin/cabin basement etc. can be undertaken in presence of the maintenance staff of the Open Line without taking over the maintenance of the station.
7	When maintenance is taken over by the construction organization all the maintenance staff for that station shall be put under the administrative control of the construction organization for the period during which the maintenance is taken over.
8	During the period the maintenance of the station is with the construction organization, it shall be responsible for safe working of gears. All safety instructions issued from time to time shall be scrupulously followed by the construction staff. Dy.CSTE/C shall analyses all the signal failures at the stations under his maintenance and keep the Sr.DSTE of the division informed about the causes of failures and corrective action taken.

9	The Open Line inspectors and Officers shall associate themselves during the construction stage of the new works with a view to acquaint themselves with the works and also point out any deficiency that is required to be removed in the interest of proper functioning of the gears. The observations shall be jointly agreed and recorded in the “Site Inspection Register” to be kept at the station for this purpose.
10	No work shall be done in the existing location boxes without the presence of Open Line maintenance staff.
11	The cable route plan should be got approved from the division before undertaking he work of trenching & cable laying. Instructions laid down in the joint procedure order issued by the Board vide letter No. 2004/Sig/G/7 dtd. 17/12/2004 and Rly. Bds Circular no.17/2013 dtd. 24.06.2013 should be followed meticulously.
12	Power supply equipment proposed to be installed by construction organization or Rly PSUs for the works shall be in accordance with the latest policy/circular/instructions issued from time to time. For any deviation, approval of PCSTE/OL should be taken.
13	The open line organization shall carry out the cable insulation test, power supply auditing and the functional tests before permitting the construction organization or Rly PSUs to start the non interlocking working. In case any alteration is done in the wiring diagram by the construction staff or Rly PSUs after carrying out the final function test by the open line staff, this shall be brought to the notice of open line officials for conducting the test again on the altered portion.
14	Construction organization or Rly PSUs shall complete the requisite work in all respect and carry out the continuity test, wire count, break down test and the functional tests before offering the installation to the open line in writing for joint tests.
15	Sr.DSTE shall plan the positioning of the maintenance staff immediately after the construction organization or Rly PSUs gives the programme of commissioning for the year. It will help positioning of staff well in time so that maintenance of the station can be taken over immediately after commissioning.
16	Record of functional test carried out on a selection table/locking table/route control charts, contact break down test and continuity test shall be submitted to Open line immediately after commissioning. Open line official shall keep these documents at the respective station in a sealed packet carrying the details of the documents and signature of Sr. DSTE/DSTE of the division.
17	The details of the works undertaken at the stations shall be advised by the construction organization or Rly PSUs to the concerned Sr.DSTE of the division in advance. Dy.CSTE/C or Rly PSUs in-charge of work and Sr.DSTE of the division shall have periodical meetings to review the progress of ongoing works and decide the dates of joint inspection of the works nearing completion. Minutes of such meetings shall be recorded jointly & issued to all concerned for guidance & follow up.
18	Within 15 days of commissioning of Signalling work, another joint inspections both at supervisor and at officers level by Open Line and Construction organization or Rly PSUs shall be undertaken. The deficiencies noted during this inspections shall be listed in two categories as under:- Category-A: Items shall be complied before handing over. Category-B: Items that shall be complied after handing over with the target dates mutually agreed in writing.

19	Sr. DSTE of the division shall ensure that Open line staff actively associate with the construction activities particularly in the following areas:- (i) Site plan and size of RRI/Panel (ii) Cable Route Plan and cable laying (iii) Signal locations (iv) Power supply sources and equipment plan (v) Cable testing.
20	While handing over the maintenance of the works executed by the construction organization or Rly PSUs, the following completion documents and maintenance accessories shall be handed over to the division. All the drawings should be in standard size as laid down in Para 8.3.4 of Signal Engineering Manual Part-I, Addendum of Corrigendum Slip No-4. i) Approved SIP of the station ii) Completion wiring diagrams iii) Selection/Locking tableRoute control chart and Dog-Chart iv) Panel diagram/VDU diagrams (in case of panel or EI installations) v) Sighting Committee certificate on proper Performa as per SEM. vi) Approved as well as updated 'As Laid' Cable route plan. vii) Cable termination plan/diagrams in relay room and location boxes, power supply room. viii) Wire counts details n case of RRI. ix) Power supply arrangement plan. x) Cable insulation test record. xi) Signal infringements at the station. xii) Track circuit bonding plan. xiii) Battery history card, track circuit test card and earth resistance measurement record. xiv) The tools and plants and spares. xv) 3 sets of CDs containing all completion drawings. xvi) Floor Plans of Relay room, IPS room, Battery room, ASM roon, OFC/Telecom cable room, Maintainer room. xvii) Earthing Plan of Relay room, IPS room & Panel/VDU ASM's room xviii) Earthing value joint measurements records. One laminated copy of the documents mentioned at i, ii, iii, vi, vii, ix, x, xi, xii, xiii, xvi, xvii above shall be kept at the station for reference of the maintenance staff.
21	The safety certificate shall be made over to the Division at the time of handing over of the maintenance of the station. In addition, technical documents of the new equipment, if installed, shall also be handed over.

Annexure '9/8'

Application for PCSTE Sanction

No…… Office……

From, Dated……..

The ……..

Designation……..

To,

The Principal Chief Signal & Telecommunication Engineer,

……..

Sir,

I hereby apply for your sanction to_____ _____ being commissioned and opened for public carriage of passengers when ready as CRS sanction is not required as per para 9.2.1.(b) of SEM Pt-I.

The following documents are enclosed :-

(a) Description of proposed works.
(b) Signalling and interlocking plan no.
(c) Panel Front PlateVDU Diagram No. (If it is Panel or VDU)
(d) Selection Table/Locking Table/Route Control Chart no.
(e) Station Working Rule diagram. no.
(f) Station Working Rule for traffic working.
(g) List of infringement to Schedule of Dimensions.
(h) List of deviations from the manuals of instruction for signaling and interlocking and Block signaling.
(i) List of deviations from General and Subsidiary Rules.
(j) Restrictions.
(k) The work will be executed by the following office.

Modus Operandi

The work……..shall be carried out after disconnection/non-interlock & traffic block if required at site. After completion of the work reconnection memo shall be issued. The work will be commissioned by…… on receipt of the PCSTE sanction and after signing the safety certificate and when required, also dispatch a telegram to your address intimating that the work has been opened and the safety certificate signed and issued.

Yours faithfully

Signature……

Designation…….

Dated……..

Annexure '9/9'

PCSTE Sanction

…….Railway

No…… Dated……..

From,

Principal Chief Signal& Telecommunication Engineer,
……..

To,

The ……..

Designation……..

Sub:……..

Ref: Your Application No…….dtd…….…

In response to your application No……..dated…….Sanction is accorded to the work……being carried out.

I do not propose to inspect the work prior to its opening for the carriage of passengers. When ready, it may be opened on a Safety Certificate which should be submitted to me direct without any dely.

I propose to inspect the work prior to its opening for the public carriage of passengers. Advice of the date, when work will be ready for inspection should be intimated at least 14 days before it is proposed to open it.

Principal Chief Signal & Telecommunication Engineer
……..……..

No…… Dated…….

Strike out paragraph not applicable.

Annexure '9/10'

Check List

for CRS/PCSTE application for sanction

Provision of …….at station on ……..….section of ……division.

List of documents enclosed:-

(a) Description of proposed works.
(b) Signalling and Interlocking plan no.
(c) Panel Front Plate /VDU diagrams Diagram No. (If it is Panel or EI installations)
(d) Route Control Chart/Selection Table/Locking Table no.
(e) Station Working Rule diagram. no.
(f) Station Working Rule for traffic working.
(g) List of infringement to Schedule of Dimensions.
(h) List of deviations from the manuals of instruction for signaling and interlocking and Block signaling.
(i) List of deviations from General and Subsidiary Rules.
(j) Restrictions.
(k) The work will be executed by the following office.

Yours faithfully

Signature……
Designation…….
Dated…….

CHAPTER – 10 : CONTRACTS FOR WORKS

Section 1 : General

10.1.1.0 **Code references for contracts :-** The procedure and rules for the calling for tenders and entering into contracts for execution of works and supply of materials are contained in Chapter XII of the Indian Railway Code for the Engineering Department and Chapters III, IV and VII of the Indian Railway Code for the Stores Department. (Relevant extracts at Annexure '10/1').

10.1.2.0 **Lists of approved contractors and Registration of applications :-**

10.1.2.1 Lists of approved contractors shall be maintained in the offices of the Sr Divisional Signal and Telecommunication Engineers for their Divisions and the Chief Signal and Telecommunication Engineer for the whole Railway. As per Para 1216E, these lists shall be deemed to be confidential records and examined and revised periodically. They shall be maintained in manuscript form under the following categories of work :-

(a) For works to be executed under Open line conditions affecting safety of running trains :-

(i) Provision of Electrical Signalling/IB Signalling works/LC gate works

(ii) Provision of Relay Interlocking, Electronic Interlocking, Automatic Signalling, CTC, ATP etc

(iii) Laying of cables.

(iv) Any other signalling work

(b) For works to be executed under Open Line conditions not involving safety of running train or on new lines or in areas away from track opened for traffic.

(i) Provision of Electrical Signalling/IB Signalling works/LC gate works

(ii) Provision of Relay Interlocking, Electronic Interlocking, Automatic Signalling, CTC, ATP etc.

(iii) Laying of Cables.

(iv) Any other Signalling work.

10.1.2.2 The Principal Chief Signal and Telecommunication Engineer shall authorise the inclusion in the lists of approved contractors the names of applicants considered capable of executing works under one or more of the categories in para 10.1.2.1 above. On receipt of an application from a contractor desirous of having his name included in the lists of approved contractors, the Sr Divisional Signal and Telecommunication Engineer shall satisfy himself on the following points (as per para 1215) and forward the application with his remarks to the Principal Chief Signal and

Telecommunication Engineer :-

(a) His/her position as an independent contractor.

(b) His/her capacity to undertake and carry out works satisfactorily, as vouched for by a responsible official or (firm).

(c) His/her previous experience on works similar to that to be contracted for, in proof of which original certificates or testimonials may be called for and their genuineness verified, if need be, by reference to the signatories thereof.

(d) His/her knowledge from actual personal investigation of the resources of the division in which he offers to work.

(e) His/her ability to supervise the work personally or by competent and duly authorised agents. It shall be ensured that each application for registration is accompanied by

i) A certificate showing the financial standing of the applicant,

ii) An authorised copy of the Income-tax Clearance certificate,

iii) Original certificates (or authorised copies thereof) regarding past experience of works.

10.1.3.0 **Basis for inviting tenders and entering into contracts:**

10.1.3.1 The documents which form a contract are as follows:-

(a) Instructions to parties tendering.

(b) Tender form, if any.

(c) Specifications — Standard and special,

(d) Schedule of items and quantities, rates, etc.

(e) Conditions of contract — Standard and Special,

(f) Agreement Form.

(g) Scope of the work.

(h) location particulars of the work.

Note:- Items (a) to (h) above shall be comprehensive and explicit so that there is the least possibility of conflicting, even differing, interpretations being placed on the intentions of the contract.

10.1.3.2 Each Sr Divisional Signal and Telecommunication Engineer shall be in possession of copies of items & para 10.1.3.1 and acquaint himself with the provisions contained therein. SSE/JE (Signal) may be supplied with

copies of these where necessary

10.1.3.3 Copies of items vide paras 10.1.3.1 shall be available in the Office of the Chief Signal and Telecommunication Engineer and the Senior Divisional Signal and Telecommunication Engineer for information to contractors who shall be advised to make themselves conversant with the provisions contained there in before tendering for works or supply of materials. This information may be given in Electronic Form for e - Tenders.

10.1.3.4 Amendments when considered necessary to any para, clause or items of the regulations for tenders and contract, the general conditions of contract, the specifications for materials and works and the schedule of rates shall be authorised solely by the Principal Chief Signal and Telecommunication Engineer and intimated to the Executive Officers and supervisors who shall acknowledge receipt of the same.

Section 2 : Tenders/E – Tenders

10.2.4.0 **Calling for tenders :-** Tenders/e - Tenders may be invited for any of the works required to be carried out under contract. When calling for tenders for works, the provisions contained in para 1211. shall be observed. Tenders may be invited by the Sr Divisional Signal and Telecommunication Engineer, Deputy Chief Signal and Telecommunication Engineer/Construction, Divisional Railway Manager or the Chief Signal and Telecommunication Engineer depending on the contract value or works (s) according to the procedure that may be prescribed by the Administration for the purpose. The common methods of obtaining tenders are by advertisement (Open Tenders), by direct invitation to a limited number of contractors (Limited Tender) and in exceptional cases by invitation to one contractor (Single Tender).

10.2.5.0 **Tender Notices/ e – Tender Notices**

10.2.5.1 Notice for inviting tenders (NIT) shall be to the Form prescribed by the Administration and shall embody the stipulations contained in para 1239.

10.2.5.2 When limited tenders are invited, tender notices shall be issued to Contractors on the list of approved contractors.

10.2.5.3 When open tenders are invited, Tender Notices shall be available for participation for all subject to eligibility criteria mentioned. In addition, if so prescribed by the Administration, tender notices shall be published in the newspapers on the approved list of the Government of India, and/or on official web portal for on line tendering, it being left to the authority inviting enders to select the newspapers and the number of insertions for this purpose.

10.2.6.0 **Tender forms :-** Tender forms shall embody the contents of the contract document either directly or by reference. Documents pertaining to Technical & financial eligibility criteria if any and special conditions should also be available. Tender forms shall be down loaded from the official web portal by paying prescribed fees on line by the tenderers. Tenderers shall be required to submit evidence regarding their financial status,

previous experience and ability to execute the works and an authorised copy of the Audited Balance Sheet duly certified by the Chartered Accountant along with other documents mentioned in Tender Notice, at the time of submitting their bid on line without which their tenders shall not be considered.

10.2.7.0 **Drawings and Specifications**

10.2.7.1 Unless otherwise agreed to specifically, the execution of works and supply of materials on contract shall be according to the Railway's Drawings and the Specifications.

10.2.7.2 If no specification exists for any item of work or supply, a complete specification of the item and of the materials to be used shall be prepared and embodied in the contract documents or made available for reference by intending tenderers.

10.2.8.0 **Earnest Money, Security Deposit and Performance Guarantee**

10.2.8.1 The tenderer shall be required to deposit online with the tender, earnest money at the rates prescribed by the Administration as initial security. In case of acceptance of the tender, the earnest money will be retained by the Railway as a part of the security for the due and faithful fulfilment of the contract and the balance to make up the security deposit at the rates stipulated by the Administration or will be recovered by percentage deductions from the Contractor "on account" bills.

10.2.8.2 The earnest money of all unsuccessful tenderers shall be refunded as expeditiously as possible after completion of tender process.

The successful bidder will have to submit a Performance Guarantee (PG) as per conditions laid down in GCC

10.2.9.0 **Tender committees :-** The Administration will specify

The Administration will specify

(a) The constitution and functions of Tender Committees, Convenor for the purpose of evaluation and submission of recommendations to competent authority, should be as per latest Railway board guidelines/SOP.

(b) The powers of acceptance of tenders and entering in to contract and signing of contract agreement etc., shall be exercised by the competent authority as per latest Railway board guidelines/SOP.

(c) The conditions in regarding to acceptance of single tenders.

Note(i) :- When a single tender is received in response to a call for limited or open tenders, the urgency of the work may be the criterion for its acceptance, provided the tendered rates are considered reasonable.

Note(ii) :- If the rates in a single tender are considered inordinately high, negotiations may be carried out by the Tender Committee with the

tenderer in the interest of work.

10.2.10.0 **Opening of Tenders :- At the advertised time and place Tenders/e - Tender shall be opened**

10.2.11.0 **Acceptance of Tenders**

10.2.11.1 Tenders opened and duly initialed at the Tender Committee Meeting shall be valued and will be adjudged by the Tender Committee and recommendations made to the authority empowered to accept the tender.

10.2.11.2 Ordinarily the lowest eligible tender may be accepted by the Railway unless such acceptances would not be in the public interest. The acceptance or rejection of any tender is left entirely to the discretion of the authority empowered to deal with the matter and no explanation can be demanded by any tenderer as to the cause of rejection of his tender.

10.2.12.0 **Entering into Contracts :-** No contract shall be entered into unless authority exists for commencement of the work. No authority shall enter into a contract beyond its own powers of sanction.

10.2.13.0 **Execution of contract documents.**

10.2.13.1 The tenderer whose tender is accepted shall be required to appear at the office of the General Manager, Principal Chief Signal and Telecommunication Engineer, Divisional Railway Manager or Sr Divisional Signal and Telecommunication Engineer/Dy. Chief Signal and Telecommunication Engineer (Con) as the case may be, in person or if a firm or corporation, a duly authorised representative shall so appear, and execute the contract within stipulated time mentioned in LOA after notice that the contract has been awarded to him. Failure to do so shall constitute a breach of the agreement effected by the acceptance of the tender in which case the earnest money accompanying the tender may be forfeited.

10.2.13.2 In the event of any tenderer, whose tender is accepted, refusing to execute the contract documents the authority may determine that such Tenderer has abandoned the contract and thereupon his tender and the acceptance thereof shall be null and void and the earnest money accompanying the tender shall be forfeited.

10.2.14.0 **Form of Contract Documents.**

10.2.14.1 Every contract shall be complete in respect of the documents it constitutes.

10.2.14.2 The contract agreement required to be executed by the successful tenderer shall be in the form prescribed by the Administration for the purpose. The draft of the contract agreement shall be vetted by the Accounts Department

10.2.15.0 **Copies of Contract documents.**

10.2.15.1 Sufficient number of copies of the contract documents shall be available for use in the offices of the Sr Divisional Signal and Telecommunication Engineer and the in-charge supervisor. If materials are to be inspected by an agency other than the consignee, a copy of the contract documents shall be supplied to the inspecting agency also. No deviations in the contract documents shall be permitted without the sanction of the competent authority.

10.2.15.2 The executive Officers and in-charge supervisor shall thoroughly study the contract documents for works to be carried out and shall adhere to the provisions contained therein

10.2.16.0 **Items of works not included in contract documents.**

10.2.16.1 Officer in-charge & supervisor in-charge of the work shall ensure that no item of work that is not included in the contract is carried out without the sanction of the competent authority.

10.2.16.2 Items of work not included in a contract may, if deemed expedient, be executed under the contract at the rates mutually agreed on by the contracting parties

10.2.17.0 **Modifications to contract :-** In the event of any of the provisions of the contract requiring to be modified after the contract documents have been signed, the modifications shall be made in writing and signed by the Railway and the contractor and no work shall proceed under such modifications until this has been done. Any verbal or written arrangement abandoning modifying, extending, reducing or supplementing the contract or any of the terms thereof shall be deemed conditional and shall not be binding on the Railway unless and until the same is incorporated in a formal instrument signed by the Railway and the contractor, and till then the Railway shall have the right to repudiate such arrangement

10.2.18.0 **Issue of Departmental materials to contractors :-** When so specified in the contract, cables and other S&T materials shall be issued to contractor and receipts obtained for the same, It shall be ensured that the Contractors takes reasonable care of all materials made over to him and that on completion of the works the unused balance of the same are handed over by him in good order.

Section 3 : Measurements and Measurement Books

10.3.19.0 **Code References :-**

The rules regarding the recording of measurements in measurement books, care and custody of measurement books and the preparation of contract bills are embodied in Para 1313 E. (Enclosed as annexure 10/1)

10.3.20.0 **Measurement Books**

10.3.20.1 Each measurement book shall have :-

(a) Instructions printed at the commencement,

(b) An Index which should be posted up-to-date. A form of measurement book is attached as Annexure '10/1'.

10.3.20.2 Measurement books shall be issued to the SSE/JE in-charge of the work, by the Sr.Divisional Signal and Telecommunication Engineer as required duly numbered thus M. B. No ..... of yyyy. As few books shall be issued as can conveniently meet the requirements.

10.3.20.3 A register of measurement books shall be maintained by the Sr.Divisional Signal and Telecommunication Engineer wherein the contents and the movement of each measurement book shall be indicated.

10.3.20.4 Measurement books shall be sent from one Office to another in the personal custody of a responsible person.

10.3.20.5 Completed measurement books and those not in use although not completely written up, shall if they are no longer required, be sent to the Sr.Divisional Signal and Telecommunication Engineer to be filed in his Office.

10.3.20.6 If a measurement book is lost, immediately after the loss is discovered, the matter shall be reported to the Sr.Divisional Signal and Telecommunication Engineer, who will obtain sanction to write off the book from the registers of Measurement Books.

10.3.21.0 **Recording Measurements.**

10.3.21.1 Every entry in a measurement book shall be commenced with a Statement as to how it is to be billed. The instructions for recording measurements contained in Annexure '10/1' shall be followed.

10.3.21.2 The pages of every measurement book shall be machine numbered; no page on any account shall be torn out, nor shall any entry be erased or defaced so as to be illegible.

10.3.21.3 Entries shall be recorded continuously in the measurement book. No blank pages shall be left. Any page left blank inadvertently shall be cancelled by diagonal lines drawn across the pages, the cancellation being attested.

10.3.21.4 When any measurements are cancelled, the cancellation shall be supported by the initials with date of the Officer ordering the cancellation or by the Official who made the measurements. The reasons for cancellation shall be recorded.

10.3.21.5 Should it be found necessary to make any additions to measurements already taken owing to omission at the time of the measurements, such additions shall be detailed in a subsequent page of the book, giving reference to the page containing the original measurements and explaining the reasons for the same.

10.3.21.6 Entries of the measurements shall be made direct into the measurement book at the site of the work in the presence of the contractor or his agent. The copying of entries from a rough note book or other record is forbidden.

10.3.21.7 The Contractor or his authorised agent shall be present at the time of all on account and 'final' measurements and should sign the measurements recorded in the measurement book as acknowledgement of his acceptance of the accuracy of the measurements and the classification of materials. The contractor shall be given due notice of the time and date on which the measurements are to be taken. Failing his attendance, the work may be measured up in his absence and such measurements shall notwithstanding such absence, be binding upon him whether or not be shall have signed the measurement book provided hat any objection to any measurement that is made by him in writing within seven days of the date of such measurements shall be duly investigated and considered.

10.3.21.8 After the contractor's dated signature has been obtained, the official recording the measurements shall attest his full signature and designation at the end of the measurements. RTGS/NEFT – mandate Form must be collected.

10.3.22.0 **'On Account' Measurements**

10.3.22.1 For measurements against which on account bills are prepared and which are subject to adjustment in final bills, the official in charge may use percentage measurements for the part the work completed.

10.3.22.2 Quantities for payment' on account' shall never exceed the estimated quantities of actual work done upto the time of payment. In order to ensure this, the following certificate shall be recorded by the official in charge in the measurement book for all' on account' measurements.

"I hereby certify that not less than the quantity of work paid for has actually been done and that the measurements are from plans/approximately estimated ".

10.3.23.0 **Responsibility of the Assistant/ Divisional Signal and Telecommunication Engineer for measurements of works.** The Assistant/Divisional Signal and Telecommunication Engineer shall be responsible for the correctness of the measurements for all works in his charge. He shall ensure that the measurements are made in the specified manner and shall either record them himself or have them recorded by the Inspector Supervisor in charge of the work.

10.3.24.0 **Computation of quantities**

After measurements for a work are recorded, the quantities shall be computed and entered in the 'contents' column of the measurement book. This may be done by the official recording the measurements himself or in the office.

10.3.25.0 **Preparation of abstracts in Measurements Books**

10.3.25.1 After the quantities are computed, an abstract shall be prepared in the measurement book in the Sr.Divisional Signal and Telecommunication Engineer's Office. The abstract shall be headed thus "Abstract of pages.....to....."

The abstract shall show the correct description of the item, total quantity done up-to-date, rate and the total value for each item measured/checked, in connection with the work. The total of the values of all the items will be the total value of the work done up-to-date.

10.3.25.2 In the case of a first and final bill this amount will not be subject to any deduction and will be paid in full to the contractor. Except Security Deposit and other taxes.

In the case of an 'on account' bill, the prescribed percentage of the value of the work done since the last certificate shall be deducted as security deposit towards the due fulfilment of the contract. The balance will be the amount payable to the Contractor and this amount must be written clearly in words as well as in figures.

10.3.25.3 On satisfactory completion of the work the security deposit will be refunded to the contractor after the final payment for the work is made. Part of the security deposit may be kept with the Railway in specified cases whenever it is necessary to watch the results of the work and to ensure that there will be no failure of the work for the period stipulated in the contract.

10.3.25.4 The abstract shall be signed and dated by the Assistant/ Divisional Signal and Telecommunication Engineer.

10.3.25.5 Each paragraph of the measurement book to which the abstract pertains shall be crossed in Red Ink diagonally thus.

See Abstract pages ……

10.3.25.6 At the foot of each abstract the relevant bill number and date of preparation of the bill shall be entered.

10.3.26.0 **Submission of bills**

After the abstract is prepared in the measurement book, the bill shall be prepared in the Sr.Divisional Signal and Telecommunication Engineer's/ Dy. Chief Signal and Telecommunication Engineer's Office on the prescribed form or submitted by firm.

10.3.27.0 **Checking of bills in Sr.Divisional Signal and Telecommunication Engineer's/Dy.Chief Signal and Telecommunication Engineer's Office.**

10.3.27.1 The quantities in measurement books submitted by the Assistant Signal and Telecommunication Engineer/Divisional Signal and Telecommunication Engineer shall be checked for arithmetical accuracy in the Sr.Divisional Signal and Telecommunication Engineer's/Dy. Chief Signal and Telecommunication Engineer Office where it will be verified that-

(a) the description of items is according to the contract schedule.

(b) there are no delays in recording measurements,

(c) the quantities executed are according to sanctioned plans and estimates,

(d) where measurements are not recorded, correct reference to the standard measurements is clearly given, and

(e) the quantities recorded in the bill agree with those shown in the measurement book and rates higher than those sanctioned are not allowed.

10.3.27.2 An endorsement shall be made at the end of Assistant/Divisional Signal and Telecommunication Engineer's abstract in the measurement book under the signature of Sr.Divisional Signal and Telecommunication Engineer/ Dy. Chief Signal and Telecommunication Engineer giving reference to the particulars of the bill passed. RTGS/NEFT mandate form to be obtained from contractor for enabling e – payments.

10.3.27.3 The Railway Administration may lay down a time schedule for preparation and checking of bills.

10.3.28.0 **Bill Registers :-**

The Sr.Divisional Signal and Telecommunication Engineer's/Dy. Chief Signal and Telecommunication Engineer’s Office shall maintain Bill Registers in which all bills prepared in their office shall be recorded. The Register shall be kept up-to-date and complete information with reference to agreements, work orders, abstracts in measurement books and amounts

10.3.29.0 **Disputes with Contractors.**

10.3.29.1 On the completion of every work, 'No claims certificate' shall be obtained from the Contractor according to the relevant clause or clauses in the condition of contract. Every month or when 'on account' bills are made out, the contractor shall be instructed to submit a list of outstanding claims, if any.

10.3.29.2 If there is any likelihood of a dispute with a contractor, all documents, measurements and letters bearing on the case shall be at once collected and stored in safe custody with the officer responsible for the case. In all such cases care shall be taken to commence all correspondence with the contractor with head line 'without prejudice'. The contractor shall be instructed to submit complete details of his claims in writing with a certificate to the effect that these are final.

10.3.29.3 Inspite of all reasonable efforts to effect agreement, if the contractor refuses to sign the final contract certificate, the final contract certificate shall be forwarded to the Accounts Officer to tender payment. The contractor may accept payment if he wishes under protest.

10.3.29.4 Disputes arising out of the contract for a work, between the contractor on the one hand and the Railway Administration on the other, shall be referred for arbitration in accordance with the relevant clause in the General Conditions of Contract.

10.3.30.0 **Completion Reports**

10.3.30.1 Completion report shall be drawn by the Sr DSTE/ Dy.CSTE in-charge of the work after completion of all required works. Expendires of stores, contract, adjustment memos, work shop debits, D&G charges etc., all shall properly accounted.

10.3.30.2 Completion estimate shall be drawn and obtain Finance concurrence as per rules/ SOP.

10.3.30.3 Works shall be deleted from list of the works and same shall be intimated to PCSTE.

☞ *Note :- This Chapter has under mentioned Annexures for further study*		
S.no	*Annexure no*	*Description*
1	*10/1*	*Instructions For Securing Measurements*

Annexure '10/1'

INSTRUCTIONS FOR SECURING MEASUREMENTS

1. The measurement book is the basis of all accounts of quantities of work done by contract and shall be so kept that the transactions may be readily traceable into the accounts by the entry of the number and date of bill in this book when the bill has been prepared and the entry of the number and page if the measurements affected shall then be crossed red ink diagonally thus

2. All measurements are to be taken down in ink in this book and in no others. The description of work must be lucid so as to admit of easy identification and check .The measurements shall show the quantities of work done as par last measurement and the total to date. A reference to the pages of the measurement book, where the previous measurements are to be found, also be given.

3. No erasures arc allowed. If a mistake is made, the wrong wordings or figures shall be neatly scored out and correct entries made. Every such correction shall be initialled.

4. For large works, a separate measurement book may be specially set apart, or if found convenient, even two or more books may be set apart for different classes of works.

5. The measurements book must be looked upon as important records; they shall be carefully checked by the Engineer-in-charge to set that they are kept up as complete records of each kind of work done for which certificates have been granted. The eventual return of all books to the Sr Divisional Signal and Telecommunication Engineer's Office or Dy. Chief signal and Telecom Engineer’s office for record shall be insisted upon.

6. Whenever an Divisional/Assistant Signal and Telecommunication Engineer is required to submit his measurement book to the Sr Divisional Signal and Telecommunication Engineer's Office or Dy. Chief signal and Telecom Engineer’s office a Second book may be provided for his use where necessary.

7. Whenever progress on a work is reported in lump sum quantities under sub-heads of works, the number and page of the measurement books shall invariably be quoted.

8. It will be found convenient to keep the measurement relating to one work together and to effect this, the number of pages likely to be wanted for a work for sub-head shall be estimated and set apart for the purpose of recording the measurements consecutively.

9. Before detailing the measurements relating to a work the following information shall invariably be given at the top of the first page of such measurements, each item being underlined in red ink :

1. Final/on account contract certificate No.
2. For previous on account bill see page.....of measurement Book No.
3. Name of work.
4. Situation of work.
5. Agency by which work executed.
6. Reference to Agreement of work order.
7. Date commenced.

8. Date completed.

9. Date of measurement.

10. For facility of reference and to assist in carrying out the instructions given in para 8 above, the index shall be kept up-to-date.

CHAPTER 11: RELIABILITY, AVAILABILITY, MAINTAINABILITY & SAFETY (RAMS)

Section 1: Introduction

11.1.1.0 In any system output depends on input. In Railways punctuality is an output. It depends on how subsystems are available for service. Availability depends on Reliability and how good/fast the maintenance (maintainability). Hence R, A, M, S parameters are important. Punctuality is directly related to equipment, failures and Reliability in the science of failures.

11.1.2.0 RAMS Assurance can help stakeholders in balancing performance requirements and LCC to improve customer satisfaction along with compliance of National/International regulatory requirements. Many standards have suggested that RAMS characteristics can be visualized as the Key Performance Indicator (KPI) at System Level.

11.1.3.0 RAMS concept is relatively new to the Railways as compared to other industries such as nuclear, power, chemical, avionics, etc.

11.1.4.0 The key objective of a railway system is to achieve a defined level of rail traffic in a given time safely. The objective can be fulfilled by implementing RAMS principle and process as suggested by EN50126 / 128 / 129.

11.1.5.0 What will happen if there is a lack of RAMS Assurance?

(a) Loss of KPI

(b) Customers satisfaction, project credibility will be on doubt

(c) Global Certification?

(d) Development as well as operational cost will be more.

Section 2: Terms and Definitions

11.2.6.0 **Assurance**: Confidence in achieving a goal being pursued. Declaration intended to give confidence.

11.2.7.0 **Audit**: Systematic, independent, documented process for obtaining records, statements of fact or other relevant information and assessing them objectively to determine the extent to which specified requirements are fulfilled.

11.2.8.0 **Availability**: Ability of an item to be in a state to perform a required function under given conditions at a given instant of time or over a given time interval, assuming that the required external sources are provided.

11.2.9.0 **Corrective Maintenance**: Maintenance carried out after fault detection to effect restoration.

11.2.10.0 **Failure mode**: Manner in which failure occurs.

11.2.11.0 **Failure Rate**: Limit of the ratio of the conditional probability that the instant of time T, of a failure of a product falls within a given time interval ($t, t+ \Delta t$) and the duration of this interval Δt, when Δt tends towards zero, given that the items is in an up state at the start of the interval.

11.2.12.0 **Functional Safety**: Part of overall safety that depends on functional and physical units operating correctively in response to their inputs.

11.2.13.0 **Hazard**: Condition that could lead to an accident

11.2.14.0 **Hazard Analysis**: Process of identifying hazards and analysing their causes and the derivation of requirements to limit the likelihood and consequences of hazards to a tolerable level.

11.2.15.0 **Hazard Rate**: Rate of occurrence of hazard.

11.2.16.0 **Independent Safety Assessment**: Process to determine whether the system/ product meets the specified safety requirements and to form a judgement as to whether the system/product is fit for its intended purpose in relation to safety.

11.2.17.0 **Life Cycle**: Series of identified stages through which all items go, from its conception to disposal.

11.2.18.0 **Maintainability**: Ability to be retained in, or restored to, a state to perform as required, under given conditions of use and maintenance.

11.2.19.0 **Preventive Maintenance**: Maintenance to carry out to mitigate degradation and reduce the probability of failure.

11.2.20.0 **RAM Plan**: Documented set of time scheduled activities, resources and events serving to implement the organisational structure, responsibilities, procedures, activities, capabilities and resources that together ensure that an item will satisfy RAM requirements relevant to a given contract or project.

11.2.21.0 **Reliability**: Ability to perform as required, without failure for a given time interval, under given conditions.

11.2.22.0 **Risk**: Combination of expected frequency of loss and the expected degree of severity of that loss.

11.2.23.0 **Safety**: Freedom from unacceptable risk.

11.2.24.0 **Safety Case**: Documented demonstration that the product (eg. a system, subsystem or equipment) complies with the specified safety requirements.

11.2.25.0 **Safety Function**: Function whose sole purpose is to ensure safety.

11.2.26.0 **Safety Integrity**: Ability of a safety related system to achieve its required safety functions under all the stated conditions within a stated operational environment and within a stated duration.

11.2.27.0 **Safety Integrity Level**: One of a number of defined discrete levels for specifying the Safety Integrity requirements for safety related functions to be allocated to the safety related systems.

11.2.28.0 **Safety Related Systems**: A system is safety related if its failure can cause harm to life and property eg. Interlocking System, ATP or carries responsibility for safety.

11.2.29.0 **Safety Plan**: Documented set of time scheduled activities, resources and events serving to implement the organisations, responsibilities, procedures, activities, capabilities and resources that together ensure that an item will satisfy given safety requirements relevant to a given contract or project.

11.2.30.0 **Validation**: Confirmation, through the provision of objective evidence that the requirement for a specific intended use or applications have been fulfilled.

11.2.31.0 **Verification**: Confirmation through the provision of objective evidence that specified requirements have been fulfilled.

11.2.32.0 **System Assurance**: Is an analytical branch of engineering technology about ensuring high values of confidence, reliability, availability, maintainability and safety of complex engineering systems. System Assurance = RAM Assurance + Safety Assurance

11.2.33.0 Safety Assurance: Positive declaration intended to give confidence that Risk Process and behaviours are being managed to acceptable levels through appropriate measures.

11.2.34.0 Safety Assurance Acts & Standards:

(a) Indian Railways Act

(b) Metro Railway Act

(c) International Standards (EN50126, 50128, 50129, yellow book)

(d) ROGS Legislation 2006 (Railway and other Guided Transport Systems)

Section 3: Examples of RAMS Parameters for Railways

11.3.35.0 Reliability Parameters:

Failure Rate	$\lambda(t)$
Mean Up Time	*MUT*
Mean Operating Time To Failure (for non-repairable items)	*MTTF*
Mean Operating Time Between Failures (for repairable items)	*MTBF*
Reliability	*R(t)*
Failure probability	*F(t)*

11.3.36.0 Maintainability Parameters:

Mean Down Time	MDT
Mean Operating Time Between Maintenance	MTBM
MTBM (Corrective or Preventive)	MTBM(C), MTBM(P)
Mean Time To Maintain	MTTM
MTTM (Corrective or Preventive)	MTTM(C), MTTM(P)
Mean Time to Restore	MTTR

11.3.37.0 Availability Parameters:

Availability	A
Inherent	Ai
Operational	Ao

A_i = MTTF/(MTTF+MTTR) for non-repairable items

Ai = MTBF/(MTBF+MTTR) for repairable items

11.3.38.0 Safety Parameters:

Hazard Rate	h(t)
Probability of Wrong Side Failure	PWSF
Tolerable Hazard Rate	*THR*

11.3.39.0 Failure Rate and Mean Time Between Failures (MTBF)

Consider a batch of N items and that, at any time to, a number to have failed. The cumulative time T is given by

T = Nt, if it's assured that each failure is replaced when it occurs.

T = [t1+t2+t3…...+tk+(N-k) t]

Where, t1 is the occurrence of first failure, etc. Failed items are not replaced.

The observed failure rate is given by

Gama = k/T

The observed MTBF of the N items

MTBF = T/k _____(for repairable items)

MTTF = T/k _____(for non-repairable items)

The only difference between MTBF & MTTF is that the value of T will be calculated as indicated above.

Reliability equation is as under

$R(t) = e^{-\lambda t}$ (for constant λ)

Section 4: Safety Integrity Level (SIL)

11.4.40.0 **Functional Safety**: Functional Safety is part of the overall safety which depends on the correct functioning of a safety related system.

11.4.41.0 What is a Safety Related System?

A system is safety related if its failure can cause harm to life and property.

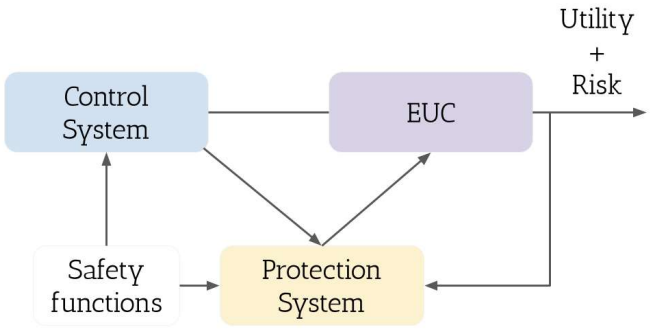
A safety related system performs one or more safety functions.

11.4.42.0 **Safety Function**: Function which is intended to achieve or maintain a safe state of the Equipment Under Control (EUC).

11.4.43.0 **Equipment Under Control (EUC)**: Equipment, Machinery, Apparatus or Plant used for manufacturing, process, transportation, medical or other activities, eg: Railway station points & signals, train, etc.

11.4.44.0 **Equipment Under Control Control System (EUCCS)**: System which responds to input signals from the process and or from an operator and generates output signals causing the EUC to operate in the designed manner, eg. Interlocking system, train driver.

11.4.45.0 Safety System Diagram:



11.4.46.0 Requirement of Safety Related System:

(a) Safety function requirements i.e., what is function supposed to do

(b) Safety integrity requirements i.e., what is the likelihood of the correct operation of the safety function

11.4.47.0 **Safety Integrity:** The ability of a safety related system is to achieve its required safety function under all stated conditions within stated operational environment and within a stated period of time (EN 50129).

If you introduce a protection system to reduce risk you are now “Trusting” it, to do the job. Safety Integrity is a measure of that Trust.

11.4.48.0 **Safety Functions in Railway Signalling :-**

11.4.49.0 Important elements of Signalling System:

(a) Interlocking System

(b) Train Detection System

(c) Point Operation

(d) Signal

(e) ATC Infrastructure

The functions performed by the above systems can be considered as safety functions.

11.4.50.0 **Tolerable Hazard Rate**: A hazard rate which guarantees that resulting risk does not exceed a target individual risk.

SIL is associated with safety functions and NOT equipment.

11.4.51.0 SIL is used for two purposes:

a) SIL is used to give an interval for the rate of safety critical failures. Applies to random faults mostly due to intrinsic physical processes such as aging.

b) SIL defines measures to be applied in the design and during the manufacturing process to keep systematic failures low.

11.4.52.0 SIL Table according to EN 50129:

Tolerable Hazard Rate (THR)	Safety Integrity Level (SIL)
$10^{-9} \leq THR < 10^{-8}$	SIL 4
$10^{-8} \leq THR < 10^{-7}$	SIL 3
$10^{-7} \leq THR < 10^{-6}$	SIL 2
$10^{-6} \leq THR < 10^{-5}$	SIL 1

11.4.53.0 **Reliability Vs. Quality**

Quality is a static measure. It is confirmation to specification one time.

Reliability is a dynamic measure. Reliability is Quality in time domain (over a period of time). Reliability is Quality changing over time. It is a motion picture instead of a snapshot.

CHAPTER - 12: INSTALLATION , TESTING & MAINTENANCE OF MECHANICAL SIGNALLING EQUIPMENT

Section 1- Installation

12.1.1.0 **Lever Frame**

Conformity to drawings and specifications: Interlocking frames shall conform to approved drawings and specifications

12.1.2.0 **Marking of Parts**

12.1.2.1 Standards, quadrants, quadrant supports, levers shoes, locking boxes and covers, shafts and intermediate supports shall be marked with identifying letters or figures painted on them.

12.1.2.2 Plungers, interlocking bars, mechanical interlocking locks and other parts constituting the 'Locking' shall be stamped as under:

(a) Plungers shall be marked with their relevant lever numbers and number of the 'locking' box;

(b) Interlocking bars shall bear the number beginning from the first left hand side lock to last right hand side lock, the number of the channel of the locking box, and its position in the channel; and

(c) Mechanical interlocking locks (popularly called 'dogs') shall be marked with their respective lever number and the number of the channel of the locking box.

12.1.3.0 **Assembling Mechanical Locking:** Mechanical locking shall be assembled in accordance with approved Locking Diagram. Approval shall be obtained for any alteration, constructional or technical, found essential during the course of assembling locking. Some important instructions for assembling locking are detailed below:

12.1.3.1 Locking boxes shall be correctly aligned.

12.1.3.2 All pin joints shall be made a proper fit with no slackness. This is of particular importance in catch handle type of locking where a number of pin joints are provided to work the locking through the catch handle. For newly fitted locking frames there shall be practically no lost motion in connections or locking. Levers shall be straightened where required to maintain proper alignment. It shall be ensured that all parts of the lever frames including spare levers and their tappets are intact.

12.1.3.3 (a) Notches shall be marked off in accordance with the locking diagrams. Marking shall be done with the help of templates and the outline punch marked.

(b) In the case of catch handle locking, it is most important that the catch rod and links are in correct position before marking.

(c) It is a good practice to operate the catch handle several times, both in the normal and the reverse positions, in the case of catch handle locking and operate the lever several times in the case of Direct Lever Locking before the notches are actually marked. While doing so, the catch block shall be properly held against the quadrant.

12.1.3.4 Notches shall be cut to proper sizes and shapes as marked. Normal and reverse notches for point levers and those signal levers which have back tails or long leads may be cut with a small clearance up to 1 mm. Notches of straight edge locks may have a clearance up to 3 mm for the straight edge of the lock.

12.1.3.5 Joints in interlocking bars shall be avoided as far as possible. Where necessary, the joints shall be made smithy welding. The welded bars shall be kept truly straight and perfect in shape.

12.1.3.6 All locks and driving pieces shall be securely fixed square to the interlocking bar. Locks (dogs) shall have free movement in the channels of the locking box.

12.1.3.7 Unsupported and long interlocking bars spanning more than 10 levers in SA1101 Lever Frame and 7 levers in SA 530 Lever Frame shall be provided with a carrier lock or dummy pieces.

12.1.3.8 Any top piece fixed in the plunger shall be ordinarily rivetted except when it is accessible from outside, it shall be welded on to the plunger.

12.1.3.9 Each assembled interlocking bar shall be checked and tested to see that -

(a) It is in accordance with approved Locking Diagram.

(b) the locking provided through each bar is in accordance with approved interlocking table.

(c) the lost motion of a lever, which is locked, shall be such that the lever block cannot be moved more than 12 mm over the lever quadrant in the case of locking operated directly by the lever. In the case of catch handle operated locking, it shall not be possible to raise the catch handle block more than 10 mm.

(d) The locking shall be strong, durable and accurate. It shall not be possible for any mechanical lock to enter a notch other than that for which it is intended or to force the locking.

(e) The locking in the lever frame shall be effected just at the commencement of the movement of the lever or the catch handle as the case may be.

(f) The release locking shall not be effected before the completion of the movement of the lever.

(g) The locked lever shall not be free before the releasing lever has completed its travel proving thereby that there are no conflicting notches.

12.1.3.10 Plungers, interlocking bars and locks (dogs) shall be marked as detailed in Para 12.1.2.2.(c)

12.1.3.11 Locking strips shall be in accordance with approved plans.

12.1.3.12 Locking box covers shall be securely fixed so that it is not possible to force the locking by displacing the locks or bars. The covers shall be padlocked and sealed.

12.1.3.13 After assembling locking, it shall be carefully checked and compared with the approved Locking Diagram.

12.1.4.0 **Checking of Components:** The JE/SSE (Signal) shall carefully check the components of a new interlocking frame on receipt against plans and take necessary action.

12.1.5.0 **Erection:** Interlocking frames shall be erected in accordance with approved plans and locking fitted in accordance with the approved Locking Diagram. The locking shall then be carefully checked and compared with the approved Locking Diagram and tested by pulling of individual levers against the approved Interlocking Table as per instructions laid down in para 12.26.144

12.1.6.0 **Subsequent Additions and Alterations**

12.1.6.1 Any additions and alterations to interlocking frames shall be carried out only in accordance with the approved revised plans and interlocking tables.

12.1.6.2 The JE/SSE (Signal) shall requisition all requirements of materials in accordance with the additions and alterations shown on the approved revised plan and locking diagram and when requisitioning new plungers shall quote correct drawing number and if that is not available, state the type of frame and length of plunger required. When requisitioning new mechanical interlocking locks (dogs), the locks (dogs) required must be specified by quoting the type number shown against each lock on the standard drawings.

12.1.6.3 After carrying out additions and alterations, locking shall be checked completely as detailed in para 12.26.144

12.1.7.0 **Information to the Divisional Signal & Telecommunication Engineer :** The JE/SSE (Signal) shall inform the DSTE/ADSTE in charge of the section when a new interlocking frame or an alteration to the existing interlocking frame is ready for his test and introduction.

Section 2: Point Rodding and Fittings

12.2.8.0 **Point Rodding:** Point rodding shall be solid and shall be in accordance with I.R.S Specification No. S 5.

12.2.9.0 **Alignment:** Point rodding shall be straight before being laid. Runs of rodding shall normally be in a straight alignment.

12.2.10.0 **Distance from the Centre of Track :** The distance from the nearest rod to the centre of the nearest track shall not be less than 1905 mm on BG lines and 1370 mm on MG/NG lines except when platform walls or other structures render this impossible.

12.2.11.0 **Branching Off:** Where practicable, rods in a main run shall be arranged so that they branch off on the track side in regular order.

12.2.12.0 **Rodding Running Under Track:** When practicable, rodding running under tracks shall be arranged to permit the standard spacing and packing of sleepers. The top of the rodding shall not be less than 25 mm below the bottom of the rails. Whenever possible, the running of rods under rail joints shall be avoided.

12.2.13.0 **Fixing Roller Standard:**The top of roller standards adjacent to tracks shall not be fixed more than 64 mm above rail level.

12.2.14.0 **Spacing of Rollers:** Runs of rodding shall be supported on rollers spaced not more than 2.2 metre apart for solid rodding and 1.85 metre apart for tubular rodding. On curves the distance apart shall be 1.85 metre for both types.

12.2.15.0 **Offsets in Roddings :** Offsets in rodding shall be limited, wherever possible to 60 mm. Offsets of 90 mm may, however, be made where solid rodding is used. The minimum length of the offset shall not be less than twice the amount of the offset. Where more than one offset is necessary, a guide shall be provided. When making offsets in joints, the pin shall be kept in place to keep the pin holes true.

Section 3 - Point Operation in Mechanical Signalling

12.3.16.0 **Essential requirements before Interlocking of a Point:** Before any newly laid point is connected to the lever frame or a point machine and interlocked, it shall be ensured that the JE/SSE (P-Way) has taken necessary steps as detailed in Annexure-12/1

12.3.17.0 **Adjustment of Point Switches :** Point switches and switch and lock movements shall be so adjusted that the lever operating them is sprung equally in the normal and reverse positions. This springing shall normally be 12 mm on the lever quadrant.

12.3.18.0 **Permissible Distances :** The distance at which Points may be worked by rodding must not exceed 320 metres except where the stroke at the lever tail is not less than 200 mm, in which case the above distance may be increased to 460 metres. For special type of switches, the Chief Signal and Telecommunication Engineer may stipulate the permissible distance.

Section 4 - Facing Point Locks, Switch & Lock Movements and Spring Point Levers

12.4.19.0 **Layout**

12.4.19.1 Facing Point layouts shall be installed in accordance with approved plans.

12.4.19.2 Facing Point lock shall be fitted on the gauge tie plate where steel sleepers are not provided.

12.4.19.3 A key lock shall be so placed as to be inaccessible during the passage of a train unless the arrangement is such that a key is used to release the signals and cannot be brought back to the points until such signals have been put to the 'ON' position.

12.4.19.4 The FPL may be fitted nearer to the closed switch rail leading to higher speed movement.

12.4.20.0 **Notches:** Notches in stretcher blades shall be cut clean and square and to correct size

12.4.21.0 **Facing Point Lock - Plunger Type**

12.4.21.1 Facing Point lock plunger shall be square ended and installed 12 to 20 mm clear of the stretcher blades when the points are unlocked.

12.4.21.2 The plunger shall have a stroke of at least 150 mm.

12.4.22.0 **OBSTRUCTION TEST :** Facing point locks shall be tested by 'obstruction test' with a test piece of 5 mm thickness placed between the switch and stock rails at 150 mm from the toe. With the obstruction in, it shall not then be possible for the lever working the facing point lock to be latched.

12.4.23.0 Switch and lock layouts are not permitted in new installations. Spring point levers self-reversing shall be installed in accordance with approved plan.

Section 5 - Location of Signals in Mechanical Signalling

12.5.24.0 **Placement of Signal**

12.5.24.0 Signals shall be so located that they do not infringe the Schedule of Dimensions. Deviations shall be sanctioned by the competent authority.

12.5.25.0 Signal posts shall normally be on the left side and adjacent to the line to which they refer, and signals shall be located so that they are normally on the left of, or above the line to which they apply, unless authorised by special instructions.

12.5.26.0 Signals shall be inspected by a Signal sighting committee consisting of officials of Signal, Traffic and Mechanical/ Electrical Departments before being brought into use.

12.5.27.0 **Fixed Green Light of Warner Signal :-** The fixed green light above a Warner Signal on a post by itself shall be provided and shall be repeated.

12.5.28.0 **Fixing Signal arms :-** All signal arms shall be fixed on the left hand side of the post.

12.5.29.0 **Placement of Warner Signal :-** A warner Signal may be placed, either

12.5.29.1 on a post by itself with a fixed green light by night 1.5 metres to 2 metres above it at an adequate distance (not less than 1200 metres unless otherwise permitted by approved special instructions) outside the First Stop Signal or Gate Stop Signal ; or

12.5.29.2 on the same post, but 1.5 metres to 2 metres below the arm of the outer Signal ; and

12.5.29.3 on the same post as, but 1.5 metres to 2 metres below the Last Stop Signal of a station in rear.

12.5.30.0 **Warner Signal below a Stop Signal :-** When the Warner Signal is placed below a Stop Signal, the variable light of the Stop Signal shall take the place of the fixed green light of the Warner Signal and the arrangements shall be such that the Warner Signal cannot be taken ' OFF' while the Stop signal above it is ' ON '.

12.5.31.0 **Stop Signals:**

12.5.31.1 **Semaphore arm:** The Semaphore arm of a Stop Signal shall be square ended. The front of the arm shall be red with a white bar. The back of the arm shall be white with a black band. The bars shall be parallel to the end of the arm.

12.5.31.2 **Day and night indications:** By day the arm of a Stop Signal shall be horizontal in the ' ON ' position and 45° to 60° below the horizontal in the ' OFF ' position. By night, the signal shall display a red light in the ' ON ' position and a green light in the ' OFF ' position.

12.5.32.0 **Stop Signals Location of**

12.5.32.1 **Outer Signals:** The outer signal where provided or the Home Signal, where an outer Signal is not provided, shall be placed not less than 400 metres in rear of the Points up to which the line may be obstructed after Line Clear has been given to the Station in rear.

12.5.32.2 **Home Signals:** .The Home Signal shall be placed in rear of all connections, if any, on the line to which it refers.

12.5.32.3 **Routing Signal:** A Routing Signal shall be placed in the rear of the Points, which it protects.

12.5.32.4 **Starter Signals-**

(a) A starter Signal shall be placed at not less than 400 metres in advance of the Home Signal.

(b) Where a Starter Signal is provided for each converging line, it shall be

placed as to protect the adjacent running line or lines.

(c) Where one starter Signal only is provided for two or more converging lines, it shall be placed outside the connections on the line to which it applies.

12.5.32.5 **Intermediate Starter:** An Intermediate Starter Signal shall be placed clear of fouling marks in rear of the points, if any, which it protects.

12.5.32.6 **Advanced Starter:** Unless approved under special instructions, an Advanced Starter Signal shall be placed outside all connections on the line to which it applies. It shall be placed at not less than 180 metres from the outermost point in the case of single line section. This distance shall be reckoned from the starter in the case of double line section. On special nominated sections, the portion between the fouling mark to block section limit shall be track circuited when

(a) visibility of the portion between fouling marks to block section limit is obscured from the place of operation; or

(b) frequent shunting involving main line takes place, or

(c) Advanced Starter is placed at a distance of full train length beyond the trailing points.

12.5.33.0 Where Signals are carried on gantries, left hand signal in each case shall refer to the left hand line, the second signal from the left shall refer to the next line from the left and so on. The Signal for the main line shill be placed at a higher level than the signal or signals referring to the other running line or lines.

12.5.34.0 **Semaphore arm:** The Semaphore arm of a Calling-on Signal shall be a short square ended arm. The front of the arm shall be white with a red bar. The back of the arm shall be white with a black bar. The bars shall be parallel to the end of the arm.

12.5.34.1 **Day indication-By day, the arm shall be-**

(a) Horizontal in the ' ON ' position ;

(b) 45° to 60° below the horizontal in the ' OFF ' position.

12.5.34.2 Calling-on Signal shall be placed below a Stop Signal governing the approach of a train. Under approved special instructions, a Calling-on Signal may be provided below any other Stop Signal except the last Stop Signal.

12.5.35.0 **Shunt Signal**

12.5.35.1 Under special instructions, a shunt Signal may be a miniature Semaphore arm.

12.5.35.2 **Miniature Semaphore Signal:** (a) The Semaphore arm of a shunt Signal shall be square ended. The front of the arm shall be red with a white bar. The back of the arm shall be white with a black bar. The bars shall be

parallel to the end of the arm.

(b) By day the arm of a Stop Signal shall be horizontal in the ' ON ' position and 45° to 60° below the horizontal in the ' OFF ' position. By night, the signal shall display a red light in the ' ON ' position and a green light in the ' OFF ' position.

12.5.35.3 **Diverging routes :-** More than one shunt signal may be placed on the same post and when so placed, the top most Shunt Signal shall apply to the extreme left-hand line and the second shunt signal from the top shall apply to the next line from the left and so on. One shunt Signal with or without a route indicator may also be provided for a number of diverging routes.

12.5.36.0 **Repeating Signal**

12.5.36.1 A repeating signal is a signal placed in rear of a fixed signal for the purpose of repeating to the driver of an approaching train aspects of the fixed signal in advance. It shall be provided with a marker consisting of a white-enameled disc with letter ' R ' in black.

12.5.36.2 A repeating signal shall be of a square ended Semaphore arm

12.5.36.3 (a) The arm of a semaphore repeating signal shall be square ended. The front of the arm shall be yellow with a black bar. The back of the arm shall be white with a black bar. The bars shall be parallel to the end of the arm.

(b) By day, the arm shall be horizontal in the ' ON ' position and 45° to 60° below the horizontal in the ' OFF ' position. By night the signal shall exhibit a yellow light in the ' ON ' position and a green Light in the ' OFF ' position.

12.5.37.0 **Co-acting Signals:-** Co-acting signals are duplicate signals fixed below Stop Signals and are provided where, in consequence of the height of the signal post or of there being an over-bridge or other obstacle, the main arm or light is not in view of the Driver during the whole time that he is approaching it.

12.5.38.0 **Distant Signals**

12.5.38.1 **Electric Lighting of Distant Signal :-** Distant Signals shall be electrically lit on all the Trunk routes and important main lines

12.5.38.2 **Location:-** On both double and single line sections, the Distant Signal shall be placed at an adequate distance in rear of the First Stop Signal, the adequate distance being not less than 1 kilometre.

12.5.39.0 **Multiple Aspect Semaphore Signalling**

12.5.39.1 **Semaphore arm:** The Semaphore arm of a Distant Signal shall be fish-tailed. The front of the arm shall be yellow with a fish-tailed black bar. The back of the arm shall be white with a fish-tailed black bar.

12.5.39.2 **Day and night indications:** (a) By day, the arm of a Distant Signal shall be horizontal in the ' ON ' position displaying the ' caution ' aspect. It shall

be 40° to 45° above the horizontal in the ' OFF ' position displaying ' Attention ' aspect. It shall be 85° to 90° above the horizontal in the 'OFF ' position displaying the ' Proceed ' aspect.

(b) By night, the signal shall display one yellow light for the ' Caution' aspect, two yellow lights in a vertical line 1.5 metres apart for the ' Attention ' aspect and one green light for the ' Proceed ' aspect.

12.5.40.0 **Stop Signals:**

12.5.40.1 The Semaphore arm of a Stop Signal shall be square ended. The front of the arm shall be red with a white bar. The back of the arm shall be white with a black band. The bars shall be parallel to the end of the arm.

12.5.40.2 **Day and night indications:** (a) By day, the arm of a Stop Signal shall be horizontal in the ' ON ' position displaying the ' STOP ' aspect. It shall be 40° to 45° above the horizontal in the ' OFF ' position displaying the ' Caution' aspect. It shall be 85° to 90° above the horizontal in the ' OFF ' position displaying the ' Proceed ' aspect.

12.5.41.0 **Stop Signals-Location of**

12.5.41.1 **Home Signal:** The Home Signal shall be placed in rear of all connections, if any, on the line to which it refers. The Home Signal shall be placed not less than 180 metres in rear of the points upto which the line may be obstructed after line clear has been granted to the station in rear.

12.5.41.2 **Routing Signal:** A Routing Signal shall be placed as in Para 12.5.32.3.

12.5.41.3 **Starter Signals:** When one Starter Signal is provided for each converging line, it shall be so placed as to protect the adjacent running line or lines.

12.5.41.4 **intermediate Starter Signal.:** An Intermediate Starter Signal shall be placed clear of fouling marks in rear of the points, if any, which it protects.

12.5.41.5 **Advanced Starter Signal-** Unless approved under special instructions, an Advanced Starter Signal shall be placed outside all connections on the line to which it applies. It shall be located at a distance of not less than 120 metres from the outermost points in the case of Single Line Section. This distance shall be reckoned from the starter in case of Double Line Section. On special nominated sections the portion between the fouling mark to block section limit shall be track circuited when

(a) visibility of the portion between fouling mark to block section limit is obscured from the place of operation; or

(b) frequent shunting involving main lines takes place; or

(d) Advanced Starter is placed at a distance of full train length beyond the trailing points.

12.5.42.0 **Calling On signal**

12.5.42.1 **The Semaphore arm of a Calling-on Signal :-** By day, the Semaphore arm of a Calling-on Signal shall be-

(a) Horizontal in the ' ON ' position, and

(b) 40° to 45° above the horizontal in the ' OFF ' position.

12.5.43.0 **Holding Route :-**

12.5.43.1 Signals governing movements over Points shall be placed as close as possible to the Points. Where a signal is more than 180 metres from the Facing points it controls, arrangements shall be made to keep the Points locked until the train has passed them. Similar arrangements shall also be made to hold consecutive Points, where the distance between them should be more than 180 metres.

12.5.43.2 At a station where trains run through at speeds more than 50 kilometres per hour, such arrangements to hold the route are also required in case of trailing points situated more than 180 meters from the signal controlling them. However, such arrangements are not required if the Points are locked in either position by the signal in advance.

12.5.43.3 Route holding arrangements for facing or trailing points are however not necessary, if due to the manipulations required in the system of control, it is impossible under normal working conditions for the Points to be operated before the train has passed.

Section 6 - Signal Wire Run

12.6.44.0 **SINGLE WIRE/ROPE**

12.6.44.1 Galvanised steel wire strand No.7/1.40 mm (dia) to IS Specification No. 2265, or galvanized solid steel wire 3.35 mm (dia) to IRS Specification No. Sl 62, shall be used for signal connections.

12.6.44.2 Coils of wire shall be run out from a creel. Twists, kinks and nicks shall be avoided.

12.6.45.0 **Galvanized wire rope :** Galvanized steel wire rope for signalling purposes ordinarily used for single wire runs shall be to IRS Specification No. S-3.

12.6.46.0 **Wire rope round wheels :** Wire rope when used for making turns on wheels shall not be less than 2 metre in length.

12.6.47.0 **Wire rope on cabin wire adjusters :** Wire rope when used on cabin wire adjusters and draft wheels shall be in a continuous length.

Section 7 - Painting of Mechanical Signalling

12.7.48.0 **Equipment to be painted :** The following are the main items of mechanical signalling equipment that need regular painting:

12.7.48.1 Signal arms, Point Indicators, boards and markers;

12.7.48.2 Signal posts and Signal fittings;

12.7.48.3 Interlocking frames, their levers & fittings;

12.7.48.4 Ground and miscellaneous gear, such as rodding, guide roller assembly, cranks, compensators, turnout wheels;

12.7.48.5 lifting barriers, if maintained by Signal and Telecommunication department.

12.7.49.0 **Colouring Scheme:** Items detailed in Para 12.7.48 shall be painted in accordance with colouring scheme shown for the respective item as in Anexure -12/2

12.7.50.0 **Interval of Painting :** The general principles for fixing intervals shall be the following :

12.7.50.1 Signal arms, indicators, etc., which are required to be painted to conventional shapes and/or colours shall be painted once a year.

12.7.50.2 Signal posts which are to be kept conspicuous and cabin levers which are to be distinguished by colours, shall be painted once in three years;

12.7.50.3 Rodding, guide roller assembly, etc., which only need a protective covering, shall be painted once in six years.

12.7.51.0 **Note :** The intervals mentioned above may be reduced in areas where, on account of weather or other local conditions painting at shorter intervals is necessary

12.7.52.0 **Record of Painting**

12.7.52.1 Each JE/SSE (Signal) shall maintain a Painting Register, one page or more being allotted to each station or cabin on his section. Record of the painting work done with dates of commencing and completing the work shall be regularly entered in this register.

12.7.52.2 Each JE/SSE (Signal) shall also record the date of painting with particulars of the programme carried out in the Signal Incidences Book maintained at each station. This should be done on the very first inspection by him after painting work has been completed.

12.7.53.0 **General Instructions on Paints/Painting**

12.7.54.0 **Stores**

12.7.54.1 Paint shall be stored in a cool dry place away from flame or naked light.

All containers shall be kept securely closed when not in use, to avoid loss of material due to skinning and contamination. Paints, especially quick drying paints, shall not be left open to the atmosphere.

12.7.55.0 **Mixing of Paints:**

12.7.55.1 Paints from pigments and pastes shall be prepared in accordance with standard specifications

12.7.55.2 Linseed oil or turpentine shall not be added to ready mixed paints. Mixing of kerosene with paints is forbidden.

12.7.56.0 **Painting of Steel Works**

12.7.56.1 The surface of metal shall be clean and free from dirt, scale, deteriorated old paint and rust and shall be perfectly dry before application of paint. Painting shall be done preferably during dry weather.

12.7.56.2 For a new steel work, three coats of paints shall be applied. First coat shall be 'Red Lead', the second 'Red Oxide' and the third of the specific paint recommended for the structure. When repainting a structure, if the old paint does not show any signs of blistering, scaling or cracking, it need not be scraped off. It may be used as foundation for the new coat.

12.7.56.3 When there are patches of blistering, scaling or cracking, these patches shall be cleaned down to the steel and treated in a manner as recommended for a new structure.

12.7.56.4 The surface of metal shall be cleaned by scraping, chipping or scrubbing with brickbats. No chemical of any kind shall be used. Old paint may be burnt, if necessary and then scraped off.

12.7.56.5 Paint shall be applied with brushes and spread evenly, smoothly and as thin as possible. Rags or waste cotton shall not be used for the application of paint.

12.7.56.6 Brushes shall be thoroughly cleaned in proper driers after use.

12.7.57.0 **Enamel Signal Arms**

12.7.57.1 Enamel signal arms shall be scrubbed with soap to clean them. If corroded in places, these shall be touched up with enamel paints.

12.7.57.2 Signal arms which are not enamelled, shall be painted with enamel paints.

12.7.58.0 **Numbering of Points and Signals :**

12.7.58.1 The cabin lever number of points and facing point Lockbars shall be painted in 50 mm letters on the web and close to the toe of the switch rail.

12.7.58.2 The number of the cabin lever operating the signal shall be painted in 50 mm letters at the back of the signal arm blade

12.7.58.3 **Numbers shall be written in codes as under:** (table)

i)	Points – 20 W	'20 W' stands for lever No. 20 West Cabin
ii)	Points & Facing Point lock (P32 L33E)	'P32" stands for points lever No. 32 and 'L33E' stands for lock bar lever No. 33, East Cabin
iii)	Signals – 51A - 52B - 53 GFA	'51A' stands for lever No. 51 'A' Cabin, '52B' stands for lever No. 52 'B' Cabin '53 GFA' stands for lever No. 53 Ground Frame 'A'.

12.7.58.4 Numbers of interlocking frame levers shall also be painted on the front and back side of the main girder supporting the frame opposite each lever in 50 mm white letters

12.7.59.0 **Dates of Painting :** The year and the month of painting work shall be clearly and neatly shown on all interlocking frames, signal posts, block instruments, Station master's slide control frames and instruments. Suitable entries shall also be made in the Register maintained for this purpose.

Section 8 – Maintenance of Mechanical Signalling

12.8.60.0 **Cabin equipment :-** All cabins shall be equipped with

12.8.60.1 A Cabin Diagram showing the location and normal position of all points. Facing point locks, Signals and Level Crossings, with their respective cabin lever numbers;

12.8.60.2 A lever pull chart suitably exhibited;

12.8.60.3 Lever collars;

12.8.60.4 A suitable staging with ladder for attending to interlocking frame;

12.8.60.5 Telephones, as necessary;

12.8.60.6 All equipments as required to be provided by concerned Departments including a clock, and

12.8.60.7 A name board of sufficient size.

Note:- *The lever pull chart, referred in (2) above, need not be provided in Cabins where the number of release lever concerned are indicated on the lever name plates to guide the Cabin man in following the correct sequence for pulling the levers.*

12.8.61.0 **Numbering of levers:-** All levers, including spares, shall be numbered consecutively through the frame from left to right. Each lever shall be provided with a number plate. This will show the function in the order of operation which must precede to release when no lever description board is provided.

12.8.62.0 **Cabins - Cleanliness, Repair and Fittings:**

12.8.62.1 Ground floor of cabins shall be kept clean and tidy.

12.8.62.2 Cabins including battery room and repair room shall be kept in good condition and periodically painted/distempered. The roof shall be so maintained as to prevent leakage. "Ghara" stands shall be kept properly drained off. Wash basins, where provided, shall be kept in perfect repair. Any repair required shall be promptly notified to the JE/SSE (Woks).

12.8.62.3 JE/SSE (Signal) shall bring to the notice of his Sr.DSTE or DSTE or ADSTE incharge of the section when repairs to cabin are delayed.

12.8.62.4 Windows of cabins shall be kept clean. This is the duty of the operating staff.

12.8.62.5 Each JE/SSE (Signal) shall see that the cabin diagram, pull chart, Cabin man's instructions and Maintained programme provided in the Cabin are up-to-date and their glazed frames are in proper repair.

Section 9 - Maintenance of Lever Frame

12.9.63.0 **Responsibility:** Each JE/SSE (Signal) is personally responsible for efficient maintenance of the locking of the interlocking frames under his charge. Locking box covers shall be kept padlocked and sealed with his personal seal supplied to him for the purpose.

12.9.64.0 **Instructions for Technicians**

12.9.64.1 All accessible working parts of interlocking frames shall be thoroughly lubricated and cleaned.

12.9.64.2 All split pins shall be checked to ensure that they are intact and properly open. Also all nuts and bolts shall be checked and tightened wherever necessary.

12.9.64.3 Handles of levers of interlocking frames shall be kept clean and bright by the Transportation Staff.

12.9.65.0 **Instructions for JE/SSE** (Signal):

12.9.65.1 The JE/SSE (Signal) shall ensure that the interlocking frames are kept clean and lubricated

12.9.65.2 Locking of all interlocking frames shall be tested as per instructions contained in para 12.26.144

12.9.65.3 During the testing of interlocking frames, careful note shall be taken of slack locking and if the slack locking exceeds the limits laid down in Para no's.12.1.3.9.(c), overhauling shall be taken up forthwith

12.9.65.4 Locking of all interlocking frames shall be overhauled at least once every three years or earlier, if necessary, as per instructions laid down in para

12.25.133

Section 10 - Maintenance of Lead outs (Rodding)

12.10.66.0 All lead outs shall be examined regularly and kept clean and well oiled.

12.10.67.0 All connections, especially the down rods, shall be examined to detect cracks and worn out threads. The whole of the apparatus shall be kept rigidly and solidly connected.

Section 11 - Maintenance of Point Rodding and Fittings

12.11.68.0 **Examination**

12.11.68.1 All runs of rodding shall be regularly examined to see that all joints make good and solid connections.

12.11.68.2 All couplings, adjusting screws and adjustable joints shall be examined for looseness, worn out threads or slack nuts.

12.11.68.3 Rodding shall be examined for detecting weak places due to rust, corrosion or other causes.

12.11.69.0 **Alignment:** The rods shall be kept in true alignment and free from all obstructions.

12.11.70.0 **Rollers:** Rollers shall fit properly in the standards and move freely. Broken roller standards, missing rollers and pins shall be replaced.

12.11.71.0 **Roller Foundations:** Roller foundations shall be realigned if found to have shifted or sunk and shall be renewed when necessary.

12.11.72.0 In all new installations, Points, Locks and Bars shall not be operated by wire but by rodding or by electrical means.

12.11.73.0 **Operation of Facing Point lock :-** The operation of the facing point lock shall depend on the correct operation of the Lockbar. This Lockbar shall form a part of the transmission and be in series with it.

Section 12 - Maintenance of Cranks, Compensators, Screws, Bolts, Nuts, Pins and Cotters

12.12.74.0 **Examination**

12.12.74.1 All cranks and compensators shall be examined regularly and working parts well oiled.

12.12.74.2 The position of the compensators shall be carefully checked when a point or Lockbar does not work properly or where the stroke at the adjusting crank has to be unduly increased.

12.12.75.0 **Replacement of cracked or broken parts :** Cracked or broken parts of cranks, compensators shall be repaired or renewed.

12.12.76.0 **Cleaning of oil holes :** Oil holes shall be cleaned before oil is applied. Oil shall not be used in excess as it collects dust.

12.12.77.0 **Screws, bolts and nuts :** All screws, bolts and nuts shall be in place and shall be kept tight.

12.12.78.0 **Adjusting Screws :** Threads of point adjusting screws shall be oiled regularly and kept free from dust. Lock nuts of adjusting screws and set screws of adjusting cranks shall be kept tight.

12.12.79.0 **Pins**

12.12.79.1 All pins of cranks, wheels, etc., shall be in place and kept oiled regularly. All pins shall be raised slightly where possible, and oil applied underneath so as to ensure proper distribution.

12.12.79.2 All badly worn pins shall be replaced by new pins.

12.12.79.3 Tight pins can often be loosened with kerosene; if this is not possible, pins shall be driven out and thoroughly cleaned. Pin holes shall never be filed.

12.12.80.0 **Keys and Cotters:** All keys and cotters shall be examined regularly. Keys and cotters shall be in place and properly spread, split ends shall be spread not less than 10 mm. Keys and cotters shall be of proper size and in good condition, old nails or pieces of wire shall not be used.

Section 13 - Maintenance of Points

12.13.81.0 Visual inspection of the condition of insulation of insulated rod joints shall be done during maintenance inspections.

12.13.82.0 **Responsibility of JE/SSE (P-Way) & JE/SSE (Signal):**

12.13.82.1 Maintenance of switches, gauge tie plates and bolts and cotters for pad locks is the responsibility of the JE/SSE (P-Way). In addition, all non-interlocked switches, leading and following stretcher bars connecting rods, self reversing levers, hand points, spring loaded points are also maintained by him. Connections to switches other than those specified above such as facing point locks and bars, detectors, point and trap indicators are maintained by the JE/SSE (Signal).

12.13.82.2 S&T Staff shall not take on themselves the responsibility of disconnecting the permanent way fittings or making adjustments thereof.

12.13.82.3 The primarily responsibility of points layout and its allied equipments binding with the concerned departments who own the equipment/gears.

12.13.83.0 **Damaged Switches and Packing of Sleepers**

12.13.83.1 Advice of switches found chipped or damaged shall be promptly given in writing to the JE/SSE (P way).

12.13.83.2 Sleepers under interlocked points and cross overs shall be kept well packed. Advice of loose packing shall be given promptly to the JE/SSE (P-

Way). When packing of sleepers of interlocked points is being done by the permanent way staff, signal technician shall also be present there.

12.13.84.0 **Testing and Adjustment**

12.13.84.1 All switches shall be frequently tested to see if they are in correct adjustment. For testing, the switches shall be moved slowly and it shall be ensured that-

(a) the switches are set properly against the stock rail, and

(b) the lever operating the points springs a little, equally in the normal and the reverse positions before being latched.

12.13.84.2 When two or more switches are operated by a single lever, all switches shall be adjusted correctly.

12.13.84.3 The switches get out of adjustment on account of wear in fittings and pins, creep of track and poor track maintenance. When the slackness in fittings becomes excessive, slack fittings shall be replaced. Where creep and maintenance of track is affecting the adjustment of signalling gear at the points, it shall be promptly reported to the JE/SSE (P-Way).

Section 14 - Maintenance of Facing Point Locks, Switch & Lock Movements and Spring Point Levers

12.14.85.0 **Facing point locks :** Facing point lock plunger shall be kept lubricated and free from rust. The end of the plunger shall be square. It shall be adjusted to be at least 12 mm clear from the stretcher blades, when the points are unlocked. The edges of notches in stretcher blades shall be kept square and sharp.

12.14.86.0 **Switch & Lock Movement for existing installations:**

12.14.86.1 All sliding parts of switch and lock movements shall be thoroughly lubricated.

12.14.86.2 Switch and lock movements shall be kept adjusted so that the Lockbar shall rise to two-thirds of its full height before the switch points start to open.

12.14.87.0 **Loose or cracked parts:** Switch and lock movements and facing point locks and bars shall be examined closely to detect loose or cracked parts. Loose or cracked parts shall be replaced at the earliest.

12.14.88.0 **Cross Slides :** In the case of facing point bolt locks provided with cross slides or plunger detection, there shall be no undue play between the cross slide and the facing point detector connections.

12.14.89.0 **Testing Switch and Lock movements and facing point locks:** Switch and lock movements and facing point locks shall be given an obstruction test by inserting a test piece of 5mm between the switch and stock rails at 150 mm from the toe of the switch. The obstruction shall

12.14.89.1 In the case of facing point bolt lock, prevent the lever working the facing point bolt lock from being latched;

12.14.89.2 In the case of facing point lock, key type, prevent the extraction of the key and

12.14.89.3 In the case of switch and lock movement, prevent the signal governing the movement over the switch and lock movement from being cleared.

12.14.90.0 Spring Points if used in the facing direction must be fitted with facing point locks. They may be operated from a Cabin or from a ground lever. In the later case, the ground lever shall be of such a type that it will not move when the Points are being trailed through.

Section 15 - Maintenance of Lockbars

12.15.91.0 **Length of Lockbars: -** The length of Lockbar shall exceed the greatest distance between any two adjacent axles of any vehicle likely to be used on the section.

12.15.92.0 Lockbars shall not be less than 12800 mm long for Broad Gauge and 12200 mm for Metre Gauge and Narrow Gauge lines.

12.15.93.0 **Adjustment**

12.15.93.1 Lockbars shall be so installed that with the operating lever normal

(a) An outside Lockbar lies practically level with the top of the rail.

(*NOTE : In new installations, outside Lockbars shall not be used*);

(b) An inside Lockbar lies 38 mm below the top of the rail and the bar does not lean away from the rail.

(c) Whenever practicable, the end of the Lockbar shall not be more than 25 mm clear of the toe of the switch, when the points are bolted.

12.15.93.2 Each Lockbar shall be provided with not less than twelve clips and three stops located in such a manner that the maximum inter clip distance is 1220 mm. A Lockbar clip shall be fixed as near the end of the bar as possible but not more than 225 mm.

12.15.93.3 Lockbar shall not be installed over a check rail portion as a matter of course. Under special circumstances and only if other electrical means cannot be provided, Lockbars can be installed over check rail portion when they are fitted with a special type of check rail joint

12.15.93.4 Lockbars shall preferably be installed on the rail associated with straight track.

12.15.94.0 **Lockbar General**

12.15.94.1 Lockbars shall be kept straight and true both vertically and horizontally, driving pieces shall be examined to see if there is any lost motion.

12.15.94.2 Lockbar clips, stops and guides shall be tightly attached to the rails. Arms and pins of clips shall be kept in good order so as to ensure that the Lockbars fit close up to the rail. Bearings of clips shall be kept well oiled. Lockbar clips shall not be bent to prevent leaning away. Care shall be taken to ensure that there is adequate clearance between clip hanger and the top studs when fully tightened so as to permit easy working.

12.15.94.3 All Lockbars, clips, stops and guides shall be overhauled and thoroughly cleaned at least once a year. Renewals shall be made where necessary.

12.15.94.4 If rail creep is affecting the working of Lockbars, Technician shall make a report in writing to the JE/SSE (P-Way), giving copy to the JE/SSE (Signal).

12.15.95.0 **Testing Lockbars :** Lockbars shall be tested to see that with the operating lever normal or reverse-

12.15.95.1 Outside Lockbars lie practically level with the top of the rail;

12.15.95.2 Inside Lockbars lie 38 mm below the top of the rail. They shall lie flush with the top of the rail when the lever is in the mid-stroke.

Section 16 - Maintenance of Mechanical Detectors

12.16.96.0 All detectors shall be examined regularly and the slides kept well lubricated.

12.16.97.0 **Detector notches :** Detector notches shall be kept square and not eased beyond the standard size. Notches in point slides shall be kept properly adjusted.

12.16.98.0 **Cleaning of detector shoes and angle slides :** Detector shoes and their angle slides should be regularly cleaned and graphited to permit free movement of the detector shoe.

12.16.99.0 **Testing of detectors:** Detectors shall be regularly tested by obstruction test with a test piece of 5mm thickness placed between the switch and stock rail at 150mm from the toe, with the obstruction in, the signal slide shall get obstructed.

12.16.100.0 **Replacement of cracked or broken parts** : Cracked or broken parts of detectors shall be repaired or renewed.

Section 17 - Maintenance of Signal Wire Run (Single Wire)

12.17.101.0 **Alignment :** All wire runs shall be maintained in proper level and alignment.

12.17.102.0 **Wires Generally**

12.17.102.1 Damaged wire or wire rope showing signs of corrosion or rusting shall be renewed. Kinked wire shall be straightened and bad joints removed.

12.17.102.2 To prevent excessive wear over pulley wheels at one point, the wire shall be periodically moved and worn out portion shifted off the pulley wheel.

12.17.103.0 **Wires under track :** Wires running under the track shall neither drag on the ground nor rub against ballast, base of rails or side of sleepers.

Section 18 - Maintenance of Pulley Stakes and Pulleys

12.18.104.0 **Pulleys and Pulley Stakes**

12.18.104.1 Brackets of pulley wheels shall be examined to see that they are securely fixed and that the wheels revolve freely.

12.18.104.2 All damaged, broken and missing pulley wheels or brackets shall be renewed promptly.

12.18.104.3 Bent pulley stakes shall be straightened and worn-out ones replaced.

Section 19 – Maintenance of Electrical Repeater and Signal Lamps/LEDs

12.19.105.0 The JE/SSE (Signal) shall test the working of the signals fitted with electric repeaters when making their periodical inspection and see that they are in proper adjustment.

12.19.106.0 **Examination and Replacement of Signal & Indicator Lamps**

12.19.106.1 All Signals (oil lit and electrically lit) and Point and Trap Indicator lamps shall be examined regularly and any lamps or founts found defective shall be replaced promptly. Cases of excessive damage to lamps or neglect in cleaning shall be reported to the Sr.DSTE or DSTE or ADSTE incharge of the section. Cleaning of signal lamps is the responsibility of the operating staff.

12.19.106.2 **Overhauling and Repairs :** All lamps shall be overhauled once a year and the date of overhaul marked on each lamp.

12.19.106.3 Only petty repairs to Signal lamps are to be carried out by JE/SSE (Signal). Lamps requiring heavy repairs shall be carefully packed and sent to the Signal Workshops.

Section 20 - Maintenance of Double Wire Installations

12.20.107.0 **Transmissions Generally:**

12.20.107.1 All transmissions shall work freely and loss of stroke at the end of transmission shall be avoided.

12.20.107.2 There shall be no binding in any part of transmissions and all wires and ropes shall lie properly in the grooves of pulleys and wheels respectively. All pulleys shall be parallel to the wire run.

12.20.107.3 All transmissions shall be kept taut; there shall be no appreciable sag between pulleys.

12.20.107.4 Care shall be taken to see that wires and ropes do not scrape against platform and other objects as rails, chairs, covers and lids.

12.20.107.5 All wheels and mechanisms shall be thoroughly lubricated with a grease gun and made to work easily. All cam paths shall be kept well lubricated with grease.

12.20.108.0 **Maintenance of Compensators:** Compensators shall be maintained as per the instructions for 'Installation and Maintenance of Double Wire Compensators'. Some of the salient features are-

12.20.108.1 All moving parts shall be lubricated every fortnight with axle oil medium to IS:1628 and where nipples are provided Grease to IS:507 shall be used with a grease gun. The compensator wheels grooves and the wire rope lengths working in them shall be lubricated with graphited grease grade I to IS:508.

12.20.108.2 It shall be checked that all bolts are properly tightened and all split pins and nuts are in proper positions.

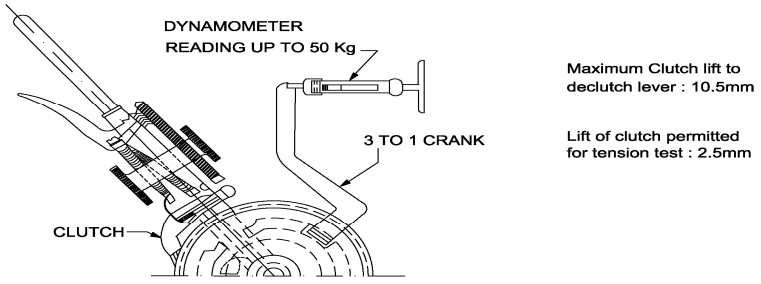
12.20.108.3 All broken parts shall be replaced immediately.

12.20.108.4 It shall be checked that both the weight levers move independently. Weight levers shall not be bound together either by wire or by chain.

12.20.108.5 It shall be maintained that locking pawl is in a position normally free from the ratchet rod but during lever operation they shall be engaged properly with the ratchet rod teeth.

12.20.108.6 In case of hard working of Double Wire transmission during cold weather, it shall be checked by night that weight levers are not binding against stop pin at the bottom of the ratchet rod and adjustments made where required.

12.20.109.0 **Testing Point, Lock, Detector and Signal Levers :** The SSE (Signal) incharge of Double Wire transmission shall test each point, lock, detector and signal lever by means of dynamometer as shown in Diagram once every 6 months. It shall be ascertained that-



DOUBLE WIRE INTERLOCKING FRAME -TESTING CLUTCH LEVER WITH DYNAMOMETER

12.20.109.1 the force required to operate a lever does not ordinarily exceed 35kg. In the case of a signal lever without detectors, it shall not ordinarily exceed 27 kg.

12.20.109.2 Point, lock or detector lever clutch does not trip after completion of lever movement. Should it do so, it indicates that the coupling spring adjustment is not correct or that somewhere in the transmission, usually at point or lock mechanism or detector, tight working or an obstruction exists. The cause shall always be carefully traced and rectified.

12.20.110.0 **Broken wire locks:** Broken wire locks on Point and Lock mechanisms shall be inspected to see that they are properly adjusted and are working correctly.

12.20.111.0 Reconnection or repairs after wire breakage

12.20.111.1 After each test, or after wire breakage, compensator weights shall first be lifted by means of a pulley block to enable reconnection or repairs to be made.

12.20.111.2 The drum of a trailable or clutch lever shall be reset by an auxiliary lever

12.20.112.0 **Responsibility and Periodicity of Broken Wire Tests :** Broken wire tests shall be carried out by each Senior Section Engineer (Signal) incharge of maintenance of Double Wire installations atleast once a year and ensure that equipment failure is on safe side, unless otherwise specified.

12.20.113.0 Necessary steps shall be taken to shorten the wire wherever necessary in a newly laid Double Wire transmission for the wire stretch occuring in the course of time.

Section 21 - Maintenance of Signals and Fittings

12.21.114.0 **Signal arms and its Fittings:**

12.21.114.1 Signal arms on a straight track shall be at right angles to the track concerned when sufficient approach is on the straight. On curves, signal arms shall be at right angles to an imaginary line drawn from the signal to the point where the best view is to be provided to the Loco Pilot.

12.21.114.2 Signal arms shall be kept securely fastened to the spectacles casting and maintained at the correct level when in the 'ON' position.

12.21.114.3 Signal arms shall be kept clean and properly painted.

12.21.114.4 It shall be ensured that spectacles are securely fixed on their spindles or studs, bearing brackets are firmly attached to the post and back light blinders are in their correct position.

12.21.114.5 Arm spindles or studs and counterweight lever connections shall be oiled regularly and kept working freely. It shall be ensured that counterweights are in their correct position and firmly secured and that the lock nuts on

adjustable joints are tight.

12.21.114.6 Spectacle roundels or glasses found loose, cracked or broken shall be attended to and replaced promptly.

12.21.115.0 **Slotting**

12.21.115.1 Where signals are slotted, any of the controlling agencies shall be independently capable of returning the signal to 'ON'.

12.21.115.2 When a Warner arm is placed below a Stop arm, the slot lever shall be installed so that the Warner cannot be taken 'OFF' unless the Stop Signal above is at 'OFF'. When the Stop Signal is returned to 'ON', the Warner shall return to 'ON'.

12.21.116.0 **Shunt Signals:** Where a Shunt Signal is placed on the same post as a main line signal, it shall be placed not less than 2.5 metres below the main arm.

12.21.117.0 **Balance Levers :** Where practicable, balance levers shall be placed out of reach from the ground and parallel to the line. Counterweights shall be securely fastened to the levers and safety bolts provided at the end of the levers.

12.21.118.0 **Back Lights**

12.21.118.1 **Indications :** Except when the front light can be seen from the Cabin or interlocking frame location, back lights of mechanical signals shall show a white light when the arm is in the 'ON' position and no light in any other position.

12.21.118.2 **Warner Signal Fixed green light:** The fixed green light of a Warner Signal shall also show a back light.

12.21.118.3 **Distant Signals :** Distant Signals in multiple aspect area shall show two back lights in the 'ON' position and no light in any other position. However, where electric lighting of Distant Signals is provided, the Distant Signals in Multiple aspect area may show atleast one back light in the 'ON' position if the arrangement is such that the bottom light is lit only when the arm is displaying the 'attention' aspect

12.21.119.0 **Trimming of Trees Obstructing Visibility:** Where trees interfere with the Loco Pilot's or cabin man's sighting of signals, the JE/SSE (Signal) shall arrange with the JE/SSE (Works) for Trimming of tree branches or cutting of trees within the Railway boundary. If the trees are in private land, the Sr.DSTE or DSTE or ADSTE incharge of the section shall be informed so that he may arrange through Sr.DEN or DEN or ADEN for the removal or cutting of the trees.

12.21.120.0 **Artificial backgrounds**

12.21.120.1 Signals may be provided with an artificial background painted white, if found necessary to improve the sighting or when the background is a bridge or structure, such portion as necessary to form a background, may

be painted white or be white washed.

12.21.120.2 Arms provided with an artificial background shall project 150 mm beyond or be provided with a tell tale arm so that it can be seen by the Station Master.

12.21.121.0 **Posts :** Signal posts shall be kept plumb. Signal post guys, where provided shall be kept tight and in good repair.

12.21.122.0 **Deck Boards :** Deck boards on bracket posts and gantries shall be inspected to detect decayed or loose boards. Missing or defective boards shall be replaced.

12.21.123.0 **Provision of Signal Dubbers and Kerosene :** Whenever kerosene lit signals are provided, the signal dubbers and kerosene shall be provided by Traffic staff.

12.21.124.0 **Cleaning of Signal Glasses and Lamp Lenses :** The Traffic staff are responsible for wiping and cleaning of signal glasses, roundels, and lenses of kerosene lit signals. Roundels of electrically lit signals shall be cleaned by signal staff.

Section 22 - Maintenance of Point, Trap and Shunting Permitted Indicators

12.22.125.0 **Type, Provision , Placing and Indication**

12.22.125.1 Point, Trap and Shunting Permitted Indicators shall be provided in accordance with the approved plans.

12.22.125.2 Type, Provision, Placing and Indication of these indicators are covered in detail in Chapter 7.7.36.0

Section 23 - Maintenance Schedule

12.23.126.0 The maintenance schedule for various items of 'Mechanical signalling equipment shall be as in Annexure -12/3. At stations having dense traffic and high speeds, the Railways may prescribe more frequent inspections, if considered necessary.

Section 24 - Overhauling and Testing of Interlocking Mechanical / Electromechanical

12.24.127.0 **Interlocking Frames**

12.24.127.1 All interlocking frames, interlocking key boxes and Station Master's Slide Control frames shall be overhauled once in three years and tested periodically once in a year. The records of the same shall be maintained.

12.24.127.2 The Electromechanical Frames shall be overhauled and tested periodically as per requirement contained in the above para

12.24.128.0 **System and Equipment Testing**

12.24.128.1 Relay interlocking and Electronic interlocking installations shall be tested as indicated in Chapter 21

12.24.128.2 Periodical testing and overhauling of mechanical signalling equipment used in signalling installations shall be done as per requirements contained in para 12.25.133&12.26.144

Section 25 - Overhauling of Interlocking Frames, Interlocking Key Boxes and Station Master's Slide Control Frames

12.25.129.0 **Periodicity**

12.25.129.1 All interlocking frames, interlocking key boxes and Station Master's Slide Control frames shall be overhauled once in every three years or earlier if necessary.

12.25.130.0 **Programme of overhauling**

12.25.130.1 The Sr.DSTE/ DSTE shall draw out a programme of overhauling of interlocking frames, interlocking key boxes and Station Master's slide control frames on the Division on a monthly basis spread preferably over the first 9 months of a calendar year. Any overhauling work that may fall into arrears due to unforeseen circumstances shallbe completed during the remaining three months

12.25.130.2 **The programme shall indicate:**

(a) Particulars of Interlocking Frames, Interlocking Key Boxes and Station Master’s slide control frames to be overhauled month by month;

(b) Dates of last overhauling.

12.25.130.3 A Copy of the programme shall be supplied to each JE/SSE (Signal) in charge who has to carry out the overhauling accordingly.

12.25.131.0 **Requirement of materials for overhauling**

12.25.131.1 The Supervisory JE/SSE (Signal) incharge shall check before undertaking the work that the material needed for overhauling of interlocking frames, interlocking key boxes and Station Master's slide control frames is available. If not, he shall requisition the same sufficiently in advance so that it is available for the work in time.

12.25.132.0 **Interlocking Frame**

12.25.132.1 Instructions for working of Traffic during overhauling When a lever frame, Station Master's control frame or interlocked key box or any other interlocking frame is being overhauled, the station shall be worked in accordance with special instructions issued by each Railway to ensure full compliance with G.R.3.38 and 3.39 and the Station Master shall be

responsible for ensuring that all the facing points over which the train will pass, are correctly set, clamped and padlocked and that all trailing points over which the train will pass are correctly set before taking 'OFF' signals. The manner in which Station Master will ensure this shall be clearly laid down. Detailed temporary working instructions shall be prepared for every station by the DSO for each phase of the work in consultation with concerned DSTE and supplied to the station before overhauling is commenced.

12.25.132.2 A notification showing the date and time when the overhauling work would be taken in hand, its probable duration and instructions for the station Master to issue Caution Order to Loco pilots and for Loco Inspector to advise Loco pilots to observe the temporary speed restriction shall be issued jointly by the Sr.DSTE / DSTE and the DSO.

12.25.132.3 **For the portion of the yard, which is controlled by the Cabin to be overhauled:-**

(a) Before undertaking overhauling work wherever required, suitable communication facilities shall be provided.

(b) The Loco pilots shall be issued a Caution order at the last stopping station instructing him/her of the speed restriction of 30 Km/h at the station where overhauling of interlocking frame is in progress.

(c) Special instructions for reception/dispatch of trains and shunting operations, if any, should be drawn keeping in view GR & SR 15.09 (2) and annexed to the notification referred to in 12.25.132.1

(d) Distant/warner signal and main home signal proceed aspect shall be made inoperative. Distant/warner signal shall be kept at 'ON' position and first auto signal in automatic block sections to be kept inoperative for ‘Attention’ and ‘Proceed’ aspects.

(e) Wherever work is required to be done during night hours, adequate lighting arrangements shall be provided.

(f) Section of the track, on which caution order is to be imposed, must be specified.

12.25.132.4 Only after the Station Master on duty has assured himself that the line has been correctly set and facing points locked for the required movement, he may permit the signals to be taken 'OFF' for the reception or dispatch of a train.

12.25.132.5 Disconnection Memo on Form No. S&T/DN (Annexure-3/11) shall be given by the JE/SSE (Signal) in charge to the Station Master on duty and his signature shall be obtained before starting the work and reconnection memo shall be given after the work is completed.

12.25.132.6 The issue of caution orders and the imposition of speed restriction of 30 KMPH shall remain in force until interlocking frame has been tested by the authorised official and normal working is restored.

12.25.133.0 **Detailed Instructions for Overhauling**

12.25.133.1 All interlocking frames shall be overhauled under the personal supervision of the JE/SSE (Signal) In charge.

12.25.133.2 Before taking up the overhauling of a lever frame, the JE/SSE (Signal) in charge shall examine the lever frame and collect the necessary materials to carry out repairs/rectify deficiencies noticed.

12.25.133.3 Quadrants, Quadrant supports, Levers, Lever shoes, Locking Boxes, Locking Box supports and Holding down bolts shall be examined to see that they are rigidly fixed and properly secured. The levers shall be aligned in straight line in case any misalignment exists. It shall be ensured that all the parts of the Lever Frame including the spare levers and their tappets are intact. It shall also be ensured that proper padlocking and sealing facility exists.

12.25.133.4 All notches in tappets, locks and the lock rivets shall be examined to ensure that they have not become unduly worn or slack. False (spare) notches in tappets shall be filled up or the tappets replaced.

12.25.133.5 All interlocking bars shall be examined to ensure that they are rigid, truly straight and not twisted or buckled. Spare holes in interlocking bars shall be filled up.

12.25.133.6 All tappets and connecting link pins shall be examined to see that they are not unduly worn out and do not permit excessive lost motion.

12.25.133.7 All worn out/cracked parts must be replaced wherever necessary.

12.25.133.8 The inside of the locking box as also the bars, the tappets and the locks shall be thoroughly cleaned with a stiff brush and kerosene. Afterwards, they shall be polished with graphite mixed with thin oil and finally dusted with graphite.

12.25.133.9 Each interlocking bar shall be tested before refitting to ensure that the locking provided through it is correct and the lost motion of the locked levers or their catch handles is within the permissible limit, i.e. the lever block cannot be moved more than 12mm over the lever quadrant in the case of locking operated directly by the lever. In case of catch handle operated locking, it shall not be possible to raise the catch handle block more than 10mm. Defects, if any, shall be rectified.

12.25.133.10 Locking shall be refitted carefully. Before replacing the covers, the locking shall be again powdered with graphite. Thin mineral oil shall be used for lubrication when graphite is found unsuitable owing to climatic conditions.

12.25.133.11 It shall be ensured that waste cotton pieces or any other foreign materials are not left inside the locking boxes.

12.25.133.12 Before replacing the covers, the locking shall be carefully checked with the approved locking diagram.

12.25.134.0 **Resumption of normal working**

12.25.134.1 It will be personal responsibility of the Supervisory JE/SSE (Signal) to test all interlocking frames up to 20 working levers after completion of overhauling and to introduce normal working. As regards interlocking frames with 21 to 30 working levers, normal working can be introduced after completion of overhauling and testing of interlocking by the Supervisory JE/SSE (Signal). The DSTE/ADSTE must, however test the locking as soon as possible after normal working has been resumed by the Supervisory JE/SSE (Signal).

12.25.134.2 When the lever frame to be overhauled is more than 20 working levers, as soon as the overhauling work has been completed, it will be the personal responsibility of the JE/SSE (Signal) in charge to intimate the Signal Officer concerned that the interlocking frame is ready for his test.

12.25.134.3 In regard to interlocking frames with more than 30 working levers, it will be the personal responsibility of the Signal Officer concerned to test all such interlocking frames after completion of overhauling and to introduce normal working.

12.25.134.4 The testing referred to in para 12.26.141 shall be carried out as per approved locking table.

12.25.134.5 The Sr.DSTE/DSTE/ADSTE must keep in close touch with the progress of the overhauling work and see that the duration of the work is reduced to what is actually necessary to avoid undue continuance of the restrictions imposed on the traffic working.

12.25.135.0 **Overhauling of Interlocking Key Boxes**

12.25.135.1 Interlocking Key Boxes must be overhauled under the personal supervision of the JE/SSE (Signal) in charge.

12.25.135.2 Disconnection Memo on Form S&T/DN (Annexure 3/11) shall be issued by the JE/SSE (Signal) to the Station Master on duty and his signature obtained before starting and reconnection memo issued after completion of the overhauling work.

12.25.135.3 The responsibility for the safe custody of the interlocking keys released from the interlocking key box as long as the locking is disconnected shall rest with the JE/SSE (Signal) in-charge.

12.25.136.0 **Overhauling of Station Master's Slide Control Frame**

12.25.136.1 Station Master's Slide Control Frames shall be overhauled under the personal supervision of the JE/SSE (Signal) incharge.

12.25.136.2 Instructions for working of Traffic during overhauling as laid down in para 12.25.132.1 shall be applicable during overhauling of Station Master's Slide Control Frames.

12.25.136.3 The Supervisory JE/SSE (Signal) incharge shall issue disconnection memo on Form S&T/DN Annexure-3/11 to the Station Master on duty and

his signature obtained before starting and reconnection memo issued after completion of overhauling work. The following precautions should also be taken by SM on duty to ensure the safety of the train movements.

(a) The Cabin man on duty should be instructed to see that no signals are taken 'OFF' by him/her unless permitted by the Station Master on duty.

(b) It should be ensured that no electric slot or key is transmitted in an unauthorised manner and that no conflicting slot or key is transmitted.

12.25.137.0 **Certificate of Overhauling**

12.25.137.1 Locking overhauling certificates on Form S&T/LO Annexure-12/4 for all interlocking frames, interlocking key boxes and Station Master's slide control frames shall be promptly submitted by the JE/SSE (Signal) to the Sr.DSTE/DSTE , in duplicate. The Sr.DSTE/DSTE shall, after necessary scrutiny forward a consolidated statement of the overhauling carried out every quarter along with one copy of the Certificate to the PCSTE for his record.

Section 26 - Testing of Interlocking Frames, Interlocking Key Boxes and Station Master's Slide Control Frames

12.26.138.0 **Periodicity**

12.26.138.1 Locking of interlocking frames, interlocking key boxes and Station Master's slide control frames shall be tested once in a year or earlier if required.

12.26.138.2 Locking shall also be tested before opening a new installation, after every addition and alteration and immediately after overhauling.

12.26.139.0 **Programme of Testing**

12.26.139.1 The testing of interlocking frames, interlocking key boxes and Station Master's slide control frames shall be done in a programmed basis on the same lines as laid down in Section '25' for overhauling. The testing programme shall be arranged as follows:-

12.26.139.2 First at the time of overhauling, second a year later and third 2 years after overhauling. At the end of the third year, a test should be carried out separately only if overhauling is not proposed to be done within two months of the date on which periodical testing is due.

12.26.140.0 **Essential requirements of Locking to be tested**

12.26.140.1 It is essential that the interlocking of all signals with the points is so effected as to ensure the following conditions :-

(a) It shall not be possible to take 'OFF' a running signal until all points on the running line are correctly set and the facing points locked, all points giving access to the running lines from interlocked siding and goods lines are set against the running lines and interlocked level crossing gates on the routes including overlap are closed and locked across the road way. A signal lever when pulled shall lock or back lock as necessary the levers

operating the points and gate locks referred to above.

(b) It shall not be possible to take 'OFF' conflicting signals at the same time.

(c) When signals are at 'ON', all points, which would be locked by the taking 'OFF' of such signals, shall be free.

(d)It shall not be possible to take 'OFF' a Warner Signal until all the relative Stop Signals in advance have first been taken 'OFF' and when 'OFF' it must back lock such signals.

(e)It shall not be possible to take 'OFF' main home signal to proceed aspect unless all advance signals like advanced starter, intermediate starter and mainline starter are taken 'OFF'.

12.26.140.2 The locking in the frame shall correspond with that shown in the approved Interlocking Table.

12.26.140.3 The Locking shall be strong, durable and accurate and it should not be possible –

(a) for any mechanical lock to enter a notch other than that for which it is intended;

(b) to force the locking.

12.26.140.4 The locking in the lever frame shall act on the commencement of the movement of the lever in the case of a direct type lever frame. In the case of a catch handle type of lever frame, the locking shall act on the commencement of the movement of the catch handle.

12.26.140.5 The release locking shall not be effected before the completion of the movement of lever and releasing the catch handle in the case of catch handle type lever frame.

12.26.140.6 Lever collars and slide collars shall be tested so that it is not possible to operate a lever or slide with a collar placed on it.

12.26.141.0 **Methods of testing**

12.26.141.1 There are two methods of testing the locking :-

(a) against the Interlocking Table, and

(b) against the signalling Plan.

12.26.142.0 **Testing by the JE/SSE (Signal) incharge:-**

12.26.142.1 TheJE/SSE(Signal) incharge shall test locking against the interlocking table and do spot checks from the Signalling Plan.

12.26.143.0 **Testing by the DSTE/ADSTE**

12.26.143.1 DSTE/ADSTE shall test locking against the interlocking table as well as make spot checks from the Signalling Plan.

12.26.144.0 **Instructions for testing**

12.26.144.1 All levers shall be placed in their normal position before testing is started. When testing against the interlocking table, levers shall ordinarily be tested consecutively beginning from lever number-1.

12.26.144.2 In each case where a lever is released by a combination of other levers, it is necessary to have all the levers of the group pulled over to test the back locks, then to put back each lever separately, leaving the other levers pulled over, to test each individual lock.

12.26.144.3 Some examples of testing locking are as follows

(a) Warner Signal

(i) Try the Warner lever. It should be locked.

(ii) Pull over the relative Loop Homes and Outer Levers and also Loop starters if worked from the same cabin. Try the Warner lever again, it should be locked.

(iii) Pull over the relative Main Home and Outer levers and try the Warner lever again. It should be locked.

(iv) Pull over the Main line Starter lever and once more try the Warner lever. It should be free if there is no other running signal in advance of the Starter signal.

(v) After pulling the Warner lever, see if the Outer, the Home and the Starter signal levers are back locked. If Advanced starter exist then should be back locked.

(vi) Replace the Warner, the Outer and the Home signal levers, but leave the starter lever pulled over. Try again to pull the Warner lever. The Warner lever should be locked.

(b) Stop Signal

(i) Set up each route separately, and if the stop signal is released by another signal (e.g., the Outer by the Home), prove that the releasing and back locking is correct as in the case of Warner signal levers.

(ii) See that all conflicting point and signal levers, both in advance and in rear of the signal under test, are locked.

(iii) Replace the signal lever under test and pull over each conflicting point lever or combination of conflicting point levers in the route separately and see if the signal lever is locked.

(iv) Try each conflicting signal with the signal lever under test in the normal position, pulling such levers as required releasing each

conflicting signal.

(v) Points in rear of the signal under test should be tried both in the normal and reverse positions with the signal lever reversed to see if the both ways locking is correct.

(vi) The extent of this both ways locking in rear of each signal varies according to local condition.

(c) Junction Signal Test

(d) In Figure 1, when lever No.2 is tested, it is not sufficient to pull 3 and try if 4 is locked normal because 3 locks 4 both ways. The methods of testing where the facing points stand normally unlocked, should be as follows :-

(i) Try 1 and 2. Both should be locked.

(ii) Pull 3 and try 1, 2 and 4,2 and 4 should be locked and 1 free.

(iii) Pull 1 and see that 3 is back locked.

(iv) Replace all levers and pull 4 and 3,1 should be locked, 4 locked in reverse and 2 free.

(v) Pull 2 and see that 3 is back locked.

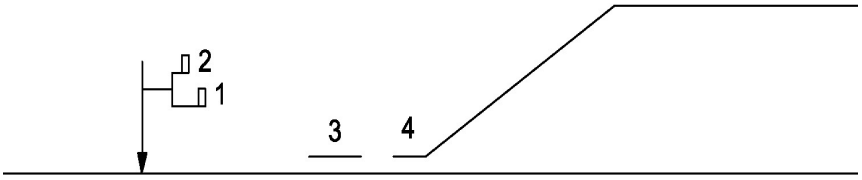


Fig : 1

(e) Point Locking

(i) The locking between point levers is tested on the same principles as the locking between signal levers, care being taken that each point lever is free to be tested by pulling the releasing levers, if any, and also that points are not being held by facing point locks.

(f) Special/Conditional Locking

(g) Figure 2 is an example of special locking, where 1 is released by 3 when 2 is normal. The method should be as follows :-

(i) Try 1,1 should be locked,

(ii) Pull 2 and try 1,1 should be free,

(iii) Pull 1,3 should be locked and 2 backlocked.

(iv) Replace all levers. Pull 3 and try 1,1 should be free,

(v) Pull 1,2 should be locked and 3 backlocked.

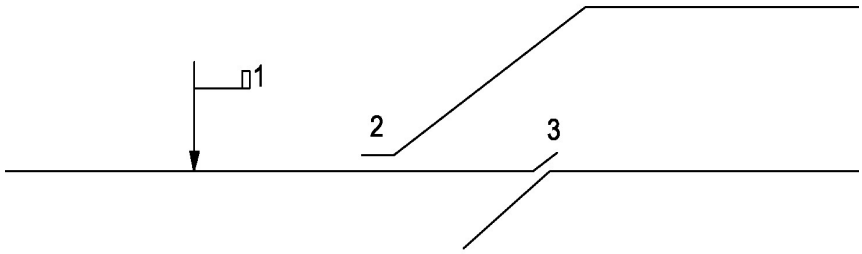


Fig : 2

(vi) In each case, where a lever is released by a special combination, as in Fig.3, where 1 is released by 4 with 2 and 3 normal, each part of the combination must be tested on the same principle as before. Either 2 reverseor 3 reverse must "break-down" the release on 1 by 4.

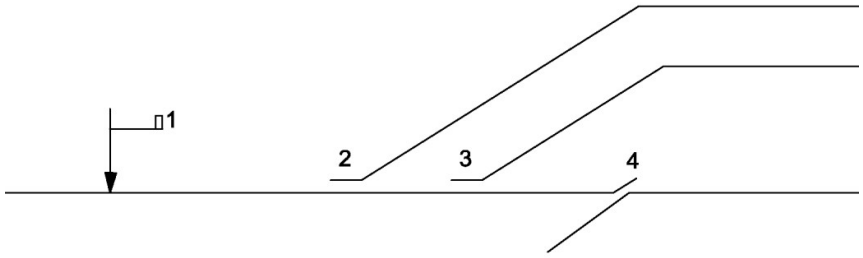


Fig : 3

(h) Reciprocal Locks and Checking

(i) As each lock is tested, its reciprocal should, whenever possible, be tested at the same time. As and when they are dealt with, the figures must be ticked off on the interlocking table so that no lock is missed.

(j) Locked Levers

(k) While testing a locked lever, a steady pressure should be applied so as to ensure that it is not free before the releasing lever has completed its travel. This will prove that there are no conflicting notches.

12.26.145.0 **Test Charts**

12.26.145.1 Test Charts showing the manipulation of various levers to check each locking as detailed in para 12.26.144 should preferably be prepared to

facilitate testing of large interlocking frames.

12.26.146.0 **Lost Motion Test**

12.26.146.1 When a lever or catch handle which should be locked can be moved more than the allowance stipulated in para 12.1.3.9.(c). it should be considered as having too much lost motion, which must be removed.

12.26.147.0 **Certificate of testing locking**

12.26.147.1 Certificate of Testing locking on Form S&T/LT (Annexure-12/5) for all interlocking frames, interlocking Key Boxes and Station Master's slide control frames shall be promptly submitted by the JE/SSE(Signal) to the Sr.DSTE/DSTE in duplicate. The Sr.DSTE/DSTE shall, after necessary scrutiny, forward a consolidated statement along with a copy of the certificate to the PCSTE for his/her record.

Section 27 - Records of Testing and Overhauling

12.27.148.0 **Records of testing and overhauling**

12.27.148.1 Record of testing and overhauling of interlocking frames, interlocking key boxes and Station Master's slide control frames shall be maintained as follows :-

(a)At individual stations : In the Signal incidences and Inspection register.

(b)With the JE/SSE (Signal): In a separate Register, a separate page shall be used for each interlocking frame, interlocking key box and Station Master's slide control frame.

(c) With the Sr.DSTE / DSTE and the PCSTE on cards. One card shall be used for each interlocking frame, interlocking key box and Station Master's slide control frame. These cards shall be suitably filled to indicate -

(i) Testing or overhauling due "current period".

(ii) Testing or overhauling "over -due".

(iii) Testing or overhauling "done during the period".

(iv) Testing or overhauling "done during previous period and not requiring immediate attention".

(d) A site verified copy of approved locking table and locking diagram, which was basis for testing on completion of overhauling shall each be maintained with the JE/SSE (signal) and the Divisional Office.

12.27.149.0 **Date of last overhauling and testing**

12.27.149.1 Date of last overhauling and testing shall be painted on the lever frames,

interlocking key boxes and Station Master's slide control frames.

Section 28 – Testing and Overhauling of Electromechanical Installations

12.28.150.0 **Mechanical Locking**

12.28.150.1 Instructions contained in the preceding paragraphs for the installation, maintenance, overhauling and testing of locking of mechanical interlocking frames shall also apply to electromechanical installations with mechanical locking frames wherever relevant.

12.28.150.2 The mechanical locking of interlocking frames shall be tested for lost motion, as per para 12.1.3.9(c) and lost motion, when in excess of the prescribed limits, shall be removed.

12.28.151.0 **Electrical Locking:**

12.28.151.1 Inelectromechanical installations electric lever locks, key transmitters, circuit controllers, Detectors etc shall be provided as per approved plan.

12.28.151.2 Diagram of electrical circuits for each such gears shall be prepared and shall show all the necessary controls which shall be required to release a particular lever.

12.28.151.3 Where lever locks have been provided in the mechanical lever frames for controlling operation of Electric Point Machine, Electric Signal Machine or for route holding etc the functioning of the same shall be tested.

(a) Points Lever: - In points lever ABDE locks are provided for Track & Indication locking. Lever shall be tested to ensure that the lever lock is energised and the lever is free to be operated in A/E position only when the controlling tracks are clear. The lever shall again be locked in D position when the point is being operated from normal to reverse and lever shall be free only when the points has operated fully and locked and detected in reverse position. Similarly, while operating the lever from reverse to normal the lever shall be first locked in E position to ensure track locking and subsequently will be held at B position for point operation. The lever lock shall not energise at B position unless the points have operated, set, locked and detected in normal position.

(b) Signal Lever: - While pulling the signal lever from normal to reverse the lever shall be locked in A position and will be free only when route for the reception of the train has been correctly set, all facing points locked and all the controlling tracks are clear i.e. all conditions are fulfilled for taking 'OFF' the signal. While putting back the lever from reverse to normal position the lever shall be locked at B position and will be free only when the intended movement is completed and it is free from back locking as well as approach locking where provided. Normal indication locking must also be effective in B position in case of semaphore signal.

12.28.152.0 **Testing and overhauling of Electrical Locking**

12.28.152.1 Mechanical interlocking frames provided with electrical signalling equipment shall be tested and overhauled as per requirements contained in paras.12.25.133 & 12.26.144

12.28.152.2 The testing of electrical locking shall be carried out in accordance with the approved release circuit of each lever as shown on the approved diagram of electrical circuits.

12.28.152.3 For testing the electrical locking provided on a lever, the other levers required to release that lever shall be pulled and it must be ensured that the lever is free to be pulled mechanically.

12.28.152.4 All the conditions for the release of the electrical lock of the lever under test shall be set up. The lever shall now be pulled from normal to reverse for testing of A&D locks and reverse to normal for E&B locks and proper functioning of the lever locks checked.

12.28.152.5 A test lamp/voltmeter shall now be connected on the electrical lock of the lever under test. The lever shall now be pulled from normal to reverse and vice versa. The test lamp should lit up at lock positions. The lever shall be held at these positions one after the other and all the lock release conditions shall be broken one by one. It shall be ensured that feed to the lever lock is disconnected each time. While doing this test, care shall be taken to test each circuit individually one after the other duly isolating other parallel circuits.

12.28.153.0 **Testing of other electrical equipment**

12.28.153.1 Instructions contained in Chapter 19 for testing of electrical signalling installations shall also apply to electromechanical installations wherever relevant.

12.28.153.2 All electrical equipments such as electric lever locks, circuit controllers etc., as well as wiring shall be examined and tested for insulation. The insulation must not be less than 10 Mega Ohm. Defective equipments shall be replaced.

12.28.153.3 Moving parts of circuit controllers, their mechanical connections and contact makers shall be checked. Repairs and replacements should be carried out wherever necessary.

12.28.154.0 **Date of last overhauling and testing:**

12.28.154.1 Date of last overhauling and testing should be painted on the interlocking frame.

12.28.155.0 **Certificate of overhauling:**

12.28.155.1 Locking overhauling certificate on Form S&T/LO Annexure -12/4 (para 12.25.137.0) for all mechanical interlocking frames shall be promptly submitted by the JE/SSE (Signal) to the Sr.DSTE/DSTE, who shall after necessary scrutiny forward a consolidated certificate to the PCSTE.

☞ *Note:- This Chapter has undermentioned Annexures for further study*		
S.no	*Annexure no*	*Description*
1	*12/1*	*Essential requirements before interlocking of a Point*
2	*12/2*	*Mechanical Signalling Gears - Colouring scheme*
3	*12/3*	*Maintenance Schedule for Mechanical Signalling*
4	*12/4*	*Certificate of Overhauling interlocking Frames*
5	*12/5*	*Certificate of Testing locking*

ESSENTIAL REQUIREMENTS BEFORE INTERLOCKING OF A POINT

Before the interlocking work at points is undertaken, it must be ensured that the JE/SSE(P-Way) has :-

i) brought the track to correct level and alignment.

ii) eased off rail joints on either side of points to be interlocked and closed the stock rail joints associated with lock bars.

iii) fully ballasted and packed all points which are to be interlocked and taken adequate measures to prevent lateral and longitudinal movement of points.

iv) provided creep and level pillars.

v) arranged the sleepers on adjacent tracks in alignment, where rods and wires have to cross.

vi) seen that the gauge is correct.

vii) provided and fixed special timbers as required.

viii) provided means to prevent creep in the vicinity of points.

ix) fitted gauge tie plates correctly.

x) made the stretchers of such a length so that the throw of switches is as per approved drawings.

xi) adjusted loose heel switches so that:-

a) they can be thrown both ways with ease and can be housed against the stock rail by hand and remain there when the pressure is removed.

b) the planed surface of the switch rail fully houses against the stock rail as per approved drawings.

xii) adjusted fixed heel switches so that:-

a) they normally lie in the mid-position and flex equally in the normal and reverse positions.

b) the planed surface of the switch rails fully houses against the stock rails as per approved drawings.

xiii) fitted flexible stretchers so that they flex equally in the normal and reverse positions,

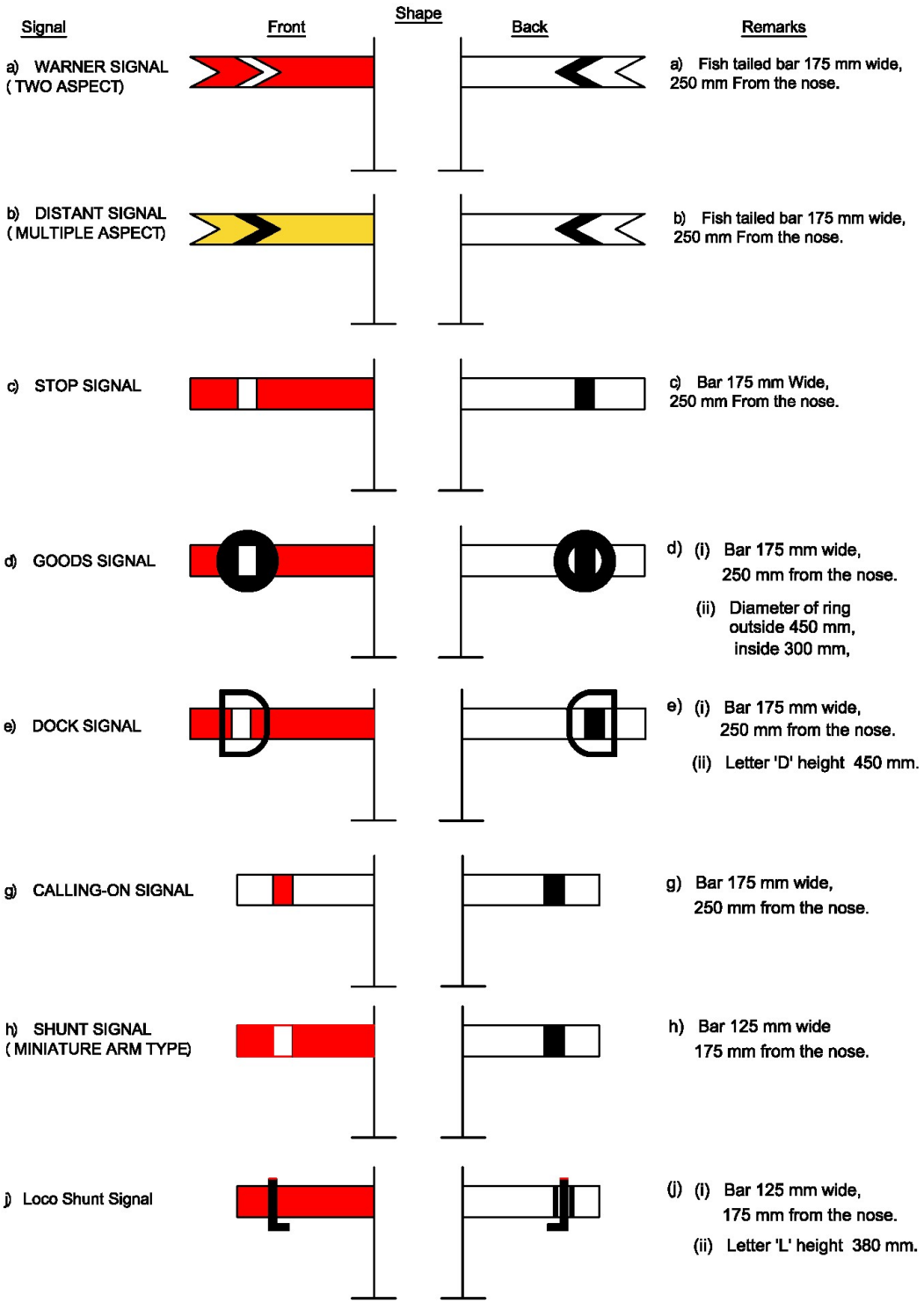
xiv) provided a stop for the open position of a single switch layout.

MECHANICAL SIGNALLING GEARS

Annexure : 12/2

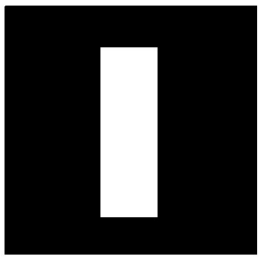
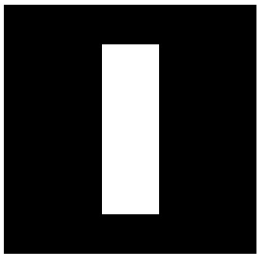
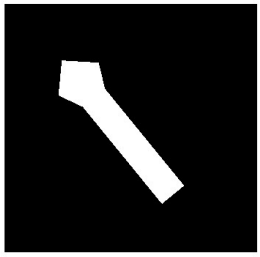
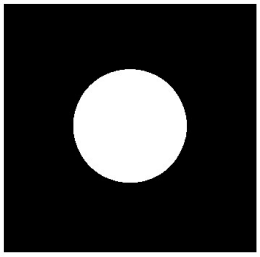
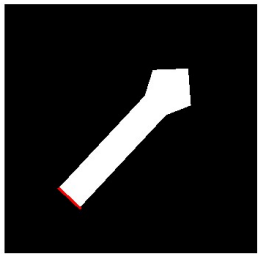
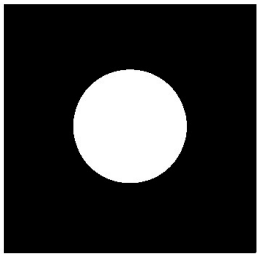
SEMAPHORE SIGNAL ARMS, POINT INDICATORS, BOARDS & MARKERS

Sheet No : 1



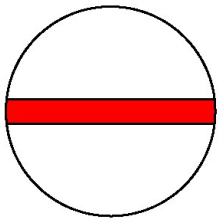
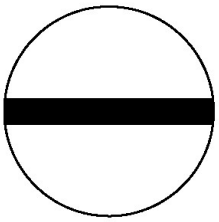
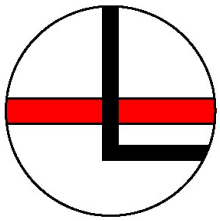
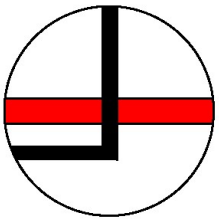
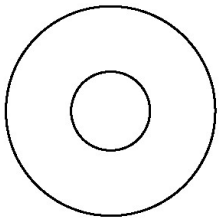
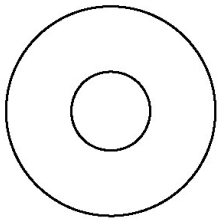
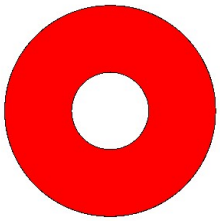
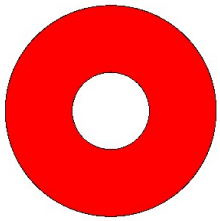
Annexure : 12/2

Sheet No : 2

Signal	Shape		Remarks
	Point Facing	Point Trailling	
			p) (i) Points set for straight.
(P) Point Indicator (Directional Type)			(ii) Points set for left hand turnout.
			(iii) Points set for right hand turnout.

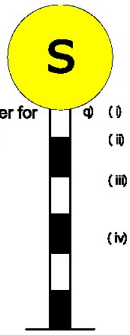
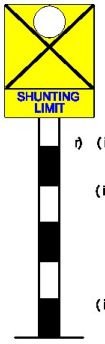
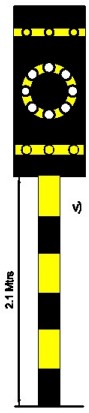
Annexure : 12/2

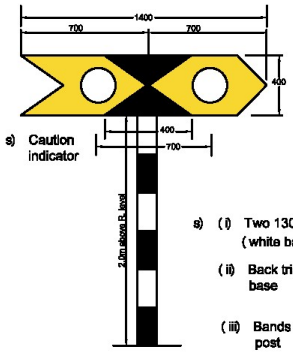
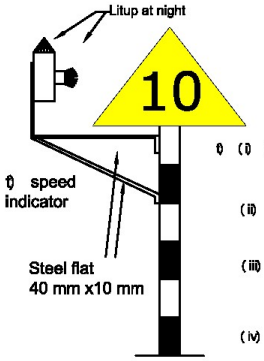
Sheet No : 3

Signal	Shape		Remarks
	Front	Back	
k) SHUNT SIGNAL (DISC TYPE)			k) BAR 150mm WIDE.
l) Loco Signal (Disc type)			l) (i) Bar 150 mm wide (ii) Letter 'L' height 380 mm.
	POINT FACING	POINT TRALLING	
m) POINT INDICATOR (TAGET TYPE)			m) POINT SET FOR STRAIGHT.
n) POINT INDICATOR (TAGET TYPE)			n) TRAP OPEN OR DERAIL ON THE LINE.

Annexure : 12/2

Sheet No : 4

Signal	Shape	Remarks
q) `S' Marker for outlying sidings		q) (i) Disc diameter : 1 metre. (ii) Letter 'S' height : 300 mm. : 40 mm thick (iii) Height of centre of disc from rail level : 2000 mm (iv) Bands on the post : 300 mm wide, Black & White bands alternately
r) shunting limit board		r) (i) Rectangular board : 600 mm x 1 metre (ii) Height from rail level to the underside of the board containinng the cross : 2000 mm (iii) Bands on the post : 300 mm wide Black & White bands alternately
v) Goods warning board		v) (i) Rectangular board : 53 x 1800 mm (ii) Yellow band & circle as shown : 140 mm (iii) Plastic reflector as shown (iv) Bands on the post : 300 mm wide, Black & Yellow bands alternately.

Signal	Shape	Remarks
s) Caution indicator		s) (i) Two 130mm. Yellow lights (white back lights) (ii) Back triangle base : 400 mm (iii) Bands on the post : 300 mm wide, Black & White bands alternately (Ref.-Annexure 8/4-para 807 & 808 of P-Way Manual)
t) speed indicator		t) (i) Equilateral traingular board-side : 1 metre (ii) letters indicating speed-hight : 300 mm, 40 mm thick (iii) Height from rail level to bottom of the board : 2000 mm (iv) Bands on the post : 300 mm wide Black & White bands alternately (Ref.-Annexure 8/4-para 807 & 808 of P-Way Manual)

COLOURING SCHEME		
S.No.	**EQUIPMENT**	**COLOUR**
1	Signal arms, point indicators, boards & markers	As per shape &colour as shown in figures.
2	**Signal post & signal fitting**	
a)	**Signal Post**	
	i) Up to a height of 2 meters above rail level	Black
	ii) Reminder of the post	White
b)	Signal spectacle & other fittings on the post	Black
c)	Shunt Signal counter weight, side facing cabin man	White
3	**Interlocking frames, their levers and fittings** :	
	Levers above quadrants shall be painted with enamels in colours as under :-	
	i) Warner Lever	Green
	ii) (a) Distant Signal lever 45 degree aspect	Yellow
	(b) -do- 90 degree aspect	Green
	(iii) Other Signal levers	Red
	iv) (a) Slot lever mechanical	Same colour as of the lever slotted with a 150 mm wide blue band in the middle.
	(b) Slot lever electrical	Same colour as of the lever slotted with a 150 mm wide yellow band in the middle.
	(v) Point lever	Black

S. No.	EQUIPMENT	COLOUR
	(vi) Facing point lock lever	Blue
	(vii) Economical Facing point lock lever	Upper Half-Black
		Lower Half-Blue
	viii) Station Master's Control lever	Upper Half-White
		Lower Half-Black
	(ix) Level crossing Gate control lever	Chocolate
	(x) Release Lock Lever	Black with a 150mm wide Blue band in the middle.
	(xi) Detector (Double Wire)	Red and Blue bands 150mm wide alternately.
	(xii) Route lever	Upper half-Red Lower half – Black
	(xiii) Siding key control lever	Black
	(xiv) King lever	Red & White bands 150 mm wide alternately.
	(xv) Spare lever	White
(b)	**Interlocking frame parts and fittings except levers above quadrants.**	.
	(i) Interlocking frame supports, Quadrants, Levers below Quadrants, locking troughs, Drop Block, Catch Handle connections, Indicator plates.	Black
	(ii) Down rods between lever tail and pedestal crank	Black
4	**Ground and Miscellaneous-** Cranks, compensators, wheels, stakes facing point locks, lock bars detectors, cabin wire adjusters, interlocking key boxes and foot. Rests.	Black

Maintenance Schedule for Mechanical Signalling

Annexure 12/3

S.No.	Gears	Maintenance work to be done	Periodicity		
			Technician/ Signal	JE/SSE Signal	SSE Signal In charge
1	**2**	**3**	**4**	**5**	**6**
1	**CABIN : General**				
1.1		Cleaning the ground floor & locking.	F	M	Q
1.2		Checking Cabin diagram, pull chart, station working instructions, maintainer's programme up to date and warning board (DW Cabin)		M	Q
1.3		Test locking against interlocking table and yard diagram and broken wire test (if D.W)		Random Check	Y
1.4		Check the sealing and locking of interlocking tray, resetting handle box, etc.	M	M	Q
1.5		Check the lever frame is due for overhauling.		M	Q
1.6		Check the installation as per current plan.		M	Q
1.7		Check the inter changeability of keys of padlocks where available.	F	M	Q
1.8		Check from the operator any defect or discrepancy of the gears.	F	M	Q

F = Fortnightly; M =Monthly; Q = Quarterly; Y = Yearly:

S.No.	Gears	Maintenance work to be done	Periodicity		
			Technician/ Signal	JE/SSE Signal	SSE Signal In charge
1	**2**	**3**	**4**	**5**	**6**
1.10		Check the signals are visible from place of operation.	F	M	Q
2	**LEVER FRAME**				
2.1		Check bolts and nuts for tightness and opening of split pins including the safety bolts of D.W.compensator.	F	M	Q
2.2		Lubricate all accessible parts.	F	M	Q
2.3		Operate lever and check that full stroke is transmitted.	M	Q	HY
2.4		Check the painting of levers.		Q	HY
2.5		Check the tipping of D.W. Clutch levers.		M	Q
2.6		All connections including down rods must be examined to detect cracks, slipped threads and spacing.	F	M	Q
2.7		Check the pulley block in D.W. cabin is in order.	F	M	Q
2.8		Check the double wire compensator that when the transmission is at rest, both the grips of locking pawl are disengaged with the teeth of Ratched Rod. The weights must be floating freely and in level with each other during lever operation check that engaged ratched rod teeth.	M	Q	HY
2.9		Check adjust weight lever in such a way that during the hottest period of the day, the lever must not reach the breakage mark on the ratchet rod.	M	Q	HY

F = Fortnightly; M =Monthly; Q = Quarterly; HY = Half Yearly ; Y = Yearly:

S.No.	Gears	Maintenance work to be done	Periodicity		
			Technician/ Signal	JE/SSE Signal	SSE Signal In charge
1	2	3	4	5	6
2.10		Check the lubrication of compensator, Wheel, grooves and wire rope with graphite grease Gr.I to IS 508. Check the lubrication of moving parts of axle oil. Check the lubrication when nipple are provided with grease IS-507.	F	M	Q
2.11		Check the drum lock fitted on the coupled clutch/direct lever for effectiveness.	F	M	Q
3.	Leadout and cranks	Examine clean oils and observe cranks and wheels for lost motion or worn out parts and check the foundations are rigid. All worn out pins are replaced.	F	M	Q
4.1		Check the rodding run in true alignment. Spacing of roller trestles free from vegetation growth.	F	M	Q
4.2		Check the roller are moving freely and replace the missing or broken parts and check whether alignment requires earthwork.	M	Q	HY
4.3		Check joints are good and make solid connection.	F	M	Q
4.4		Check the rodding run under the track is clear rails and obstructions.	M	Q	HY
4.5		Check the coupling adjusting screws and joints looseness stripped threads are slack nuts and lubricate threads.	F	M	Q
4.6		Check the rodding run for any weak place due to rust, corrosion etc.	M	Q	HY

F = Fortnightly; M =Monthly; Q = Quarterly; HY = Half Yearly ; Y = Yearly:

S.No.	Gears	Maintenance work to be done	Periodicity		
			Technician/ Signal	JE/SSE Signal	SSE Signal In charge
1	2	3	4	5	6
4.7		Check the position of compensator when a point or lock doesn't work properly.	As & when required	-	-
4.8		Check whether the roller trestles requires realignment due to shifting or sinking.	Q	HY	Y
4.9		Check the adjusting sleeve of the adjusting crank and check the loss for track.	Q	HY	Y
5	Wire transmission (Single wire)				
5.1		Check the alignment level, Corrosion and for kinks.	M	Q	HY
5.2		Check the wear pulley damage or broken pulleys to be replaced.	M	Q	HY
5.3		Check the wire run underneath the track for rubbing against blast bottom rails or sides or sleeper and also for corrosion and kinks in the troughing underneath level crossing.	M	Q	HY
5.4		Check the wire joints, split links and adjacent joints do not get entangled.	M	Q	HY
5.5		Check for the spacing of stakes (not more than 10m is allowed).	M	Q	HY
5.6		Check wheels are rigidly fixed on foundation and replace worn out wheels.	M	Q	HY

F = Fortnightly; M =Monthly; Q = Quarterly; HY = Half Yearly ; Y = Yearly:

S.No.	Gears	Maintenance work to be done	Periodicity		
			Technician/ Signal	JE/SSE Signal	SSE Signal In charge
1	2	3	4	5	6
6	**DOUBLE WIRE**				
6.1		Check all transmission work freely and loss of stroke at the end of the transmission must be avoided.	F	M	Q
6.2		Check all pulleys are parallel to the wire run. Check that there is no binding in any part of transmission.	F	M	Q
6.3		Check the spacing of the stakes is not more than 15 mt. for points and detector & 20 mt. for signals.	M	Q	HY
6.4		Check disconnecting pins are available at the end of transmission for conducting broken wire test.	M	Q	HY
6.5		Check pull wire and return wire are properly identified by painting.	M	Q	HY
6.6		Check the wheel guides are intact and avoid wire slipping from turnout wheels.	M	Q	HY
7.1		Cleaning and lubricating of moving parts. Also check point chairs are cleaned regularly by Nominated departments	F	M	Q
7.2		Check the gauge of the points, opening of the switches and squaring and packing condition of sleepers under gauge tie plate and slide chair fixing bolts through P.Way staff	F	M	Q
7.3		Make visual check regarding the condition of switches, sleepers,gauge tie plate.	F	M	Q (jointly with JE/SSE/(P.Way)

F = Fortnightly; M =Monthly; Q = Quarterly; HY = Half Yearly ; Y = Yearly:

S.No.	Gears	Maintenance work to be done	Periodicity		
			Technician/ Signal	JE/SSE Signal	SSE Signal In charge
1	2	3	4	5	6
7.4		Check the switches are housed properly against stock rail and check spring on the switches equally in the normal & reverse positions.	F	M	Q
7.5		Check the points for Obstruction test	F	M	Q
7.6		Check the ends of plunger of the lock and notches are square.	F	M	Q
7.7		Check the lock plunger is 12 mm clear from the strecher blades when unlocked. In case of EFP locks,check locking Dog protrudes 3 mm (at least)after locking both the strecher.	F	M	Q
7.8		Check for tightness of Bolts & Nuts and arrange for tightening / replacing missing bolts and nuts of Flexible strechers.	F	M	Q
7.9		Check lock bars are straight and examine the driving pieces for looseness.	F	M	Q
7.10		Check the lock bar clips and stops for tightness and lubricate the bearing of clips.	F	M	Q
7.11		Badly wornout clips to be replaced and check if creep is affecting the working of lock bar.	F	M	Q
7.12		Check that lock bars lie 38mm below the top of the rail.	F	M	Q
7.13		Check the broken wire locks & test for easy movement & also check the point mechanism butting against its stop in normal & reverse.	F	M	Q
7.14		Check that no two similar wards exist for conflicting train movements for hand plunger lock, lever lock.	-	M	Q

S.No.	Gears	Maintenance work to be done	Periodicity		
			Technician/ Signal	JE/SSE Signal	SSE Signal In charge
1	2	3	4	5	6
8	**DETECTORS**				
8.1		Check the signal wires to the detectors are in alignment & check when the signal is returned to 'ON' all signal slides are travelling back to their stops.	F	M	Q
8.2		Check for obstruction test.	F	M	Q
8.3		Lubricate the detector slides thoroughly – Check the cross slides for any undue play. Also check the detectors are fixed rigidly.	F	M	Q

9	**SIGNAL**				
9.1		Check the condition of the post, fittings, level of arms and the posts are properly plumbed and lubricate working parts.	F	M	Q
9.2		Inspect the platforms where provided for decayed or loose boards.	F	M	Q
9.3		Cleaning of the roundels, glasses and lenses.	Applicable for Traffic Staff		
9.4		Check the founts of signal lamps are in good repair. Check for damaged flame guard, leaky founts and broken burners and non-standard wick.	F	M	Q
9.5		Check the focussing of Signals.	M	M	Q
9.6		Testing of Signals.	M	M	Q
9.7		Where double wire mechanism is used, check lubrication of cam path.	F	M	Q
9.8		Check all signal lamps are overhauled.	Y	Y	Y
9.9		Cleaning of Signal arms. Painting to be got done if required.	Q	HY	Y
9.10		Check adjustment of back light.	M	Q	HY

Annexure 12/4

.....RAILWAY

SIGNAL AND TELECOMMUNICATION DEPARTMENT

CERTIFICATE OF OVERHAULING INTERLOCKING FRAMES

No.....

To

The Divisional Signal and Telecommunication Engineer,
.....Division

This is to certify that I have personally supervised the overhauling of the interlocking frames/ Interlocking key boxes/Station Master's slide control frames at the station/cabins on my section shown below, and that they have been thoroughly examined and cleaned, also repaired or refitted where necessary and all parts are in good working order. The mechanical locking is in accordance with the approved interlocking charts and interlocking tables.

Station /Cabin name	Apparatus overhauled	Date of last overhaul	Date overhauled	Reference to plans		Remarks
.	.	.	.	Locking table, locking diagram	Signalling plan	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.

Dated:

Signature.... .....

Designation.....

Station.....

Annexure 12/5

Form No. S&T/LT

.....RAILWAY

SIGNAL AND TELECOMMUNICATION DEPARTMENT

CERTIFICATE OF TESTING LOCKING

for the month ending.....20..

No.....

To,

The Divisional Signal & Telecommunication Engineer,

.....Division

This is to certify that I have personally tested the interlocking frames, interlocking key boxes and Station Master's slide control frames at the stations on my section shown below and that they are in good working order and that the mechanical locking/electrical locking is in accordance with the approved interlocking tables.

Name of Station /Cabin	Apparatus tested	Date of last test	Date tested	Tested against		Remarks
				Signalling plan	Locking table/ selection table	
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.

Dated:

Signature.... .....

Designation.....

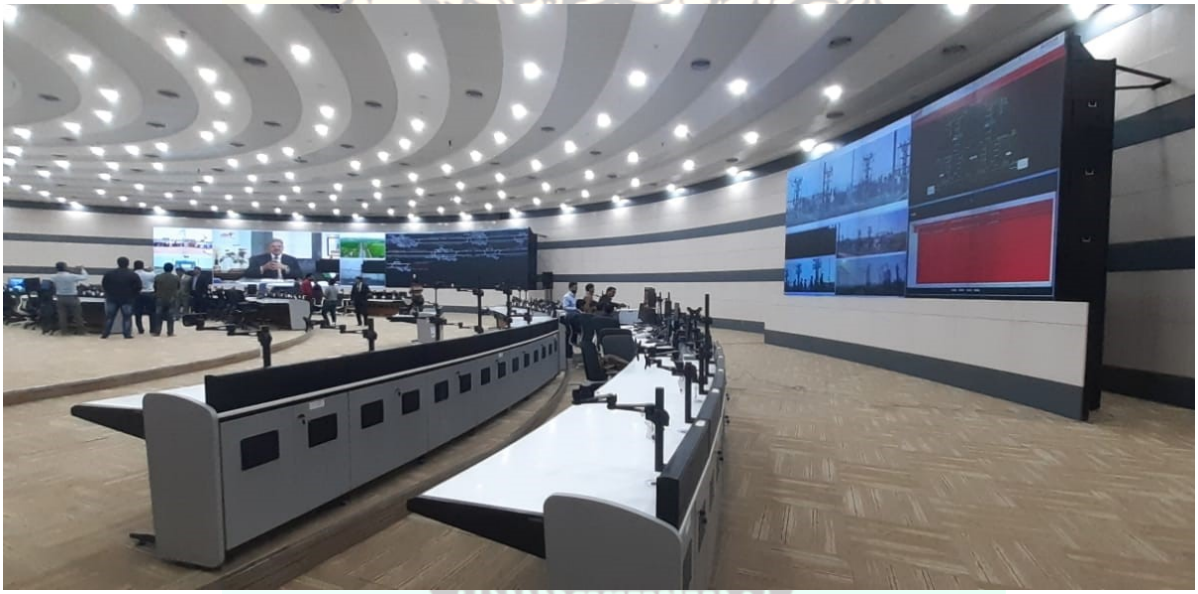
Station.....

LIST OF ADDENDUM AND CORRIGENDUM

A & C No.	Date of A & C	Subject Matter	Chapter	Reference to Para and Page in which correction or Addition has been made	
				Para No.	Page No.

BIBILOGRAPHY

Intentionally left blank now



GALUDH
GALUDH